



AOS-CX 10.18.xxxx Fundamentals Guide (5420, 6200 Switch Series)

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About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

Latest version available online

Command syntax notation conventions

About the examples

Identifying switch ports and interfaces

Applicable products

This document applies to the following products:

- HPE Aruba Networking 5420 Switch Series (S0U67A, S0U55A, S0U63A, S0U64A, S0U65A, S0U75A, S0U72A, S0U78A, S0U58A, S0U73A, S0U74A, S0U71A, S0U76A, S0U70A, S0U77A, S0U60A, S0U61A, S0U62A, S0U66A, S0U68A)
- HPE Aruba Networking 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A, R8Q67A, R8Q68A, R8Q69A, R8Q70A, R8Q71A, R8V08A, R8V09A, R8V10A, R8V11A, R8V12A, R8Q72A, JL724B, JL725B, JL726B, JL727B, JL728B, S0M81A, S0M82A, S0M83A, S0M84A, S0M85A, S0M86A, S0M87A, S0M88A, S0M89A, S0M90A, S0G13A, S0G14A, S0G15A, S0G16A, S0G17A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention

Usage

example - text	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example - text	In code and screen examples, indicates text entered by a user.
Any of the following: • <example - text> • <example - text> • example - text • example - text	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: • In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch (CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch (config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch (config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch (config-vlan-<VLAN-ID>) #
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port**.

On the HPE Aruba Networking 6200 Switch Series

- member: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.

- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

On the HPE Aruba Networking 6400 and 5420 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/1 and 1/2.
 - Line modules are on the front of the switch starting in slot 1/3.
- port: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

About AOS-CX

AOS-CX is a new, modern, fully programmable operating system built using a database-centric design that ensures higher availability and dynamic software process changes for reduced downtime. In addition to robust hardware reliability, the AOS-CX operating system includes additional software elements not available with traditional systems, including:

- Automated visibility to help IT organizations scale: The HPE Aruba Networking Network Analytics Engine allows IT to monitor and troubleshoot network, system, application, and security-related issues easily through simple scripts. This engine comes with a built-in time series database that enables customers and developers to create software modules that allow historical troubleshooting, as well as analysis of historical trends to predict and avoid future problems due to scale, security, and performance bottlenecks.
- Programmability simplified: A switch that is running the AOS-CX operating system is fully programmable with a built-in Python interpreter as well as REST-based APIs, allowing easy integration with other devices both on premise and in the cloud. This programmability accelerates IT organization understanding of and response to network issues. The database holds all aspects of the configuration, statistics, and status information in a highly structured and fully defined form.
- Faster resolution with network insights: With legacy switches, IT organizations must troubleshoot problems after the fact, using traditional tools like CLI and SNMP, augmented by separate, expensive monitoring, analytics, and troubleshooting solutions. These capabilities are built in to the AOS-CX operating system and are extensible.

- High availability: For switches that support active and standby management modules, the AOS-CX database can synchronize data between active and standby modules and maintain current configuration and state information during a failover to the standby management module.
- Ease of roll-back to previous configurations: The built-in database acts as a network record, enabling support for multiple configuration checkpoints and the ability to roll back to a previous configuration checkpoint.

Subtopics

AOS-CX system databases

Network Analytics Engine introduction

AOS-CX CLI

Aruba CX mobile app

HPE Aruba Networking NetEdit

Ansible modules

AOS-CX Web UI

AOS-CX REST API

In-band and out-of-band management

SNMP-based management support

User accounts

AOS-CX system databases

The AOS-CX operating system is a modular, database-centric operating system. Every aspect of the switch configuration and state information is modeled in the AOS-CX switch configuration and state database, including the following:

- Configuration information
- Status of all features
- Statistics

The AOS-CX operating system also includes a time series database, which acts as a built-in network record. The time series database makes the data seamlessly available to HPE Aruba Networking Network Analytics Engine agents that use rules that evaluate network conditions over time. Time-series data about the resources monitored by agents are automatically collected and presented in graphs in the switch Web UI.

Network Analytics Engine introduction

The HPE Aruba Networking Network Analytics Engine is a first-of-its-kind built-in framework for network assurance and remediation. Combining the full automation and deep visibility capabilities of the AOS-CX

operating system, this unique framework enables monitoring, collecting network data, evaluating conditions, and taking corrective actions through simple scripting agents.

This engine is integrated with the AOS-CX system configuration and time series databases, enabling you to examine historical trends and predict future problems due to scale, security, and performance bottlenecks. With that information, you can create software modules that automatically detect such issues and take appropriate actions.

With the faster network insights and automation provided by the HPE Aruba Networking Network Analytics Engine, you can reduce the time spent on manual tasks and address current and future demands driven by Mobility and IoT.

AOS-CX CLI

The AOS-CX CLI is an industry standard text-based command-line interface with hierarchical structure designed to reduce training time and increase productivity in multivendor installations.

The CLI gives you access to the full set of commands for the switch while providing the same password protection that is used in the Web UI. You can use the CLI to configure, manage, and monitor devices running the AOS-CX operating system.

Aruba CX mobile app

The Aruba CX mobile app enables you to use a mobile device to configure or access a supported AOS-CX switch. You can connect to the switch through Bluetooth or Wi-Fi.

You can use this application to do the following:

- Connect to the switch for the first time and configure basic operational settings—all without requiring you to connect a terminal emulator to the console port.
- View and change the configuration of individual switch features or settings.
- Manage the running configuration and startup configuration of the switch, including the following:
 - Transferring files between the switch and your mobile device
 - Sharing configuration files from your mobile device
 - Copying the running configuration to the startup configuration
- Access the switch CLI.

Refer to the Aruba CX mobile app [**datasheet**](#) for more information.

HPE Aruba Networking NetEdit

HPE Aruba Networking NetEdit enables the automation of multidevice configuration change workflows without the overhead of programming.

The key capabilities of NetEdit include the following:

- Intelligent configuration with validation for consistency and compliance
- Time savings by simultaneously viewing and editing multiple configurations
- Customized validation tests for corporate compliance and network design
- Automated large-scale configuration deployment without programming
- Ability to track changes to hardware, software, and configurations (whether made through NetEdit or directly on the switch) with automated versioning

For more information about HPE Aruba Networking NetEdit, search for NetEdit at the following website:

www.hpe.com/support/hpesc

Ansible modules

Ansible is an open-source IT automation platform.

HPE Aruba Networking publishes a set of Ansible configuration management modules designed for switches running AOS-CX software. The modules are available from the following places:

- The **arubanetworks.aoscx_role** role in the Ansible Galaxy at: **https://galaxy.ansible.com/arubanetworks/aoscx_role**
- The **aoscx-ansible-role** at the following GitHub repository: **<https://github.com/aruba/aoscx-ansible-role>**

AOS-CX Web UI

The Web UI gives you quick and easy visibility into what is happening on your switch, providing faster problem detection, diagnosis, and resolution. The Web UI provides dashboards and views to monitor the status of the switch, including easy to read indicators for: power supply, temperature, fans, CPU use, memory use, log entries, system information, firmware, interfaces, VLANs, and LAGs. In addition, you use the Web UI to access the Network Analytics Engine, run certain diagnostics, and modify some aspects of the switch configuration.

AOS-CX REST API

Switches running the AOS-CX software are fully programmable with a REST (REpresentational State Transfer) API, allowing easy integration with other devices both on premises and in the cloud. This programmability—combined with the HPE Aruba Networking Network Analytics Engine—accelerates network administrator understanding of and response to network issues.

The AOS-CX REST API enables programmatic access to the AOS-CX configuration and state database at the heart of the switch. By using a structured model, changes to the content and formatting of the CLI output do not affect the programs you write. And because the configuration is stored in a structured database instead of a text file, rolling back changes is easier than ever, thus dramatically reducing a risk of downtime and performance issues.

The AOS-CX REST API is a web service that performs operations on switch resources using HTTPS **POST**, **GET**, **PUT**, **DELETE**, and **PATCH** methods.

A switch resource is indicated by its Uniform Resource Identifier (URI). A URI can be made up of several components, including the host name or IP address, port number, the path, and an optional query string. The AOS-CX operating system includes the AOS-CX REST API Reference, which is a web interface based on the Swagger UI. The AOS-CX REST API Reference provides the reference documentation for the REST API, including resources URIs, models, methods, and errors. The AOS-CX REST API Reference shows most of the supported read and write methods for all switch resources.

In-band and out-of-band management

Management communications with a managed switch can be either of the following:

In band

In-band management communications occur through ports on the line modules of the switch, using common communications protocols such as SSH and SNMP.

When you use an in-band management connection, management traffic from that connection uses the same network infrastructure as user data. User data uses the `data plane`, which is responsible for moving data from source to destination. Management traffic that uses the data plane is more likely to be affected by traffic congestion and other issues affecting the user network.

Out of band

OOBM (out-of-band management) communications occur through a dedicated serial or USB console port or through a dedicated networked management port.

OOBM operates on a `management plane` that is separate from the `data plane` used by data traffic on the switch and by in-band management traffic. That separation means that OOBM can continue to function even during periods of traffic congestion, equipment malfunction, or attacks on the network. In addition, it can provide improved switch security: a properly configured switch can limit management access to the management port only, preventing malicious attempts to gain access through the data ports.

Networked OOBM typically occurs on a management network that connects multiple switches. It has the added advantage that it can be done from a central location and does not require an individual physical cable from the management station to the console port of each switch.

SNMP-based management support

The AOS-CX operating system provides SNMP read access to the switch. SNMP support includes support of industry-standard MIB (Management Information Base) plus private extensions, including SNMP events, alarms, history, statistics groups, and a private alarm extension group. SNMP access is disabled by default.

User accounts

To view or change configuration settings on the switch, users must log in with a valid account. Authentication of user accounts can be performed locally on the switch, or by using the services of an external TACACS+ or RADIUS server.

Two types of user accounts are supported:

- Operators: Operators can view configuration settings, but cannot change them. No operator accounts are created by default.
- Administrators: Administrators can view and change configuration settings. A default locally stored administrator account is created with username set to **admin** and no password. You set the administrator account password as part of the initial configuration procedure for the switch.

Initial Configuration

Perform the initial configuration of a factory default switch using one of the following methods:

- Load a switch configuration using zero-touch provisioning (ZTP). When ZTP is used, the configuration is loaded from a server automatically when the switch booted from the factory default configuration.
- Connect to the switch wirelessly with a mobile device through Bluetooth, and use the Aruba CX Mobile App to deploy an initial configuration from a provided template. The template you choose during the deployment process determines how the management interface is configured. Optionally, as the final deployment step, you can select to import the switch into NetEdit through a WiFi connection to the NetEdit server.

Alternatively, you can use the Aruba CX Mobile App to manually configure switch settings and features for a subset of the features you can configure using the CLI. You can also access the CLI through the mobile application.

- Connect the management port on the switch to your network, and then use SSH client software to reach the switch from a computer connected to the same network. This requires that a DHCP server is installed on the network. Configure switch settings and features by executing CLI commands.
- Connect a computer running terminal emulation software to the console port on the switch. Configure switch settings and features by executing CLI commands.

**NOTE**

Reserved ports or ports used by other applications/services within the system are not recommended to be used for other services. When two services use the same port there is a chance of unexpected behaviors from these services. Best practice is to use a unique port for each service across the system.

Subtopics

Initial configuration using ZTP

Initial configuration using the Aruba CX mobile app

Initial configuration using the CLI

Configuring banners

Using the Web UI

Configuring the management interface

Restoring the switch to factory default settings

Limitations and considerations

REST API profile switching procedure

Management interface commands

NTP commands

Initial configuration using ZTP

Zero Touch Provisioning (ZTP) configures a switch automatically from a remote server.

Prerequisites

- The switch must be in the factory default configuration.
Do not change the configuration of the switch from its factory default configuration in any way, including by setting the administrator password.
- Your network administrator or installation site coordinator must provide a Category 6 (Cat6) cable connected to the network that provides access to the servers used for Zero Touch Provisioning (ZTP) operations.

Procedure

1. Connect the network to a data port.
See the Installation Guide for switch to determine the location of the switch ports.
2. If the switch is powered on, power off the switch.
3. Power on the switch. During the ZTP operation, the switch might reboot if a new firmware image is being installed. ZTP goes to "Failed" state if the switch receives DHCP IP for vlan1 and does not receive any ZTP options within 60 seconds.

Initial configuration using the Aruba CX mobile app

This procedure describes how to use your mobile device to connect to the Bluetooth interface of the switch to connect to the switch for the first time so that you can configure basic operational settings using the Aruba CX mobile app.

Prerequisites

- You have obtained the USB Bluetooth adapter that was shipped with the switch. Information about the make and model of the supported adapter is included in the information about the HPE Aruba Networking Installer Mobile App in the Apple Store or Google Play.
- The HPE Aruba Networking Installer Mobile App must be installed on your mobile device.
- Bluetooth must be enabled on your mobile device.
- Your mobile device must be within the communication range of the Bluetooth adapter.
- If you are planning to import the switch into NetEdit, your mobile device must be able to use a Wi-Fi connection—not Bluetooth—to access the NetEdit server.

If your mobile device does not support simultaneous Bluetooth and Wi-Fi connections, you must use the NetEdit interface to import the switch at a later time. You can use the **Devices** tab to display the IP address of the switches you configured using your mobile device.

- The switch must be installed and powered on, with the network operating system boot sequence complete.

For information about installing and powering on the switch, see the Installation Guide for the switch.

Because you are using this mobile application to configure the switch through the Bluetooth interface, it is not necessary to connect a console to the switch.

- Bluetooth and USB must be enabled on the switch. On switches shipped from the factory, Bluetooth and USB are enabled by default.

Procedure

1. Install the USB Bluetooth adapter in the USB port of the switch.
For switches that have multiple management modules, you must install the USB Bluetooth adapter in the USB port of the active management module. Typically, the active management module is the module in slot 5.

Switches shipped from the factory have both USB and Bluetooth enabled by default.

For information about the location of the USB port on the switch, see the Installation Guide for the switch.

2. Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device. If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

```
Switch_model - Serial_number
```

For example: 8325-987654X1234567 or 8320-AB12CDE123

A switch supports one active Bluetooth connection at a time.

On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.



3. Open the Aruba CX mobile app on your mobile device.
The application attempts to connect to the switch using the switch Bluetooth IP address and the default switch login credentials. The **Home** screen of the application shows the status of the connection to the switch:
 - If the login attempt was successful, the Bluetooth icon is displayed and the status message shows the Bluetooth IP address of the switch. In addition, the connection graphic is green. You can continue to the next step.
 - If the login attempt was not successful, but a response was received, the Bluetooth icon is displayed, but the status message is: **Login Required**. You can continue to the next step. When you tap one of the tiles, you will be prompted for login credentials.
 - If the login attempt did not receive a response, the Bluetooth icon is not displayed, and the status message is: **No Connection**.
4. Create the initial switch configuration:

- You can deploy an initial configuration to the switch. Through this process, you supply the information required by a configuration template that you choose from a list of templates provided by the application. Then you deploy the configuration to the switch and, optionally, import the switch into NetEdit.



CAUTION

When you deploy a switch configuration, it becomes the running configuration, replacing the entire existing configuration of the switch. All changes previously made to the factory default configuration are overwritten.

If you plan to both deploy a switch configuration and customize the configuration of switch features, deploy the initial configuration first.

To deploy an initial switch configuration, tap: **Initial Config** and follow the instructions in the application.

- Alternatively, you can complete the initial configuration of the switch by tapping **Modify Config** and then selecting the features and settings to configure.
- You can also use the **Modify Config** feature to configure some switch features after the initial configuration is complete. For more information about what you can configure using the Aruba CX mobile app, see the online help for the application.

Subtopics

Troubleshooting Bluetooth connections

Troubleshooting Bluetooth connections

Subtopics

Bluetooth connection IP addresses

Bluetooth is connected but the switch is not reachable

Bluetooth is not connected

Bluetooth connection IP addresses

The Bluetooth connection uses IP addresses in the 192.168.99.0/24 subnet.

Switch

192.168.99.1

Mobile device

192.168.99.10

Bluetooth is connected but the switch is not reachable

Symptom

The mobile device settings indicate that the device is connected to the switch through Bluetooth. However, the mobile application indicates that the switch is not reachable.

Solution 1

Cause

The mobile device is paired with a different nearby switch.

Action

1. Verify the model number and serial number of the switch to which you are attempting to connect.
2. Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device. If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

`Switch_model-Serial_number`

For example: `8325-987654X1234567` or `8320-AB12CDE123`

A switch supports one active Bluetooth connection at a time.

On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 2

Cause

The mobile device is connected to a different network—such as through a Wi-Fi connection—that conflicts with the subnet used for the switch Bluetooth connection.

Action

Disconnect the mobile device from the network that is using the conflicting subnet.

For example, use the mobile device settings to turn off or disable Wi-Fi. If you choose to disable Wi-Fi on the mobile device, and you are not able to access cellular service, you will not be able to connect to the NetEdit server to import the switch, but you can still deploy a switch configuration.

Bluetooth is not connected

Symptom

Your mobile device cannot establish a Bluetooth connection to the switch.

Solution 1

Cause

Bluetooth is not enabled on your mobile device.

Action

- Use your mobile device settings application to enable Bluetooth.
- Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device. If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

Switch_model-Serial_number

For example: **8325-987654X1234567** or **8320-AB12CDE123**

A switch supports one active Bluetooth connection at a time.

On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 2

Cause

Your mobile device is not within the broadcast range of the Bluetooth adapter.

Action

Move closer to the switch.

Devices can communicate through Bluetooth when they are close, typically within a few feet of each other.

Solution 3

Cause

Your mobile device is not paired with the switch.

Action

1. Use your mobile device settings application to enable Bluetooth.

2. Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device. If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

Switch_model-Serial_number

For example: **8325-987654X1234567** or **8320-AB12CDE123**

A switch supports one active Bluetooth connection at a time.

3. On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 4

Cause

Bluetooth is not enabled on the switch.

New switches are shipped from the factory with the USB port and Bluetooth enabled. However, an installed switch might have been configured to disable Bluetooth or disable the USB port, which the USB Bluetooth adapter uses.

Action

Use a different CLI connection to enable Bluetooth on the switch.

- Use the **show bluetooth** CLI command to show the Bluetooth configuration and the status of the Bluetooth adapter.
- To enable the USB port, enter the CLI command: **usb**
- An inserted USB drive must be mounted each time the switch boots or fails over to a different management module. To mount the drive, enter the CLI command: **usb mount**
- To enable Bluetooth, enter the CLI command: **bluetooth enable**

Solution 5

Cause

Another mobile device has already connected to the switch through Bluetooth. This cause is likely if your device is repeatedly disconnected within 1-2 seconds of establishing a connection.

Action

1. Use a different CLI connection to see if there is another device connected:
Use the **show bluetooth** CLI command to show the Bluetooth configuration and the status of the Bluetooth adapter.
2. Either disconnect the other device or use that device to communicate with the switch.
A switch can use Bluetooth to connect to one mobile device at a time.

Solution 6

Cause

The switch has been restarted since the mobile device was last paired with the switch, and the device is having difficulty establishing the Bluetooth connection.

Action

1. Use the Bluetooth mobile device settings to forget the switch device.
2. Use your mobile device settings application to disable Bluetooth. Use your mobile device settings application to enable Bluetooth.

If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

Switch_model-Serial_number

For example: **8325-987654X1234567** or **8320-AB12CDE123**

A switch supports one active Bluetooth connection at a time.

On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 7

Cause

The USB Bluetooth adapter is not installed in the switch.

If the switch has multiple management modules, the USB Bluetooth adapter might be installed in the management module that is not the active management module.

Action

Install the USB Bluetooth adapter in the USB port of the switch.

For switches that have multiple management modules, you must install the USB Bluetooth adapter in the USB port of the active management module. Typically, for new switches, the active management module is the module in slot 5 (HPE Aruba Networking 8400 switches) or slot 1 (HPE Aruba Networking 6400 switches).

For information about the location of the USB port on the switch, see the Installation Guide for the switch.

Solution 8

Cause

A problem occurred with the Bluetooth feature on the switch. For example, the software daemon was stopped and then restarted.

Action

1. Use a different connection to the switch CLI to disable and then enable Bluetooth.

```
switch(config)# bluetooth disable  
switch(config)# bluetooth enable
```

2. Use the Bluetooth mobile device settings to forget the switch device.
3. Use your mobile device settings application to disable Bluetooth.
4. Use your mobile device settings application to enable Bluetooth.
5. Use your mobile device settings application to enable Bluetooth.
If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

Switch_model-Serial_number

For example: **8325-987654X1234567** or **8320-AB12CDE123**

A switch supports one active Bluetooth connection at a time.

On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 9

Cause

A switch that is member of a stack (but is not the conductor switch), has a USB Bluetooth adapter installed, but mobile application has lost contact with that switch.

Action

Remove and then reinstall the USB Bluetooth adapter.

Do not remove the USB Bluetooth adapter from the conductor switch.

Initial configuration using the CLI

This procedure describes how to connect to the switch for the first time and configure basic operational settings using the CLI. In this procedure, you use a computer to connect to the switch using the either the console port or management port.

Procedure

1. Connect to the console port or the management port.
2. Log into the switch for the first time.
3. Configure switch time using the NTP client.

Subtopics

[Connecting to the console port](#)

[Connecting to the management port](#)

[Logging into the switch for the first time](#)

[Setting switch time using the NTP client](#)

Connecting to the console port

Prerequisites

- A switch installed as described in its hardware installation guide.
- A computer with terminal emulation software.
- A USB-C cable.

Procedure

1. Connect the USB-C port on the switch to the USB-C port on the computer using a USB-C cable.
2. Start the terminal emulation software on the computer and configure a new serial session with the following settings:
 - Speed: 115200 bps
 - Data bits: 8
 - Stop bits: 1
 - Parity: None
 - Flow control: None
3. Start the terminal emulation session.
4. Press **Enter** once. If the connection is successful, you are prompted to log in.

Optional console port speed setting

If desired, the console port speed can be set with the **console baud-rate** command. For example, setting the console port speed to 9600 bps:

```
switch(config)# console baud-rate 9600
```

This command will configure the baud rate immediately for the active serial console session. After the command is executed the user will be prompted to re-login. The serial console will be inaccessible until the terminal client settings are updated to match the baud rate of the switch.


```
Continue (y/n)? y
```

Showing the console port current speed:

```
switch# show console  
Baud Rate: 9600
```

For details on the **console baud-rate** and **show console** commands, see [Switch system and hardware commands](#).

Connecting to the management port

Prerequisites

- Two Ethernet cables
- SSH client software

Procedure

1. By default, the management interface is set to automatically obtain an IP address from a DHCP server, and SSH support is enabled. If there is no DHCP server on your network, you must configure a static address on the management interface:
 - a. Connect to the [console port](#)
 - b. Configure the [management interface](#).
2. Use an Ethernet cable to connect the management port to your network.
3. Use an Ethernet cable to connect your computer to the same network.
4. Start your SSH client software and configure a new session using the address assigned to the management interface. (If the management interface is set to operate as a DHCP client, retrieve the IP address assigned to the management interface from your DHCP server.)
5. Start the session. If the connection is successful, you are prompted to log in.

Subtopics

[Configure using DHCP or static IP](#)

Configure using DHCP or static IP



NOTE

Users can use any data ports for in-band management purposes. IP DHCP is supported on interface VLAN 1 only. All switch ports are part of access VLAN 1 by default. Static IP address and IP DHCP configuration can co-exist on VLAN 1, however static addresses take precedence whenever configured.

DHCP Configuration

```
switch#: config
switch(config)#: vlan 1
Switch(config-vlan-1)#: description Management VLAN
Switch(config-vlan-1)#: end
Switch#
!
Switch(config)#: interface 1/1/1
Switch(config-if)#: description IN-BAND Management Port
Switch(config-if)#: vlan access 1
Switch(config-if)#: no shutdown
Switch(config-if)#: end
Switch#
!
Switch(config)#: interface vlan 1
Switch(config-if-vlan)#: description IN-BAND Management Interface
Switch(config-if-vlan)#: ip dhcp
Switch(config-if-vlan)#: no shutdown
Switch(config-if-vlan)#: end
Switch#
!
```

Without DHCP Configuration

```
switch#: config
switch(config)#: vlan 1
Switch(config-vlan-1)#: description Management VLAN
Switch(config-vlan-1)#: end
Switch#
!
Switch(config)#: interface 1/1/1
Switch(config-if)#: description IN-BAND Management Port
Switch(config-if)#: vlan access 1
Switch(config-if)#: no shutdown
Switch(config-if)#: end
Switch#
```

```
!  
Switch(config)#: interface vlan 1  
Switch(config-if-vlan)#: description IN-BAND Management Interface  
Switch(config-if-vlan)#: no ip dhcp  
Switch(config-if-vlan)#: ip address 192.168.10.1/24  
Switch(config-if-vlan)#: no shutdown  
Switch(config-if-vlan)#: end  
Switch#
```

Logging into the switch for the first time

The first time you log in to the switch you must use the default administrator account. This account has no password, so you will be prompted on login to define one to safeguard the switch.

Procedure

1. When prompted to log in, specify **admin**. When prompted for the password, press **ENTER**. (By default, no password is defined.)

For example:

```
switch login: admin  
password:
```

2. Define a password for the **admin** account. The password can contain up to 32 alphanumeric characters in the range ASCII 32 to 127, which includes special characters such as asterisk (*), ampersand (&), exclamation point (!), dash (-), underscore (_), and question mark (?).

For example:

```
Please configure the 'admin' user account password.  
Enter new password: *****  
Confirm new password: *****  
switch#
```

3. You are placed into the manager command context, which is identified by the prompt: `switch#`, where `switch` is the model number of the switch. Enter the command `config` to change to the global configuration context `config`.

For example:

```
switch# config  
switch(config)#
```

Setting switch time using the NTP client

Prerequisites

- The IP address or domain name of an NTP server.
- If the NTP server uses authentication, obtain the password required to communicate with the NTP server.

Procedure

1. If the NTP server requires authentication, define the authentication key for the NTP client with the command **ntp authentication**.
2. Configure an NTP server with the command **ntp server**.
3. By default, NTP traffic is sent on the default VRF. If you want to send NTP traffic on the management VRF, use the command **ntp vrf**.
4. Review your NTP configuration settings with the commands **show ntp servers** and `show ntp status`.
5. See the current switch time, date, and time zone with the command **show clock**.

Example

This example creates the following configuration:

- Defines the authentication key **1** with the password **myPassword**.
- Defines the NTP server **my-ntp.mydomain.com** and makes it the preferred server.
- Sets the switch to use the management VRF (**mgmt**) for all NTP traffic.

```
switch(config)# ntp authentication-key 1 md5 myPassword
switch(config)# ntp server my-ntp.mydomain.com key 10 prefer

switch(config)# ntp vrf mgmt
```

Configuring banners

Procedure

1. Configure the banner that is displayed when a user connects to a management interface. Use the command `banner motd`. For example:

```
switch(config)# banner motd ^
```

Enter a new banner. Terminate the banner with the delimiter you have chosen.

```
> > This is an example of a banner text which a connecting user
> > will see before they are prompted for their password.
```

```
> >
```

```
> > As you can see it may span multiple lines and the input
> > will be terminated when the delimiter character is
> > encountered.^
```

Banner updated successfully!

2. Configure the banner that is displayed after a user is authenticated. Use the command `banner exec`. For example:

```
switch(config)# banner exec &
```

Enter a new banner. Terminate the banner with the delimiter you have chosen.

```
> > This is an example of a different banner text. This time
> > the banner entered will be displayed after a user has
> > authenticated.
```

```
> >
```

```
> > & This text will not be included because it comes after the '&'
```

Banner updated successfully!

Using the Web UI

The Web UI is disabled by default. Follow these steps to enable it on the management port and log in.

The Web UI is enabled by default on the default VRF.

Prerequisites

- A connection to the switch CLI.

Procedure

1. Log in to the CLI.
2. Switch to `config` context and enable the Web UI on the management port VRF with the command `https-server vrf mgmt`.

For example:

```
switch# config
switch(config)# https-server vrf mgmt
switch# config
switch(config)# https-server vrf default
```

3. Start your web browser and enter the IP address of the management port in the address bar,
For example: **https://192.168.1.1**
4. The Web UI starts and you are prompted to log in.

Configuring the management interface

Prerequisites

A connection to the console port.

Procedure

1. Switch to the management interface context with the command `interface mgmt`.
2. By default, the management interface on the management port is enabled. If it was disabled, re-enable it with the command `no shutdown`.
3. Use the command `ip dhcp` to configure the management interface to automatically obtain an address from a DHCP server on the network (factory default setting). Or, assign a static IPv4 or IPv6 address, default gateway, and DNS server with the commands `ip address`, `ipv6 address`, `ip static`, `default-gateway`, and `nameserver`.
4. SSH is enabled by default on the management VRF. If disabled, enable SSH with the command `ssh server vrf mgmt`.

Examples

This example enables the management interface with dynamic addressing using DHCP:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# no shutdown
switch(config-if-mgmt)# ip dhcp
```

This example enables the management interface with static addressing creating the following configuration:

- Sets a static IPv4 address of **198.168.100.10** with a mask of **24** bits.

- Sets the default gateway to **198.168.100.200**.
- Sets the DNS server to **198.168.100.201**.

```
switch(config)# interface mgmt
switch(config-if-mgmt)# no shutdown
switch(config-if-mgmt)# ip static 198.168.100.10/24
switch(config-if-mgmt)# default-gateway 198.168.100.200
switch(config-if-mgmt)# nameserver 198.168.100.201
```

Restoring the switch to factory default settings

Prerequisites

You are connected to the switch through its Console port.



NOTE

This procedure erases all user information and configuration settings. Consider backing up your running configuration first.

1. Optionally, back up the running configuration with either **copy running-config <REMOTE-URL>** or **copy running-config <STORAGE-URL>**. The **json** storage format is required for later configuration restoration.
2. Switch to the configuration context with the command **config**.
3. Erase all user information and configuration, restoring the switch to its factory default state with the command **erase all zeroize**. Enter **Y** when prompted to continue. The switch automatically restarts.
4. Optionally restore your saved configuration (it must be in **json** format) with either **copy <REMOTE-URL> running-config** or **copy <STORAGE-URL> running-config** followed by **copy running-config startup-config**.

Example

Backing up the running configuration to a file on a remote server (using TFTP), resetting the switch to its factory default state, and then restoring the saved configuration.

```
switch# copy running-config tftp://192.168.1.10/backup_cfg json vrf mgmt

% Total      % Received % Xferd  Average Speed   Time    Time       Time
Current                                  Dload  Upload   Total     Spent    Left
Speed
```

```

100 10340      0      0 100 10340      0 1329k --:--:-- --:--:-- --:--:--
1329k
100 10340      0      0 100 10340      0 1313k --:--:-- --:--:-- --:--:--
1313k
switch#
switch#
switch# erase all zeroize
This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
This should take several minutes to one hour to complete.
Continue (y/n)? y

The system is going down for zeroization.
[ OK ] Stopped PSPO Module Daemon.
[ OK ] Stopped AOS-CX Switch Daemon for BCM.
...
[ OK ] Stopped Remount Root and Kernel File Systems.
[ OK ] Reached target Shutdown.
reboot: Restarting system
Press Esc for boot options
ServiceOS Information:
    Version:          GT.01.03.0006
    Build Date:       2018-10-30 14:20:44 PDT
    Build ID:         ServiceOS:GT.01.03.0006:8ee0faaa52da:201810301420
    SHA:              xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
...
##### Preparing for zeroization #####
##### Storage zeroization #####
##### WARNING: DO NOT POWER OFF UNTIL #####
##### ZEROIZATION IS COMPLETE #####
##### This should take several minutes #####
##### to one hour to complete #####
##### Restoring files #####
Boot Profiles:
0. Service OS Console
1. Primary Software Image [XL.10.02.0010]
2. Secondary Software Image [XL.10.02.0010]
Select profile(primary):
Bootting primary software image...
Verifying Image...
Image Info:
    Name: AOS-CX

```



```

Version: XL.10.02.0010
Build Id: AOS-CX:XL.10.02.0010:feaf5b9b7f09:201901292014
Build Date: 2019-01-29 12:43:50 PST
Extracting Image...
Loading Image...
Done.
kexec_core: Starting new kernel
System is initializing
fips_post_check[5473]: FIPS_POST: Cryptographic selftest started...SUCCESS
[ OK ] Started Login banner readiness check.
...
8400X login: admin
Password:
switch#
switch#
switch# copy tftp://192.168.1.10/backup_cfg running-config json vrf mgmt

% Total      % Received % Xferd  Average Speed   Time    Time       Time
Current
                        Dload  Upload  Total    Spent    Left
Speed
100 10340  100 10340    0     0  2858k      0  --:--:--  --:--:--  --:--:--
2858k
100 10340  100 10340    0     0  2804k      0  --:--:--  --:--:--  --:--:--
2804k
Large configuration changes will take time to process, please be patient.
switch#
switch#
switch# copy running-config startup-config
Large configuration changes will take time to process, please be patient.
switch#

```

Limitations and considerations

Profile switching between VSX and VSF

- VSF and VSX are mutually exclusive features. Before switching between profiles, ensure the device is in standalone mode.
- Configuration data is not compatible between VSF and VSX profiles. Back up your configurations before changing profiles (switching will erase startup configurations and reboot the device).

- Never apply the VSX profile on an active VSF setup. Applying VSX profile on an active VSF commander may cause undesired behavior and potentially require a factory reset.
- Checkpoints created in one profile mode should not be applied to a switch in a different profile mode. For example, do not apply VSX checkpoints on a switch configured for VSF profile, or vice versa.
- Configuration downloads should match the current profile of the switch. Do not download VSX configurations to a switch in VSF profile mode, or vice versa.

Software downgrade considerations

- When performing a software downgrade on a device configured with either VSX or VSF profiles, ensure that the startup configuration is erased prior to the downgrade process. This prevents configuration incompatibilities that could arise between different software versions.
- Failure to erase the startup configuration before downgrade may result in unexpected behavior, including potential boot failures or feature inconsistencies when the device initializes with the older software version which does not support VSX.

REST API profile switching procedure

About this task

Complete the following steps when changing profiles via REST API:

Procedure

1. Ensure the device is in standalone mode.
2. Execute **erase startup-config**.
3. Set the **configured_profile** in System table to the desired profile (VSX or Basic).
4. Save the configuration.
5. Reboot the switch to activate the desired profile.

Results

It is best practice to always save the running configuration after changing profiles and before rebooting. If the running configuration is not saved after a profile change, a subsequent reboot may return the device to the Basic (default) profile.

Management interface commands

Select a command from the list in the left navigation menu.

Subtopics

- [default-gateway](#)
- [ip static](#)
- [nameserver](#)
- [show interface mgmt](#)

default-gateway

Syntax

```
default-gateway <IP-ADDR>  
no default-gateway <IP-ADDR>
```

Description

Assigns an IPv4 or IPv6 default gateway to the management interface. An IPv4 default gateway can only be configured if a static IPv4 address was assigned to the management interface. An IPv6 default gateway can only be configured if a static IPv6 address was assigned to the management interface. The default gateway should be on the same network segment.

The **no** form of this command removes the default gateway from the management interface.

Parameter	Description
<code><IP-ADDR></code>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:x xxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

Examples

Setting a default gateway with the IPv4 address of **198.168.5.1**:

```
switch(config)# interface mgmt  
switch(config-if-mgmt)# default-gateway 198.168.5.1
```

Setting an IPv6 address of **2001:DB8::1**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# default-gateway 2001:DB8::1
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
5420 6200	config-if-mgmt	Administrators or local user group members with execution rights for this command.

ip static

Syntax

```
ip static <IP-ADDR>/<MASK>
no ip static <IP-ADDR>/<MASK>
```

Description

Assigns an IPv4 or IPv6 address to the management interface.

The **no** form of this command removes the IP address from the management interface and sets the interface to operate as a DHCP client.

Parameter	Description
<IP-ADDR>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:x xxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

Parameter	Description
<MASK>	Specifies the number of bits in an IPv4 or IPv6 address mask in CIDR format (x), where x is a decimal number from 0 to 32 for IPv4, and 0 to 128 for IPv6.

Examples

Setting an IPv4 address of **198.51.100.1** with a mask of **24** bits:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip static 198.51.100.1/24
```

Setting an IPv6 address of **2001:DB8::1** with a mask of **32** bits:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip static 2001:DB8::1/32
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-if-mgmt	Administrators or local user group members with execution rights for this command.

nameserver

Syntax

```
nameserver <PRIMARY-IP-ADDR> [ <SECONDARY-IP-ADDR> ]
no nameserver <PRIMARY-IP-ADDR> [ <SECONDARY-IP-ADDR> ]
```

Description

Assigns a primary or secondary IPv4 or IPv6 DNS server to the management interface. IPv4 DNS servers can only be configured if a static IPv4 address was assigned to the management interface. IPv6 DNS servers can only be configured if a static IPv6 address was assigned to the management interface. The default gateway should be on the same network segment.

The **no** form of this command removes the DNS servers from the management interface.

Parameter	Description
<code><PRIMARY-IP-ADDR></code>	Specifies the IP address of the primary DNS server. Specify the address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxx x:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
<code><SECONDARY-IP-ADDR></code>	Specifies the IP address of the secondary DNS server. Specify the address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxx x:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

Examples

Setting primary and secondary DNS servers with the IPv4 addresses of **198.168.5.1** and **198.168.5.2**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# nameserver 198.168.5.1 198.168.5.2
```

Setting primary and secondary DNS servers with the IPv6 addresses of **2001:DB8::1** and **2001:DB8::2**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# nameserver 2001:DB8::1 2001:DB8::2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420	config-if-mgmt	Administrators or local user group members with execution rights for this command.

show interface mgmt

Syntax

```
show interface mgmt
```

Description

Shows status and configuration information for the management interface.

Example

```
switch# show interface mgmt
Address Mode                : static
Admin State                 : up
Mac Address                 : 02:42:ac:11:00:02
IPv4 address/subnet-mask    : 192.168.1.10/16
Default gateway IPv4        : 192.168.1.1
IPv6 address/prefix         : 2001:db8:0:1::129/64
IPv6 link local address/prefix: fe80::7272:cfff:fe4d:e485/64
Default gateway IPv6        : 2001:db8:0:1::1
Primary Nameserver          : 2001::1
Secondary Nameserver        : 2001::2
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

NTP commands

Select a command from the list in the left navigation menu.

Subtopics

[ntp authentication](#)

[ntp authentication-key](#)

[ntp disable](#)

[ntp enable](#)

[ntp server](#)

[ntp trusted-key](#)

[ntp vrf](#)

[show ntp associations](#)

[show ntp authentication-keys](#)

[show ntp servers](#)

[show ntp statistics](#)

[show ntp status](#)

ntp authentication

Syntax

```
ntp authentication
no ntp authentication
```

Description

Enables support for authentication when communicating with an NTP server.

The **no** form of this command disables authentication support.

Examples

Enabling authentication support:

```
switch(config)# ntp authentication
```

Disabling authentication support:

```
switch(config)# no ntp authentication
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp authentication-key

Syntax

```
ntp authentication-key <KEY-ID> {md5 | sha1}
    [{ <PLAINTEXT-KEY> [trusted] | ciphertext <ENCRYPTED-KEY> }]
no ntp authentication-key <KEY-ID> {md5 | sha1}
    [{ <PLAINTEXT-KEY> [trusted] | ciphertext <ENCRYPTED-KEY> }]
```

Description

Defines an authentication key that is used to secure the exchange with an NTP time server. This command provides protection against accidentally synchronizing to a time source that is not trusted.

The **no** form of this command removes the authentication key.



NOTE

Using "#" in the NTP password will prevent the NTP to synchronize when authentication is enabled

Parameter	Description
<code><KEY-ID></code>	Specifies the authentication key ID. Range: 1 to 65534.
<code>md5</code>	Selects MD5 key encryption.
<code>sha1</code>	Specifies SHA1 key encryption.
<code><PLAINTEXT-KEY></code>	Specifies the plaintext authentication key. Range: 8 to 40 characters. The key may contain printable ASCII characters excluding "#" or be entered in hex. Keys longer than 20 characters are assumed to be hex. To use an ASCII key longer than 20 characters, convert it to hex.
<code>trusted</code>	Specifies that this is a trusted key. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.
<code>ciphertext <ENCRYPTED-KEY></code>	Specifies the ciphertext authentication key in Base64 format. This is used to restore the NTP authentication key when copying configuration files between switches or when uploading a previously saved configuration.



NOTE

When the key is not provided on the command line, plaintext key prompting occurs upon pressing Enter, followed by prompting as to whether the key is to be trusted. The entered key characters are masked with asterisks.

Examples

Defining key 10 with MD5 encryption and a provided plaintext trusted key:

```
switch(config)# ntp authentication-key 10 md5 F8245a0b trusted
```

Defining key 5 with SHA1 encryption and a prompted plaintext trusted key:

```
switch(config)# ntp authentication-key 5 sha1
Enter the NTP authentication key: *****
Re-Enter the NTP authentication key: *****
Configure the key as trusted (y/n)? y
```

Removing key 10:

```
switch(config)# no ntp authentication-key 10
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp disable

Syntax

```
ntp disable
```

Description

Disables the NTP client on the switch. The NTP client is disabled by default.

Examples

Disabling the NTP client.

```
switch(config)# ntp disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp enable

Syntax

```
ntp enable
no ntp enable
```

Description

Enables the NTP client on the switch to automatically adjust the local time and date on the switch. The NTP client is disabled by default.

The **no** form of this command disables the NTP client.

Examples

Enabling the NTP client.

```
switch(config)# ntp enable
```

Disabling the NTP client.

```
switch(config)# no ntp enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp server

Syntax

```
ntp server <IP-ADDR> [key <KEY-NUM>] [minpoll <MIN-NUM>] [maxpoll <MAX-NUM>]
[burst | iburst][prefer] [version <VER-NUM>]
no ntp server <IP-ADDR>
               <IP-ADDR> [key <KEY-NUM>] [minpoll <MIN-NUM>] [maxpoll <MAX-
NUM>][burst | iburst] [prefer] [version <VER-NUM>]
```

Description

Defines an NTP server to use for time synchronization, or updates the settings of an existing server with new values. Up to eight servers can be defined.

The **no** form of this command removes a configured NTP server.



NOTE

The default NTP version is 4; it is backwards compatible with version 3.

Parameter	Description
server <IP-ADDR>	Specifies the address of an NTP server as a DNS name, an IPv4 address (x.x.x.x), where x is a decimal number from 0 to 255, or an IPv6 address (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

Parameter	Description
	When specifying an IPv4 address, you can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100. When specifying an IPv6 address, you can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.
<code>key <KEY-NUM></code>	Specifies the key to use when communicating with the server. A trusted key must be defined with the command ntp authentication - key and authentication must be enabled with the command ntp authentication . Range: 1 to 65534.
<code>minpoll <MIN-NUM></code>	Specifies the minimum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 6 (64 seconds).
<code>maxpoll <MAX-NUM></code>	Specifies the maximum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 10 (1024 seconds).
<code>burst</code>	Send a burst of packets instead of just one when connected to the server. Useful for reducing phase noise when the polling interval is long.
<code>iburst</code>	Send a burst of six packets when not connected to the server. Useful for reducing synchronization time at startup.
<code>prefer</code>	Make this the preferred server.
<code>version <VER-NUM></code>	Specifies the version number to use for all outgoing NTP packets. Range: 3 or 4. Default: 4.



NOTE

NTP is backwards compatible.

Usage

For features such as Activate and ZTP, a switch that has a factory default configuration will automatically be configured with **pool.ntp.org**. NTP server configurations via DHCP options are supported. The DHCP server can be configured with maximum of two NTP server addresses which will be supported on the switch. Only IPV4 addresses are supported.

NTP uses a stratum to describe the distance between a network device and an authoritative time source:

- A stratum 1 time server is directly attached to an authoritative time source (such as a radio or atomic clock or a GPS time source).

- A stratum 2 NTP server receives its time through NTP from a stratum 1 time server.

When using multiple servers with same stratum setting, the best practice to configure a preferred server, so NTP will attempt to use the preferred server as the primary NTP connection. If a preferred server is not manually set when NTP is enabled, the configured server with the lowest stratum will automatically be set as the preferred server. If there are servers with the same stratum, this auto prefer status will prevent AOS-CX from toggling between different servers as the primary server. Auto prefer selection of servers with same stratum (if not manually selected) may change after reconfiguring the switch, or after executing the **reboot** command.

Examples

Defining the ntp server pool.ntp.org, using iburst, and NTP version 4.

```
switch(config)# ntp server pool.ntp.org iburst version 4
```

Removing the ntp server pool.ntp.org.

```
switch(config)# no ntp server pool.ntp.org
```

Defining the ntp server my-ntp.mydomain.com and makes it the preferred server.

```
switch(config)# ntp server my-ntp.mydomain.com prefer
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ntp trusted-key

Syntax

```
ntp trusted-key <KEY-ID>
no ntp trusted-key <KEY-ID>
```

Description

Sets a key as trusted. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.

The **no** form of this command removes the trusted designation from a key.

Parameter	Description
<code><KEY-ID></code>	Specifies the identification number of the key to set as trusted. Range: 1 to 65534.

Examples

Defining key 10 as a trusted key.

```
switch(config) # ntp trusted-key 10
```

Removing trusted designation from key 10:

```
switch(config) # no ntp trusted-key 10
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

ntp vrf

Syntax

```
ntp vrf <VRF-NAME>
no ntp vrf <VRF-NAME>
```

Description

Specifies the VRF on which the NTP client communicates with an NTP server.

The **no** form of the command returns to default VRF.

Parameter	Description
<VRF-NAME>	Specifies the name of a VRF.

Example

Setting the switch to use the default VRF for NTP client traffic.

```
switch(config) # ntp vrf default
```

Setting the switch to use the default management VRF for NTP client traffic.

```
switch(config) # ntp vrf mgmt
```

Returning the switch to use the default VRF for NTP client traffic.

```
switch(config) # no ntp vrf
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show ntp associations

Syntax

```
show ntp associations
```

Description

Shows the status of the connection to each NTP server. The following information is displayed for each server:

- Tally code : The first character is the Tally code:
 - (blank): No state information available (e.g. non-responding server)
 - x : Out of tolerance (discarded by intersection algorithm)
 - . : Discarded by table overflow (not used)
 - - : Out of tolerance (discarded by the cluster algorithm)
 - + : Good and a preferred remote peer or server (included by the combine algorithm)
 - # : Good remote peer or server, but not utilized (ready as a backup source)
 - * : Remote peer or server presently used as a primary reference
 - o : PPS peer (when the prefer peer is valid)
- ID: Server number.
- NAME: NTP server FQDN/IP address (Only the first 24 characters of the name are displayed).
- REMOTE: Remote server IP address.
- REF_ID: Reference ID for the remote server (Can be an IP address).
- ST: (Stratum) Number of hops between the NTP client and the reference clock.

- LAST: Time since the last packet was received in seconds unless another unit is indicated.
- POLL: Interval (in seconds) between NTP poll packets. Maximum (1024) reached as server and client sync.
- REACH: 8-bit octal number that displays status of the last eight NTP messages (377 = all messages received).

Example

```
switch# show ntp associations
-----
ID          NAME          REMOTE          REF-ID ST LAST POLL REACH
-----
1          192.0.1.1          192.0.1.1          .INIT. 16    -   64    0
* 2  time.apple.com    17.253.2.253          .GPSs.  2   70  128   377
-----
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ntp authentication-keys

Syntax

```
show ntp authentication-keys
```

Description

Shows the currently defined authentication keys.

Examples

```
switch# show ntp authentication-keys
```

```
-----  
Auth key   Trusted   MD5 password  
-----  
10         No       *****  
20         Yes      *****
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show ntp servers

Syntax

```
show ntp servers
```

Description

Shows all configured NTP servers, including any DHCP servers, default pool servers or any server with the status **auto prefer**.

Example

```
switch# show ntp servers
```

```
-----  
NTP SERVER KEYID MINPOLL MAXPOLL OPTION VER  
-----  
192.0.1.18      -         5        10 iburst  3  
192.0.1.19      -         6        10  none   4
```

```
192.0.1.20 - 6 8 burst 3 prefer
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ntp statistics

Syntax

```
show ntp statistics
```

Description

Shows global NTP statistics. The following information is displayed:

- Rx-pkts: Total NTP packets received.
- Current Version Rx-pkts: Number of NTP packets that match the current NTP version.
- Old Version Rx-pkts: Number of NTP packets that match the previous NTP version.
- Error pkts: Packets dropped due to all other error reasons.
- Auth-failed pkts: Packets dropped due to authentication failure.
- Declined pkts: Packets denied access for any reason.
- Restricted pkts: Packets dropped due to NTP access control.
- Rate-limited pkts: Number of packets discarded due to rate limitation.

- KOD pkts: Number of Kiss of Death packets sent.

Examples

```
switch(config)# show ntp statistics
Rx-pkts 100
Current Version Rx-pkts 80
Old Version Rx-pkts 20
Err-pkts 2
Auth-failed-pkts 1
Declined-pkts 0
Restricted-pkts 0
Rate-limited-pkts 0
KoD-pkts 0
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ntp status

Syntax

```
show ntp status
```

Description

Shows the status of NTP on the switch.

Examples

Displaying the status information when the switch is not synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.
Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds
Not synchronized with an NTP server.
```

Displaying the status information when the switch is synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.
Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds
Synchronized to NTP Server 17.253.2.253 at stratum 2.
Poll interval = 1024 seconds.
Time accuracy is within 0.994 seconds
Reference time: Thu Jan 28 2016 0:57:06.647 (UTC)
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Client probe and configuration

Client probing allows the switch to wake up silent clients it is connected to. A silent client is one that doesn't send any packet to allow a switch to authenticate the client until probed. These clients may send packets on a link up but will stop sending packets once they enter power saving mode (i.e., sleep mode).

Client probing provides the following benefits:

- The switch port can be the source of the probe packets that can wake up silent clients. This eliminates the need for dedicated device in the network to generate the probe packets. It also reduces the amount of BUM traffic in the network otherwise used to wake up silent clients.
- Control the type of probe packets to be sent, how often to send them and when to stop.
- The data to be used in the probe packets. Some of the clients such as industrial weighing scales do not respond to any random probe packet. They must be probed with the broadcast IP address of the subnet associated with the IP assigned to them.

Subtopics

Client probing commands

Client probing commands

Select a command from the list in the left navigation menu.

Subtopics

(remote) associate profile
clear client probe statistics
client probe
client-probe autogen-probe-file force-regen
client-probe profile force-download
client probe enable
entry
probe-duration
probe-interval
probe-type
(local) profile
remote-server refresh interval
show client probe statistics
show client probe profile
show running-config
vlan-list

(remote) associate profile

Syntax

```
associate profile {sftp user <username> password [ciphertext <CIPHERTEXT-PASSWORD> plaintext <PLAINTEXT PASSWORD> | tftp]} {host <IP-ADDR | FQDN>} [port <PORT>] {file <FILE-PATH> [vrf<VRF-NAME>]}  
no associate profile {sftp user <username> password [ciphertext <CIPHERTEXT-PASSWORD> plaintext <PLAINTEXT PASSWORD> | tftp]} {host <IP-ADDR | FQDN>} [port <PORT>] {file <FILE-PATH> [vrf<VRF-NAME>]}
```

Description

Configures how profile information used in probe packets is obtained. The default setting obtains information from a remote file via tftp or sftp.

The **no** form sets to default configuration.

Usage

The switch uses the name of the remote file to create a profile. Make sure the remote file follows proper naming requirements.

Parameter	Description
USERNAME	Specifies the username, which must be between 2 and 64 characters in length.
CIPHERTEXT-PASSWORD	Specifies the password. Maximum of 64 characters.
PLAINTEXT-PASSWORD	Specifies the password. Maximum of 64 characters.
IP-ADDR FQDN	Specifies the remote server IP address or hostname.
PORT	Sets the sftp or tftp port number. sftp default: 22 tftp default: 69
FILE-PATH	Specifies the file name to be copied.
VRF-NAME	(Default) Specifies the VRF to be used for reaching the remote server.

Examples

Configuring the client-probe profile to an SFTP location with standard SFTP port 22:

```
switch(config)# client probe  
switch(config-cp)# associate profile sftp user johndoe password plaintext  
hello host 20.20.20.1 file client-info.json
```

Configuring the client-probe profile to an SFTP location hosted on custom SFTP port 3333:

```
switch(config)# client probe  
switch(config-cp)# associate profile sftp user johndoe password plaintext  
hello host 20.20.20.1 port 3333 file client-info.json
```

To configure a remote profile, provide a JSON profile to a **remote_server**, using the following format:

```
[  
  {  
    "vlan": "<VLAN-ID>",  
    "enable": true|false,  
    "source_mac": "<MAC-ADDR>",  
    "source_ip": "<IP-ADDR>",  
    "broadcast_ip": "<IP-ADDR>",  
    "directed_ip_range": {  
      "start_ip": "<IP-ADDR>",  
      "end_ip": "<IP-ADDR>"  
    },  
    "directed_ip_list": [  
      "<IP-ADDR>",  
      ...  
      ...  
      "<IP-ADDR>"  
    ],  
    },  
    ...  
    ...  
  ]  
]
```

Configuring a remote profile:

```
[  
  {  
    "enable": true,  
    "vlan": "10",  
    "source_mac": "aa:bb:cc:dd:ee:10",  
    "source_ip": "10.0.10.1",  
    "broadcast_ip": "10.0.10.255"  
  },  
  {  
    "enable": true,  
    "vlan": "30",  
    "source_mac": "aa:bb:cc:dd:ee:30",  
  }  
]
```

```

"source_ip": "10.0.30.1",
"directed_ip_range": {
  "start_ip": "10.0.30.2",
  "end_ip": "10.0.30.30"
},
{
  "enable": true,
  "vlan": "50",
  "source_mac": "aa:bb:cc:dd:ee:50",
  "source_ip": "10.0.50.1",
  "broadcast_ip": "10.0.50.255",
  "directed_ip_list": [
    "10.0.50.2",
    "10.0.50.7"
  ],
  "directed_ip_range": {
    "start_ip": "10.0.50.11",
    "end_ip": "10.0.50.30"
  }
}
]

```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-cp	Administrators or local user group members with execution rights for this command.

clear client probe statistics

Syntax

```
clear client probe statistics [interface [<IFRANGE>]]
```

Description

Clears the statistics associated with client probing.

Parameter	Description
-----------	-------------

<IFRANGE>	Specifies the interface identifier range.
-----------	---

Example

Clearing client probe statistics on all enabled interfaces:

```
switch# clear client probe statistics
```

Clearing client probe statistics on a given interface:

```
switch# clear client probe statistics interface 1/1/1
```

Command History

Release	Modification
---------	--------------

10.16.1000	Command introduced
------------	--------------------

Command Information

Platforms	Command context	Authority
-----------	-----------------	-----------

5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.
--------------	---------------	--

client probe

Syntax

```
client probe
no client probe
```

Description

Sets the client probe to global context.

The **no** form of this command removes the global context from client probe.

Examples

Setting the client probe to global context:

```
switch(config) # client probe
switch(config-cp) #
```



NOTE

For more information on features that use this command, refer to the Multicast Guide for your switch model.

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-if	Administrators or local user group members with execution rights for this command.

client-probe autogen-probe-file force-regen

Syntax

```
client-probe autogen-probe-file force-regen
```

Description

Forces the switch to regenerate the probe file. By default, the probe file is regenerated every 30 seconds when enabled.

Examples

Forcing the switch to regenerate the probe file:

```
switch# client probe autogen-probe-file force-regen
```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
	Manager (#)	Administrators or local user group members with execution rights for this command.

client-probe profile force-download

Syntax

```
client-probe profile force-download
```

Description

Enables a user to immediately force download a remote profile rather than waiting until the next cycle of **remote-server refresh-interval** timer.

Example

Force-downloading a remote profile:

```
switch# client probe profile force-download
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

client probe enable

Syntax

```
client probe enable
no client probe enable
```

Description

Enables client probing on a port. Client probing must also be enabled globally for it to be operational.

Example

Enabling client probing on the switch:

```
switch(config)# client probe
switch(config-cp)# enable
```

Enabling client probe on a physical interface:

```
switch(config)# interface 1/1/1
switch(config-if)# client probe enable
```

Enabling client probe on a LAG interface:

```
switch(config)# interface lag 1
switch(config-lag-if)# client probe enable
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-if	Administrators or local user group members with execution rights for this command.

entry

Syntax

```
entry <ENTRY-ID> {vlan <VLAN-ID> | source-mac <MAC-ADDR> | source-ip <IPV4-ADDR> | broadcast-ip <IPV4-ADDR> | directed-ip-range <START-IPV4-ADDR> <END-IPV4-ADDR> | directed-ip-list <IPV4-ADDR>}
no entry <ENTRY-ID> {vlan <VLAN-ID> | source-mac <MAC-ADDR> | source-ip <IPV4-ADDR> | broadcast-ip <IPV4-ADDR> | directed-ip-range <START-IPV4-ADDR> <END-IPV4-ADDR> | directed-ip-list <IPV4-ADDR>}
```

Description

Configures entries in the profile.

Parameter	Description
vlan	(Mandatory) Specifies the VLAN ID associated with the entry. By default the switch will send probe packets for all the entries in the JSON. But this can be restricted to only certain VLANs using the vlan - list configuration.
source-mac	Specifies the source MAC address to use in the probe packets. If omitted, the system MAC address is used in the probe packets.

Parameter	Description
source-ip	(Mandatory) Specifies the source IP address to use in the probe packets.
broadcast-ip	Specifies the broadcast IP address to use when ping - broadcast probe is enabled.
directed-ip-range	Specifies the range of destination IP address to use when ping - sweep probe is enabled. Only IPv4 subnets from /8 to /32 are supported.
directed-ip-list	Specifies the list of destination IP addresses to use when the ping - unicast probe is enabled. A maximum of 255 addresses can be used. Only a single address can be specified at a time and the command must be repeated for every address that must be added to the list. If the limit is exceeded, the switch will discard the profile.
<ENTRY-ID>	Specifies the entry ID.
<VLAN-ID>	Specifies the VLAN ID.
<MAC-ADDR>	Specifies the source MAC address.
<IPV4-ADDR>	Specifies the selected IP address.
<START-IP-ADDR>	First IP address in the range.
<END-IP-ADDR>	Last IP address in the range



NOTE

For a profile to be deemed valid, it must contain one of the following keys: broadcast-ip, directed-ip-range, or directed-ip-list.



NOTE

If any of the mandatory attributes are not configured, the entry will be treated as incomplete and the profile will not be used.

Example

Configuring local profile entries:

```
switch(config)# client probe
switch(config-cp)# profile WakeUp-Clients
switch(config-cp-profile)# entry 1
```

```

switch(config-cp-profile-entry)# vlan 30
switch(config-cp-profile-entry)# source-mac aa:bb:cc:dd:ee:30
switch(config-cp-profile-entry)# source-ip 10.0.30.1
switch(config-cp-profile-entry)# broadcast-ip 10.0.30.255
switch(config-cp-profile-entry)# directed-ip-range 10.0.30.2 10.0.30.30
switch(config-cp-profile)# entry 2
switch(config-cp-profile-entry)# vlan 50
switch(config-cp-profile-entry)# source-ip 10.0.50.1
switch(config-cp-profile-entry)# directed-ip-list 10.0.50.2
switch(config-cp-profile-entry)# directed-ip-list 10.0.50.7

```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-cp- profile-entry	Administrators or local user group members with execution rights for this command.

probe-duration

Syntax

```

probe-duration [periodic | client-detection]
no probe duration [periodic | client-detection]

```

Description

Configures how long the switch will probe for clients. The switch stops probing once the client is detected on a port, or continues probing if one or more clients are detected on the port. The timer default is periodic.

Example

Configuring the switch to stop probing once a client is detected on the port:

```
switch(config)# client probe
switch(config-cp)# probe-duration client-detection
```



NOTE

When probe duration is configured as "client-detection", the probe packets will stop after the first client is detected on the interface. If there are other silent clients on the same interface, they will no longer see any probe packets and may not awaken. It is recommended to use the probe duration as "periodic" if multiple silent clients can be connected on the same interface.

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-cp	Administrators or local user group members with execution rights for this command.

probe-interval

Syntax

```
probe-interval <ID>
no probe-interval <ID>
```

Description

Configures the time period, in seconds, for the switch to wait before probing for clients again.

Parameter	Description
<ID>	Specifies the interval between consecutive probes, in seconds. Default: 60 seconds.

Example

Configuring the switch to send probe packets every 30 seconds:

```
switch(config)# client probe
switch(config-cp)# probe-interval 30
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-cp	Administrators or local user group members with execution rights for this command.

probe-type

Syntax

```
probe-type {ping-broadcast | ping-unicast | ping-sweep | garp}
```

Description

Configures up to multiple packet types to use for probing. The switch sends a packet of each configured type per probe cycle.

Example

Configuring the switch to send probe of all packet types:

```
switch(config)# client probe
switch(config-cp)# probe-type ping-broadcast ping-unicast ping-sweep garp
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-cp	Administrators or local user group members with execution rights for this command.

(local) profile

Syntax

```
profile <PROFILE-NAME>  
no profile <PROFILE-NAME>
```

Description

Creates a local or remote profile and enters the profile context.

Parameter	Description
<i><PROFILE-NAME></i>	Specifies the name of the profile.

Example

Configuring a new local profile:

```
switch(config)# client probe  
switch(config-cp)# profile WakeUp-Clients  
switch(config-cp-profile)#
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-cp	Administrators or local user group members with execution rights for this command.

remote-server refresh interval

Syntax

```
remote-server refresh interval <ID>  
no remote-server refresh interval <ID>
```

Description

Configures the time period, in seconds, when the switch will periodically obtain the latest version of the probe profile file from the remote location.

Parameter	Description
<ID>	Time interval, in seconds, to wait before refreshing the client probing data from the source. Default: 3600 seconds

Example

Configuring the switch to refresh the probe data from the remote location every 3600 seconds:

```
switch(config)# client probe  
switch(config-cp)# remote-server refresh-interval 3600
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-cp	Administrators or local user group members with execution rights for this command.

show client probe statistics

Syntax

```
show client probe statistics [interface <IFRANGE>]
```

Description

Displays the statistics associated with client probing.

Parameter	Description
<IFRANGE>	Specifies the interface identifier range.

Example

Showing client probe statistics on all enabled interfaces:

```
switch# show client probe statistics
Client-Probe Statistics
Interface 2/1/10
-----
Ping Unicast      : 0
Ping Broadcast    : 3441
Ping Sweep        : 225482321
GARP              : 225502494
Total Probe Cycles : 107
```

```

Interface 2/1/11
-----
Ping Unicast      : 0
Ping Broadcast    : 3441
Ping Sweep        : 225482321
GARP              : 225502494
Total Probe Cycles : 107
Interface 2/1/11
-----
Ping Unicast      : 0
Ping Broadcast    : 3441
Ping Sweep        : 225482321
GARP              : 225502494
Total Probe Cycles : 107

```

Showing client probe statistics on a given interface:

```

switch# show client probe statistics interface 2/1/10
Client-Probe Statistics
Interface 2/1/10
-----
Ping Unicast      : 0
Ping Broadcast    : 3442
Ping Sweep        : 225550786
GARP              : 225568028
Total Probe Cycles : 107

```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

show client probe profile

Syntax

```
show client probe profile [PROFILE-NAME]
```

Description

Displays the information associated with the client probe profile.

Parameter	Description
<PROFILE-NAME>	Specifies the name of the profile.

Example

Showing all profiles on the switch:

```
switch# show client probe profile
Client-Probe Profile Information
Entries that are not valid are indicated with a '*'.
Name                : probe-info
Type                : Downloaded
Download Status     : Completed
Last Download Attempt : Thu Aug 22 09:08:47 UTC 2024
Last Successful Download : Thu Aug 22 09:08:47 UTC 2024
Profile Status      : In Use
Entry ID: 1
-----
VLAN                : 50
Source MAC          : aa:bb:cc:dd:ee:50
IPv4
-----
Source IP           : 10.0.50.1
Broadcast IP        : 10.0.50.255
Directed IP List    : -
Directed IP Range   : -
Entry ID: 2
-----
VLAN                : 30
Source MAC          : aa:bb:cc:dd:ee:50
IPv4
-----
Source IP           : 10.0.50.1
Broadcast IP        : -
```

```

Directed IP List : 10.0.50.2, 10.0.50.7, 10.0.50.8, 10.0.50.7, ...
Directed IP Range : -
Entry ID: 3
-----
VLAN : 40
Source MAC : aa:bb:cc:dd:ee:50
IPv4
-----
Source IP : 10.0.40.1
Broadcast IP : -
Directed IP List : -
Directed IP Range : 10.0.40.11 - 10.0.40.30

```

Showing a local profile on the switch:

```

switch# show client probe profile
Client-Probe Profile Information
Entries that are not valid are indicated with a '*'.
Name : WakeUp-Clients
Type : Local
Download Status : N/A
Last Download Attempt : N/A
Last Successful Download : N/A
Profile Status : In Use
Entry ID: 1
-----
VLAN : 10
Source MAC : aa:bb:cc:dd:ee:10
IPv4
-----
Source IP : 10.0.10.1
Broadcast IP : 10.0.10.255
Directed IP List : -
Directed IP Range : -
Entry ID: 2
-----
VLAN : 30
Source MAC : aa:bb:cc:dd:ee:30
IPv4
-----
Source IP : 10.0.30.1
Broadcast IP : 10.0.30.255
Directed IP List : -
Directed IP Range : 10.0.30.2 - 10.0.30.30
Entry ID: 3

```

```

-----
VLAN                      : 50
Source MAC                 : aa:bb:cc:dd:ee:50
IPv4
-----
Source IP                  : 10.0.50.1
Broadcast IP               : 10.0.50.255
Directed IP List           : 10.0.50.2, 10.0.50.7
Directed IP Range          : 10.0.50.11 - 10.0.50.30

```

Showing a local profile with invalid entries on the switch:

```

switch# show client probe profile
Client-Probe Profile Information
Entries that are not valid are indicated with a '*'.
Name                      : WakeUp-Clients
Type                      : Local
Download Status           : N/A
Last Download Attempt     : N/A
Last Successful Download  : N/A
Profile Status            : Error
Entry ID: 1
-----
VLAN                      : 10
Source MAC                 : aa:bb:cc:dd:ee:10
IPv4
-----
Source IP                  : 10.0.10.1
Broadcast IP               : 10.0.10.255
Directed IP List           : -
Directed IP Range          : -
Entry ID: 2*
-----
VLAN                      : 30
Source MAC                 : aa:bb:cc:dd:ee:30
IPv4
-----
Source IP                  : 10.0.30.1
Broadcast IP               : -
Directed IP List           : -
Directed IP Range          : -
Entry ID: 3
-----
VLAN                      : 50
Source MAC                 : aa:bb:cc:dd:ee:50

```

```
IPv4
-----
Source IP           : 10.0.50.1
Broadcast IP        : 10.0.50.255
Directed IP List    : 10.0.50.2, 10.0.50.7
Directed IP Range   : 10.0.50.11 - 10.0.50.30
Entry ID: 4*
-----
VLAN                : -
Source MAC          : aa:bb:cc:dd:ee:70
IPv4
-----
Source IP           : 10.0.70.1
Broadcast IP        : 10.0.70.255
Directed IP List    : 10.0.70.2, 10.0.70.7
Directed IP Range   : 10.0.70.11 - 10.0.70.30
```

Showing a specific profile on the switch:

```
switch# show client probe profile remote
Client-Probe Profile Information
Entries that are not valid are indicated with a '*'.
Name           : remote
Type           : Downloaded
Download Status : Failed, VRF not operational
Last Download Attempt : Thu Aug 22 08:08:47 UTC 2024
Last Successful Download : N/A
Profile Status  : Pending
switch# show client probe profile remote
Client-Probe Profile Information
Entries that are not valid are indicated with a '*'.
Name           : remote
Type           : Downloaded
Download Status : Completed
Last Download Attempt : Thu Aug 22 08:10:47 UTC 2024
Last Successful Download : Thu Aug 22 08:10:47 UTC 2024
Profile Status  : Error, File parsing failed
switch# show client probe profile remote
Client-Probe Profile Information
Entries that are not valid are indicated with a '*'.
Name           : remote
Type           : Downloaded
Download Status : Pending, VRF not available
Last Download Attempt :
Last Successful Download :
```

```

Profile Status          : Pending
switch# show client probe profile remote
Client-Probe Profile Information
Entries that are not valid are indicated with a '*'.
Name                    : remote
Type                    : Downloaded
Download Status         : Failed, Connection timeout
Last Download Attempt   : Thu Aug 22 09:02:47 UTC 2024
Last Successful Download : Thu Aug 22 08:35:22 UTC 2024
Profile Status          : In Use

```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

show running-config

Syntax

```
show running-config
```

Description

Shows the current running configuration.

Example

Showing the current running configuration for a local profile:

```

switch# show running-config
Current configuration:
!

```

```

...
client probe
  probe-type ping-broadcast ping-unicast ping-sweep garp
  probe-interval 50
  probe-duration client-detection
  vlan-list unsecure port secure port
  profile WakeUp-Clients
    entry 1
      vlan 30
      source-mac aa:bb:cc:dd:ee:30
      source-ip 10.0.30.1
      broadcast-ip 10.0.30.255
      directed-ip-range 10.0.30.2 10.0.30.30
    entry 2
      vlan 50
      source-ip 10.0.50.1
      directed-ip-list 10.0.50.2
      directed-ip-list 10.0.50.7
enable
...
interface 1/1/1
  no shutdown
  no routing
  vlan access 1
  client probe enable
...

```

Showing the running configuration for a remote profile:

```

switch# show running-config
Current configuration:
!
...
client probe
  associate profile sftp user admin password plaintext xyz host 20.20.20.1
port 3333 file      client-info.json vrf cp
  remote-server refresh-interval 80
  probe-type ping-broadcast ping-unicast ping-sweep garp
  probe-interval 50
  probe-duration client-detection
  vlan-list unsecure port secure port
  enable
...
interface 1/1/1
  no shutdown

```

```
no routing
vlan access 1
client probe enable
```

Showing running configuration for autogen client probe profile:

```
switch# show running-config client-probe
client probe
  autogen-probe-file
    name 8m-VSF
    vlans 33,3000-3002
  autogen-enable
enable
```

Showing running configuration for all:

```
switch# show running-config all
Current configuration:
!
...
client probe
  no associate profile
  remote-server refresh-interval 300
  no probe-type
  probe-interval 60
  probe-duration periodic
  no enable
  ...
interface lag 1
  no shutdown
  no routing
  vlan access 1
  ...
  no client probe enable
interface 1/1/1
  no shutdown
  no routing
  vlan access 1
  ...
  no client probe enable
  ...
...
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

vlan-list

Syntax

```
vlan-list [unsecure {port|all}] [secure {port|all}]  
no vlan-list [unsecure {port|all}] [secure {port|all}]
```

Description

Configures VLANs in the profile to use when probing on the port. The VLANs can be configured for two modes of the port. The first mode, "unsecured" is when the port is not secured by port-access (i.e., no authenticated client on the port). the second mode, "secured" is when the port is secured by port-access (i.e., there is at least one authenticated client on the port). The VLANs can be configured differently for each mode. Use "all" if the switch must send probes for all the VLANs in the profile and "port" when the switch must send probes for only those VLANs of which the port is a member.

The **no** form disables VLANs in the profile to use when probing on the port.

Example

Configuring the switch to use VLANs associated with the port for probing in both secure and unsecure mode:

```
switch(config) # client probe  
switch(config-cp) # vlan-list unsecure port secure port
```

Configuring the switch to use all the VLANs associated with the port for probing when not secured, and switch to VLANs associated with the port once secured:


```
switch(config)# client probe
switch(config-cp)# vlan-list unsecure all secure port
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-cp	Administrators or local user group members with execution rights for this command.

Telnet access

Telnet server enables switches to accept Telnet connections from clients to manage the switch. The user authentication is password based authentication (RADIUS, TACACS+ or locally stored password). The Telnet server can be enabled on a desired VRF using the `telnet server` command. The maximum number of Telnet sessions per VRF is five.

In the default configuration, Telnet access is disabled.

Subtopics

Telnet commands

Telnet commands

Select a command from the list in the left navigation menu.

Subtopics

show telnet server sessions

show telnet server

telnet server

show telnet server sessions

Syntax

```
show telnet server sessions [vrf <VRF-NAME> | all-vrfs]
```

Description

Shows all active Telnet sessions for the specified VRF or all VRFs. If no VRF is provided, the Telnet sessions on the **default** VRF are shown.

Parameter	Description
vrf <VRF-NAME>	Specifies the Telnet sessions for a specific VRF.
all-vrfs	Specifies the Telnet sessions for all VRFs

Examples

Showing the Telnet session on the **default** VRF:

```
switch(config)# show telnet server sessions
TELNET sessions on VRF default:
  IPv4 TELNET Sessions:
    Server IP      : 10.1.1.1
    Client IP      : 10.1.1.2
    Client Port    : 58835
```

The following example shows the Telnet sessions on all VRFs. If only the **default** VRF is supported by the switch, only the **default** VRF will display in the output of this command.

```
switch(config)# show telnet server sessions all-vrfs
TELNET sessions on VRF mgmt:
  IPv4 TELNET Sessions:
    Server IP      : 10.1.1.1
    Client IP      : 10.1.1.2
    Client Port    : 58835
TELNET sessions on VRF default:
  IPv4 TELNET Sessions:
    Server IP      : 20.1.1.1
    Client IP      : 20.1.1.2
    Client Port    : 58837
```

Command History

Release	Modification
10.08.1021	Command introduced on the 6200, 6300, 6400 Switch Series.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show telnet server

Syntax

```
show telnet server
```

Description

Shows the Telnet server configuration.

Examples

Showing the Telnet server configuration:

```
switch(config)# show telnet server
TELNET Server Configuration:
  IP Version       : IPv4
  TCP Port         : 23
  Enabled VRFs     : default, vrf1, vrf2,
                   red, green
```

Command History

Release	Modification
10.08.1021	Command introduced on the 6200, 6300, 6400 Switch Series.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

telnet server

Syntax

```
telnet server vrf <VRF-NAME>
no telnet server vrf <VRF-NAME>
```

Description

Enables the Telnet server on the desired VRF. Telnet server is disabled by default.

The **no** form of this command disables the Telnet server.

Parameter	Description
vrf <VRF-NAME>	Specifies the VRF on which the Telnet server will be enabled or disabled.

Examples

Configuring the Telnet server on the **default** VRF:

```
switch(config)# telnet server vrf default
```

Configuring the Telnet server on the **mgmt** VRF:

```
switch(config)# telnet server vrf mgmt
```

Command History

Release	Modification
10.08.1021	Command introduced on the 6200, 6300, 6400 Switch Series.

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

Interface configuration

Subtopics

[Configuring a layer 2 interface](#)

[Single source IP address](#)

[Unsupported transceiver support](#)

[Configuring an interface persona](#)

[Monitor mode](#)

[Interface commands](#)

Configuring a layer 2 interface

Procedure

1. Change to the interface configuration context for the interface with the command `interface`.
2. Set the interface MTU (maximum transmission unit) with the command `mtu`.
3. Review interface configuration settings with the command `show interface`.

Example

```
switch(config) # interface 1/1/1
switch(config-if) # no routing
switch(config-if) # mtu 1900
switch(config) # interface 1/1/1
switch(config-if) # mtu 1900

switch(config) # interface 1/1/1
switch(config-if) # mtu 1900
```

On the 6400 switch series:

```
switch(config) # interface 1/3/1  
switch(config-if) # mtu 1900
```

Single source IP address

Certain IP-based protocols used by the switch (such as RADIUS, sFlow, TACACS, and TFTP), use a client-server model in which the client's source IP address uniquely identifies the client in packets sent to the server. By default, the source IP address is defined as the IP address of the outgoing switch interface on which the client is communicating with the server. Since the switch can have multiple routing interfaces, outgoing packets can potentially be sent on different paths at different times. This can result in different source IP addresses being used for a client, which can create a client identification problem on the server. For example, it can be difficult to interpret system logs and accounting data on the server when the same client is associated with multiple IP addresses.

To resolve this issue, you can use the commands `ip source-interface` and `ipv6 source-interface` to define a single source IP address that applies to all supported protocols (RADIUS, sFlow, TACACS, and TFTP), or an individual address for each protocol. This ensures that all traffic sent by a client to a server uses the same IP address.

Unsupported transceiver support

Transceiver products (optical, DAC, AOCs) that are listed as supported by a switch model are detailed in the Transceiver Guide. Transceiver products that are not listed, are considered unsupported; this would include transceivers that are:

- Non-HPE Aruba Networking branded products
- HPE branded products that were designed for non-AOS-CX switch models (e.g. Comware)
- HPE branded products designated for use in HPE Compute Servers or Storage
- Transceivers originally designated for use in HPE Aruba Networking WLAN controllers or former Mobility Access Switch (MAS) products
- End-of-life HPE Aruba Networking Transceivers

The unsupported transceiver mode (UT-mode) is designed to allow the possible use of these unsupported products. Not all unsupported products can be recognized and enabled; they may be unable to be identified (do not follow the proper MSA standards for identification). These unsupported transceiver products are enabled only on a best-effort basis and there are no guarantees implied for their continued operation.

This feature is enabled by default. A periodic system log will be generated by default at an interval of 24 hours listing the ports on which unsupported transceivers are present. The log interval is configurable and can be disabled by setting the log-interval to **none**.

Configuring an interface persona

A persona is a template interface. It is a mechanism intended to ease the process of configuring one or multiple interfaces that behave in the same manner. For example, a persona could be an uplink interface with a well-known collection of VLANs to which it belongs. Instead of manually configuring each interface one-by-one, the persona collects the common configuration settings and allows them to be applied on the interface or a collection of interfaces.

Persona configuration is similar to configuring other interfaces. Most of the commands in the interface context are available. Because of the nature of the commands, several of them do not apply to the persona (for example, applying the same IP address configuration to several ports) and have been removed from the available list.

Configuration AAA and port-access commands is not supported within the interface context when configuring a persona:

```
switch(config)# interface persona access
switch(config-if)#port-access <command>
switch(config-if)#aaa <command>
```

In addition, the following commands are not supported:

```
aaa authentication port-access dot1x authenticator macsec
aaa authentication port-access dot1x supplicant
aaa authentication port-access dot1x authenticator mka cak-length
arp ipv4 <IPV4_ADDR> mac <MAC_ADDR>
downshift-enable
energy-efficient-ethernet
error-control { auto | none | base-r-fec | rs-fec }
ip bootp-gateway <IPV4-ADDR>
ip forward-protocol udp <IPV4-ADDR> {<PORT-NUM> | <PROTOCOL>}
ip helper-address <IPV4-ADDR> [vrf <VRF-NAME>]
ip igmp router-alert-check [enable | disable]
ip igmp snooping [auto vlan <VLAN-LIST>]
ip igmp snooping [blocked vlan <VLAN-LIST>]
ip igmp snooping [fastleave vlan <VLAN-LIST>]
ip igmp snooping [forced-fastleave vlan <VLAN-LIST>]
ip igmp snooping [forward vlan <VLAN-LIST>]
ip ospf <PROCESS-ID> area <AREA-ID>
ip ospf passive
ip rip <PROCESS-ID> {all-ip | ip-address}
ip rip all-ip disable
ip rip all-ip enable
ip rip all-ip receive disable
ip rip all-ip send disable
```

```

    ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>}
egress <PORT-NUM>
    ipv6 mld snooping [auto vlan <VLAN-LIST>]
    ipv6 mld snooping [blocked vlan <VLAN-LIST>]
    ipv6 mld snooping [fastleave vlan <VLAN-LIST>]
    ipv6 mld snooping [forced-fastleave vlan <VLAN-LIST>]
    ipv6 mld snooping [forward vlan <VLAN-LIST>]
    ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
    ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>
    ipv6 ospfv3 passive
    ipv6 ripng <PROCESS-ID>
    lacp port-id <PORT-ID>
    lacp port-priority <PORT-PRIORITY>
    lag <ID>
    link-poe
    lldp dot3 poe
    lldp med poe
    lldp med poe [priority-override]
    persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
    port-access port-security sticky-learn mac-address
    power-over-ethernet
    power-over-ethernet allocate-by {usage | class}
    power-over-ethernet assigned-class {3 | 4 | 6 | 8}
    power-over-ethernet pd-class-override
    power-over-ethernet power-pairs {alt-a|alt-a-and-alt-b}
    power-over-ethernet pre-std-detect
    power-over-ethernet priority {critical|high|low}
    ptp lag-role {primary | secondary}
    spanning-tree cost <PORT-COST>
    spanning-tree instance <INSTANCE-ID> cost <PORT-COST>
    spanning-tree instance <INSTANCE-ID> port-priority <PRIORITY-MULTIPLIER>
    spanning-tree port-priority <PRIORITY-MULTIPLIER>
    spanning-tree vlan <A:1-4094> cost <0-200000000>
    spanning-tree vlan <A:1-4094> port-priority <0-15>
    split [confirm]
    track by <OBJECT-ID>
    vsx-sync {access-lists | qos | rate-limits | vlans | policies | irdp}

```

Modes

There are three supported modes:

1. Copy—The **copy** mode is a one-step configuration that copies the persona configuration to an interface. Further changes to the persona will not be applied to the interfaces using that mode.

2. **Attach**—Unlike the **copy** mode, besides applying the configuration to the interface immediately, the **attach** mode also follows the persona configuration. It means that the subsequent changes to the persona will be applied to the interfaces attached to it.

3. **Label**—The **label** mode follows the interface persona configuration. The two exceptions are **copy** and **attach** modes. The **label** mode classifies the purpose of an interface and maintains functionality without requiring a persona to exist.

Predefined and custom persona names

There are two predefined interface persona names:

- uplink
- access

These names have no predefined configuration. To use them, they must be manually configured as needed. You can also create personas with a custom name. These custom personas can be created and configured in the same manner as the predefined ones. The only difference is the command used to apply them to the interface. The procedure below provides the details.

Creating and configuring an interface persona

About this task

Follow these steps to create and configure an interface persona:

Procedure

1. Create the interface **persona** with the command **interface persona < PERSONA-NAME >**.
2. Set the interface configuration as any other physical interface.
3. Review the configuration with the command **show running-configuration current-context**.
4. Switch to an interface context with the command **interface < PORT >**.
5. Apply the persona configuration to the interface and set the mode with the command **persona custom <PERSONA-NAME> <mode>**. Note that **<custom>** is listed among the three arguments required for configuration.

Results

For information on this feature, see the related video on the [**AirHeads Broadcasting Channel**](#).



NOTE

All interface configurations related to Device-Profile are supported persona interface.

Examples

To copy a predefined persona name configuration to an interface:

1. Configure the interface persona:

```
switch(config)# interface persona uplink
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# vlan access 100
switch(config-if)# exit
```

2. Apply the configuration with **copy** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona uplink copy
switch(config-if)# exit
```

To attach a custom persona name to several interfaces simultaneously:

1. Configure the interface persona:

```
switch(config)# interface persona mytemplate
switch(config-if)# no shutdown
switch(config-if)# vrf attach upstream
switch(config-if)# exit
```

2. Apply the configuration with **attach** mode:

```
switch(config)# interface 1/1/1-1/1/24
switch(config-if)# persona custom mytemplate attach
switch(config-if)# exit
```

To label a custom persona name to an interface:

1. Apply the configuration with **label** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona custom mypersona
switch(config-if)# exit
link-poe
lldp dot3 poe
lldp med poe
lldp med poe [priority-override]
persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
port-access port-security sticky-learn mac-address
power-over-ethernet
power-over-ethernet allocate-by {usage | class}
power-over-ethernet assigned-class {3 | 4 | 6 | 8}
power-over-ethernet pd-class-override
power-over-ethernet power-pairs {alt-a|alt-a-and-alt-b}
power-over-ethernet pre-std-detect
power-over-ethernet priority {critical|high|low}
```

```

ptp lag-role {primary | secondary}
spanning-tree cost <PORT-COST>
spanning-tree instance <INSTANCE-ID> cost <PORT-COST>
spanning-tree instance <INSTANCE-ID> port-priority <PRIORITY-MULTIPLIER>
spanning-tree port-priority <PRIORITY-MULTIPLIER>
spanning-tree vlan <A:1-4094> cost <0-200000000>
spanning-tree vlan <A:1-4094> port-priority <0-15>
split [confirm]
track by <OBJECT-ID>
vsx-sync {access-lists | qos | rate-limits | vlans | policies | irdp}

```

Modes

There are three supported modes:

1. Copy—The **copy** mode is a one-step configuration that copies the persona configuration to an interface. Further changes to the persona will not be applied to the interfaces using that mode.
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There are two predefined interface persona names:

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Creating and configuring an interface persona

About this task

Follow these steps to create and configure an interface persona:

Procedure

1. Create the interface **persona** with the command **interface persona < PERSONA-NAME >**.
2. Set the interface configuration as any other physical interface.

3. Review the configuration with the command **show running-configuration current-context**.
4. Switch to an interface context with the command **interface < PORT >**.
5. Apply the persona configuration to the interface and set the mode with the command **persona custom <PERSONA-NAME> <mode>**. Note that **<custom>** is listed among the three arguments required for configuration.

Results

For information on this feature, see the related video on the [AirHeads Broadcasting Channel](#).



NOTE

All interface configurations related to Device-Profile are supported persona interface.

Examples

To copy a predefined persona name configuration to an interface:

1. Configure the interface persona:

```
switch(config)# interface persona uplink
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# vlan access 100
switch(config-if)# exit
```

2. Apply the configuration with **copy** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona uplink copy
switch(config-if)# exit
```

To attach a custom persona name to several interfaces simultaneously:

1. Configure the interface persona:

```
switch(config)# interface persona mytemplate
switch(config-if)# no shutdown
switch(config-if)# vrf attach upstream
switch(config-if)# exit
```

2. Apply the configuration with **attach** mode:

```
switch(config)# interface 1/1/1-1/1/24
switch(config-if)# persona custom mytemplate attach
switch(config-if)# exit
```

To label a custom persona name to an interface:

1. Apply the configuration with **label** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona custom mypersona
switch(config-if)# exit
```

Monitor mode

Monitor mode displays interface statistics in real time, updating frequently. The update interval varies by product, number of visible ports, and display options in use. View the current update interval in the help menu.

Formatting options can be supplied either at command launch or while monitor mode is active. Press **?** to see the available formatting options available in the current command context. Exit the help menu with **q**.

Monitor mode detects terminal size to adjust the number of interfaces and statistics viewable on screen. Additional interfaces or statistics columns can be viewed with the navigation keys: Arrow keys, PageUp, PageDown, Home, and End. Serial console does not automatically detect terminal setting changes, resulting in unexpected output. Recommended recovery steps are to exit and restart.

Monitor mode refreshes data automatically until exited with **q**.

Interface commands

Select a command from the list in the left navigation menu

Subtopics

allow-unsupported-transceiver

default interface

description

energy-efficient-ethernet

flow-control

interface

interface loopback

interface vlan

interpacket-gap

ip address

ip mtu

ip tcp mss

ip unnumbered

ipv6 address

ipv6 source-interface

ipv6 tcp mss

mtu
persona
rate-interval
routing
show allow-unsupported-transceiver
show interface
show interface dom
show interface energy-efficient ethernet
show interface flow-control
show interface link-diagnostics
show interface statistics
show interface transceiver
show interface supported-transceivers
show interface utilization
show ip interface
show ip source-interface
show ipv6 interface
show ipv6 source-interface
shutdown
speed

allow-unsupported-transceiver

Syntax

```
allow-unsupported-transceiver [confirm | log-interval {none | <INTERVAL>}]  
no allow-unsupported-transceiver
```

Description

Allows unsupported transceivers to be enabled or establish connections. Transceivers with speeds up to 100G are enabled by this command.



NOTE

This command is enabled by default, allowing the use of third party transceiver products without adding the command in the configuration. Disabling this command with the **no** form will now disable the command in the running and stored configurations.

The **no** form of this command disallows using unsupported transceivers.

Parameter	Description
<code>confirm</code>	Specifies that unsupported transceiver warnings are to be automatically confirmed.
<code>log-interval none</code>	Disables unsupported transceiver logging.
<code>log-interval <INTERVAL></code>	Sets the unsupported transceiver logging interval in minutes. Default: 1440 minutes. Range: 1440 to 10080 minutes.

Usage

When none of the parameters are specified it will display a warning message to accept the warranty terms. With **confirm** option the warning message is displayed but the user is not prompted to **(y/n)** answering. Warranty terms must be agreed to as part of enablement and the support is on best effort basis.

Examples

Allowing unsupported transceivers with follow-up confirmation:

```
switch(config)# allow-unsupported-transceiver
Warning: The use of unsupported transceivers, DACs, and AOCs is at your
own risk and may void support and warranty. Please see HPE Warranty terms
and conditions.
Do you agree and do you want to continue (y/n)? y
```

Allowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# allow-unsupported-transceiver confirm
Warning: The use of unsupported transceivers, DACs, and AOCs is at your
own risk and may void support and warranty. Please see HPE Warranty terms
and conditions.
```

Configuring unsupported transceiver logging with an interval of every 48 hours:

```
switch(config)# allow-unsupported-transceiver log-interval 2880
```

Disabling unsupported transceiver logging:

```
switch(config)# allow-unsupported-transceiver log-interval none
```

Disallowing unsupported transceivers with follow-up confirmation:

```
switch(config)# no allow-unsupported-transceivers
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,
which could impact network connectivity. Use 'show allow-unsupported-
transceiver'
to identify unsupported transceivers, DACs, and AOCs.
```

```
Ccontinue (y/n)? y
```

Disallowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# no allow-unsupported-transceiver confirm  
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,  
which could impact network connectivity. Use 'show allow unsupported-  
transceiver'  
to identify unsupported transceivers, DACs, and AOCs.  
switch(config)#
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

default interface

Syntax

```
default interface <INTERFACE-ID>
```

Description

Sets an interface (or a range of interfaces) to factory default values.

Parameter	Description
	Specifies the ID of a single interface or range of interfaces. Format: member/slot/port or member/slot/port - member/slot/port to specify a range.

Parameter	Description
<INTERFACE-ID>	

Examples

Resetting an interface:

```
switch(config) # default default interface 1/1/1
```

Resetting an range of interfaces:

```
switch(config) # default default interface 1/1/1-1/1/10
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

description

Syntax

```
description <DESCRIPTION>
no description
```

Description

Associates descriptive information with an interface to help administrators and operators identify the purpose or role of an interface.

The **no** form of this command removes a description from an interface.

Parameter	Description
<code><DESCRIPTION></code>	Specify a description for the interface. Range: 1 to 64 ASCII characters (including space, excluding question mark).

Examples

Setting the description for an interface to **DataLink 01**:

```
switch(config-if) # description DataLink 01
```

Removing the description for an interface.

```
switch(config-if) # no description
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-if</code>	Administrators or local user group members with execution rights for this command.

energy-efficient-ethernet

Syntax

```
energy-efficient-ethernet
no energy-efficient-ethernet
```

Description

Enables auto-negotiation of Energy-Efficient Ethernet (EEE) on an interface. EEE Negotiation is established only on auto-link negotiation with supported link partners.

This command is only available on ports that support Energy-Efficient Ethernet, which includes 1GBASE-T ports, but not SmartRate ports.

The **no** form of this command disables the auto-negotiation.

Examples

Configuring an interface:

```
switch(config) # interface 1/1/1  
switch(config-if) # energy-efficient-ethernet
```

Disabling Energy Efficient Ethernet on an interface:

```
switch(config) # interface 1/1/1  
switch(config-if) # no energy-efficient-ethernet
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

flow-control

Syntax

```
flow-control rxtx  
no flow-control rxtx
```

Description

Command **flow-control** enables negotiation of IEEE 802.3x link-level flow control on the current interface. The switch advertises link-level flow control support to the link partner. The final configuration is determined based on the capabilities of both partners.

Each invocation of this command replaces the previous configuration.

The **no** form of these commands disables any configured flow control on the selected interface.

Parameter	Description
<code>rxtx</code>	Enables the ability to honor received and to transmit IEEE 802.3x LLFC pause frames to the remote device.

Usage (flow control)

- For interfaces that auto-negotiate, link-level flow control is subject to negotiation, plus speed and other parameters. Both ends of the link must negotiate the same flow control mode for it to be applied.
- For interfaces that do not auto-negotiate, the configured link-level flow control mode is always applied and the user is responsible for ensuring that both ends of the link are configured for the same mode.
- All members of a LAG must have the same flow control configuration.
- Lossless flow control is only supported for single destination unicast traffic. Replicated traffic (for example, broadcast, multicast, mirroring) cannot be guaranteed to be lossless.
- Lossless behavior is not supported when operating in a VSF stack configuration.
- Lossless flow control will only operate correctly when both the ingress and egress interfaces have flow control enabled.

Examples

Enabling support for RXTX flow control:

```
switch(config) # interface 1/1/1
switch(config-if) # flow-control txrx
```

Disabling support for RXTX flow control:

```
switch(config) # interface 1/1/1
switch(config-if) # no flow-control txrx

switch(config) # interface 1/1/1
switch(config-if) # flow-control rxtx pool 2
```

```
switch(config)# interface 1/1/1
switch(config-if)# no flow-control rxtx pool 2

switch(config)# interface 1/1/1
switch(config-if)# flow-control priority rx 3,4

switch(config)# interface 1/1/1
switch(config-if)# no flow-control priority rx 3,4
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

interface

Syntax

```
interface <PORT-NUM>
```

Description

Switches to the **config-if** context for a physical port. This is where you define the configuration settings for the logical interface associated with the physical port.

Parameter	Description
	Specifies a physical port number. Format: member/slot/port .

Parameter	Description
<PORT-NUM>	

Examples

Configuring an interface:

```
switch(config) # interface 1/1/1
switch(config-if) #
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

interface loopback

Syntax

```
interface loopback <ID>
no interface loopback <ID>
```

Description

Creates a loopback interface and changes to the **config-loopback-if** context. Loopback interfaces are layer 3.

The **no** form of this command deletes a loopback interface.

Parameter	Description
<INSTANCE>	Specifies the loopback interface ID. Range: 1 to 256

Examples

```
switch# config
switch(config)# interface loopback 1
switch(config-loopback-if)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

interface vlan

Syntax

```
interface vlan <VLAN-ID>
no interface vlan <VLAN-ID>
```

Description

Creates an interface VLAN also known as an SVI (switched virtual interface) and changes to the **config-if-vlan** context. The specified VLAN must already be defined on the switch.

The **no** form of this command deletes an interface VLAN.

Parameter	Description
<VLAN-ID>	Specifies the loopback interface ID.
none	Do not reserve any internal VLANs.

Examples

```
switch# config
switch(config)# vlan 10
switch(config-vlan-10)# exit
switch(config)# interface vlan 10
switch(config-if-vlan)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

interpacket-gap

Syntax

```
interpacket-gap [high average]
no interpacket-gap
```

Description

Configures the interpacket gap (IPG) between transmitted packets for the interface.

The **no** form of this command sets IPG as default.

Parameter	Description
<code>high average</code>	Use a higher average IPG at certain speeds.

Examples

Allow the interface to use a higher IPG.

```
switch(config)# interface 1/1/1
switch(config-if)# interpacket-gap high-average
```

Use default IPG.

```
switch(config)# interface 1/1/1
switch(config-if)# no interpacket-gap
```

Output for **show interface** when **high-average** mode is configured and the port is linked at 1 Gbps.

```
switch(# show interface 1/1/1
Interpacket gap: high-average
```

Command History

Release	Modification
10.13.1060	Command introduced.

Command Information

Platforms	Command context	Authority
6200	<code>config-if</code>	Administrators or local user group members with execution rights for this command.

ip address

Syntax

```
ip address <IP-ADDR>/<MASK>
           [secondary]
no ip address <IP-ADDR>/<MASK>
           [secondary]
```

Description

Sets an IPv4 address for the current layer 3 interface.

The **no** form of this command removes the IPv4 address from the interface.

Parameter	Description
<code><IP-ADDR></code>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100 .
<code><MASK></code>	Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.
<code>secondary</code>	Specifies a secondary IP address. (Supported on 6300, except for S3L75A, S3L76A, S3L77A.)

Examples

```
switch(config)# interface 1/1/1
switch(config-if)# ip address 192.168.100.1/24

switch(config)# interface 1/1/1
switch(config-if)# No ip address 192.168.100.1/24

switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 192.168.20.1/24

switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# ip address 192.168.100.1/24

switch(config)# interface loopback 1
switch(config-loopback-if)# routing
```

```
switch(config-loopback-if) # ip address 192.168.20.1/24
```

Assigning the IP address **192.168.199.1** with a mask of **24** bits to interface VLAN **10**:

```
switch(config) # interface vlan 10
switch(config-if-vlan) # ip address 192.168.199.1/24
```

Removing the IP address **192.168.199.1** with a mask of **24** bits from interface VLAN **10**:

```
switch(config) # interface vlan 10
switch(config-if-vlan) # no ip address 192.168.199.1/24
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip mtu

Syntax

```
ip mtu <VALUE>
no ip mtu
```

Description

Sets the IP MTU (maximum transmission unit) for an interface. This defines the largest IP packet that can be sent or received by the interface. This value should be less than or equal to the overall MTU for the interface.

The **no** form of this command sets the IP MTU to the default value 1500.

Parameter	Description
<VALUE>	Specifies the IP MTU in bytes. Range: 68 to 9198. Default: 1500.

Usage

The IP MTU value for subinterface must be less than or equal to the parent MTU for the subinterface. The subinterface uses its IP MTU value and not the parent IP MTU value.

Examples

Setting the IP MTU to 576 bytes:

```
switch(config-if-vlan)# ip mtu 576

switch(config)# interface 1/1/1.10
switch(config-subif)# ip mtu 6000
```

Command History

Release	Modification
10.08	Subinterface support added.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip tcp mss

Syntax

```
ip tcp mss <VALUE>
no ip tcp mss <VALUE>
```

Description

Sets the TCP (Transmission Control Protocol) MSS (Maximum Segment Size) for an interface. When configured, the MSS option in the TCP header is set in the TCP handshake for negotiation during connection establishment. Once the TCP session is established, changing the TCP MSS value on the interface will not affect the existing session.

The **no** form of this command removes the TCP MSS configuration from the interface.



NOTE

TCP MSS is supported on ROP, SVI, subinterface, and L3 GRE tunnel interface.

Parameter	Description
<VALUE>	Specifies the IPv4 TCP MSS in bytes. Range: 68 to 65495.

Examples

Configuring the IPv4 TCP MSS for the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip tcp mss 1460
```

Removing the IPv4 TCP MSS configuration from the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip tcp mss 1460
```

Command History

Release	Modification
10.14.1000	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	<code>config-if-vlan</code>	Administrators or local user group members with execution rights for this command.

ip unnumbered

Syntax

```
ip unnumbered <ifname>
no ip unnumbered <ifname>
```

Description

The IP unnumbered feature allows the switch to process IP packets without configuring a unique IP address on an interface by telling the router to use the IP address of another selected interface as the address for that link. Once set, this port will borrow the user-configured primary IPv4 address from an lender interface and use that IP address in its L3 control plane exchange.

The **no** version of this command removes IP unnumbered support for this port. Once unset, this port will remove borrowed IPv4 address.

Parameter	Description
<code><Ifname ></code>	Name of the interface

Usage

The port where IP unnumbered is configured is known as the borrower interface. Only route-only ports can borrow an address.

The port the address is borrowed from is known as the lender interface. Only a loopback interface can be used as a lender interface. The unnumbered interface and lender interface must belong to the same VRF. Additionally, the lender interface for an IP unnumbered interface cannot itself configured as unnumbered.

The following caveats apply to this feature:

- Only the primary IPv4 address can be borrowed on an IP unnumbered interface. You cannot configure a secondary IP address for an interface that is already configured as an IP unnumbered interface.

- If an IP address is configured on an interface that is already configured as an IP unnumbered interface, it will function as a normal IP interface.
- An IPv6 unnumbered configuration on the interface is not supported. If **ip unnumbered** is enabled on the interface, only the IPv4 address is borrowed. However, Any IPv6 configurations can co-exist with an ip unnumbered interface.
- An IP unnumbered interface cannot have multiple lender interfaces for the same address family.
- The **ip unnumbered** command requires that the interface borrowing the address is a routed port. The **IP unnumbered** feature is only supported on devices connected with point-to-point interfaces.
- An unnumbered IP address can not be used as tunnel source.
- The unnumbered interface and lender interface must belong to the same VRF.
- Deleting a lender interface removes all the associated IP unnumbered references.

Examples

Configuring IP unnumbered for a route-only port:

```
switch(config)# interface 1/1/1
switch(config-if)# ip unnumbered loopback1
```

The following examples removes IP unnumbered support for this port. Once unset, this port will remove the borrowed IPv4 address.

```
switch(config)# interface 1/1/1
switch(config-if)# no ip unnumbered loopback1
```

If an IP address is configured on an interface that is already configured as an IP unnumbered interface (or vice versa), both configurations can coexist on the interface. The switch's arbitration logic ensures that static IP configuration is given priority over unnumbered.

```
switch(config)# interface 1/1/1
switch(config-if)# ip unnumbered interface loopback 1
switch(config-if)# ip address 10.16.1.1/24
switch(config-if)# ip address 192.168.1.2/24 secondary
or
switch(config)# interface 1/1/1
switch(config-if)# ip address 10.16.1.1/24
switch(config-if)# ip unnumbered interface loopback 1
switch(config-if)# ip address 192.168.1.2/24 secondary
switch(config)# sh running-config interface
switch(config-if)# do sh running-config interface 1/1/1
interface 1/1/1
  no shutdown
  ip address 10.16.1.1/24
  ip address 192.168.1.2/24 secondary
```

```
ip unnumbered interface loopback 1
! ip unnumbered is ignored when static ip is configured
```

If loopback lender interface does not exist during IP unnumbered configuration, a prompt will be displayed in CLI to allow the user to create that interface.

```
switch(config)# interface 1/1/1
switch(config-if)# ip unnumbered interface loopback 1
Lender port not found.
Do you want to create (y/n)? y
switch(config-if)#
Failure case:
switch(config)# ip unnumbered interface loopback 1
Lender port not found.
Do you want to create (y/n)? y
Lender port: loopback1 creation failed
switch(config-if)
```

Command History

Release	Modification
10.14.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-if	Administrators or local user group members with execution rights for this command.

ipv6 address

Syntax

```
ipv6 address <IPV6-ADDR>/<MASK>{eui64 | [tag <ID>]}
no ipv6 address <IPV6-ADDR>/<MASK>
```

Description

Sets an IPv6 address on the interface.

The **no** form of this command removes the IPv6 address on the interface.



NOTE

This command automatically creates an IPv6 link-local address on the interface. However, it does not add the **ipv6 address link-local** command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the **ipv6 address link-local** command.

Parameter	Description
<code><IPV6-ADDR></code>	Specifies the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55 .
<code><MASK></code>	Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.
<code>eui64</code>	Configure the IPv6 address in the EUI - 64 bit format.
<code>tag <ID></code>	Configure route tag for connected routes. Range: 0 to 4294967295. Default: 0.

Examples

Setting the IPv6 address **2001:0db8:85a3::8a2e:0370:7334** with a mask of 24 bits:

```
switch(config-if) # ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Removing the IP address **2001:0db8:85a3::8a2e:0370:7334** with mask of 24 bits:

```
switch(config-if) # no ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

ipv6 source-interface

Syntax

```
ipv6 source-interface {sflow | tftp | radius | tacacs | ntp | syslog |  
ubt |  
  
dhcp-relay|  
simplivity | http | sftp-scp | ssh-client  
| ipfix  
| dns | all} {interface <IFNAME> | <IPV6-ADDR>} [vrf <VRF-NAME>]  
no ipv6 source-interface {sflow | tftp | radius | tacacs | ntp | syslog |  
ubt |  
  
dhcp-relay|  
simplivity | http | sftp-scp | ssh-client  
| ipfix  
| dns | all} [interface <IFNAME> | <IPV6-ADDR>] [vrf <VRF-NAME>]
```

Description

Sets a single source IPv6 address for a feature on the switch. This ensures that all traffic sent the feature has the same source IPv6 address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IPv6 addresses: either by specifying a static IPv6 address, or by using the address assigned to a switch interface. If you define both options, then the static IPv6 address takes precedence.

The `no` form of this command deletes the single source IPv6 address for all supported protocols, or a specific protocol.

Parameter	Description
sflow tftp radius tacacs ntp syslog ubt dhcp-relay simplivity http sftp-scp ssh-client ipfix dns all	Sets a single source IPv6 address for a specific protocol. The <code>all</code> option sets a global address that applies to all protocols that do not have an address set.
interface <IFNAME>	Specifies the name of the interface from which the specified protocol obtains its source IPv6 address.
<IPv6-ADDR>	Specifies the source IPv6 address to use for the specified protocol. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the <code>default</code> VRF, if the <code>vrf</code> option is not used). Specify the IP address in IPv6 format (<code>xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx</code>), where <code>x</code> is a hexadecimal number from 0 to F.
vrf <VRF-NAME>	Specifies the name of the VRF from which the specified protocol sets its source IPv6 address.

Examples

Configuring the IPv6 address `2001:DB8::1` as the global single source address:

```
switch# config
switch(config)# ipv6 source-interface all 2001:DB8::1/32
```

Configuring the IPv6 address `2001:DB8::1` on VRF `sflow-vrf` on interface `1/1/2` as the single source address for sFlow:

```
switch(config)# vrf sflow-vrf
switch(config-vrf)# exit
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# vrf attach sflow-vrf
switch(config-if)# ipv6 address 2001:DB8::1/32
switch(config-if)# exit
switch(config)# ipv6 source-interface sflow interface 1/1/2 vrf sflow-vrf
```

Stop the source IPv6 address from using the IP address on interface `1/1/1` on VRF `one`.

```
switch(config)# no ipv6 source-interface all interface 1/1/1 vrf one
```

Clear the source IPv6 address `2001:DB8::1`.

```
switch(config) # no ipv6 source-interface all 2001:DB8::1
```

Command History

Release	Modification
10.17.1000	The ipfix protocol parameter was introduced.
10.16.1000	Updated to include ubt support.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ipv6 tcp mss

Syntax

```
ipv6 tcp mss <VALUE>  
no ipv6 tcp mss <VALUE>
```

Description

Sets the TCP (Transmission Control Protocol) MSS (Maximum Segment Size) for an interface. When configured, the MSS option in the TCP header is set in the TCP handshake for negotiation during connection establishment. Once the TCP session is established, changing the TCP MSS value on the interface will not affect the existing session.

The **no** form of this command removes the TCP MSS configuration from the interface.



NOTE

TCP MSS is supported on ROP, SVI, subinterface, and L3 GRE tunnel interface.

Parameter	Description
<VALUE>	Specifies the IPv6 TCP MSS in bytes. Range: 68 to 65495.

Examples

Configuring the IPv6 TCP MSS for the interface:

```
switch(config) # interface 1/1/1
switch(config-if) # ipv6 tcp mss 1440
```

Removing the IPv6 TCP MSS configuration from the interface:

```
switch(config) # interface 1/1/1
switch(config-if) # no ipv6 tcp mss 1440
```

Command History

Release	Modification
10.14.1000	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

mtu

Syntax

```
mtu <VALUE>
no mtu
```

Description

Sets the MTU (maximum transmission unit) for an interface. This defines the maximum size of a layer 2 (Ethernet) frame. Frames larger than the MTU (1500 bytes by default) are dropped and cause an ICMP fragmentation-needed message to be sent back to the originator.

To support jumbo frames (frames larger than 1522 bytes), increase the MTU as required by your network. A frame size of up to 9198 bytes is supported.

The largest possible layer 1 frame will be 18 bytes larger than the MTU value to allow for link layer headers and trailers.

The **no** form of this command sets the MTU to the default value 1500.

Parameter	Description
<code><VALUE></code>	Specifies the MTU in bytes. Range: 46 to 9198. Default: 1500.

Examples

Setting the MTU on interface **1/1/1** to 1000 bytes:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# mtu 1000
```

Setting the MTU on interface **1/1/1** to the default value:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# no mtu
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

persona

Syntax

```
persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
no persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
```

Description

Associates one of three persona types with an interface to classify the purpose or role of an interface. On the 10000 Switch Series, “access” persona ports are typically connected to workloads / VMs, and the “uplink” (fabric) persona ports are connected to the core / spine.

The **no** form of this command removes the interface persona.

Parameter	Description
access	Selects the access persona type.
uplink	Selects the uplink persona type.
custom <PERSONA-NAME>	Selects the custom persona type with a user - provided name. Range: 1 to 64 printable ASCII characters including space.
copy	Specifies the mode: copies settings from the persona interface of the same name.
attach	Specifies the mode: attaches the specified interface to the persona interface of the same name.

Usage

- If the mode is specified, either copy or attach, the interface configuration is dependent on the interface template whose name is "access", "uplink", or "<PERSONA-NAME>". On the other hand, if the mode is not specified, then the persona is just a label in the interface, and its configuration is not modified even if the interface persona exists. When configuring the mode, one of the following options is possible:

- The **copy** option performs a one-time copy of the template interface. Subsequent changes to the template are not copied and the 'persona' setting is just a label. If the mode is set to **copy** and the interface persona does not exist, then the CLI command fails with the message "Interface persona not found".
- The **attach** option performs a copy of the template interface, and subsequent changes to the template interface configuration are immediately applied to all attached interfaces. The template interface does not need to exist before attaching other interfaces to it. After attaching a template, the copied settings can be modified for an individual interface. However, any change in the attached template will overwrite the modified values with the new template values.
- When a mode is specified, it should match an interface created with the command `interface persona <PERSONA-NAME>`. The only exception to this rule is when the mode is set to `attach` and the persona does not already exist.
- The mode is only available to be configured for an interface that meets the following conditions:
 - IS a physical interface
 - IS NOT a LAG member
 - IS NOT a persona interface

Examples

Configuring an access persona:

```
switch(config) # interface 1/1/1
switch(config-if) # persona access
```

Configuring an uplink persona:

```
switch(config) # interface 1/1/1
switch(config-if) # persona uplink
```

Configuring a custom persona named "mypersona":

```
switch(config) # interface 1/1/1
switch(config-if) # persona custom mypersona
```

Removing the persona setting.

```
switch(config-if) # no persona
```

Copying a predefined persona name configuration to an interface:

1. Configuring the interface persona:


```
switch(config)# interface persona uplink
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# vlan access 100
switch(config-if)# exit
```

2. Applying the configuration from the persona named "mypersona" with **copy** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona custom mypersona copy
switch(config-if)# exit
```

Attaching a custom persona name named "mypersona" to several interfaces simultaneously:

1. Configuring an interface persona named "mypersona":

```
switch(config)# interface persona mypersona
switch(config-if)# no shutdown
switch(config-if)# vrf attach upstream
switch(config-if)# exit
```

2. Applying the "mypersona" configuration with **attach** mode:

```
switch(config)# interface 1/1/1-1/1/24
switch(config-if)# persona custom mypersona attach
switch(config-if)# exit
```

Command History

Release	Modification
10.10	Added optional parameters: attach, copy.
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

rate-interval

Syntax

```
rate-interval <VALUE>
no rate-interval
```

Description

This command sets the time interval to calculate interface rates. Lower intervals are more useful for detecting traffic bursts, but may increase computation load to the overall system. Intervals must be a multiple of five seconds. The command-line interface will not accept a rate interval value that is not a multiple of five.

The **no** form of this command sets the rate collection interval to the default value of 300 seconds.

Parameter	Description
<VALUE>	The statistics rate collection interval in seconds. The supported range is 5 - 300 seconds, where the number of seconds is a multiple of five.

Examples

Setting the rate collection interval to 50 seconds

```
switch(config)# interface 1/1/1
switch(config-if)# rate-interval 50
```

Setting the rate collection interval to the default value:

```
switch(config-if)# no rate-interval
```

The following example shows the command-line interface warning that appears while configuring an invalid rate-interval.

```
switch(config)# interface 1/1/1
switch(config-if)# rate interval 6
The interval must be a multiple of 5.
```

Usage

The rate collection interval must be configured in the multiples of 5. Any other value will be rejected and the CLI will display the error message, **The interval must be a multiple of 5.**

Command History

Release	Modification
10.12.1000	Command supported on all platforms.
10.12	Command Introduced on 6300 and 8360 Switch series.

Command Information

Platforms	Command context	Authority
All platforms	<code>config-if</code>	Administrators or local user group members with execution rights for this command.

routing

Syntax

```
routing  
no routing
```

Description

Enables routing support on an interface, creating a L3 (layer 3) interface on which the switch can route IPv4/IPv6 traffic to other devices.

By default, routing is disabled on all interfaces.

The **no** form of this command disables routing support on an interface, creating a L2 (layer 2) interface.



CAUTION

If you enable this configuration, collection of flow tracking statistics is disabled.

Examples

Enabling routing support on an interface:

```
switch(config-if) # routing
```

Disabling routing support on an interface:

```
switch(config-if)# no routing
```

Command History

Release

Modification

10.07 or earlier

- -

Command Information

Platforms

Command context

Authority

5420

config-if

Administrators or local user group members with execution rights for this command.

6200

show allow-unsupported-transceiver

Syntax

```
show allow-unsupported-transceiver
```

Description

Displays configuration and status of unsupported transceivers.

Examples

Showing unallowed unsupported transceivers:

```
switch(config)# show allow-unsupported-transceiver
Allow unsupported transceivers : no
Logging interval               : 1440 minutes
-----
Port          Type          Status
-----
1/1/31        SFP-SX                    unsupported
1/1/32        SFP-1G-BXD                unsupported
1/1/2         SFP28DAC3                 unsupported
```

Showing allowed unsupported transceivers:

```

switch# show allow-unsupported-transceiver
Allow unsupported transceivers : yes
Logging interval               : 1440 minutes
-----
Port          Type          Status
-----
1/1/31        SFP-SX          unsupported-allowed
1/1/32        SFP-1G-BXD      unsupported-allowed
1/1/2         SFP28DAC3       unsupported

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

show interface

Syntax

```

show interface [<IFNNAME>|<IFRANGE>] [brief | physical]
show interface [<IFNNAME>|<IFRANGE>] [extended [non-zero] | [human-readable]]
show interface [<IFNNAME>] monitor [human-readable]
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief]
show interface lag [<LAG-ID>] [extended [non-zero] | [human-readable]]
show interface lag [<LAG-ID>] monitor [human-readable]

```

Description

Shows active configurations and operational status information for interfaces.

Parameter	Description
<code><IFNAME></code>	Specifies a interface name.
<code><IFRANGE></code>	Specifies the port identifier range.
<code>brief</code>	Shows brief info in tabular format.
<code>physical</code>	Shows the physical connection info in tabular format.
<code>extended</code>	Shows additional statistics, including the tx filtered and rx filtered counters. - Rx filter packets are protocol packets received when the protocol is disabled on the switch and there is only one port in the VLAN. Protocols include OSPF, PIM, RIP, LACP, and LLDP. - An example of a Tx filtered packet would be a multicast packet being filtered from going out of the ingress port.
<code>human-readable</code>	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G. This is available only in the CLI interface output.
<code>non-zero</code>	Shows only non zero statistics.
<code>LAG</code>	Shows LAG interface information.
<code>monitor</code>	Continuously monitor interface statistics.
<code>LOOPBACK</code>	Shows loopback interface information.
<code>TUNNEL</code>	Shows tunnel interface information.
<code>VLAN</code>	Shows VLAN interface information.
<code><LAG-ID></code>	Specifies the LAG number. Range: 1 - 256

Parameter	Description
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0 - 255
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1 - 255
<VLAN-ID>	Specifies the VLAN ID. Range: 1 - 4094
VXLAN	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1

Examples

```
switch# show interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Link state: up for 2 days (since Sun Jun 21 05:30:22 UTC 2020)
Link transitions: 1
Description: backup data center link
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 1500
Type 1GbT
Full-duplex
qos trust none
Speed 1000 Mb/s
Auto-negotiation is on
Flow-control: off
Error-control: off
                Energy-Efficient Ethernet is enabled
                MDI mode: MDIX
L3 Counters: Rx Enabled, Tx Enabled
Rate collection interval: 300 seconds
Rates                RX                TX                Total (RX+TX)
```

Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00
Statistics	RX	TX	Total
Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
L3 Packets	0	0	0
L3 Bytes	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0
Other	0	0	0

Showing information when interface 1/1/1 is configured:

MDI mode: MDIX

VLAN Mode: native-untagged

Native VLAN: 1

Allowed VLAN List: all

Rate collection interval: 300 seconds

Rates	RX	TX	Total (RX+TX)
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00
Statistics	RX	TX	Total
Packets	0	0	0

Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

Showing information when the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

Showing information when the interface is currently linked with energy-efficient-ethernet negotiated:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Energy-Efficient Ethernet is enabled and active
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is down
Admin state is up
State information: Disabled by VSX
Link state: down for 3 days (since Tue Mar 16 05:20:47 UTC 2021)
Link transitions: 0
Description:
Hardware: Ethernet, MAC Address: 04:09:73:62:90:e7
MTU 1500
Type SFP+DAC3
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: 1502-1505
Rate collection interval: 300 seconds
```

Rate		RX		TX		Total
(RX+TX)						
-----				-----		

Mbits / sec		0.00		0.00		
0.00						
KPkts / sec		0.00		0.00		
0.00						
Unicast		0.00		0.00		
0.00						
Multicast		0.00		0.00		
0.00						
Broadcast		0.00		0.00		
0.00						
Utilization		0.00		0.00		
0.00						
Statistic		RX		TX		
Total						
-----				-----		

Packets		0				
0	0					
Unicast		0				
0	0					
Multicast		0				
0	0					
Broadcast		0				
0	0					
Bytes		0				
0	0					
Jumbos		0				
0	0					
Dropped		0				
0	0					
Pause Frames		0				
0	0					
Errors		0				
0	0					
CRC/FCS		0				
n/a	0					
Collision		n/a				
0	0					
Runts		0				

```
n/a          0
Giants          0
n/a          0
```

Showing information when the interface is configured with EEE and the EEE has auto-negotiated:

```
switch(config-if) # show interface 1/1/1 physical
```

Control	EEE	Link	Admin	Speed	Flow-
Port	Type	PoE Power	Config	Status Config	Port
Config	Status Config	Status (Watts)	State	Information	Description
1/1/1	1GbT	up	up	1G	auto
off	on	on	--	10M/100M/1G	--

Showing the monitor information:



NOTE

In monitor mode, the CLI refreshes data automatically until it is exited by entering **q**. Pressing **?** opens the help menu to display which options are available in this context.

```
Interface 1/1/1 is up
```

Rate	RX	TX	Total
(RX+TX)			
MBits / sec	30196.43	30196.43	
60392.85			
MPkts / sec	58977.39	58977.40	
117954.79			
Unicast	0.00	0.00	
0.00			
Multicast	58977.39	58977.40	
117954.79			
Broadcast	0.00	0.00	
0.00			
Utilization %	75.49	75.49	
150.98			
Statistic	RX	TX	Total
(RX+TX)			

```

Packets                4756527649                4756527865
9513055514
Unicast                0
0                      0
Multicast              4756527649                4756527865
9513055514
Broadcast              2
0                      2
Bytes                 304417778668                304417795428
608835574096
Jumbos                0
0                      0
Dropped                0                19028847730
19028847730
Pause Frames          0
0                      0
Errors                0
0                      0
CRC/FCS               0
n/a                   0
help: ?, quit: q
Help for Interface Monitor
h  Toggle human-readable mode
c  Clear interface statistics
Does not apply to rates
Arrows, PgUp, PgDn, Home, End
Navigate interface statistics
Delay: 2
help: ?, quit: q

```

Showing the output for interface 1/1/1 in human-readable format:



NOTE

In human-readable format, the **< 1** symbol for **Utilization** indicates that the amount of packets is between zero and one. This is true in cases where the number of bytes increases but the number of packets and the **Utilization** value is not displayed even in the normal output, where the human-readable parameter is not included in the command.

```

switch(config-if)# show interface 1/1/1 human-readable
Interface 1/1/1 is up
Rate                RX                TX                Total
(RX+TX)
-----
-----

```

Bits / sec		3M	
3M	6M		
Pkts / sec		316	316
633			
Unicast		319	319
638			
Multicast		0	
0	0		
Broadcast		0	
0	0		
Utilization %		< 1	< 1
< 1			
Statistic		RX	TX
Total			

Packets		577K	
577K	1M		
Unicast		577K	
577K	1M		
Multicast		0	
51	51		
Broadcast		0	
15	15		
Bytes		744M	
745M	1G		
Jumbos		0	
0	0		
Dropped		0	
0	0		
Filtered		0	
0	0		
Pause Frames		0	
0	0		
Errors		0	
0	0		
CRC/FCS		0	
n/a	0		
Collision		n/a	
0	0		
Runts		0	
n/a	0		

Giants	0
n/a	0

Showing information about extended counters:



NOTE

The output of the `show interface extended` command varies depending on the switch model and configuration.

```
switch(config-if)# show interface 1/1/17 extended
```

```
-----  
Interface 1/1/17  
-----
```

Statistics	Value
Dot1d Tp Port In Frames	547
Dot1d Tp Port Out Frames	608
Dot3 In Pause Frames	0
Dot3 Out Pause Frames	0
Ethernet Stats Broadcast Packets	19
Ethernet Stats Bytes	40162
Ethernet Stats Packets	342

```
...
```

Error-Statistics	Value
Dot1d Base Port MTU Exceeded Discards	0
Dot3 Control In Unknown Opcodes	0
Dot3 Stats Alignment Errors	0
Dot3 Stats FCS Errors	0
Dot3 Stats Frame Too Longs	0
Dot3 Stats Internal Mac Transmit Errors	0
Ethernet RX Oversize Packets	0

```
...
```

Showing interface link-status:

```
switch# show interface link-status
```

```
-----  
Port                Type           Physical      Link          Last  
                   Link State    Transitions   Change  
-----  
1/1/1                1G-BT         down          0             --  
1/1/2                1G-BT         up            1             1 minute ago (Fri  
Mar 09 12:36:56 UTC 2018)
```

1/1/3	1G-BT	up	1	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/4	--	down	0	--
1/1/5	--	down	0	--

Showing interface loopback 1 link-status:

Port	Type	Physical Link State	Link Transitions	Last Change
loopback1	--	up	--	--

Showing interface 1/1/2-1/1/3 link-status:

Port	Type	Physical Link State	Link Transitions	Last Change
1/1/2	1G-BT	up	1	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/3	1G-BT	up	1	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)

Showing interface link-status:

```
switch# show interface link-status
```

Port	Type	Physical Link State	Link Transitions	Link Flaps Ignored	Last Change
1/1/1	1G-BT	down	0	0	--
1/1/2	1G-BT	up	1	0	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/3	1G-BT	up	1	0	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/4	--	down	0	0	--
1/1/5	--	down	0	0	--

Showing state information when interface is blocked:

```
8360(config-if)# show interface 1/1/1
Interface 1/1/1 is up (Blocked)
Admin state is up
State information: Blocked by UDLD
Link state: up for 1 minute (since Mon Jun 10 09:25:27 UTC 2024)
Link transitions: 1
Description:
Persona:
Hardware: Ethernet, MAC Address: 00:fd:45:67:85:91
```

```

MTU 1500
Type 10G-LR / 10G SFP+ LR
Full-duplex
qos trust none
Speed 10000 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rate collection interval: 300 seconds

```

Rate (RX+TX)	RX	TX	Total
Mbits / sec	0.00	0.00	
0.00			
KPkts / sec	0.00	0.00	
0.00			
Unicast	0.00	0.00	
0.00			
Multicast	0.00	0.00	
0.00			
Broadcast	0.00	0.00	
0.00			
Utilization %	0.00	0.00	
0.00			
Statistic	RX	TX	
Total			
Packets	15		
15	30		
Unicast	12		
12	24		
Multicast	3		
3	6		
Broadcast	0		
0	0		
Bytes	1350	1350	
2700			
Jumbos	0		
0	0		

Dropped		0
0	0	
Pause Frames		0
0	0	
Errors		0
0	0	
CRC/FCS		0
n/a	0	
Collision		n/a
0	0	
Runts		0
n/a	0	
Giants		0
n/a	0	

Command History

Release	Modification
10.15	Added state information when port goes into down state.
10.11	Added <code>monitor</code> parameter.
10.10	Added <code>human-readable</code> parameter.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (<code>></code>) or Manager (<code>#</code>)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (<code>></code>) only.

show interface dom

Syntax

```
show interface [<INTERFACE-ID>] dom [detail]
```

Description

Shows diagnostics information and alarm/warning flags for the optical transceivers (SFP, SFP+, QSFP+). This information is known as DOM (Digital Optical Monitoring). DOM information also consists of vendor determined thresholds which trigger high/low alarms and warning flags.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port .
detail	Show detailed information.

Example

```
switch# show interface dom
-----
-----
Port      Type      Channel  Temperature Voltage  Tx Bias  Rx Power      Tx
Power
              (Celsius)  (Volts)  (mA)      (mW/dBm)  (mW/dBm)
-----
-----
1/1/1     SFP+SR           47.65      3.31      8.40      0.08, -10.96
0.63, -2.49
1/1/2     SFP+SR           n/a        n/a       n/a       n/a          n/a
1/1/3     SFP+DA3          42.10      3.24      n/a       n/a          n/a
1/1/4     QSFP+SR4  1      44.46      3.30      6.12      0.08, -10.96
0.63, -1.95
2          44.46      3.30      6.04      0.08, -10.96 0.63, -2.00
3          44.46      3.30      6.51      0.08, -10.96 0.60, -2.16
4          44.46      3.30      6.19      0.08, -10.96 0.63, -1.94
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface energy-efficient ethernet

Syntax

```
show interface [<IFNAME>|<IFRANGE>] energy-efficient-ethernet
```

Description

Displays Energy-Efficient Ethernet information for the interface.

Parameter	Description
<i><IFNAME></i>	Specifies the name of an interface on the switch. Use the format member/slot/port (for example, 1/1/1).
<i><IFRANGE></i>	Specifies the port identifier range of an interface on the switch. Use the format member/slot/port (for example, 1/1/1).

Example

The following example shows when the interfaces are Energy-Efficient Ethernet capable.

```
switch# show interface energy-efficient-ethernet
```

Port	Enabled	Negotiated	Speed (MB/s)	TX Wake Time (us)	RX Wake Time (us)
1/1/1	no	no	--	--	--
1/1/2	yes	yes	100	36	36
1/1/3	yes	yes	1000	17	17

```

1/1/4    no        no        --        --        --
1/1/5    yes       no        1000     --        --
switch# show interface 1/1/1 energy-efficient-ethernet
-----
Port      Enabled      Negotiated      Speed      TX Wake      RX Wake
              (Mb/s)        Time (us)      Time (us)
-----
1/1/1     no          no             1000       --           --
switch#

```

The following example shows when the interface is not Energy-Efficient Ethernet capable :

```

switch# show interface 1/1/1 energy-efficient-ethernet
Port 1/1/1 does not support Energy-Efficient-Ethernet

```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
5420	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.
6200		

show interface flow-control

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] flow-control [detail]
```

Description

Shows the flow control configuration, status, and statistics of the specified interface for interfaces on which flow control is enabled.



NOTE

If **detail** is not specified, this command shows a summary of all flow controlled interfaces with one interface per line. If **detail** is specified, this command shows flow control detailed statistics.



NOTE

As of AOS-CX 10.10, the separate **show flow-control** command has been removed, with it being effectively replaced by this command.

Parameter	Description
<code><IFNNAME> <IFRANGE></code>	Specifies the interface (port) name or range. When no interface range is specified, only interfaces with flow control enabled in the configuration or status are shown.
<code>detail</code>	Shows detailed information.

Examples

Showing summary flow control information:

```
switch# show interface flow-control
```

```
-----  
Port      Flow  
          Control  
-----  
1/1/1     config: llfc rx  
          status: llfc rx  
1/1/2     config: llfc rx  
          status: none
```

Showing summary flow control information with PFC:

```
switch# show interface flow-control
```

```
-----  
Port      Flow  
          Control  
-----  
1/1/1     config: pfc rxtx-1,2
```

```

status: pfc rxtx-1,2
1/1/2    config: pfc rxtx-5
status: none

```

Showing summary flow control information with PFC:

```

switch# show interface flow-control
Flow Control Watchdog Settings
  Trigger Timeout: 100 milliseconds
  Resume Time:    100 milliseconds

```

Port	Flow Control	Watchdog Status	Watchdog Timeouts
1/1/1	config: llfc rx status: llfc rx		
1/1/2	config: llfc rx status: llfc rx	incompatible	0
1/1/10	config: pfc rxtx-1,2 status: pfc rxtx-1,2	enabled	1234
1/1/12	config: pfc rxtx-1,2 status: pfc rxtx-1,2	error	0
1/1/32:4	config: pfc rxtx-5 status: pfc rxtx-5		

Showing summary flow control information where the configuration does not match status due to a reboot required to apply PFC configuration in hardware:

```

switch# show interface flow-control
Flow Control Watchdog Settings
  Trigger Timeout: 100 milliseconds (actual: not applied)
  Resume Time:    100 milliseconds (actual: not applied)

```

Port	Flow Control	Watchdog Status	Watchdog Timeouts
1/1/1	config: llfc rx status: llfc rx		
1/1/2	config: llfc rx status: llfc rx	incompatible	0
1/1/10	config: pfc rxtx-1,2 status: none	pending	1234
1/1/12	config: pfc rxtx-1,2 status: none	pending	0
1/1/32:4	config: pfc rxtx-5 status: none		

Showing detailed flow control information with RX flow control enabled:

```

switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: llfc rx
Statistics                               RX
-----
Dot3 Pause Frames                        0

```

Showing detailed flow control information with RX flow control enabled:

```

switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: llfc rx
Flow-control watchdog: disabled
Statistics                               RX
-----
Dot3 Pause Frames                        0

```

Showing detailed flow control information with RXTX flow control enabled:

```

switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: llfc rxtx
Statistics                               RX                               TX
-----
Dot3 Pause Frames                        0                               0

```

Showing detailed flow control information with PFC enabled:

```

switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: pfc rxtx-4,5
Statistics                               RX                               TX
-----
Priority 0 Pauses                        0                               0
Priority 1 Pauses                        0                               0
Priority 2 Pauses                        0                               0
Priority 3 Pauses                        0                               0
Priority 4 Pauses                        0                               0
Priority 5 Pauses                        0                               0
Priority 6 Pauses                        0                               0

```

Priority 7 Pauses	0	0
Total Pause Frames	0	0

Showing detailed flow control information with PFC enabled and flow control watchdog disabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: pfc rxtx-4,5
Flow-control watchdog: disabled
Statistics
```

	RX	TX
Priority 0 Pauses	0	0
Priority 1 Pauses	0	0
Priority 2 Pauses	0	0
Priority 3 Pauses	0	0
Priority 4 Pauses	0	0
Priority 5 Pauses	0	0
Priority 6 Pauses	0	0
Priority 7 Pauses	0	0
Total Pause Frames	0	0

```
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: pfc rxtx-4,5
Flow-control watchdog: disabled
Statistics
```

	RX	TX
Priority 0 Pauses	0	0
Priority 1 Pauses	0	0
Priority 2 Pauses	0	0
Priority 3 Pauses	0	0
Priority 4 Pauses	0	0
Priority 5 Pauses	0	0
Priority 6 Pauses	0	0
Priority 7 Pauses	0	0
Total Pause Frames	0	0

Showing detailed flow control information with both PFC and flow control watchdog enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: pfc rxtx-4,5
Flow-control watchdog: enabled
```


Statistics	RX	TX
-----	-----	-----
Priority 0 Pauses	0	0
Priority 1 Pauses	0	0
Priority 2 Pauses	0	0
Priority 3 Pauses	0	0
Priority 4 Pauses	0	0
Priority 5 Pauses	0	0
Priority 6 Pauses	0	0
Priority 7 Pauses	0	0
Total Pause Frames	0	0
Queue Watchdog Timeouts		
-----	-----	
Queue 0	0	
Queue 1	0	
Queue 2	0	
Queue 3	0	
Queue 4	0	
Queue 5	0	
Queue 6	0	
Queue 7	0	

Showing detailed flow control information when flow control watchdog is enabled in the configuration but it could not be applied because the configured flow control mode is not compatible with watchdog:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: llfc rx
Flow-control watchdog: incompatible
```

Showing detailed flow control information when flow control watchdog is enabled in the configuration but could not be applied because a compatible flow control mode first requires a reboot:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
Admin state is up
Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
Flow-control: off
Flow-control watchdog: pending
```

Command History

Release	Modification
10.10	Examples updated with new and changed output elements.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface link-diagnostics

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] link-diagnostics
```

Description

Shows information about select link diagnostics, including MAC, PHY, and Transceiver diagnostics.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
link - diagnostics	Shows link diagnostics information.

Examples



NOTE

AOS-CX 10.18 adds Link Delay Information:

- **Link-Up Failure:** The number of times a link-up was detected, but was lost when the link-up delay time expired.
- **Link-Down Recovery:** The number of times a link-down was detected, but was recovered when the link-down delay time expired.
-

Showing interface information for 1/1/12 link diagnostics:

```
switch# show interface 1/1/12 link-diagnostics
```


Interface 1/1/12	Current State	Changes	Last
Change			

Port Status	Waiting for link		
Link Down Reason	Local fault		
Transceiver			
No transceiver	--	0	--
TX Fault	--	0	--
Transceiver error	--	0	--
System error	--	0	--
DOM error	--	0	--
Configuring transceiver	--	0	--
Transceiver disabled	--	0	--
Transceiver unsupported	--	0	--
MAC and PHY			
TX Disabled	--	0	--
Autoneg Incomplete	--	0	--
No Signal Detected	--	0	--
No RX Lock	True	1	22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)			
Remote Fault	--	0	--
Local Fault	True	1	22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)			
Link State	Down	1	16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)			

```

Link Delay Information                                Count      Last
Change
  Link-Up Failure                                    2           22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
  Link-Down Recovery                                1           16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)

switch# show interface 1/1/12 link-diagnostics
-----
-----
Interface 1/1/12      Current State      Changes      Last
Change
-----
-----
Port Status          Link up
Link Down Reason      --
Transceiver
No transceiver        --                0           --
TX Fault              --                0           --
Transceiver error     --                0           --
System error          --                0           --
DOM error             --                0           --
Configuring transceiver --                0           --
Transceiver disabled  --                0           --
Transceiver unsupported --                0           --
MAC and PHY
TX Disabled           --                2           22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Autoneg Incomplete    --                0           --
No Signal Detected    --                2           22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
No RX Lock             --                2           22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Remote Fault          --                2           21
seconds ago (Tue Nov 14 14:30:49 UTC 2023)
Local Fault           --                2           22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
  Link State          Up                1           16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)
Link Delay Information                                Count      Last
Change
  Link-Up Failure                                    2           22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)

```

```
Link-Down Recovery 1 16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)
```

Showing interface information for 1/1/7-1/1/8 link-diagnostics:

```
switch# show interface 1/1/7-1/1/8 link-diagnostics
-----
----
Interface 1/1/7          Current State          Changes          Last
Change

-----
---
Port Status              Link up
Link Down Reason         --
Transceiver
No transceiver           --                0                --
TX Fault                 --                0                --
Transceiver error        --                0                --
  System error           --                0                --
DOM error                --                0                --
Configuring transceiver  --                0                --
Transceiver disabled     --                0                --
Transceiver unsupported  --                0                --
MAC and PHY
TX Disabled              --                2                22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Autoneg Incomplete       --                0                --
No Signal Detected       --                2                22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
  No RX Lock             --                2                22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Remote Fault             --                2                21
seconds ago (Tue Nov 14 14:30:49 UTC 2023)
  Local Fault            --                2                22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
  Link State              Up                1                16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)
Link Delay Information          Count          Last
Change
  Link-Up Failure            2            22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
  Link-Down Recovery        1            16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)
-----
```

Interface 1/1/8	Current State	Changes	Last
Change			

Port Status	Link up		
Link Down Reason	--		
Transceiver			
No transceiver	--	0	--
TX Fault	--	0	--
Transceiver error	--	0	--
System error	--	0	--
DOM error	--	0	--
Configuring transceiver	--	0	--
Transceiver disabled	--	0	--
Transceiver unsupported	--	0	--
MAC and PHY			
TX Disabled	--	2	22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)			
Autoneg Incomplete	--	0	--
No Signal Detected	--	2	22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)			
No RX Lock	--	2	22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)			
Remote Fault	--	2	21
seconds ago (Tue Nov 14 14:30:49 UTC 2023)			
Local Fault	--	2	22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)			
Link State	Up	1	16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)			
Link Delay Information		Count	Last
Change			
Link-Up Failure		2	22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)			
Link-Down Recovery		1	16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)			

Command History

Release	Modification
10.18	Introduces Link Delay Information.
10.14	Command introduced.

Command Information

Platforms	Command context	Authority
9300P	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface statistics

Syntax

```
show interface [<IFNAME>|<IFRANGE>] statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] statistics monitor [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics monitor [non-zero] [human-readable]
show interface vxlan <VXLAN-ID> statistics [non-zero] [human-readable]
```

Description

Shows statistics for switch interfaces such as packets transmitted and received, bytes transmitted and received, broadcast and multicast packets.

Parameter	Description
<code><IFNAME></code>	Specifies a interface name.
<code><IFRANGE></code>	Specifies the port identifier range.

Parameter	Description
LAG	Shows LAG interface information.
<LAG-ID>	Specifies the LAG number. Range: 1 - 256
VXLAN	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1
monitor	Continuously monitor interface statistics.
human-readable	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G.
non-zero	Shows only non zero statistics.

Examples

Showing statistics of all interfaces:

```
show interface statistics
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/1	2727136	1975	0	17796	195	0	82	1788	96	54	0	0
1/1/10	0	0	0	0	0	0	0	0	0	0	0	0
1/1/11	0	0	0	0	0	0	0	0	0	0	0	0
1/1/12	0	0	0	0	0	0	0	0	0	0	0	0
...												
1/1/30 - lag1	0	0	0	11271	92	0	0	0	0	51	0	0
1/1/31 - lag2	2360	25	50	2732119	2040	0	0	0	178	1839	0	0
1/1/32 - lag2	0	0	0	11373	93	0	0	0	0	51	0	0
vlan1	0	0	0	0	0	0	0	0	0	0	0	0

Showing statistics of all interfaces with only non-zero statistics:

```
show interface statistics non-zero
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/1	2727136	1975	0	17796	195	0	82	1788	96	54	0	0
1/1/30 - lag1	0	0	0	11271	92	0	0	0	0	51	0	0
1/1/31 - lag2	2360	25	50	2732119	2040	0	0	0	178	1839	0	0
1/1/32 - lag2	0	0	0	11373	93	0	0	0	0	51	0	0

Showing statistics of all interfaces in the human-readable format:

```
show interface statistics human-readable
```

Interface	RX Bytes	RX Pkts	RX Drops	TX Bytes	TX Pkts	TX Drops	RX Bcast	RX Mcast	TX Bcast	TX Mcast	RX Pause	TX Pause
1/1/1	744M	577K	0	745M	578K	0	0	0	73	287	0	0
1/1/2	474M	367K	0	475M	369K	0	0	0	73	288	0	0
1/1/3	0	0	0	0	0	0	0	0	0	0	0	0

Showing statistics of a single interfaces:

```
show interface 1/1/2 statistics
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/2	2725080	1931	0	25877	253	0	21	1788	65	55	0	0

Showing statistics of all members of a LAG interface:

```
show interface lag1 statistics
```

Interface	RX Bytes	RX Packets	RX Drops	TX Bytes	TX Packets	TX Drops	RX Broadcast	RX Multicast	TX Broadcast	TX Multicast	RX Pause	TX Pause
1/1/3 - lag1	2424	26	0	2734082	2062	0	0	0	191	1848	0	0
1/1/30 - lag1	0	0	0	12383	100	0	0	0	0	59	0	0
lag1	2424	26	0	2746465	2162	0	0	0	191	1907	0	0

Showing error statistics of all interfaces:

```
show interface error-statistics
```

Interface	RX Errors	TX Errors	Giants	Runts	CRC/FCS	Collisions
1/1/1	190	20	100647	0	0	0
1/1/10	0	0	100	290	7165	949
1/1/11	0	0	0	0	0	0
1/1/12	0	0	0	0	0	0
...						
1/1/30 - lag1	1500	500	45800	0	0	0
1/1/31 - lag2	0	0	11	27	0	0
1/1/32 - lag2	0	0	0	0	6	18

Showing monitor statistics:



NOTE

The rows and columns of show interface monitor statistics depends on the length of width of the client terminal. The CLI can be navigated using the arrow keys as well as the PageUp, PageDown, Home, and End keys.

```
show interface statistics monitor
```

Interface	RX Bytes	RX Packets >>
1/1/1	3440525421984	53758209526
1/1/2	3440526607008	53758228042
1/1/3	3440527785312	53758246453
1/1/30	3440559671264	53758744653
1/1/31	3440560851680	53758763098
1/1/32	3440562028704	53758781489

help: ?, quit: q

```
Help for Interface Monitor

f    Toggle full statistics
h    Toggle human-readable mode
n    Toggle non-zero mode
r    Toggle rate display

c    Clear interface statistics
      Does not apply to rates

Arrows, PgUp, PgDn, Home, End
      Navigate interface statistics

Delay:2

help: ?, quit: q
```

Showing monitor error statistics in human-readable format:

```
show interface 1/1/1-1/1/3,1/1/30-1/1/32 error-statistics monitor human-readable
```

Interface	RX Errors	TX Errors	RX Giants	RX Runts	CRC/FCS	Collisions
1/1/1	0	0	0	0	0	0
1/1/2	0	0	0	0	0	0
1/1/3	0	0	0	0	0	0
1/1/30	0	0	0	0	0	0
1/1/31	0	0	0	0	0	0
1/1/32	0	0	0	0	0	0

```
Human-readable

help: ?, quit: q
```

```
Help for Interface Monitor

h    Toggle human-readable mode
n    Toggle non-zero mode

c    Clear interface statistics
      Does not apply to rates

Arrows, PgUp, PgDn, Home, End
      Navigate interface statistics

Delay:2

help: ?, quit: q
```

Command History

Release	Modification
10.11	Added <code>moitor</code> parameter.
10.10	Added <code>human-readable</code> parameter.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface transceiver

Syntax

```
show interface [<INTERFACE-ID>] transceiver [detail | threshold-violations]
```

Description

Displays information about transceivers present in the switch. The information shown varies for different transceiver types and manufacturers. Only basic information is shown for unsupported HPE and third-party transceivers installed in the switch and they are also identified with an asterisk in the output.

Parameter	Description
<INTERFACE-ID>	Specifies the name or range of an interface on the switch. Use the format member/slot/port (for example, 1/3/1).
detail	Show detailed information for the interfaces.
threshold-violations	Show threshold violations for transceivers.

Example

Showing summary transceiver information with identification of unsupported transceivers:

```
switch(config)# show interface transceiver
```

Port Number	Type Number	Product Number	Serial	Part
1/1/1	SFP+SR	J9150A	MYxxxxxxxxx	1990-3657
1/1/2	SFP+ER*	--	--	--

```

1/2/1    QSFP+SR4      JH233A      MYxxxxxxxxx  2005-1234
1/2/2    QSFP+ER4*    --          --          --
1/3/1    SFP28DAC3    844477-B21  MYxxxxxxxxx  77fc-7ce7

```

* unsupported transceiver

```
switch(config)# show interface transceiver
```

```

-----
Port      Type      Product      Serial      Part
          Number    Number      Number
-----
1/1/15    SFP+SR    J9150D      xxxxxxxxxxx  1990-4634
1/1/16    SFP+SR    J9150D      xxxxxxxxxxx  1990-4634

```

Showing detailed transceiver information:

```
switch(conf#) show interface transceiver detailing
```

Transceiver in 1/1/1

```

Interface Name      : 1/1/1
Type                : SFP+SR
Connector Type      : LC
Wavelength          : 850nm
Transfer Distance   : 0m (SMF), 30m (OM1), 80m (OM2), 300m (OM3)
Diagnostic Support   : DOM
Product Number      : J9150A
Serial Number       : MYxxxxxxxxx
Part Number         : 1990-3657

```

Status

```

Temperature : 47.65C
Voltage     : 3.31V
Tx Bias     : 8.40mA
Rx Power    : 0.08mW, -10.96dBm
Tx Power    : 0.56mW, -2.49dBm

```

Recent Alarms :

```

Rx power low alarm
Rx power low warning

```

Recent Errors :

```
Rx loss of signal
```

Transceiver in 1/1/2

```

Interface Name      : 1/1/2
Type                : unknown
Connector Type      : ??
Wavelength          : ??
Transfer Distance   : ??
Diagnostic Support   : ??
Product Number      : ??
Serial Number       : ??

```

```

Part Number      : ??
Transceiver in 1/2/1
Interface Name   : 1/2/1
Type            : QSFP+SR4
Connector Type   : MPO
Wavelength      : 850nm
Transfer Distance : 0m (SMF), 0m (OM1), 0m (OM2), 100m (OM3)
Diagnostic Support : DOM
Product Number   : JH233A
Serial Number    : MYxxxxxxx
Part Number      : 2005-1234

```

Status

```

Temperature : 44.46C
Voltage      : 3.30V

```

```

-----
          Tx Bias  Rx Power      Tx Power
Channel#  (mA)      (mW/dBm)      (mW/dBm)
-----
1          6.12      0.00, -inf    0.63, -1.95
2          6.04      0.00, -inf    0.63, -2.00
3          6.51      0.00, -inf    0.60, -2.16
4          6.19      0.00, -inf    0.63, -1.94

```

Recent Alarms :

Channel 1 :

```

Rx power low alarm
Rx power low warning

```

Channel 2 :

```

Rx power low alarm
Rx power low warning

```

Channel 3 :

```

Rx power low alarm
Rx power low warning

```

Channel 4 :

```

Rx power low alarm
Rx power low warning

```

Recent Errors :

Channel 1 :

```

Rx Loss of Signal

```

Channel 2 :

```

Rx Loss of Signal

```

Channel 3 :

```

Rx Loss of Signal

```

Channel 4 :

```

    Rx Loss of Signal
Transceiver in 1/2/2
  Interface Name      : 1/2/2
  Type                : unknown
  Connector Type      : ??
  Wavelength          : ??
  Transfer Distance   : ??
  Diagnostic Support   : ??
  Product Number      : ??
  Serial Number       : ??
  Part Number         : ??
Transceiver in 1/3/1
  Interface Name      : 1/3/1
  Type                : SFP28DAC3
  Connector Type      : Copper Pigtail
  Transfer Distance   : 0.00km (SMF), 0m (OM1), 0m (OM2), 0m (OM3)
  Diagnostic Support   : None
  Product Number      : 844477-B21
  Serial Number       : MYxxxxxxx
  Part Number         : 77fc-7ce7
switch(config)# show interface transceiver detail
Transceiver in 1/1/15
  Interface Name      : 1/1/15
  Type                : SFP+SR
  Connector Type      : LC
  Wavelength          : 850nm
  Transfer Distance   : 0.00km (SMF), 20m (OM1), 80m (OM2), 300m (OM3)
  Diagnostic Support   : DOM
  Product Number      : J9150D
  Serial Number       : xxxxxxxxxxx
  Part Number         : 1990-4634
Status
  Temperature         : 30.38C
  Voltage              : 3.26V
  Tx Bias              : 5.54mA
  Rx Power             : 0.56mW, -2.52dBm
  Tx Power             : 0.62mW, -2.08dBm
  Recent Alarms:
  Recent Errors:
Transceiver in 1/1/16
  Interface Name      : 1/1/16
  Type                : SFP+SR
  Connector Type      : LC

```

```

Wavelength      : 850nm
Transfer Distance : 0.00km (SMF), 20m (OM1), 80m (OM2), 300m (OM3)
Diagnostic Support : DOM
Product Number   : J9150D
Serial Number    : xxxxxxxxxx
Part Number      : 1990-4634
Status
Temperature : 30.62C
Voltage      : 3.28V
Tx Bias      : 5.64mA
Rx Power     : 0.61mW, -2.15dBm
Tx Power     : 0.59mW, -2.29dBm
Recent Alarms:
Recent Errors:

```

Showing detailed transceiver information with identification of unsupported transceivers:

```

Transceiver in 1/1/2
Interface Name      : 1/1/2
Type               : SFP+ER (unsupported)
Connector Type     : LC
Wavelength         : 3590nm
Transfer Distance   : 80m (SMF), 0m (OM1), 0m (OM2), 0m (OM3)
Diagnostic Support  : DOM
Vendor Name        : INNOLIGHT
Vendor Part Number  : TR-PX15Z-NHP
Vendor Part Revision: 1A
Vendor Serial number: MYxxxxxxx
Status
Temperature : 28.88C
Voltage      : 3.30V
Tx Bias      : 65.53mA
Rx Power     : 0.00mW, -inf
Tx Power     : 1.47mW, 1.67dBm
Recent Alarms:
Rx Power low alarm
Rx Power low warning
Recent Errors:

```

Showing transceiver threshold-violations:

```

switch(config)# show interface transceiver threshold-violations
-----
Port          Type          Channel  Type(s) of Recent
                                   Threshold Violation(s)
-----
1/1/1         SFP+SR                Tx bias high warning

```

```

50.52 mA > 40.00 mA
1/1/2      SFP+ER*      ??
1/2/1      QSFP+SR4      1      Tx power low alarm
          -17.00 dBm < -0.50 dBm
          2      Tx bias low warning
          3.12 mA < 4.00 mA
1/2/2      QSFP+ER4*      ??
1/3/1      SFP28DAC3      n/a
* unsupported transceiver
switch(config)# show interface transceiver threshold-violations
-----
Port      Type      Channel#  Recent Threshold Violations
-----
1/1/15    SFP+SR      none
1/1/16    SFP+SR      none
switch#

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface supported-transceivers

Syntax

```
show interface {<IFNAME>|<IFRANGE>} supported-transceivers
```

Description

Displays transceivers supported by a specific, or all, interface(s) in the switch. Information varies and is dependent on transceiver types, part number, and short description or interfaces.

Parameter	Description
IFNAME	Specifies the name of an interface on the switch. Use the following format: member/slot/port. Example: 1/3/1
IFRANGE	Specifies the name of an interface range on the switch. Use the following format: member/slot/port. Example: 1/1/1 - 1/1/2

Example

Showing a full switch interface or chassis switch with multiple line modules:

```
switch# show interface supported-transceivers
```

```
-----
Type                Product      Supported
                   Number        Interfaces
-----
1G-BT               JL747        1/1/33-1/1/34
1G-BT               J8177        1/1/33-1/1/34
1G-LH               J4860        1/1/33-1/1/34
IG-LX               J4859        1/1/33-1/1/34
8x25G-DAC2          S4B39
1/1/2,1/1/4,1/1/6,1/1/8,1/1/10,1/1/12,1/1/14,
1/1/16,1/1/18,1/1/20,1/1/22,1/1/24,1/1/26,
1/1/28,1/1/30,1/1/32
QSA28               845970       1/1/1-1/1/32
```

Note: For full product numbers (SKUs) with revision letters, refer to the transceiver guide.

```
switch# show interface supported-transceivers
```

```
-----
Type                Product      Supported
                   Number        Interfaces
-----
100M-FX             J9054        1/10/1-1/10/24,1/12/1-1/12/48
1G-BT               JL747
1/5/49-1/5/52,1/7/25-1/7/28,1/8/25-1/8/28,
1/9/49-1/9/52,1/10/1-1/10/28,1/12/1-1/12/48
10G-BT              JL563
```

```

1/5/49-1/5/52,1/6/1-1/6/32,1/7/25-1/7/28,

1/8/25-1/8/28,1/9/49-1/9/52,1/10/1-1/10/28,
                                1/12/1-1/12/48

10G-BiDi-D                R9X54
1/4/1-1/4/12,1/6/1-1/6/36,1/11/1-1/11/12,
                                1/12/1-1/12/48

10G-SR                    J9150
1/4/1-1/4/12,1/5/49-1/5/52,1/6/1-1/6/36,

1/7/25-1/7/28,1/8/25-1/8/28,1/9/49-1/9/52,

1/10/1-1/10/28,1/11/1-1/11/12,1/12/1-1/12/48
10G-SR                    JL782
1/4/1-1/4/12,1/6/33-1/6/36,1/11/1-1/11/12
25G-AOC15                R0Z21                1/6/1-1/6/32
25G-AOC7                 R0M45                1/6/1-1/6/32
25G-BiDi-10-U           S1C98
1/4/1-1/4/12,1/5/49-1/5/52,1/6/1-1/6/36,
8x50G-DAC2              S4B42
1/5/49-1/5/52,1/6/1-1/6/32,1/7/25-1/7/28,

1/8/25-1/8/28,1/9/49-1/9/52,1/10/25-1/10/28,
                                1/12/1-1/12/48

QSA28                    845970
1/4/1-1/4/12,1/6/33-1/6/36,1/11/1-1/11/12
Note: For full product numbers (SKUs) with revision letters,
refer to the transceiver guide.

```

Showing a single interface:

```

switch# show interface 1/1/1 supported-transceivers
-----
Type                Product      Description
                   Number
-----
Interface 1/1/1
10G-BiDi-D @       R9X54        10G BX-D 40km
10G-BiDi-U @       R9X55        10G BX-U 40km
100G-LR4           JL310        100G QSFP28 LR4
100G-SR1.2         S4B44        100G SR1.2 Q28 LC
100G-SR2           S1C93        100G MPO SR2
100G-SR4           JL309        100G QSFP28 SR4
QSA28              845970       --
@ This transceiver is only supported when inserted in a QSA

```

For full product numbers (SKUs) with revision letters, refer to the Transceiver guide.

Showing a range of interfaces:

```
switch# show interface 1/1/1-1/1/2 supported-transceivers
```

Type	Product Number	Description
Interface 1/1/1		
10G-BiDi-D @	R9X54	10G BX-D 40km
10G-BiDi-U @	R9X55	10G BX-U 40km
10G-ER @	S2P32	10G SFP+ ER
100G-SR2	S1C93	100G MPO SR2
100G-SR4	JL309	100G QSFP28 SR4
QSA28	845970	--
Interface 1/1/2		
10G-BiDi-D @	R9X54	10G BX-D 40km
10G-BiDi-U @	R9X55	10G BX-U 40km
10G-ER @	S2P32	10G SFP+ ER
25G-BiDi-10-D @	S1C96	25G LC BiDi 10km-D
25G-SR @	S2P33	25G SFP28 SR
40G-eSR4	JH233	QSFP+ eSR4
100G-BiDi	845972	--
100G-SR4	JL309	100G QSFP28 SR4
QSA28	845970	--

@ This transceiver is only supported when inserted in a QSA

For full product numbers (SKUs) with revision letters, refer to the Transceiver guide.

Showing a chassis switch with multiple line modules:

```
switch# show interface supported-transceivers
```

Type	Product Number	Supported Interfaces
100M-FX	J9054	1/10/1-1/10/24,1/12/1-1/12/48
1G-BT	JL747	1/5/49-1/5/52,1/7/25-1/7/28,1/8/25-1/8/28, 1/9/49-1/9/52,1/10/1-1/10/28,1/12/1-1/12/48
10G-BT	JL563	1/5/49-1/5/52,1/6/1-1/6/32,1/7/25-1/7/28, 1/8/25-1/8/28,1/9/49-1/9/52,1/10/1-1/10/28, 1/12/1-1/12/48
10G-BiDi-D	R9X54	1/4/1-1/4/12,1/6/1-1/6/36,1/11/1-1/11/12, 1/12/1-1/12/48
10G-SR	J9150	1/4/1-1/4/12,1/5/49-1/5/52,1/6/1-1/6/36,

		1/7/25-1/7/28,1/8/25-1/8/28,1/9/49-1/9/52, 1/10/1-1/10/28,1/11/1-1/11/12,1/12/1-1/12/48
10G-SR	JL782	1/4/1-1/4/12,1/6/33-1/6/36,1/11/1-1/11/12
25G-AOC15	R0Z21	1/6/1-1/6/32
25G-AOC7	R0M45	1/6/1-1/6/32
25G-BiDi-10-U	S1C98	1/4/1-1/4/12,1/5/49-1/5/52,1/6/1-1/6/36,
8x50G-DAC2	S4B42	1/5/49-1/5/52,1/6/1-1/6/32,1/7/25-1/7/28, 1/8/25-1/8/28,1/9/49-1/9/52,1/10/25-1/10/28, 1/12/1-1/12/48
QSA28	845970	1/4/1-1/4/12,1/6/33-1/6/36,1/11/1-1/11/12

For full product numbers (SKUs) with revision letters, refer to the Transceiver guide.

Showing a VSF stack:

```
switch# show interface supported-transceivers
```

Type	Product Number	Supported Interfaces
100M-FX	J9054	1/1/1-1/1/24,2/1/1-2/1/48
1G-BT	JL747	1/1/49-1/1/52,3/1/25-3/1/28,4/1/25-4/1/28
10G-BT	JL563	1/1/49-1/1/52,2/1/1-2/1/32,3/1/25-3/1/28, 4/1/25-4/1/28
10G-AOC15	721076	1/1/13,1/1/15-1/1/27,1/1/41-1/1/42
10G-AOC5	Q9S66	1/1/13,1/1/15-1/1/27,1/1/41-1/1/42
25G-AOC15	R0Z21	3/1/1-3/1/32
QSA28	845970	4/1/1-4/1/12,2/1/33-2/1/36,1/1/1-1/1/12

For full product numbers (SKUs) with revision letters, refer to the Transceiver guide.

Showing an empty chassis:

```
switch# show interface supported-transceivers
```

No available interfaces.

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface utilization

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] utilization [non-zero]
```

Description

Displays physical port throughput and utilization.

Parameter	Description
<i><IFNAME></i>	Specifies an interface name.
<i><IFRANGE></i>	Specifies the port identifier range.
utilization	Displays utilization statistics.
non-zero	Displays non - zero statistics

Examples

The following example shows port utilization of all interfaces:

```
switch# show interface utilization
-----|-----|-----|
-----|-----|-----|
Interval |           RX           |           TX           |           Total
(RX+TX)  |
```

Interface	seconds			Mbps	KPkt/s	Util %	Mbps	KPkt/s	Util %
Mbps	KPkt/s	Util %	Description						
----- ----- -----									
----- -----									
1/1/1		300	9578.02	788.70	95.78	25.70	45.89		
0.26	9603.72	834.59	96.04	Aruba-AP					
1/1/2		300	25.71	45.90	0.26	9581.09	788.96		
95.81	9606.80	834.86	96.07	Aruba2530-AP-conce...					
1/1/3	- lag123	300	0.00	0.00	0.00	0.00	0.00		
0.00	0.00	0.00	0.00	ISL: SWRTS-0064-1					
1/1/4		300	9261.79	804.52	92.62	9496.70	823.97	94.97	
18758.50	1628.48	187.58	Backup data center...						
1/1/5		300	9496.70	823.97	94.97	9261.79	804.52	92.62	
18758.50	1628.48	187.58	--						

Command History

Release Modification

10.07 or earlier - -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ip interface

Syntax

```
show ip interface <INTERFACE-ID>[{lag|loopback|vlan|tunnel <ID>}]
```

Description

Shows status and configuration information for an IPv4 interface.

Parameter	Description
<code><INTERFACE-ID></code>	Specifies the name of an interface. Format: member/slot/port .
<code>lag <ID></code>	Show LAG interface information, where <ID> is 1 - 256.
<code>loopback <ID></code>	Show loopback interface information, where <ID> is 0 - 255.
<code>tunnel <ID></code>	Show tunnel interface information, where <ID> is 1 - 255.
<code>vlan <ID></code>	Show VLAN interface information, where <ID> is 1 - 4094.

Example

```

switch# show ip interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
IP MTU 1500
IP TCP MSS 1460
IPv4 address 192.168.1.1/24
L3 Counters: Rx Enabled, Tx Enabled
Statistic                                RX                                TX
Total
-----
-----
L3 Packets                                0
0                                0
L3 Bytes                                0
0                                0
switch# show ip interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
IP MTU 1500
IP TCP MSS 1460
IP Directed Broadcast is Disabled
IPv4 address 2.2.2.2/32 [unnumbered to loopback1]
L3 Counters: Rx Disabled, Tx Disabled
switch# show ip interface loopback 1
Interface loopback1 is up

```

```
Admin state is up
Hardware: Loopback
IP Directed Broadcast is Disabled
IPv4 address 192.168.1.1/24
```

Command History

Release	Modification
10.14.1000	Command output can display information for an IP unnumbered interface and IP TCP MSS configuration.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ip source-interface

Syntax

```
show ip source-interface {sflow | tftp | radius | tacacs | all} [vrf
<VRF-NAME>]
```

Description

Shows single source IP address configuration settings.

Parameter	Description
sflow tftp radius tacacs all	Shows single source IP address configuration settings for a specific protocol. The all option shows the global setting that applies to all protocols that do not have an address set.

Parameter	Description
vrf <VRF-NAME>	Specifies the name of a VRF.

Examples

Showing single source IP address configuration settings for sFlow:

```
switch# show ip source-interface sflow
Source-interface Configuration Information
-----
Protocol          Source Interface
-----
sflow             10.10.10.1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ip source-interface all
Source-interface Configuration Information
-----
Protocol          Source Interface
-----
all               1/1/1
switch# show ip source-interface all
Source-interface Configuration Information
-----
Protocol          Src-Interface      Src-IP              VRF
-----
all               vlan2              2.2.2.2             default
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show ipv6 interface

Syntax

```
show ipv6 interface <INTERFACE-ID>
```

Description

Shows status and configuration information for an IPv6 interface.

Parameter	Description
<code><INTERFACE-ID></code>	Specifies an interface ID. Format: member/slot/port .

Examples

```
switch# show ipv6 interface 1/1/1
Interface 1/1/1 is up
  Admin state is up
  IPv6 address:
    2001:0db8:85a3:0000:0000:8a2e:0370:7334/24 [VALID]
  IPv6 link-local address: fe80::1e98:ecff:fee3:e800/64 (default) [VALID]
  IPv6 virtual address configured: none
  IPv6 multicast routing: disable
  IPv6 Forwarding feature: enabled
  IPv6 multicast groups locally joined:
    ff02::ff70:7334  ff02::ffe3:e800  ff02::1  ff02::1:ff00:0
    ff02::2
  IPv6 multicast (S,G) entries joined: none
  IPv6 MTU: 1524 (using link MTU)
  IPv6 TCP MSS 1440
  IPv6 unicast reverse path forwarding: none
  IPv6 load sharing: none
  RX
    0 packets, 0 bytes
  TX
    0 packets, 0 bytes
```

```

switch# show ipv6 interface vlan2
Interface vlan2 is up
  Admin state is up
    IPv6 address:
      2001::1/64 [VALID]
    IPv6 link-local address: fe80::883a:3080:247:c1c0/64 [VALID]
    IPv6 Forwarding feature: enabled
    IPv6 multicast groups locally joined:
      ff02::1 ff02::1:ff00:1 ff02::1:ff47:c1c0 ff02::1:ff00:0
      ff02::2
    IPv6 MTU 1500
    IPv6 TCP MSS 1440
    IPv6 unicast reverse path forwarding: none
    IPv6 load sharing: none
switch# show ipv6 interface 1/1/14.1
Interface 1/1/14.1 is up
  Admin state is up
    IPv6 address:
      30::1/64 [VALID]
    IPv6 link-local address: fe80::b86a:97c0:122:2f42/64 [VALID]
    IPv6 virtual address configured: none
    IPv6 multicast routing: disable
    IPv6 Forwarding feature: enabled
    IPv6 multicast groups locally joined:
      ff02::1 ff02::1:ff22:2f42 ff02::1:ff00:1 ff02::1:ff00:0
      ff02::2
    IPv6 multicast (S,G) entries joined: none
    IPv6 MTU 1500
    IPv6 TCP MSS 1440
    IPv6 unicast reverse path forwarding: none
    IPv6 load sharing: none
Encapsulation dot1q ID: 20
switch# show ipv6 interface lag2.1
Interface lag2.1 is up
  Admin state is up
    IPv6 address:
      40::1/64 [VALID]
    IPv6 link-local address: fe80::b86a:97c0:122:2f42/64 [VALID]
    IPv6 virtual address configured: none
    IPv6 multicast routing: disable
    IPv6 Forwarding feature: enabled
    IPv6 multicast groups locally joined:
      ff02::1 ff02::1:ff22:2f42 ff02::1:ff00:1 ff02::1:ff00:0

```

```
ff02::2
IPv6 multicast (S,G) entries joined: none
IPv6 MTU 1500
IPv6 TCP MSS 1440
IPv6 unicast reverse path forwarding: none
IPv6 load sharing: none
Encapsulation dot1q ID: 30
```

Command History

Release	Modification
10.14.1000	Command output can display information for an IP unnumbered interface and IPv6 TCP MSS configuration.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 source-interface

Syntax

```
show ipv6 source-interface <PROTOCOL> [{all-vrfs | vrf <VRF-NAME>}]
```

Description

Shows single source IPv6 address configuration settings.

Parameter	Description
	Specifies the protocol to configure.

Parameter	Description
<code><PROTOCOL></code>	• all: Selects the source for all protocols covered by this command. • central: Selects HPE Aruba Networking Central. • dhcp_relay: Selects DHCP relay. When you configure a <code>dhcp relay source interface</code>, you must also enable DHCP relay Option 82 using the dhcp - relay option 82source - interface command. • dns: Selects DNS. • http: Selects HTTP. • ipfix: Selects ipfix. Configures source interface for IPFIX. • ntp: Selects NTP. • radius: Selects RADIUS. • sflow: Selects sFlow. • sftp - scp: Selects SFTP and SCP. • ssh - client: Selects SSH Client. • syslog: Selects the source for syslog packets. • tacacs: Selects the source for TACACS packets. • tftp: Selects TFTP. • ubt: Selects UBT.
<code>all-vrfs vrf <VRF-NAME></code>	Specifies the name of a VRF.

Examples

Showing single source IPv6 address configuration settings for sFlow:

```
switch# show ipv6 source-interface sflow
Source-interface Configuration Information
-----
Protocol          Source Interface
-----
sflow             2001:DB8::1
```

Showing single source IPv6 address configuration settings for all protocols:

```
switch# show ipv6 source-interface all
Source-interface Configuration Information
-----
Protocol          Source Interface
-----
all               1/1/1
```

Command History

Release	Modification
10.16.1000	Added the ubt protocol parameter.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

shutdown

Syntax

```
shutdown  
no shutdown
```

Description

Disables an interface. Interfaces are disabled by default when created.

The **no** form of this command enables an interface.

Examples

Disabling an interface:

```
switch(config-if) # shutdown
```

Enabling an interface:

```
switch(config-if) # no shutdown
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

speed

Syntax

```
speed {<SPEED>
        [override] | <SPEED-DUPLEX> | auto [<SPEED>] [override]}
no speed
```

Description

Configures the link speed, duplex, and auto-negotiation settings for an interface.

The no form of this command removes the configurations and returns to the defaults.

Parameter	Description
Speed	Configures interface speed, duplex, and auto - negotiation.
10-full	10 Mbps, full duplex, no auto - negotiation
10-half	10 Mbps, half duplex, no auto - negotiation
100-full	100 Mbps, full duplex, no auto - negotiation
100-half	100 Mbps, half duplex, no auto - negotiation
1000-full	1000 Mbps, full duplex, no auto - negotiation
10g	10 Gbps, full duplex, no auto - negotiation

Parameter	Description
25g	25 Gbps, full duplex, no auto - negotiation
40g	40 Gbps, full duplex, no auto - negotiation
50g	50 Gbps, full duplex, no auto - negotiation
100g	100 Gbps, full duplex, no auto - negotiation
200g	200 Gbps, full duplex, no auto - negotiation
 NOTE Not applicable for override.	
400g	400 Gbps, full duplex, no auto - negotiation
 NOTE Not applicable for override.	
auto	Auto - negotiate speed and duplex. More than one speed can be set at a time.
10m	Allow interface to link at 10 Mbps.
100m	Allow interface to link at 100 Mbps.
1g	Allow interface to link at 1 Gbps.
2.5g	Allow interface to link at 2.5 Gbps.
5g	Allow interface to link at 5 Gbps.
10g	Allow interface to link at 10 Gbps.
25g	Allow interface to link at 25 Gbps.
40g	Allow interface to link at 40 Gbps.
50g	Allow interface to link at 50 Gbps.
100g	Allow interface to link at 100 Gbps.
200g	Allow interface to link at 200 Gbps.
400g	Allow interface to link at 400 Gbps.

Usage

The following options can be configured for an interface. The option available is based on the interface type.

```
speed <SPEED-DUPLEX>
```

Uses a fixed speed and duplex mode with no auto-negotiation. Half-duplex is only supported for 10 Mbps and 100 Mbps link speeds.

```
speed <SPEED>
```

Uses a fixed speed with no auto-negotiation. If the currently installed transceiver does not support the speed, the setting is ignored and the port will use the highest speed that is supported.

```
speed auto
```

Uses auto-negotiation and offers all speeds supported by the port and transceiver. This is the default. If the link technology does not support auto-negotiation this setting is ignored, and the port uses the highest possible fixed speed.

```
speed auto <SPEED>
```

Uses auto-negotiation and offers the specified speeds only. For ports that support pluggable transceivers, only speeds supported by the transceiver are offered and other speeds are ignored. If the link technology does not support auto-negotiation, this setting is ignored and the port uses the highest possible fixed speed.

Examples

Configuring an interface to operate at a fixed speed of 1000 Mbps with full duplex and no auto-negotiation:

```
switch(config) # interface 1/1/1  
switch(config-if) # speed 1000-full
```

Configuring an interface to operate at a fixed speed of 10 Gbps with no auto-negotiation:

```
switch(config) # interface 1/1  
switch(config-if) # speed 10g
```

Configuring an interface to auto-negotiate and advertise only 1 Gbps and 2.5 Gbps speeds:

```
switch(config) # interface 1/1/1  
switch(config-if) # speed auto 1g 2.5g
```

Configuring an interface to override the detected transceiver speed and use the configured speed if the installed transceiver does not support auto-negotiation:

```
switch(config) # interface 1/1/1  
switch(config-if) # speed auto 50g override
```

Configuring an interface to use default settings for speed, duplex, and auto-negotiation:

```
switch(config)# interface 1/1/1  
switch(config-if)#no speed
```

Command History

Release	Modification
10.18	speed <SPEED> override and speed auto <SPEED> override supported on the 5420 Switch Series.
10.09.0001	Speeds not supported by hardware hidden by CLI.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-if	Administrators or local user group members with execution rights for this command.

Source interface selection

The source IP address is determined by the system and is typically the IP address of the outgoing interface in the routing table. However, routing switches may have multiple routing interfaces and outgoing packets can potentially be sent by different paths at different times. This results in different source IP addresses.

AOS-CX provides a configuration model that allows the selection of an IP address to use as the source address for all outgoing traffic. This allows unique identification of the software application on the server site regardless of which local interface has been used to reach the destination server. The source interface selection supports selecting an IP address or interface name.



NOTE

If the source interface and source IP are configured, Source IP will have priority.

VLANs

VLANs are primarily used to provide network segmentation at layer 2. VLANs enable the grouping of users by logical function instead of physical location. They make managing bandwidth usage within networks possible by:

- Allowing grouping of high-bandwidth users on low-traffic segments
- Organizing users from different LAN segments according to their need for common resources and individual protocols
- Improving traffic control at the edge of networks by separating traffic of different protocol types.
- Enhancing network security by creating subnets to control in-band access to specific network resources

VLANs are generally assigned on an organizational basis rather than on a physical basis. For example, a network administrator could assign all workstations and servers used by a particular workgroup to the same VLAN, regardless of their physical locations.

Hosts in the same VLAN can directly communicate with one another. A router or a Layer 3 switch is required for hosts in different VLANs to communicate with one another.

VLANs help reduce bandwidth waste, improve LAN security, and enable network administrators to address issues such as scalability and network management.



NOTE

Refer to the Layer 2 Bridging Guide for VLAN configuration and commands.

Configuration and firmware management

Subtopics

Upgrade and downgrade scenarios

Hot-patch software

CLI Session Behavior for Standalone ISSU

Checkpoints

Checkpoint commands

Boot commands

Firmware management commands

Upgrade and downgrade scenarios

Upgrade and downgrade scenarios are provided by the Configuration Migration Framework (CMF).

Upgrades

All upgrade scenarios are supported and configuration is migrated.

Downgrades

About this task

The following scenario is supported:

Procedure

1. Create a checkpoint before upgrading.
2. Upgrade.
3. Perform config changes.
4. Set the startup configuration to the checkpoint created in step 1.
5. Downgrade



NOTE

Configuration changes that occur after the checkpoint in step 1 will be lost during the downgrade.

Limitations

The following are the limitations of upgrade and downgrade scenarios provided by CMF.

Table 1. Configuration from and to the same build

From/to	Checkpoint	Running	Startup
Checkpoint	N/A	Supported	Supported
Running	Supported	N/A	Supported
Startup	Supported	Supported	N/A
Config from URL in CLI Format	Supported	Supported	Supported
Config from URL in JSON Format	Supported	Supported	Supported

Table 2. Configuration from an older build to a newer build (upgrade)

From/to	Checkpoint	Running	Startup
Config from URL in CLI Format	Not supported	Not supported	Not supported
Config from URL in JSON Format	Not supported	Not supported	Not supported
Startup	Not supported	Supported	Supported

Table 3. Configuration from a newer build to an older build (downgrade)

From/to	Checkpoint	Running	Startup
Config from URL in CLI Format	Not supported	Not supported	Not supported
Config from URL in JSON Format	Not supported	Not supported	Not supported
Startup	Not supported	Not supported	Not supported

Hot-patch software

Overview

Hot-patching provides users of AOS-CX with a way to update running software without rebooting their system. Hot-patches are distributed as signed .patch files and are applied on top of an official AOS-CX image.

A hot-patch can be downloaded from a remote server onto an AOS-CX switch then applied without rebooting the switch. By default, the software patch remains on the switch after it is applied, and is automatically reapplied when the switch reboots. In order to revert an applied hot-patch update, the patch must be disabled. When the hot-patch is disabled, rebooting the box is not necessary, though the hot-patch will still remain on the system. The hot-patch can be removed from the system after disabling the hot-patch.

Once a downloaded hot-patch is removed, it must be downloaded again before it can be reapplied.



NOTE

Hot-patches are cumulative, meaning that each subsequent patch is a superset of the previous patch(es) for a given software image. When multiple hot-patches are required, the first should be removed after the second has been applied.

Prerequisites

- AOS-CX hot-patch software can be obtained from HPE Aruba Networking customer support, and is identified with a **.patch** extension. The name of the patch indicates the version of AOS-CX to which it can be applied. For example, the patch **FL.10.12.XXXX-YYYY.patch** indicates that the patch can be applied on top of switch image **FL.10.12.XXXX.swi** and brings software up to date with the **FL.10.12.YYYY** release.
- Hot-patch software can be applied only after it has been downloaded to the switch.

Caveats and limitations

- Hot-patch files cannot be managed when the switch is loaded with a .swi image that does not support the hot-patch feature.
- Only restartable daemons can be updated with a hot-patch file.
- Each hot-patch applies to only one release image and each subsequent hot-patch for that release image is cumulative. This means that applying the most recent hot-patch provides all the same fixes included in previous hot-patches. As such, only one hot-patch can be applied at a time.
- incompatible and un-applied hot-patches are allowed to remain on the system and in the config to support rollback to the previous version of the software, and to automatically apply any hot-patches that have been previously applied.
- A maximum of ten hot-patches are allowed to be present or preconfigured on the system at a time.
- A daemon/service may restart if it has a hard dependency on another service that gets restarted by being hot-patched. Upon rebooting, hot-patches are reapplied after daemons have started.
- The **dpsed** and **hpe-cardd** daemon cannot have a hot-patch applied while a switchover or failover is in progress.

To apply a hot-patch update

1. Request a hot-patch from HPE Aruba Networking customer support, then place the hot-patch on an FTP-SFTP server.
2. Use the **copy** command to copy the hot-patch from the remote server to the standalone switch or VSF.
3. Use the **hot-patch apply** command to apply the patch.

To unapply hot-patch update

This option only unapplies the patch, but does not remove the patch from the device. The patch can be reapplied again since it is still on the device. If an unapplied patch needs to be removed/deleted from the switch, it can be removed with the command **hot-patch remove <patch>.patch**.

```

switch(config)# show hot-patch
Name                               Status
-----
FL_10_12_0001-0002.patch          Not applied
switch(config)# hot-patch apply FL_10_12_0001-0002.patch
switch(config)#
switch(config)# show hot-patch
Name                               Status
-----
FL_10_12_0001-0002.patch          Applied
switch(config)# no hot-patch apply FL_10_12_0001-0002.patch
switch(config)# show hot-patch
Name                               Status
-----
FL_10_12_0001-0002.patch          Not applied
switch(config)#
switch(config)# hot-patch remove FL_10_12_0001-0002.patch
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.
switch(config)#
switch(config)# show hot-patch
No hot-patch found or configured
switch(config)#

```

To remove/delete a patch:

An applied hot-patch can be both unapplied and removed from the device with the command **hot-patch remove <patch-name>.patch**. The patch cannot be reapplied unless it is downloaded again.

CLI Session Behavior for Standalone ISSU

Overview

During a standalone ISSU upgrade, active CLI sessions are terminated while the upgrade is in progress. User session behavior varies depending on the access method.



NOTE

It is recommended to avoid running scripts or automated tools that repeatedly create new CLI sessions during ISSU, as these sessions may be terminated and could result in unexpected script failures.

Console Access CLI Session

Active console CLI sessions will be terminated while the ISSU upgrade is in progress, with the message shown below. After session termination, users can log in to the router at any time during the upgrade process.

```
The current session is being terminated due to an ongoing ISSU upgrade.  
Please login again to access the router.
```

Users are advised to use their existing console CLI session for running show commands and to avoid starting new sessions using the `vttysh` command from bash while the ISSU is in progress.

Remote Access CLI Session

Active remote CLI sessions will be terminated while the ISSU upgrade is in progress, and the message shown below will be displayed. SSH and Telnet access to the router will be temporarily unavailable during the upgrade.

Console login is recommended for router management until the ISSU upgrade is complete.

```
The current session is being terminated due to an ongoing ISSU upgrade.  
Please use the console to log in and access the router.
```

Login During ISSU

If a user logs in while an ISSU upgrade is still in progress, the following warning will be displayed after login:

```
WARNING: ISSU upgrade is in progress. Some CLI output may be incomplete  
or unavailable. Please logout and login again after the upgrade completes.
```

Also note that the message will be displayed if users try to configure other features during ISSU.

This warning indicates that certain daemons may still be restarting. As a result, CLI commands that rely on those services may return partial or incomplete output. Users should limit activity to urgent monitoring via show commands and are advised to log out until the upgrade has fully completed.

Caveats and limitations

- IGMP/MLD leave messages may not be processed until Standalone ISSU completes.
- Telemetry data collection is halted during Standalone ISSU.
- Standalone ISSU is intended to succeed when the network is in a **stable** state. Its successful completion is not guaranteed during major network events such as STP or routing re-convergence, significant port flapping, or when a large number of clients are connecting or disconnecting simultaneously.
- Standalone ISSU is not permitted when MACsec is enabled on any system port.
- During Standalone ISSU, LLDP temporarily stops sending the LLDP-MED Network Policy TLV.
- BFD sessions may be interrupted until Standalone ISSU completes.
- While Standalone ISSU is in progress, the following operations are not allowed:
 - Any hardware hot-swap activity

- Pressing any buttons
- Link state transitions
- Performing Standalone ISSU simultaneously on two or more devices, especially if OSPF is configured, is not recommended. It could result in possible route flaps, link failure, tunnel flapping and OSPF restarting on these devices.
- During Standalone ISSU, UDLD may cause link flapping if an aggressive UDLD interval is configured. This can result in a brief traffic outage. To minimize this risk, it is recommended to use the default UDLD interval.
- Standalone ISSU will be prevented if the feature readiness check fails. This check ensures that none of the following features are in a failed state:
 - ACLs, object groups, and their associated applications
 - Classifier policies, classes, and their applications
 - Port-access policies, classes, and their applications
 - Port-access group-based policies, classes, and their applications
 - Active PBR actions and their applications

Standalone ISSU Examples

For more information about `show issu` command, refer to: [GUID-8ECDD722-3092-4230-8299-E029D1401469.html](https://www.hpe.com/techdocs/8ECDD722-3092-4230-8299-E029D1401469.html).

Showing detailed status of an in-progress standalone ISSU on a switch operating in a standalone topology:

```
switch# show issu
ISSU Summary
=====
ISSU Status      : In Progress
Current Version  : FL.10.18.0000-4093- Upgrade Version : FL.10.18.0000-4093-
Upgrade Image    : secondary          Start Date       : 2025-10-13 09:39:01
Last ISSU Result: --
Rollback timer   : Not configured

ISSU Progress
=====
Upgrade Operation      Status      Start Date
-----
Initiate ISSU          Complete    2025-10-13 09:39:01
Validate System Readiness Complete    2025-10-13 09:39:01
Upgrade Data Plane Services In Progress 2025-10-13 09:39:15
Upgrade Control Plane Services Pending      2025-10-13 09:39:15
```

Finalize Upgrade	Pending	--
ISSU Complete	Pending	--

Showing detailed ISSU status when Standalone ISSU is aborted due to topology change from Standalone to other:

```
switch# show issu
ISSU Summary
=====
ISSU Status      : Not Ready
Current Version  : FL.10.18.0000-4093- Upgrade Version : FL.10.18.0000-4093-
Upgrade Image    : secondary          Start Date       : 2025-10-13 09:39:01
Last ISSU Result: Aborted (Topology changed from Standalone to other)
Rollback timer   : Not configured

ISSU Progress
=====
Upgrade Operation      Status      Start Date
-----
Initiate ISSU          Complete    2025-10-13 09:39:01
Validate System Readiness Error      2025-10-13 09:39:01
Upgrade Data Plane Services Aborted    --
Upgrade Control Plane Services Aborted    --
Finalize Upgrade        Aborted    --
ISSU Complete          Aborted    --
```



NOTE

The setup is considered as standalone when `show vsf` command shows the topology as **Standalone**.

Showing detailed ISSU status when Standalone ISSU is aborted due to insufficient system memory. Note that, if system memory utilization exceeds the 75% threshold, it will be unable to perform ISSU.

Current system memory utilization can be obtained from `show system resource-utilization` command.

```
switch# show issu
ISSU Summary
=====
ISSU Status      : Ready
Current Version  : FL.10.18.0000-4093- Upgrade Version : FL.10.18.0000-4093-
Upgrade Image    : secondary          Start Date       : 2025-10-13 09:39:01
Last ISSU Result: Aborted (System memory utilization above 75% threshold to
start ISSU)
Rollback timer   : Not configured

ISSU Progress
```

```
=====
```

Upgrade Operation	Status	Start Date
Initiate ISSU	Complete	2025-10-13 09:39:01
Validate System Readiness	Error	2025-10-13 09:39:01
Upgrade Data Plane Services	Aborted	--
Upgrade Control Plane Services	Aborted	--
Finalize Upgrade	Aborted	--
ISSU Complete	Aborted	--

```
switch#show system resource-utilization
```

```
System Resources:
```

```
Processes                : 154
CPU usage(%)              : 50
CPU usage(% average over 1 minute): 50
CPU usage(% average over 5 minute): 50
Memory usage(%)           : 80
Open FD's                 : 2189
...
...
```

Showing output from the debug buffer:

```
switch# show debug buffer module issu
```

```
show debug buffer
```

```
-----
-----
2026-04-10:17:04:13.002029|issud|LOG_DEBUG|SMM|1/2|ISSU|ISSU|Ignoring db
sync frozen change, to be processed in AMM only
2026-04-10:17:09:18.824108|issud|LOG_DEBUG|AMM|1/2|ISSU|ISSU|Writeable get
name: action is undefined
2026-04-10:17:09:18.824175|issud|LOG_DEBUG|AMM|1/2|ISSU|ISSU|ISSU type set
to undetermined
2026-04-10:17:09:18.824194|issud|LOG_DEBUG|AMM|1/2|ISSU|ISSU|Queued
writable to clear ISSU type
```

Checkpoints

A checkpoint is a snapshot of the running configuration of a switch and its relevant metadata during the time of creation. Checkpoints can be used to apply the switch configuration stored within a checkpoint

whenever needed, such as to revert to a previous, clean configuration. Checkpoints can be applied to other switches of the same platform. A switch is able to store multiple checkpoints.

Subtopics

Checkpoint types

Maximum number of checkpoints

User generated checkpoints

System generated checkpoints

Supported remote file formats

Rollback

Checkpoint auto mode

Testing a switch configuration in checkpoint auto mode

Checkpoint types

The switch supports two types of checkpoints:

- **System generated checkpoints:** The switch automatically generates a system checkpoint whenever a configuration change occurs.
- **User generated checkpoints:** The administrator can manually generate a checkpoint whenever required.

Maximum number of checkpoints

- Maximum checkpoints: 64 (including the startup configuration)
- Maximum user checkpoints: 32
- Maximum system checkpoints: 32

User generated checkpoints

User checkpoints can be created at any time, as long as one configuration difference exists since the last checkpoint was created. Checkpoints can be applied to either the running or startup configurations on the switch.

All user generated checkpoints include a time stamp to identify when a checkpoint was created.

A maximum of 32 user generated checkpoints can be created.

System generated checkpoints

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix `CPC` followed by a time stamp in the format `<YY YYMMDDHHMMSS>`. For example: `CPC20170630073127`.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

Supported remote file formats

You can restore a switch configuration by copying a switch configuration stored on a USB drive or a remote network device through SFTP/TFTP. The remote file formats that the switch supports depends on where you plan to restore the checkpoint.

Restoring a checkpoint to a...	File type supported
Running configuration	• CLI • JSON • Checkpoint
Startup configuration	• JSON • Checkpoint
Specified checkpoint	Specified checkpoint

Rollback

The term rollback is used to refer to when a switch configuration is reverted to a pre-existing checkpoint.

For example, the following command applies the configuration from checkpoint `ckpt1`. All previous configurations are lost after the execution of this command: `checkpoint rollback ckpt1`

You can also specify the rollback of the running configuration or of the startup configuration with a specified checkpoint, as shown with the following command: `copy checkpoint <checkpoint-name> {running-config | startup-config}`

Checkpoint auto mode

Checkpoint auto mode configures the switch with failover support, causing it to automatically revert to a previous configuration if it becomes inoperable or inaccessible due to configuration changes that are being made.

After entering checkpoint auto mode, you have a set amount of time to add, remove, or modify the existing switch configuration. To save your changes, you must execute the `checkpoint auto confirm` command before the auto mode timer expires. If you do not execute the `checkpoint auto confirm` command within the specified time, all configuration changes you made are discarded and the running configuration reverts to the state it was before entering checkpoint auto mode.

Testing a switch configuration in checkpoint auto mode

About this task

Process overview:

Procedure

1. Enable the checkpoint auto mode.
2. To save the configuration, enter the `checkpoint auto confirm` command before the specified time set in step 1.

Checkpoint commands

Select a command from the list in the left navigation menu.

Subtopics

[checkpoint auto](#)

[checkpoint auto confirm](#)

[checkpoint diff](#)

[checkpoint post-configuration](#)

[checkpoint post-configuration timeout](#)

checkpoint rename
checkpoint rollback
copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL>
copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}
copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>
copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME>
copy <REMOTE-URL> startup-config|running-config
copy {running-config | startup-config} <REMOTE-URL>
copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}
copy {running-config | startup-config} <STORAGE-URL>
copy startup-config running-config
copy <STORAGE-URL> running-config
erase
show checkpoint <CHECKPOINT-NAME>
show checkpoint <CHECKPOINT-NAME> hash
show checkpoint post-configuration
show checkpoint
show checkpoint date
show running-config hash
show startup-config hash
write memory

checkpoint auto

Syntax

```
checkpoint auto <TIME-LAPSE-INTERVAL>
```

Description

Starts auto checkpoint mode. In auto checkpoint mode, the switch temporarily saves the runtime configuration as a checkpoint only for the specified time lapse interval. Configuration changes must be saved before the interval expires, otherwise the runtime configuration is restored from the temporary checkpoint.

Parameter	Description
<TIME-LAPSE-INTERVAL>	Specifies the time lapse interval in minutes. Range: 1 to 60.

Usage

To save the runtime checkpoint permanently, run the **checkpoint auto confirm** command during the time lapse interval. The filename for the saved checkpoint is named **AUTO <YYYYMMDDHHSS>**. If the **checkpoint auto confirm** command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING Please "checkpoint auto confirm" within 2 minutes
switch# checkpoint auto confirm
checkpoint AUTO20170801011154 created
```

In this example, the runtime checkpoint was saved because the **checkpoint auto confirm** command was entered within the value set by the **time-lapse-interval** parameter, which was 20 minutes.

Not confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING Please "checkpoint auto confirm" within 2 minutes
WARNING: Restoring configuration. Do NOT add any new configuration.
Restoration successful
```

In this example, the runtime checkpoint was reverted because the **checkpoint auto confirm** command was not entered within the value set by the **time-lapse-interval** parameter, which was 20 minutes.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint auto confirm

Syntax

```
checkpoint auto confirm
```

Description

Signals to the switch to save the running configuration used during the auto checkpoint mode. This command also ends the auto checkpoint mode.

Usage

To save the runtime checkpoint permanently, run the **checkpoint auto confirm** command during the time lapse value set by the **checkpoint auto TIME-LAPSE-INTERVAL** command. The generated checkpoint name will be in the format **AUTO <YYYYMMDDHHSS>**. If the **checkpoint auto confirm** command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto confirm
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint diff

Syntax

```
checkpoint diff {<CHECKPOINT-NAME1> | running-config | startup-config}
               {<CHECKPOINT-NAME2> | running-config | startup-config | REMOTE_URL |
               STORAGE_URL} [vrf <VRF_NAME>] [deep]
```

Description

Shows the difference in configuration between two configurations. Compare checkpoints, the running configuration, or the startup configuration.

Parameter	Description
{<CHECKPOINT-NAME1> running-config startup-config}	Selects either a checkpoint, the running configuration, or the startup configuration as the baseline.
{<CHECKPOINT-NAME2> running-config startup-config}	Selects either a checkpoint, the running configuration, or the startup configuration to compare.
REMOTE_URL STORAGE-URL	Retrieves a CLI type configuration from a remote or local file. Configurations must be in CLI format.
[vrf <VRF_NAME>]	Specifies the VRF name when using a REMOTE_URL to download a checkpoint.
[deep]	Uses a deep mode comparison.

Usage



NOTE

REMOTE_URL or STORAGE_URL configurations must be in CLI format only, checkpoint format is currently not supported.

In order to get accurate diff results, ensure that the imported file has a valid configuration, correct indentation to represent context changes between CLI contexts, and the file uses the Unix end of line format (\n). The imported file should follow the same syntax and format as the output for **show running-config**. The use of 4 spaces is recommended over the use of tab.

When using the default diff mode, the output of a similar Unix **diff-u** is used. The deep option makes the comparison based on unique entries within a given context instead of based on the order of configuration lines.

The output of the **checkpoint diff** command has several symbols:

- The plus sign (+) at the beginning of a line indicates that the line exists in the comparison but not in the baseline.

- The minus sign (-) at the beginning of a line indicates that the line exists in the baseline but not in the comparison.
- An empty space () is used to represent configuration contexts that exist in both configurations (common) but contain added or removed entries.

Examples

In the following example, the configurations of checkpoints **cp1** and **cp2** are displayed before the `checkpoint diff` command, so that you can see the context of the **checkpoint diff** command.

```
switch# show checkpoint cp1
Checkpoint configuration:
!
!Version AOS-CX XL.10.00.0002
!Schema version 0.1.8
module 1/1 product-number jl363a
!
!
!
!
!
!
!
vlan 1,200
interface 1/1/1
    no shutdown
    ip address 1.0.0.1/24
interface 1/1/2
    no shutdown
    ip address 2.0.0.1/24
switch# show checkpoint cp2
Checkpoint configuration:
!
!Version AOS-CX XL.10.00.0002
!Schema version 0.1.8
module 1/1 product-number jl363a
!
!
!
!
!
!
!
vlan 1,200,300
```

```

interface 1/1/1
    no shutdown
    ip address 1.0.0.1/24
interface 1/1/2
    no shutdown
    ip address 2.0.0.1/24
switch# checkpoint diff cp1 cp2
--- /tmp/chkpt11501550258421      2017-08-01 01:17:38.420514016 +0000
+++ /tmp/chkpt21501550258421      2017-08-01 01:17:38.420514016 +0000
@@ -9,7 +9,7 @@
!
!
!
-vlan 1,200
+vlan 1,200,300
    interface 1/1/1
        no shutdown
        ip address 1.0.0.1/24
switch# checkpoint diff chkpt01 chkpt02
--- /tmp/chkpt011607564301327
+++ /tmp/chkpt021607564301353
@@ -1,7 +1,7 @@
!
!Version AOS-CX PL.10.06.0100V
!export-password: default
-hostname Switch
+hostname Switch1
    user admin group administrators password ciphertext
AQBapTyg9tpaiAaTfSVV5eNdFzOORRvZ6CMpglh1P+LQUHQLYgAAAGAhmRqFbkNvrgy2SBVk7H8C
5hvg/Iib8rWYFZLEaSCrobNP9EwMu+hLNM0xmsh45yG8dncP7WkxjwrW4p4Qra6dVfr0EW8xh/
lpQf8F/2Wki20Lc9JLXiYge7ti0H6cVn+G
    radius-server tracking interval 60
    no usb
switch#

```

Showing the difference in configuration between two configurations using the deep option.

```

switch# checkpoint diff running-config checkpoint run deep
-vlan 1, 100
+vlan 1

```

Showing the difference in configuration between two configurations using the deep option when there are warnings related to the file.

```

switch# checkpoint diff running-config sftp://user@172.17.0.1/config2.txt
vrf

```

```
mgmt
```

```
deep(user@172.17.0.1) Password:
```

```
sftp> get halon-src/hpe-config/scripts/configdiff/configdiff/config2.txt  
userconfig2.txt
```

```
-vlan 1, 100
```

```
+vlan 1
```

```
WARN: Use of tabs is not recommended. File: dest-config2.txt line: 15
```

```
WARN: Use of tabs is not recommended. File: dest-config2.txt line: 16
```

Showing the difference in configuration between two configurations using the deep option when there are errors in the file and no diff output is displayed.

```
switch# checkpoint diff running-config sftp://user@172.17.0.1/config.txt  
vrf
```

```
mgmt
```

```
deep(user@172.17.0.1) Password:
```

```
sftp> get config.txt userconfig.txt
```

```
ERROR: Duplicate entry found. File: dest-config.txt (line 25): 'vlan 2'
```

Command History

Release	Modification
10.16	Updated syntax to include REMOTE_URL, STORAGE_URL, and deep.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint post-configuration

Syntax

```
checkpoint post-configuration  
no checkpoint post-configuration
```

Description

Enables creation of system generated checkpoints when configuration changes occur. This feature is enabled by default.

The **no** form of this command disables system generated checkpoints.

Usage

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix **CPC** followed by a time stamp in the format **<YYYYMMDDHHMMSS>**. For example: **CPC20170630073127**.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

Examples

Enabling system checkpoints:

```
switch(config) # checkpoint post-configuration
```

Disabling system checkpoints:

```
switch(config) # no checkpoint post-configuration
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint post-configuration timeout

Syntax

```
checkpoint post-configuration timeout <TIMEOUT>
no checkpoint post-configuration timeout <TIMEOUT>
```

Description

Sets the timeout for the creation of system checkpoints. The timeout specifies the amount of time since the latest configuration for the switch to create a system checkpoint.

The **no** form of this command resets the timeout to 300 seconds, regardless of the value of the **<TIMEOUT>** parameter.

Parameter	Description
timeout <TIMEOUT>	Specifies the timeout in seconds. Range: 5 to 600. Default: 300.

Examples

Setting the timeout for system checkpoints to 60 seconds:

```
switch(config) # checkpoint post-configuration timeout 60
```

Resetting the timeout for system checkpoints to 300 seconds:

```
switch(config) # no checkpoint post-configuration timeout 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint rename

Syntax

```
checkpoint rename <OLD-CHECKPOINT-NAME>  
                <NEW-CHECKPOINT-NAME>
```

Description

Renames an existing checkpoint.

Parameter	Description
<OLD-CHECKPOINT-NAME>	Specifies the name of an existing checkpoint to be renamed.
<NEW-CHECKPOINT-NAME>	Specifies the new name for the checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-).



NOTE

Do not start the checkpoint name with **CPC** because it is used for system-generated checkpoints.

Examples

Renaming checkpoint **ckpt1** to **cfg001**:

```
switch# checkpoint rename ckpt1 cfg001
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

checkpoint rollback

Syntax

```
checkpoint rollback {<CHECKPOINT-NAME> | startup-config}
```

Description

Applies the configuration from a pre-existing checkpoint or the startup configuration to the running configuration.

Parameter	Description
<CHECKPOINT-NAME>	Specifies a checkpoint name.
startup-config	Specifies the startup configuration.

Examples

Applying a checkpoint named **ckpt1** to the running configuration:

```
switch# checkpoint rollback ckpt1  
Success
```

Applying a startup checkpoint to the running configuration:

```
switch# checkpoint rollback startup-config  
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

```
copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL>  
>
```

Syntax

```
copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL> [{cli | json | checkpoint}]  
[vrf <VRF-NAME>]
```

Description

Copies a checkpoint configuration to a remote location as a file. The configuration is exported in checkpoint format, which includes switch configuration and relevant metadata.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the name of a checkpoint.
<REMOTE-URL>	Specifies the remote destination and filename using the syntax:

Parameter	Description
	TFTP format: <pre>tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME></pre>
	SFTP format: <pre>sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME></pre>
	SCP format: <pre>scp://USER@{IP HOST}[:PORT]/FILE</pre>
cli json checkpoint	Specifies the transfer format: • Checkpoint - default • CLI • JSON
vrf <VRF-NAME>	Specifies a VRF name.

Examples

Copying checkpoint configuration to a remote file through TFTP:

```
switch# copy checkpoint ckpt1 tftp://192.168.1.10/ckptmeta vrf default
#####
100.0%
Success
```

Copying checkpoint configuration to a remote file through SFTP:

```
switch# copy checkpoint ckpt1 sftp://root@192.168.1.10/ckptmeta vrf default
The authenticity of host '192.168.1.10 (192.168.1.10)' can't be established.
ECDSA key fingerprint is SHA256:FtOm6Uxuxumil7VCwLnHz92H9LkY+eURbdddOETy50.
Are you sure you want to continue connecting (yes/no)? yes
root@192.168.1.10's password:
sftp> put /tmp/ckptmeta ckptmeta
Uploading /tmp/ckptmeta to /root/ckptmeta
Warning: Permanently added '192.168.1.10' (ECDSA) to the list of known
hosts.
Connected to 192.168.1.10.
Success
```

Command History

Release	Modification
10.18	Optional parameters have been integrated: CLI, JSON, and checkpoint.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

```
copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}
```

Syntax

```
copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}
```

Description

Copies an existing checkpoint configuration to the running configuration or to the startup configuration.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the name of an existing checkpoint.
{running - config startup - config}	Selects whether the running configuration or the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.

Examples

Copying **ckpt1** checkpoint to the running configuration:

```
switch# copy checkpoint ckpt1 running-config
Success
```

Copying **ckpt1** checkpoint to the startup configuration:

```
switch# copy checkpoint ckpt1 startup-config
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

```
copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>
```

Syntax

```
copy checkpoint <CHECKPOINT-NAME>
               <STORAGE-URL>
```

Description

Copies an existing checkpoint configuration to a USB drive. The file format is defined when the checkpoint was created.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the name of the checkpoint to copy. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-).

Parameter	Description
<code><STORAGE-URL></code>	Specifies the name of the target file on the USB drive using the following syntax: <code>usb: /<FILE></code> The USB drive must be formatted with the FAT file system.

Examples

Copying the **test** checkpoint to the **testCheck** file on the USB drive:

```
switch# copy checkpoint test usb:/testCheck
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

```
copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME>
```

Syntax

```
copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME> [vrf <VRF-NAME>]
```

Description

Copies a remote configuration file to a checkpoint. The remote configuration file must be in checkpoint format.

Parameter	Description
<code><REMOTE-URL></code>	Specifies a remote file using the following syntax: TFTP format: <pre>tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME></pre> SFTP format: <pre>sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME></pre> SCP format: <pre>scp://USER@{IP HOST}[:PORT]/FILE</pre>
<code><CHECKPOINT-NAME></code>	Specifies the name of the target checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (<code>_</code>) and dashes (<code>-</code>). Required.  NOTE Do not start the checkpoint name with <code>CPC</code> because it is used for system-generated checkpoints.
<code>vrf <VRF-NAME></code>	Specifies a VRF name. Default: <code>default</code> .

Examples

Copying a checkpoint format file to checkpoint **ckpt5** on the default VRF:

```
switch# copy tftp://192.168.1.10/ckptmeta checkpoint ckpt5
#####
100.0%
100.0%
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

```
copy <REMOTE-URL> startup-config|running-config
```

Syntax

```
copy <REMOTE-URL> startup-config|{running-config [overwrite]} [vrf <VRF-NAME>]
```

Description

Copies a remote file containing a switch configuration to the running configuration or to the startup configuration.

Parameter	Description
<code><REMOTE-URL></code>	Specifies a remote file with the following syntax: TFTP format: <pre>tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME></pre> SFTP format: <pre>sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME></pre> SCP format: <pre>scp://USER@{IP HOST}[:PORT]/FILE</pre>
<code>startup-config</code>	Indicates that the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.

Parameter	Description
<code>running-config</code>	Indicates that the running configuration receives the copied checkpoint configuration.
<code>overwrite</code>	By default, the CLI configuration contained in the copied file is applied on top of the running - config, so the CLI command does not clear the running - config before applying the CLI commands. In the event that a particular CLI command fails then the failure is logged in event logs and the next line in CLI configuration is processed. When you include the overwrite parameter, the configuration is only applied to the running - config if all the commands succeeded without errors. If there are any errors during the copy process, the existing running - config will remain intact. You can then inspect the event logs to check and fix the errors and retry the operation. If more than 20 errors are present, the operation stops processing any further commands. The event logs will display only those errors processed up to that point.
<code>vrf <VRF-NAME></code>	Specifies the name of a VRF. Default: default.

Usage

The switch copies only certain file types. The format of the file is automatically detected from contents of the file. The **startup-config** option only supports the JSON file format and checkpoints, but the **running-config** option supports the JSON and CLI file formats and checkpoints.

When a file of the CLI format is copied, it overwrites the running configuration. The CLI command does not clear the running configuration before applying the CLI commands. All of the CLI commands in the file are applied line-by-line. If a particular CLI command fails, the switch logs the failure and it continues to the next line in the CLI configuration. The event log (**show events -d hpe-config**) provides information as to which command failed.

Examples

Copying a JSON format file to the running configuration:

```
switch# copy tftp://192.168.1.10/runjson running-config
#####
100.0%
Configuration may take several minutes to complete according to
configuration file size

--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%-
-
Success
```

Copying a CLI format file to the running configuration with an error in the file:

```

switch# copy tftp://192.168.1.10/runcli running-config
#####
100.0%
Configuration may take several minutes to complete according to
configuration file size

--0%---10%---20%---30%---40%---50%---60%---70%---80%---90%---100%-
-
Some of the configuration lines from the file were NOT applied. Use 'show
events -d hpe-config' for more info.
#####
100.0%
Configuration may take several minutes to complete according to
configuration file size

--0%---10%---20%---30%---40%---50%---60%---70%---80%---90%---100%-
-
Some of the configuration lines from the file were NOT applied. Use 'show
events -d hpe-config' for more info.

```

Copying a CLI format file to the startup configuration:

```

switch# copy tftp://192.168.1.10/startjson startup-config
#####
100.0%
100.0%
Success

```

Copying an unsupported file format to the startup configuration:

```

switch# copy tftp://192.168.1.10/startfile startup-config
#####
100.0%
100.0%
unsupported file format

```

Command History

Release	Modification
10.15	The overwrite parameter is introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

```
copy {running-config | startup-config} <REMOTE-URL>
```

Syntax

```
copy {running-config | startup-config} <REMOTE-URL> {cli | json} [vrf <VRF-NAME>]
```

Description

Copies the running configuration or the startup configuration to a remote file in either CLI or JSON format.

Parameter	Description
{running-config startup-config}	Selects whether the running configuration or the startup configuration is copied to a remote file.
<REMOTE-URL>	Specifies the remote file using the syntax: TFTP format: <pre>tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME></pre> SFTP format: <pre>sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME></pre> SCP format: <pre>scp://USER@{IP HOST}[:PORT]/FILE</pre>
{cli json}	Selects the remote file format: P: CLI or JSON.

Parameter	Description
<code>vrf <VRF-NAME></code>	Specifies the name of a VRF. Default: default.

Examples

Copying a running configuration to a remote file in CLI format:

```
switch# copy running-config tftp://192.168.1.10/runcli cli
#####
100.0%
Success
```

Copying a running configuration to a remote file in JSON format:

```
switch# copy running-config tftp://192.168.1.10/runjson json
#####
100.0%
Success
```

Copying a startup configuration to a remote file in CLI format:

```
switch# copy startup-config sftp://root@192.168.1.10/startcli cli
root@192.168.1.10's password:
sftp> put /tmp/startcli startcli
Uploading /tmp/startcli to /root/startcli
Connected to 192.168.1.10.
Success
```

Copying a startup configuration to a remote file in JSON format:

```
switch# copy startup-config sftp://root@192.168.1.10/startjson json
root@192.168.1.10's password:
sftp>
root@192.168.1.10's password:
sftp> put /tmp/startjson startjson
Uploading /tmp/startjson to /root/startjson
Connected to 192.168.1.10.
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

```
copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}
```

Syntax

```
copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}
```

Description

Copies the running configuration to the startup configuration or to a new checkpoint. If the startup configuration is already present, the command overwrites the existing startup configuration.

Parameter	Description
startup-config	Specifies that the startup configuration receives a copy of the running configuration.
checkpoint <CHECKPOINT-NAME>	Specifies the name of a new checkpoint to receive a copy of the running configuration. The checkpoint name can be comprised of alphanumeric character, underscores (_) and dashes (-), and must be 32 characters or fewer.



NOTE

Do not start the checkpoint name with **CPC** because it is used for system-generated checkpoints.

Examples

Copying the running configuration to the startup configuration:

```
switch# copy running-config startup-config  
Success
```

Copying the running configuration to a new checkpoint named **ckpt1**:

```
switch# copy running-config checkpoint ckpt1
Success
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

```
copy {running-config | startup-config} <STORAGE-URL>
```

Syntax

```
copy {running-config | startup-config} <STORAGE-URL> {cli | json}
```

Description

Copies the running configuration or a startup configuration to a USB drive.

Parameter	Description
{running-config startup-config}	Selects the running configuration or the startup configuration to be copied to the switch USB drive.
<STORAGE-URL>	Specifies a remote file with the following syntax: usb:/ <file>
{cli json}	Selects the format of the remote file: CLI or JSON.

Usage

The switch supports JSON and CLI file formats when copying the running or starting configuration to the USB drive. The USB drive must be formatted with the FAT file system.

The USB drive must be enabled and mounted with the following commands:

```
switch(config)# usb
switch(config)# end
switch# usb mount
```

Examples

Copying a running configuration to a file named **runCLI** on the USB drive:

```
switch# copy running-config usb:/runCLI cli
Success
```

Copying a startup configuration to a file named **startCLI** on the USB drive:

```
switch# copy startup-config usb:/startCLI cli
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy startup-config running-config

Syntax

```
copy startup-config running-config
```

Description

Copies the startup configuration to the running configuration.

Examples

```
switch# copy startup-config running-config
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

```
copy <STORAGE-URL> running-config
```

Syntax

```
copy <STORAGE-URL> {running-config | startup-config | checkpoint  
<CHECKPOINT-NAME>}
```

Description

This command copies a specified configuration from the USB drive to the running configuration, to a startup configuration, or to a checkpoint.

Parameter	Description
<STORAGE-URL>	Specifies the name of a configuration file on the USB drive with the syntax: usb:/ <FILE>

Parameter	Description
<code>running-config</code>	Specifies that the configuration file is copied to the running configuration. The file must be in CLI, JSON, or checkpoint format or the copy will fail. the copy will not work.
<code>startup-config</code>	Specifies that the configuration file is copied to the startup configuration. The switch stores this configuration between reboots. The startup configuration is used as the operating configuration following a reboot of the switch. The file must be in JSON or checkpoint format or the copy will fail.
<code>checkpoint <CHECKPOINT-NAME></code>	Specifies the name of a new checkpoint file to receive a copy of the configuration. The configuration file on the USB drive must be in checkpoint format.



NOTE

Do not start the checkpoint name with **CPC** because it is used for system-generated checkpoints.

Usage

This command requires that the USB drive is formatted with the FAT file system and that the file be in the appropriate format as follows:

- **running-config**: This option requires the file on the USB drive be in CLI, JSON, or checkpoint format.
- **startup-config**: This option requires the file on the USB drive be in JSON or checkpoint format.
- **checkpoint** <checkpoint-name>: This option requires the file on the USB drive be in checkpoint format.

Examples

Copying the file **runCli** from the USB drive to the running configuration:

```
switch# copy usb:/runCli running-config
Configuration may take several minutes to complete according to
configuration
file size
--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%-
-
Success
```

Copying the file **startUp** from the USB drive to the startup configuration:

```
switch# copy usb:/startUp startup-config
Success
```

Copying the file **testCheck** from the USB drive to the **abc** checkpoint:

```
switch# copy usb:/testCheck checkpoint abc
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

erase

Syntax

```
erase
  checkpoint <checkpont-name>
  core-dump all|daemon|dsm|kernel|vsf
  startup-config
  all
```

Description

Deletes an existing checkpoint, startup configuration, or core-dump.

Parameter	Description
checkpoint <CHECKPOINT-NAME>	Specifies the name of a checkpoint.
core-dump all daemon <daemon-name>	Erase one of the following sets of core - dump files: all: Erase all core - dump files.

Parameter	Description
<code> kernel</code>	- daemon <daemon - name>: Erase daemon core - dump file s. - kerne:l Erase the kernel core - dump. - vsf Erase daemon core - dump files for VSF.
<code>startup-config</code>	Specifies the startup configuration.
<code>all</code>	Specifies all checkpoints.

Examples

Erasing checkpoint **ckpt1**:

```
switch# erase checkpoint ckpt1
```

Erasing the startup configuration:

```
switch# erase startup-config
```

Erasing all checkpoints:

```
switch# erase checkpoint all
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint <CHECKPOINT-NAME>

Syntax

```
show checkpoint <CHECKPOINT-NAME> [json]
```

Description

Shows the configuration of a checkpoint.

Parameter	Description
<code><CHECKPOINT-NAME></code>	Specifies the name of a checkpoint.
<code>[json]</code>	Specifies that the output is displayed in JSON format.

Examples

Showing the configuration of the **ckpt1** checkpoint in CLI format:

```
switch# show checkpoint ckpt1
Checkpoint configuration:
!
!Version AOS-CX PL.10.07.0000K-75-g55e5193
!export-password: default
lacp system-priority 65535
user admin group administrators password ciphertext
AQBapQjwipbev36io0jFfde7ZzrHckncal1D+3n8XFTZKQdmYgAAADetYOeHSme93xzdD0uz6Vr9
Kl+XBzB+2GB0UBxSF7rvgn2x8KSgkqv7iqXVQ0Te6LkSMnH4BdNaT3Bf25qyvOqmr4Yak01V3rg8
zAOADkPktQD8joTHXflzwomoIzcmv/uX
cli-session
    timeout 0
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface lag 1
```

```
no shutdown
vlan access 1
interface lag 128
no shutdown
vlan access 1
interface lag 129
shutdown
vlan access 1
lacp mode active
interface 1/1/1
no shutdown
lag 128
lacp port-id 65535
interface 1/1/2
no shutdown
vlan access 1
interface 1/1/3
no shutdown
vlan access 1
interface 1/1/4
no shutdown
vlan access 1
interface 1/1/5
no shutdown
vlan access 1
interface 1/1/6
no shutdown
vlan access 1
interface 1/1/7
no shutdown
vlan access 1
interface 1/1/8
no shutdown
vlan access 1
interface 1/1/9
no shutdown
vlan access 1
interface 1/1/10
no shutdown
vlan access 1
interface 1/1/11
no shutdown
vlan access 1
```

```

interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
snmp-server vrf default
!
!
!
!
!
https-server vrf default

```

Showing the configuration of the **ckpt1** checkpoint in JSON format:

```

switch# show checkpoint ckpt1 json
Checkpoint configuration:
{
  "AAA_Server_Group": {
    "local": {
      "group_name": "local"
    },
    "none": {
      "group_name": "none"
    }
  },
  ...
  ...
  ...
  ...

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint <CHECKPOINT-NAME> hash

Syntax

```
show checkpoint <CHECKPOINT-NAME> hash [cli | json]
```

Description

Shows a configuration checkpoint hash calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
<code><CHECKPOINT-NAME></code>	Specifies an existing checkpoint name.
<code>[cli json]</code>	Selects either the CLI or JSON format.

Examples

Showing a checkpoint SHA-256 hash in JSON format:

```
switch# show checkpoint ckpt1 hash json
Calculating the hash: [Success]
The SHA-256 hash of the checkpoint in JSON format, created in image
```

XX.10.08.xxxx:

cc7a57a9bbb4e6600d3b4180296a35f6af9e797ce9c439955dfe5de58b06da9e

This hash is only valid for comparison to a baseline hash if the configuration has

not been explicitly changed (such as with a CLI command, REST operation, etc.)

or implicitly changed (such as by changing a hardware module, upgrading the SW version, etc.).

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint post-configuration

Syntax

```
show checkpoint post-configuration
```

Description

Shows the configuration settings for creating system checkpoints.

Examples

```
switch# show checkpoint post-configuration
Checkpoint Post-Configuration feature
-----
Status      : enabled
Timeout (sec) : 300
```


Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint

Syntax

```
show checkpoint
```

Description

Shows a detailed list of all saved checkpoints.

Examples

Showing a detailed list of all saved checkpoints:

```
switch# show checkpoint
NAME           TYPE           WRITER  DATE (YYYY/MM/DD)  IMAGE VERSION
ckpt1          checkpoint     User    2017-02-23T00:10:02Z  XX.01.01.000X
ckpt2          checkpoint     User    2017-03-08T18:10:01Z  XX.01.01.000X
ckpt3          checkpoint     User    2017-03-09T23:11:02Z  XX.01.01.000X
ckpt4          checkpoint     User    2017-03-11T00:00:03Z  XX.01.01.000X
ckpt5          latest        User    2017-03-14T01:12:27Z  XX.01.01.000X
```

Command History

Release	Modification
10.08	Command syntax <code>show checkpoint list all</code> is replaced with <code>show checkpoint</code> .

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show checkpoint date

Syntax

```
show checkpoint date <START-DATE>
                    <END-DATE>
```

Description

Shows detailed list of all saved checkpoints created within the specified date range.

Parameter	Description
<START-DATE>	Specifies the starting date for the range of saved checkpoints to show. Format: YYYY - MM - DD .
<END-DATE>	Specifies the ending date for the range of saved checkpoints to show. Format: YYYY - MM - DD .

Examples

Showing a detailed list of saved checkpoints for a specific date range:

```
switch# show checkpoint date 2017-03-08 2017-03-12
NAME                TYPE        WRITER   DATE (YYYY/MM/DD)    IMAGE VERSION
ckpt2               checkpoint  User     2017-03-08T18:10:01Z  XX.01.01.000X
```

ckpt3	checkpoint	User	2017-03-09T23:11:02Z	XX.01.01.000X
ckpt4	checkpoint	User	2017-03-11T00:00:03Z	XX.01.01.000X

Command History

Release	Modification
10.08	Command syntax <code>show checkpoint list date <START-DATE> <END-DATE></code> is replaced with <code>show checkpoint date <START-DATE> <END-DATE></code>
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config hash

Syntax

```
show running-config hash [cli | json]
```

Description

Shows the running-config checkpoint hash, calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
[cli json]	Selects either the CLI or JSON format.

Examples

Showing the running-config checkpoint SHA-256 hash in CLI format:

```
switch# show running-config hash cli
Calculating the hash: [Success]
SHA-256 hash of the config in CLI format:
8db4e7e10f4b7f1a6ab17ad2b4efe0e72f1849103eaf43da62aa1d715075b89e
This hash is only valid for comparison to a baseline hash if the
configuration has
not been explicitly changed (such as with a CLI command, REST operation,
etc.)
or implicitly changed (such as by changing a hardware module, upgrading the
SW version, etc.).
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show startup-config hash

Syntax

```
show startup-config hash [cli | json]
```

Description

Shows the startup-config checkpoint hash, calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
[cli json]	Selects either the CLI or JSON format.

Examples

Showing the startup-config checkpoint SHA-256 hash in CLI format:

```
switch# show startup-config hash cli
Calculating the hash: [Success]
SHA-256 hash of the config in CLI format:
8db4e7e10f4b7f1a6ab17ad2b4efe0e72f1849103eaf43da62aa1d715075b89e
This hash is only valid for comparison to a baseline hash if the
configuration has
not been explicitly changed (such as with a CLI command, REST operation,
etc.)
or implicitly changed (such as by changing a hardware module, upgrading the
SW version, etc.).
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

write memory

Syntax

```
write memory
```

Description

Saves the running configuration to the startup configuration. It is an alias of the command **copy running-config startup-config**. If the startup configuration is already present, this command overwrites the startup configuration.

Examples

```
switch# write memory
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Boot commands

Select a command from the list in the left navigation menu.

Subtopics

[boot set-default](#)

[boot system](#)

[show boot-history](#)

boot set-default

Syntax

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Parameter	Description
primary	Selects the primary network operating system image.
secondary	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```



NOTE

For more information on features that use this command, refer to the Fundamentals Guide or the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

boot system

Syntax

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
<code>primary</code>	Selects the primary operating system image for this reboot and sets the configured default operating system image to primary for future reboots.
<code>secondary</code>	Selects the secondary operating system image for this reboot and sets the configured default operating system image to secondary for future reboots.
<code>serviceos</code>	Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovers OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the **show images** command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the **primary** or **secondary** optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.
You can use the **boot set-default** command to change the configured default operating system image.
- If you select **serviceos** as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the **boot system** command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot system** command is aborted.

Examples

Rebooting the system from the configured default operating system image:


```

switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
The system is going down for reboot.

```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```

switch# boot system secondary
Default boot image set to secondary.
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.

```

Canceling a system reboot:

```

switch# boot system
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show boot-history

Syntax

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.



NOTE

To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
all	Optional. Shows boot information for the active management module.
vsf member <1-10>	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
Index	The position of the boot in the history file. Range: 0 to 3.
Boot ID	A unique ID for the boot . A system - generated 128 - bit string.
Current Boot, up for <time>	For the current boot, the show boot - history command shows the number of seconds the module has been running on the current software.
<Timestamp>: boot reason	For previous boot operations, the show boot - history command shows the time at which the operation occurred and the reason.

Parameter	Description
	son for the boot. The reason for the boot is one of the following values: • <DAEMON - NAME> crash: The daemon identified by <DAEMON - NAME> caused the module to boot. • Kernel crash: The operating system software associated with the module caused the module to boot. • Uncontrolled reboot: The reason for the reboot is not known. • Reboot requested through database: The reboot occurred because of a request made through the CLI or other API.

Table 1. Description of reboots handled through the database

Boot History String	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short - press of the reset button
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.
Configuration version migration	A configuration version migration occurred which required a reboot.
Console error	The console failed to start.
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over from an unresponsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in the allotted time.

Boot History String	Description
Critical service fault (error)	A daemon critical to switch operation has stopped functioning. An extra error string may be present to describe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
VSF member renumbered	A user requested a renumber of a VSF member.
VSF switchover requested	A user requested a VSF switchover.
VSX software update	Reset triggered by a VSX software update.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis low critical temperature	Chassis temperature below the minimum operating threshold.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceeded.
ISSU SMM update	Standby management module reboot triggered by an In - Service Software Upgrade (ISSU).
ISSU switchover	Redundancy switchover triggered by an In - Service Software Upgrade.
ISSU aborted	Standby management module reset triggered by failure during an In - Service Software Upgrade.
Rollback timer expired	Reset triggered by the ISSU rollback timer expiring.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====
Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs
Index : 1
```

```

Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version:
FL.10.14.0000-1619-ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2
Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1
Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 : Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 : Switch boot count is 3
25 Aug 23 10:29:02 : Primary overtemperature fault detected with 0x2 in
PSU 1/1

```

Showing the boot history of the active management module and all line modules:

```

switch#
Management module
=====
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
Management module
=====
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2

```

```

Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
switch# show boot-history
Management module
=====
Index : 2
Boot ID : a61ad00d10864c748bc7893a5d4af2e4
15 Dec 23 19:02:02 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:02:02 : Switch boot count is 0
15 Dec 23 19:02:17 : PSU 1/1: Fault detected
Index : 1
Boot ID : 30d831bbfdfa425baf50a629ee01b185
15 Dec 23 19:01:58 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:01:58 : Switch boot count is 0

```

The following example displays the boot history for the VSF member **2**.

```

switch# show boot-history vsf member 2
Member-2
=====
Index : 0
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs
Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Firmware management commands

Select a command from the list in the left navigation menu.

Subtopics

- [copy {primary | secondary} <REMOTE-URL>](#)
- [copy {primary | secondary} <FIRMWARE-FILENAME>](#)
- [copy primary secondary](#)
- [copy <REMOTE-URL>](#)
- [copy secondary primary](#)
- [copy <STORAGE-URL>](#)
- [copy hot-patch](#)
- [hot-patch](#)
- [show hot-patch](#)

copy {primary | secondary} <REMOTE-URL>

Syntax

```
copy {primary | secondary} <REMOTE-URL> [vrf <VRF-NAME>]
```

Description

Uploads a firmware image to a TFTP or SFTP server.

Parameter	Description
{primary secondary}	Selects the primary or secondary image profile to upload. Required
	Specifies the URL to receive the uploaded firmware using SFTP, TFTP or SCP.

Parameter	Description
<code><REMOTE-URL></code>	TFTP format: <pre>tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME></pre> SFTP format: <pre>sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME></pre> SCP format: <pre>scp://USER@{IP HOST}[:PORT]/FILE</pre>
<code>vrf <VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

TFTP upload:

```
switch# copy primary tftp://192.0.2.0/00_10_00_0002.swi
#####
100.0%
Verifying and writing system firmware...
```

SFTP upload:

```
switch# copy primary sftp://swuser@192.0.2.0/00_10_00_0002.swi
swuser@192.0.2.0's password:
Connected to 192.0.2.0.
sftp> put primary.swi XL_10_00_0002.swi
Uploading primary.swi to /users/swuser/00_10_00_0002.swi
primary.swi                  100% 179MB 35.8MB/s 00:05
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

```
copy {primary | secondary} <FIRMWARE-FILENAME>
```

Syntax

```
copy {primary | secondary} <FIRMWARE-FILENAME>
```

Description

Copies a firmware image to USB storage.

Parameter	Description
{primary secondary}	Selects the primary or secondary image from which to copy the firmware. Required
<FIRMWARE-FILENAME>	Specifies the name of the firmware file to create on the USB storage device. Prefix the filename with usb:/. For example: usb/firmware_v1.2.3.swi For information on how to format the path to a firmware file on a USB drive, see USB URL.

Examples

```
switch# copy primary usb:/11.10.00.0002.swi
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy primary secondary

Syntax

```
copy primary secondary
```

Description

Copies the firmware image from the primary to the secondary location.

Examples

```
switch# copy primary secondary
The secondary image will be deleted.
Continue (y/n)? y
Verifying and writing system firmware...
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy <REMOTE-URL>

Syntax

```
copy <REMOTE-URL> {hot-patch|primary|secondary} [vrf <VRF-NAME>]
```

Description

Downloads a hot-patch or firmware image from a TFTP or SFTP server.

Parameter	Description
<code><REMOTE-URL></code>	Specifies the URL from which to download the firmware using SFTP or TFTP. TFTP format: <pre>tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME></pre> SFTP format: <pre>sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME></pre> SCP format: <pre>scp://USER@{IP HOST}[:PORT]/FILE</pre>
<code>{hot-patch primary secondary}</code>	Select a hot - patch or a primary or secondary image profile for receiving the downloaded firmware. Required.  NOTE For more information about hot - patch, see hot - patch.
<code>vrf <VRF-NAME></code>	Specifies the name of a VRF. Default: default.

TFTP usage

To specify a URL with:

- an IPv4 address: **tftp://** 192.0.2.1/a.txt
- an IPv6 address: **tftp://[** 2000::2**]/**a.txt
- a hostname: **tftp://** hpe.com/a.txt

To specify TFTP with:

- the port number of the server in the URL: **tftp:// 192.0.2.1:12/a.txt**
- the blocksize in the URL: **tftp:// 192.0.2.1 ;blocksize= 1462 / a.txt**
The valid blocksize range is 8 to 65464.
- the port number of the server and blocksize in the URL: **tftp://192.0.2.1 :12 ;blocksize= 1462 / a.txt**

To specify a file in a directory of URL: **tftp:// 192.0.2.1 / dir / a.txt**

SFTP usage

To specify:

- A URL with an IPv4 address: **sftp:// user@192.0.2.1/a.txt**
- A URL with an IPv6 address: **sftp:// user @[2000::2]/ a.txt**
- A URL with a hostname: **sftp:// user @ hpe.com / a.txt**
- SFTP port number of a server in the URL: **sftp:// user @ 192.0.2.1:12 / a.txt**
- A file in a directory of URL: **sftp:// user @ 192.0.2.1 / dir / a.txt**
- To specify a file with absolute path in the URL: **sftp:// user @ 192.0.2.1 // home/user/a.txt**

SCP Usage

To specify:

- A username with an IP address: **scp://user@192.0.2.1:12/a.txt**
- A username with a remote host: **scp://user@hpe.com/a.txt**

Examples

TFTP download for a hot-patch:

```
switch# copy tftp://192.168.1.1/FL.10.12.0001-0002.patch hot-patch vrf vrf1

Fetching /users/swuser/FL.10.10.0001-0002.patch to hotpatch.dnld.uE2YT1
FL.10.12.0001-0002.patch                               100%   62KB   12.4MB/s   00:00
Verifying and writing hot-patch...
```

TFTP download for primary software image:

```
switch# copy tftp://192.10.12.0/FL_10_12_0001.swi primary
The primary image will be deleted.
Continue (y/n)? y
#####
```

```
100.0%
Verifying and writing system firmware...
```

SFTP download:

```
switch# copy sftp://swuser@192.10.12.0/FL_10_12_0001.swi primary
The primary image will be deleted.
Continue (y/n)? y
The authenticity of host '192.10.12.0 (192.10.12.0)' can't be established.
ECDSA key fingerprint is SHA256:L64khLwlyLgXlARKRMiwcAAK8oRaQ8C0oWP+PkGBXHY.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.10.12.0' (ECDSA) to the list of known hosts.
swuser@192.10.12.0's password:
Connected to 192.10.12.0.
Fetching /users/swuser/ss.10.00.0002.swi to ss.10.00.0002.swi.dnld
/users/swuser/ss.10.00.0002.swi          100% 179MB 25.6MB/s
00:07
Verifying and writing system firmware...
```

Command History

Release	Modification
10.12	The <code>hot-patch</code> parameter is supported on all platforms.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy secondary primary

Syntax

```
copy secondary primary
```

Description

Copies the firmware image from the secondary to the primary location.

Examples

```
switch# copy secondary primary
The primary image will be deleted.
Continue (y/n)? y
Verifying and writing system firmware...
switch# copy sftp://stor@192.22.1.0/im-switch.swi primary vrf mgmt

The primary image will be deleted.
Continue (y/n)? y
The authenticity of host '192.22.1.0 (192.22.1.0)' can't be established.
ECDSA key fingerprint is SHA256:MyI1xbdKnehYut0NLfL69gDpNzCmZqBVvBaRR46m7o8.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.22.1.0' (ECDSA) to the list of known hosts.
stor@192.22.1.0's password:
Connected to 192.22.1.0.
sftp> get c8d5b9f-topflite.swi c8d5b9f-topflite.swi.dnld
Fetching /home/dr/im-switch.swi to c8d5b9f-topflite.swi.dnld
/home/dr/im-switch.swi          100% 226MB 56.6MB/s   00:04
Verifying and writing system firmware...
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy <STORAGE-URL>

Syntax

```
copy <STORAGE-URL> {hot-patch|primary|secondary}
```

Description

Copies, verifies, and installs a hot-patch or firmware image from a USB storage device connected to the active management module.

Parameter	Description
<code><STORAGE-URL></code>	Specifies the name of the firmware file to copy from the storage device. Required. USB format: <pre>usb: /<FILENAME></pre>
<code>{hot-patch primary secondary}</code>	Select a hot - patch image or a primary or secondary profile for receiving the copied firmware.  NOTE For more information about hot - patch, see hot - patch.

USB usage

To specify a file:

- In a USB storage device: **usb:/** a.txt
- In a directory of a USB storage device: **usb:/** dir / a.txt

Examples

```
switch# copy usb:/FL.10.12.0001-0002.patch
switch# copy usb:/FL.10.12.0001.swi primary
The primary image will be deleted.
Continue (y/n)? y
Verifying and writing system firmware...
```

Command History

Release	Modification
10.12	The <code>hot-patch</code> parameter is supported on all platforms.

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy hot-patch

Syntax

```
copy hot-patch <Word>
                {<REMOTE-URL>|<Storage-URL>} [vrf <VRF-NAME>]
```

Description

Copies a hot-patch from a switch to the specified remote URL or storage URL.

Parameter	Description
<Word>	Name of the hot - patch software to upload.
<REMOTE-URL>	Specifies the URL to receive the uploaded patch using SFTP or TFTP. For information on how to format the remote URL, see URL formatting for copy commands.
vrf <VRF-NAME>	[Optional] specify the VRF instance to use for upload.
<STORAGE-URL>	Specifies the name of the patch file to create on the USB storage device. Prefix the filename with usb:/ , for example, usb:/firmware_FL_10_12_0001 - 0002.patch .

Examples

```
switch# copy hot-patch FL_10_12_0001-0002.patch tftp:172.21.18.170/
FL_10_12_0001-0002.patch vrf vrf1
```


Related Commands

Command	Description
<code>copy <REMOTE-URL></code>	Downloads a hot - patch image from a TFTP or SFTP server.
<code>hot-patch apply</code>	Apply a hot - patch image or remove it from the switch.

Command History

Release	Modification
10.12	Hot - patch is now supported on all platforms.
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

hot-patch

Syntax

```
hot-patch apply|remove <name.patch>
no hot-patch apply <name.patch>
```

Description

Apply hot-patch software or remove it from the switch. The **no** form of the **hot-patch apply** command disables the hot-patch image, but does not remove it from the switch. Rebooting the system after disabling or removing the patch is not required.

Profile names	Description
<code>apply <name.patch></code>	Apply the specified hot - patch image to a standalone switch or VSF stack. AOS-CX hot - patch software images can be obtained

Profile names

Description

d from HPE Aruba Networking customer support, and are identified with a **.patch** extension.

```
remove <name.patch>
```

Disables the hot - patch image and removes the patch from the switch. This removal will also disable the patch. Once removed, a hot - patch must be downloaded again in order to be applied.

Usage

A hot-patch can be downloaded from a remote server onto a switch then applied without rebooting the switch. When the hot-patch is disabled, the hot-patch will still remain on the system. The disabled hot-patch can be removed from the system without the need for a reboot of the system.

If a checkpoint configuration that does not contain a hot-patch is restored to a running configuration that does have a hot-patch, the patch is not deleted, it remains as not applied but is present in the device memory.

Examples

```
switch(config)# hot-patch apply FL_10_12_0001-0002.patch
```

Related Commands

Command

Description

```
copy <REMOTE-URL>
```

Downloads and installs a hot - patch image from a TFTP or SFTP server.

```
copy hot-patch
```

Copies a hot - patch software image from a switch to a specified remote URL or storage URL.

Command History

Release

Modification

10.12

Hot - patch is now supported on all platforms.

10.10

Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show hot-patch

Syntax

```
show hot-patch [detail]
```

Parameter	Description
detail	Displays the detailed status of all hot - patches present on the system.

Description

the **show hot-patch** command displays the status of all hot-patches present on the system. The **show hot-patch detail** command displays detailed information for all hot patches present on the system.

Examples

```
switch# show hot-patch
Name                               Status
-----
FL_10_12_0001-0002.patch          Applied
switch# show hot-patch detail
Name                               : FL_10_12_0001-0002.patch
Status                            : Applied
Version                           : FL_10_12_0001-0002.patch
Compatible Version                 : FL.10.12.0001
Issues Fixed                       : CR1234, CR2345
Patch Date                        : 2022-03-29 20:46:15 UTC
Patch ID                           : AOS-CX:FL.10.12.0001-sp1-256-gd457e88d39:20220442009
Patch SHA                          : a4038d06a2e5fe7e0d457e868d39e526185b
```

Related Commands

Command	Description
<pre>cop y <REMOTE-URL></pre>	Downloads a hot - patch image from a TFTP or SFTP server.
<pre>hot -patch apply</pre>	Apply a hot - patch image or remove it from the switch.

Command History

Release	Modification
10.12	Command supported on all platforms.
10.10	Command introduced on 6300 Switch series.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Dynamic Segmentation

For information on dynamic segmentation, view the Security Guide.

SNMP

Simple Network Management Protocol (SNMP) is an Internet-standard protocol used for managing and monitoring the devices connected to a network by collecting, organizing and modifying information about managed devices on IP networks.

Subtopics

Configuring SNMP

Configuring SNMP

(The SNMP agent provides read-only access.)

Procedure

1. Enable SNMP on a VRF using the command `snmp-server vrf`.
2. Set the system contact, location, and description for the switch with the following commands:
 - `snmp-server system-contact`
 - `snmp-server system-location`
 - `snmp-server system-description`
3. If required, change the default SNMP port on which the agent listens for requests with the command `snmp-server agent-port`.
4. By default, the agent uses the community string **public** to protect access through SNMPv1/v2c. Set a new community string with the command `snmp-server community`.
5. Configure the trap receivers to which the SNMP agent will send trap notifications with the command `snmp-server host`.
6. Create an SNMPv3 context and associate it with any available SNMPv3 user to perform context specific v3 MIB polling using the command `snmpv3 user`.
7. Create an SNMPv3 context and associate it with an available SNMPv1/v2c community string to perform context specific v1/v2c MIB polling using the command `snmpv3 context`.
8. Review your SNMP configuration settings with the following commands:
 - `show snmp agent-port`
 - `show snmp community`

- `show snmp system`
- `show snmpv3 context`
- `show snmp trap`
- `show snmp vrf`
- `show snmpv3 users`
- `show tech snmp`

Example 1

This example creates the following configuration:

- Enables SNMP on the out-of-band management interface (VRF **mgmt**).
- Sets the contact, location, and description for the switch to: **JaniceM, Building2, LabSwitch**.
- Sets the community string to **Lab8899X**.

```
switch(config)# snmp-server vrf mgmt

switch(config)# snmp-server system-contact JaniceM
switch(config)# snmp-server system-location Building2
switch(config)# snmp-server system-description LabSwitch
switch(config)# snmp-server community Lab8899X
```

Example 2

This example creates the following configuration:

- Creates an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**.
- Associates the SNMPv3 user `Admin` with a context named `newContext`.

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword
priv des
priv-pass plaintext myprivpass
switch(config)# snmpv3 user Admin context newContext
```



NOTE

Refer to the SNMP Guide for SNMP Commands.

HPE Aruba Networking Central integration

The HPE Aruba Networking Central network management solution, a software-as-a-service subscription in the cloud, provides streamlined management of multiple network devices. AOS-CX switches are able to talk to HPE Aruba Networking Central and utilize cloud-based management functionality. Cloud-based management functionality allows for the deployment of network devices at sites with no or few dedicated IT personnel (branch offices, retail stores, and so forth). AOS-CX switches utilize secure communication protocols to connect to the HPE Aruba Networking Central cloud portal, and can coexist with corporate security standards, such as those mandating the use of firewalls.



NOTE

When HPE Aruba Networking Central manages AOS-CX switches, it functions as the single source of truth and the Web UI operates in read-only mode.

This feature provides:

- Zero-touch provisioning
- Network Management/Remote monitoring
- Events/alerts notification
- Switch Configuration using templates
- Firmware management

Subtopics

Connecting to HPE Aruba NetworkingCentral

Custom CA certificate

Support mode in HPE Aruba Networking Central

One Touch Provisioning

HPE Aruba Networking Central Location Discovery

Troubleshooting

HPE Aruba Networking Central commands

Connecting to HPE Aruba NetworkingCentral

AOS-CX switch downloads the location of HPE Aruba Networking Central server using:

- Command-line interface (CLI).
- HPE GreenLake server.
- DHCP options provided during ZTP.

DHCP servers are used to connect to Central on-premise management.



NOTE

When a switch is able to connect to HPE Aruba Networking Central, but is not registered in the HPE Aruba Networking Central inventory or does not have a proper license, the switch will get disconnected. If the HPE Aruba Networking Central feature is enabled using this command, the switch will then reconnect back to HPE Aruba Networking Central and will get disconnected again. This connect/disconnect process will continue until the switch is properly registered in HPE Aruba Networking Central. To avoid this unnecessary reconnection cycle, best practices is to disable HPE Aruba Networking Central until the switch is registered in HPE Aruba Networking Central, or a license is obtained for that device.

Custom CA certificate

To use a custom CA certificate to connect to HPE Aruba Networking Central, AOS-CX switch downloads the certificate from the HPE GreenLake server.



NOTE

- If there is no custom CA provided by HPE GreenLake, the CA certificate present in the device is used.
- Duplicate CA certificates from HPE GreenLake server will be ignored.
- If CA certificate is absent in consecutive responses from the HPE GreenLake server, the installed custom CA certificate in device will be removed.
- Switch will have only one custom CA certificate installed from the HPE GreenLake Server.
- The certificate installed from the HPE GreenLake server will not be displayed in the show commands.

Support mode in HPE Aruba Networking Central

When the AOS-CX switch is managed by HPE Aruba Networking Central, the switch configuration cannot be modified using other interfaces such as CLI or Web UI. The following command categories are blocked:

- auto-confirm
- boot
- checkpoint
- copy-in commands
- erase
- erps
- https-server
- mfgread
- mfgwrite
- port-access
- vsf
- All configuration commands except the aruba-central command

In cases where a maintenance or troubleshooting activity requires configuration updates, aruba-central support-mode can be enabled to allow these operations.



NOTE

The aruba-central support-mode enable or disable operation is effective only in the CLI session where it is executed and does not impact the other CLI sessions.

If the user tries to execute any command that is not allowed, an **Invalid input:** error message is displayed.

One Touch Provisioning

One Touch Provisioning (OTP) allows a switch to be provisioned on HPE Aruba Networking Central with the minimum user configuration.

Configuring One Touch Provisioning

To provision a switch to HPE Aruba Networking Central by using an OTP scenario, you can configure:

- The HPE Aruba Networking Central Location

Following configurations can be done optionally:

- The Domain Name System (DNS) server address.

- The HTTP proxy location.

Configuring HPE Aruba Networking Central location

1. Create the hpe-anw-central context.

```
switch(config)# hpe-anw-central
```

2. Configure the HPE Aruba Networking Central server location

```
switch(hpe-anw-central)# location-override <LOCATION> vrf <VRF>
```



NOTE

IPv4 or IPv6 address can be used.

3. Validate the connectivity status to HPE Aruba Networking Central

```
switch# show hpe-anw-central
```

Configuring DNS server address

1. Configure the DNS server address.

```
switch(config)# ip dns server-address <ADDRESS> vrf <VRF>
```



NOTE

IPv4 or IPv6 address can be used.

2. Validate the DNS server status

```
switch# show ip dns
```

Configuring HTTP proxy location

1. Configure the HTTP Proxy location

```
switch(config)# http-proxy <LOCATION> vrf <VRF>
```



NOTE

LOCATION can be either a FQDN, IPv4 address or IPv6 address

Limitations for Zero Touch Provisioning

The following are the limitations for OTP when provisioning a switch by using IPv6:

- Support for obtaining the HPE Aruba Networking Central location via DHCP.
- Support for connectivity to an HPE Aruba Networking Central on-premise instance.

Supportability

Debug logging can be enabled with the following command:



NOTE

The central debug logs are helpful to determine connectivity issues with HPE Aruba Networking Central.

```
switch# debug central all
```

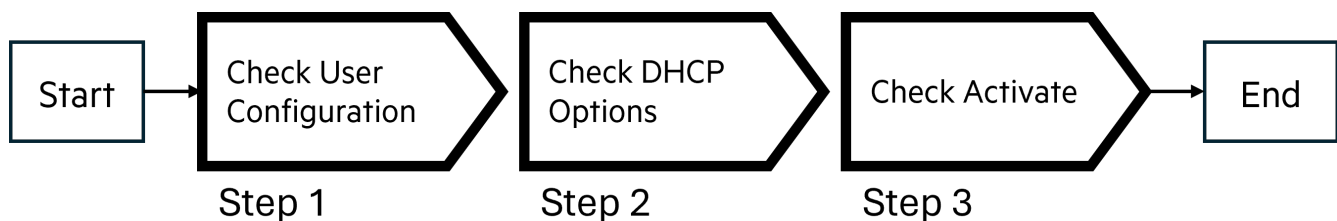
HPE Aruba Networking Central Location Discovery

The CX switch cannot obtain the HPE Aruba Networking Central location to establish connectivity. The switch attempts to retrieve the HPE Aruba Networking Central location from one of the following sources in priority order:

- User configuration via the CLI or REST API
- DHCP options from a DHCP server)
- HPE GreenLake (Activate)

If the HPE Aruba Networking Central location cannot be acquired from any of these sources, connectivity to HPE Aruba Networking Central cannot be established. The switch retries these sources using an exponential backoff with the following sequence: 0s, 2s, 4s, 8s, 16s, 32s, 64s.

Troubleshooting Workflow Diagram



Step 1: Verifying User Configuration

If the HPE Aruba Networking Central location is configured via CLI or REST API, verify the configuration as follows:

1. Check location and location alternative override

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central
location_override_location location_override_location_alternative -flist
HPE_ANW_Central table
location_override_location: "test.central.com"
```

```
location_override_location_alternative: "test2.central.com"
```

Evaluation: Verify the values match the user configuration. The location alternative may be empty if not configured.

2. Check VRF and VRF alternative override

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central location_override_vrf
location_override_vrf_alternative -flist
HPE_ANW_Central table
location_override_vrf: 9b99f005-5051-473e-a13e-029b576e9776
location_override_vrf_alternative: 9b99f005-5051-473e-
a13e-029b576e9776
```

Evaluation: The output contains UUIDs. Match them with VRF names using:

```
switch# start-shell
ovsdb-client dump VRF name _uuid
VRF table_uuid                                name
-----
-----
619e1c63-94a3-44dc-8728-3b831e77bb55 default
9b99f005-5051-473e-a13e-029b576e9776 mgmt
HPE_ANW_Central table
```

Evaluation: Verify the VRF values match the user configuration. The VRF alternative may be empty if not configured.

3. Check Central source value

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central central_source
central_source_alternative -flist
HPE_ANW_Central table
central_source: cli
central_source_alternative: cli
```

Evaluation: The `central_source` value must be set to `cli`. The central source alternative may be empty if not configured.

4. Check final location and VRF

```
switch# start-shell
```

```

        ovssdb-client dump HPE_ANW_Central location
location_alternative vrf vrf_alternative -flist
        HPE_ANW_Central table
        location: "test.central.com"
        location_alternative: "test2.central.com"
        vrf: 9b99f005-5051-473e-a13e-029b576e9776
        vrf_alternative: 9b99f005-5051-473e-a13e-029b576e9776

```

Evaluation: These are the final location and VRF values used to connect to HPE Aruba Networking Central. Verify they match the user configuration. The location alternative and VRF alternative may be empty if not configured.

Corrective Action: If values do not match the expected configuration, reconfigure the Central location using the appropriate CLI commands or REST requests.

Step 2: Verifying DHCP Options

If user configuration is not set, the switch retrieves the HPE Aruba Networking Central location from DHCP options via a DHCP server.

1. Check operational DHCP options

```

switch# start-shell
ovssdb-client dump System operational_dhcp_options
System table
operational_dhcp_options

-----
-----
-----

{central_location="test.central.com",
central_location_source=dhcp-vlan1, central_location_vrf_name=default,
central_location_alternative="test2.central.com",
central_location_alternative_source=dhcp-vlan1,
central_location_alternative_vrf_name=default}

```

Evaluation: Verify `central_location` and `central_location_alternative` match the DHCP server configuration. The location alternative may be empty if not configured. Verify `central_location_source` is set to either `dhcp-mgmt` or `dhcp-vlan`. Verify `central_location_vrf_name` matches the VRF through which the DHCP server is accessible.

2. Check central source value

```

switch# start-shell
ovssdb-client dump HPE_ANW_Central central_source
central_source_alternative -flist

```

```
HPE_ANW_Central table
central_source: dhcp
central_source_alternative: dhcp
```

Evaluation: The `central_source` value must be set to `dhcp`. The central source alternative may be empty if not configured.

3. Check final location and VRF

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central location
location_alternative vrf vrf_alternative -flist
HPE_ANW_Central table
location: "test.central.com"
location_alternative: "test2.central.com"
vrf: 619e1c63-94a3-44dc-8728-3b831e77bb55
vrf_alternative: 619e1c63-94a3-44dc-8728-3b831e77bb55
```

Evaluation: These are the final location and VRF values used to connect to Central. Verify they match the DHCP server configuration. The location alternative and VRF alternative may be empty if not configured.

Corrective Action: If values do not match, verify the DHCP server configuration includes the correct Central location options and that the switch can reach the DHCP server.

Step 3: Verifying HPE Greenlake Configuration

If neither user configuration nor DHCP options are available, the switch attempts to retrieve the HPE Aruba Networking Central location from the HPE Greenlake (Activate) server.

1. Check HPE AW Central source value

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central central_source -flist
HPE_ANW_Central table
central_source: activate
```

Evaluation: The `central_source` value must be set to `activate`.

2. Check final location and VRF

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central location vrf -flist
HPE_ANW_Central table
location: "test.central.com"
```

```
vrf: 619e1c63-94a3-44dc-8728-3b831e77bb55
```

Evaluation: These are the final location and VRF values used to connect to **Central**. Verify they match the Greenlake server configuration.

Corrective Action: If values are incorrect, verify the switch is connected to HPE Greenlake and ensure the switch can reach the HPE Greenlake server.

Troubleshooting

Subtopics

[Prerequisites](#)

[Related features](#)

[HPE Aruba Networking Central Location Discovery](#)

[HPE Aruba Networking Central Connectivity](#)

[HPE Aruba Networking Central Activate Connectivity](#)

[HPE Aruba Networking Central RCS](#)

[HPE Aruba Networking Central Disconnection Timeout](#)

Prerequisites

Prerequisites

Enable **support-mode**

Execute the following command to enable support mode on the switch. This allows the execution of CLI commands when Central is the source of truth. This way, the user can use the start-shell command to start a bash shell to run additional debugging commands:

```
switch# hpe-anw-central support-mode
```

Next, disable support mode:

```
switch# no hpe-anw-central support-mode
```

Related features

When configuring the CX device for Central connectivity, it might also be necessary to configure the following features:

Feature	Reason
<u>DNS</u>	Used to translate the Central and/or Activate host names into IPv4 and/or IPv6 addresses
HTTP Proxy	Used as a gateway between the CX Device and/or Central and Activate servers
<u>Zero Touch Provisioning</u>	Used to receive the Central location via DHCP options when the switch boots
<u>Source interface selection</u>	Used to define the outgoing interface from which the connection to Central and/or Activate is established

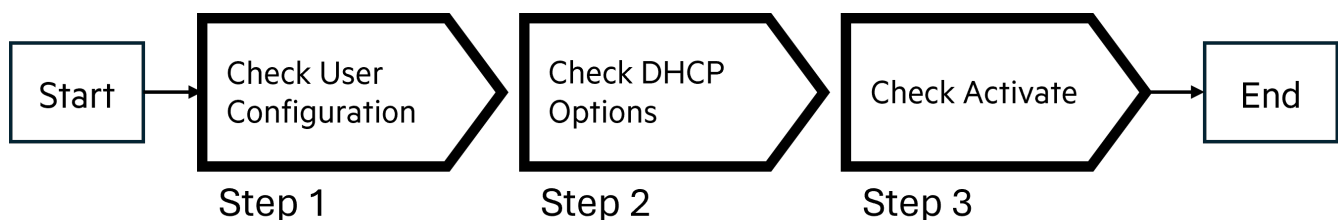
HPE Aruba Networking Central Location Discovery

The CX switch cannot obtain the HPE Aruba Networking Central location to establish connectivity. The switch attempts to retrieve the HPE Aruba Networking Central location from one of the following sources in priority order:

- User configuration via the CLI or REST API
- DHCP options from a DHCP server)
- HPE GreenLake (Activate)

If the HPE Aruba Networking Central location cannot be acquired from any of these sources, connectivity to HPE Aruba Networking Central cannot be established. The switch retries these sources using an exponential backoff with the following sequence: 0s, 2s, 4s, 8s, 16s, 32s, 64s.

Troubleshooting Workflow Diagram



Step 1: Verifying User Configuration

If the HPE Aruba Networking Central location is configured via CLI or REST API, verify the configuration as follows:

1. Check location and location alternative override

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central
```



```
location_override_location location_override_location_alternative -flist
HPE_ANW_Central table
location_override_location: "test.central.com"
location_override_location_alternative: "test2.central.com"
```

Evaluation: Verify the values match the user configuration. The location alternative may be empty if not configured.

2. Check VRF and VRF alternative override

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central location_override_vrf
location_override_vrf_alternative -flist
HPE_ANW_Central table
location_override_vrf: 9b99f005-5051-473e-a13e-029b576e9776
location_override_vrf_alternative: 9b99f005-5051-473e-
a13e-029b576e9776
```

Evaluation: The output contains UUIDs. Match them with VRF names using:

```
switch# start-shell
ovsdb-client dump VRF name _uuid
VRF table_uuid                                name
-----
-----
619e1c63-94a3-44dc-8728-3b831e77bb55 default
9b99f005-5051-473e-a13e-029b576e9776 mgmt
HPE_ANW_Central table
```

Evaluation: Verify the VRF values match the user configuration. The VRF alternative may be empty if not configured.

3. Check Central source value

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central central_source
central_source_alternative -flist
HPE_ANW_Central table
central_source: cli
central_source_alternative: cli
```

Evaluation: The `central_source` value must be set to `cli`. The central source alternative may be empty if not configured.

4. Check final location and VRF

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central location
location_alternative vrf vrf_alternative -flist
HPE_ANW_Central table
location: "test.central.com"
location_alternative: "test2.central.com"
vrf: 9b99f005-5051-473e-a13e-029b576e9776
vrf_alternative: 9b99f005-5051-473e-a13e-029b576e9776
```

Evaluation: These are the final location and VRF values used to connect to HPE Aruba Networking Central. Verify they match the user configuration. The location alternative and VRF alternative may be empty if not configured.

Corrective Action: If values do not match the expected configuration, reconfigure the Central location using the appropriate CLI commands or REST requests.

Step 2: Verifying DHCP Options

If user configuration is not set, the switch retrieves the HPE Aruba Networking Central location from DHCP options via a DHCP server.

1. Check operational DHCP options

```
switch# start-shell
ovsdb-client dump System operational_dhcp_options
System table
operational_dhcp_options

-----
-----
-----

{central_location="test.central.com",
central_location_source=dhcp-vlan1, central_location_vrf_name=default,
central_location_alternative="test2.central.com",
central_location_alternative_source=dhcp-vlan1,
central_location_alternative_vrf_name=default}
```

Evaluation: Verify `central_location` and `central_location_alternative` match the DHCP server configuration. The location alternative may be empty if not configured. Verify `central_location_source` is set to either `dhcp-mgmt` or `dhcp-vlan`. Verify `central_location_vrf_name` matches the VRF through which the DHCP server is accessible.

2. Check central source value

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central central_source
central_source_alternative -flist
HPE_ANW_Central table
central_source: dhcp
central_source_alternative: dhcp
```

Evaluation: The `central_source` value must be set to `dhcp`. The central source alternative may be empty if not configured.

3. Check final location and VRF

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central location
location_alternative vrf vrf_alternative -flist
HPE_ANW_Central table
location: "test.central.com"
location_alternative: "test2.central.com"
vrf: 619e1c63-94a3-44dc-8728-3b831e77bb55
vrf_alternative: 619e1c63-94a3-44dc-8728-3b831e77bb55
```

Evaluation: These are the final location and VRF values used to connect to Central. Verify they match the DHCP server configuration. The location alternative and VRF alternative may be empty if not configured.

Corrective Action: If values do not match, verify the DHCP server configuration includes the correct Central location options and that the switch can reach the DHCP server.

Step 3: Verifying HPE Greenlake Configuration

If neither user configuration nor DHCP options are available, the switch attempts to retrieve the HPE Aruba Networking Central location from the HPE Greenlake (Activate) server.

1. Check HPE AW Central source value

```
switch# start-shell
ovsdb-client dump HPE_ANW_Central central_source -flist
HPE_ANW_Central table
central_source: activate
```

Evaluation: The `central_source` value must be set to `activate`.

2. Check final location and VRF

```
switch# start-shell
```

```
ovsdb-client dump HPE_ANW_Central location vrf -flist
HPE_ANW_Central table
location: "test.central.com"
vrf: 619e1c63-94a3-44dc-8728-3b831e77bb55
```

Evaluation: These are the final location and VRF values used to connect to **Central**. Verify they match the Greenlake server configuration.

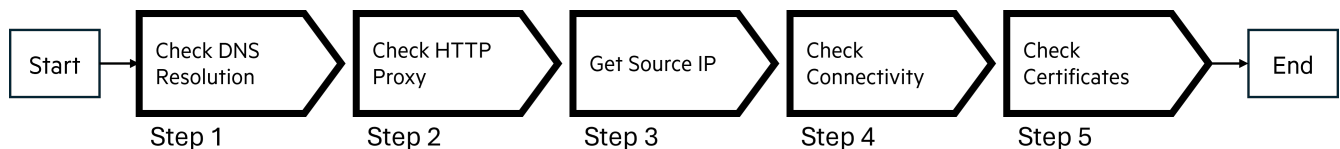
Corrective Action: If values are incorrect, verify the switch is connected to HPE Greenlake and ensure the switch can reach the HPE Greenlake server.

HPE Aruba Networking Central Connectivity

The CX switch cannot establish or maintain connectivity with Central. This can occur with both Cloud-based Central and Central on Premise (COP) deployments. Connectivity issues may manifest as:

- Failed initial connection attempts
- Intermittent disconnections
- Connection timeouts
- Certificate validation errors

Troubleshooting Workflow Diagram



Step 1: Verify DNS Resolution

If the Central location uses a fully qualified domain name (FQDN), verify that the switch can resolve the name.

Diagnostic Command:

```
switch# start-shell
sudo start_vrf_shell nslookup
```

Example:

```
switch# start-shell
sudo start_vrf_shell default
nslookup test.central.com
```

Evaluation: Verify the DNS query returns a valid IP address.

Corrective Action: If DNS resolution fails:

- Verify the Central URL is correct.
- Verify the DNS server configuration on the switch.
- Ensure the switch can reach the configured DNS server.

Step 2: Verify HTTP proxy configuration

If an HTTP proxy server is configured on the same VRF used for Central connectivity, verify connectivity to the proxy server.

1. Check for HTTP proxy configuration

```
switch# start-shell
ovsdb-client dump System operational_dhcp_options

System table
operational_dhcp_options
-----
-----
{http_proxy_location="198.51.100.33:8080",
http_proxy_location_source=user-configured,
http_proxy_location_vrf_name=default}
```

Evaluation: If `http_proxy_location` and `http_proxy_location_vrf_name` are present and the VRF matches the Central VRF, proxy connectivity must be verified.

2. Test proxy connectivity

```
switch# start-shell
sudo start_vrf_shell <Proxy VRF>
curl -v http://<Proxy URL>
```

Example:

```
switch# start-shell
sudo start_vrf_shell default
curl -v http://198.51.100.33:8080
* Trying 198.51.100.33:8080...
* Connected to 198.51.100.33 (198.51.100.33) port 8080
> GET / HTTP/1.1
> Host: 198.51.100.33:8080
> User-Agent: curl/8.7.1
> Accept: */*
>
* Request completely sent off
```

Evaluation: Verify a successful connection is established.

Corrective Action: If proxy connection fails, verify proxy server configuration and network connectivity.

Step 3: Determine source IP configuration

Verify if a source IP is configured for HPE Aruba Networking Central connectivity.

Diagnostic Command:

```
ovsdb-client dump VRF name effective_source_ip

VRF table
effective_source_ip                                name
-----
-----
{}                                                    mgmt
{all="192.0.2.1", central="192.0.2.1"}             default
```

Evaluation:

- If the central key is present, it specifies the source IP for HPE Aruba Networking Central traffic.
- If only the all key is present, it provides the source IP for all outgoing packets, including HPE Aruba Networking Central traffic.
- If neither is present, routing rules determine the source IP.

Step 4: Verify MTU Support

Test that the configured MTU is supported by the network path to Central.

Evaluation:

- If the command hangs without error messages, the MTU is supported.
- If message too long errors appear, the MTU is too large or a device in the path has a smaller MTU.

Corrective Action: If MTU issues are detected:

- Verify MTU settings on the switch.
- Verify MTU settings on intermediate network devices.

Step 5: Test basic connectivity to Central

Verify that the switch can establish an HTTPS connection to Central.

Diagnostic Command:

```
switch# start-shell
sudo start_vrf_shell <Central VRF>
curl -I -k -v [-x <Proxy URL>] [--interface <Source IP>] \
https://<Central URL>/wss
```

Example:

```
switch# start-shell
sudo start_vrf_shell default
```

```

curl -I -k -v -x 198.51.100.33:8080 --interface 192.0.2.1 \
https://test.central.com/wss

* Trying 198.51.100.33:8080...
* Name '192.0.2.1' family 2 resolved to '192.0.2.1' family 2
* Local port: 0
* Connected to 198.51.100.33 (198.51.100.33) port 443
* CONNECT tunnel: HTTP/1.1 negotiated
* allocate connect buffer
* Establish HTTP proxy tunnel to test.central.com:443
> CONNECT test.central.com:443 HTTP/1.1
> Host: test.central.com:443
> User-Agent: curl/8.7.1
> Proxy-Connection: Keep-Alive
>
< HTTP/1.1 200 Connection Established
HTTP/1.1 200 Connection Established
< Proxy-Agent: Squid/7.1
Proxy-Agent: Squid/7.1
<
* CONNECT phase completed
* CONNECT tunnel established, response 200
* ALPN: curl offers http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.2 (IN), TLS handshake, Certificate (11):
* TLSv1.2 (IN), TLS handshake, Server key exchange (12):
* TLSv1.2 (IN), TLS handshake, Request CERT (13):
* TLSv1.2 (IN), TLS handshake, Server finished (14):
* TLSv1.2 (OUT), TLS handshake, Certificate (11):
* TLSv1.2 (OUT), TLS handshake, Client key exchange (16):
* TLSv1.2 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.2 (OUT), TLS handshake, Finished (20):
* TLSv1.2 (IN), TLS handshake, Finished (20):
* SSL connection using TLSv1.2 / ECDHE-RSA-AES128-GCM-SHA256 / x25519 /
RSASSA-PSS
* ALPN: server accepted http/1.1
* Server certificate:
*  subject: C=US; ST=California; L=Santa Clara; O=Hewlett Packard
Enterprise Development; OU=Aruba; CN=test.central.com
*  start date: May  7 09:19:17 2025 GMT
*  expire date: May  6 10:19:17 2026 GMT
*  issuer: CN=Test PKI - HPE Aruba Cloud - Aruba Central CA 412; OU=Aruba;

```

```

O=Hewlett Packard Enterprise Development; L=Santa Clara; ST=California; C=US
*   SSL certificate verify result: self-signed certificate in certificate
chain (19), continuing anyway.
*   Certificate level 0: Public key type RSA (2048/112 Bits/secBits),
signed using sha256WithRSAEncryption
*   Certificate level 1: Public key type RSA (4096/152 Bits/secBits),
signed using sha256WithRSAEncryption
*   Certificate level 2: Public key type RSA (2048/112 Bits/secBits),
signed using sha256WithRSAEncryption
*   Certificate level 3: Public key type RSA (2048/112 Bits/secBits),
signed using sha256WithRSAEncryption
* using HTTP/1.x
> HEAD /wss HTTP/1.1
> Host: test.central.com
> User-Agent: curl/8.7.1
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 502 Bad Gateway
HTTP/1.1 502 Bad Gateway
< Date: Tue, 26 Aug 2025 14:46:27 GMT
Date: Tue, 26 Aug 2025 14:46:27 GMT
< Content-Type: text/html
Content-Type: text/html
< Content-Length: 150
Content-Length: 150
< Connection: keep-alive
Connection: keep-alive
< Strict-Transport-Security: max-age=3600; includeSubDomains
Strict-Transport-Security: max-age=3600; includeSubDomains
< X-Frame-Options: SAMEORIGIN
X-Frame-Options: SAMEORIGIN
< X-Request-Start: t=1756219587.032
X-Request-Start: t=1756219587.032
< X-XSS-Protection: 1; mode=block
X-XSS-Protection: 1; mode=block
< allow-cross-namespace-resources: true
allow-cross-namespace-resources: true
< allow-snippet-annotations: true
allow-snippet-annotations: true
< annotations-risk-level: Critical
annotations-risk-level: Critical

```


<

* Connection #0 to host 198.51.100.33 left intact

Evaluation: Receiving any HTTP response code (commonly 502 Bad Gateway) confirms basic connectivity. This response is expected as no information is being sent to the server, but receiving any HTTP code confirms that the connection to the server can be established.



NOTE

Include the -x flag only if an HTTP proxy is configured. Include the --interface flag only if a source IP is configured.

Corrective Action: If no response is received:

- Verify network connectivity between the switch and Central.
- Check firewall rules that may block HTTPS traffic.
- Verify the Central URL is correct and accessible.

Step 6: Verify device and server certificates

Ensure the switch has a valid device certificate and can validate the Central server certificate.

1. Verify device certificate

```
switch# show crypto pki certificate
```

Certificate Name	Cert Status	EST
Status	Associated Applications	

device-identity	installed	
n/a	none	

Evaluation: Verify the device-identity certificate is present and installed.

2. Verify server certificate and Central connectivity using the device-identity certificate

```
switch# start-shell
sudo start_vrf_shell <Central VRF>
OPENSSL_CONF=/usr/lib/ssl-3/openssl-tpm-provider.cnf \
  curl -I -v [-x <Proxy URL>] [--interface <Source IP>] \
  --cert /fs/security/certmgr/certs/device-identity.crt \
  --key /fs/security/certmgr/keys/device-identity.key \
  --ciphers "ECDHE-RSA-AES256-GCM-SHA384:ECDHE-RSA-AES128-GCM-
SHA256:AES256-GCM-SHA384:AES128-GCM-SHA256" \
  --curves "P-256:P-384:P-521" \
  --http1.1 --tls-max 1.2 \
```

```
--use-certmgr-hpe \  
"https://<Central URL>/wss"
```

**NOTE**

For Central on Premise (COP) instances, replace `--use-certmgr-hpe` with `--use-certmgr`.

Example

```
switch# start-shell  
ssudo start_vrf_shell default  
OPENSSL_CONF=/usr/lib/ssl-3/openssl-tpm-provider.cnf \  
curl -I -v -x 198.51.100.33:8080 --interface 192.0.2.1 \  
--cert /fs/security/certmgr/certs/device-identity.crt \  
--key /fs/security/certmgr/keys/device-identity.key \  
--ciphers "ECDHE-RSA-AES256-GCM-SHA384:ECDHE-RSA-AES128-GCM-  
SHA256:AES256-GCM-SHA384:AES128-GCM-SHA256" \  
--curves "P-256:P-384:P-521" \  
--http1.1 --tls-max 1.2 \  
--use-certmgr-hpe \  
https://test.central.com/wss
```

Expected output:

```
* Trying 198.51.100.33:8080...  
* Name '192.0.2.1' family 2 resolved to '192.0.2.1' family 2  
* Local port: 0  
* Connected to 198.51.100.33 (198.51.100.33) port 443  
* CONNECT tunnel: HTTP/1.1 negotiated  
* allocate connect buffer  
* Establish HTTP proxy tunnel to test.central.com:443  
> CONNECT test.central.com:443 HTTP/1.1  
> Host: test.central.com:443  
> User-Agent: curl/8.7.1  
> Proxy-Connection: Keep-Alive  
>  
< HTTP/1.1 200 Connection Established  
HTTP/1.1 200 Connection Established  
< Proxy-Agent: Squid/7.1  
Proxy-Agent: Squid/7.1  
<  
* CONNECT phase completed  
* CONNECT tunnel established, response 200  
* ALPN: curl offers http/1.1  
* Cipher selection: ECDHE-RSA-AES256-GCM-SHA384:ECDHE-RSA-AES128-GCM-  
SHA256:AES256-GCM-SHA384:AES128-GCM-SHA256
```

```

* TLSv1.2 (OUT), TLS handshake, Client hello (1):
* TLSv1.2 (IN), TLS handshake, Server hello (2):
* TLSv1.2 (IN), TLS handshake, Certificate (11):
* TLSv1.2 (IN), TLS handshake, Server key exchange (12):
* TLSv1.2 (IN), TLS handshake, Request CERT (13):
* TLSv1.2 (IN), TLS handshake, Server finished (14):
* TLSv1.2 (OUT), TLS handshake, Certificate (11):
* TLSv1.2 (OUT), TLS handshake, Client key exchange (16):
* TLSv1.2 (OUT), TLS handshake, CERT verify (15):
* TLSv1.2 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.2 (OUT), TLS handshake, Finished (20):
* TLSv1.2 (IN), TLS handshake, Finished (20):
* SSL connection using TLSv1.2 / ECDHE-RSA-AES128-GCM-SHA256 /
secp256r1 / rsaEncryption
* ALPN: server accepted http/1.1
* Server certificate:
*  subject: C=US; ST=California; L=Santa Clara; O=Hewlett Packard
Enterprise Development; OU=Aruba; CN=test.central.com
*  start date: May  7 09:19:17 2025 GMT
*  expire date: May  6 10:19:17 2026 GMT
*  issuer: CN=Test PKI - HPE Aruba Cloud - Aruba Central CA 412;
OU=Aruba; O=Hewlett Packard Enterprise Development; L=Santa Clara;
ST=California; C=US
* SSL: using certmgr API
* SSL: Peer presented certificate chain
* SSL: certificate validation successful
*  SSL certificate verify ok.
*  Certificate level 0: Public key type RSA (2048/112 Bits/secBits),
signed using sha256WithRSAEncryption
*  Certificate level 1: Public key type RSA (4096/152 Bits/secBits),
signed using sha256WithRSAEncryption
*  Certificate level 2: Public key type RSA (2048/112 Bits/secBits),
signed using sha256WithRSAEncryption
*  Certificate level 3: Public key type RSA (2048/112 Bits/secBits),
signed using sha256WithRSAEncryption
*  using HTTP/1.x
> HEAD /wss HTTP/1.1
> Host: test.central.com
> User-Agent: curl/8.7.1
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 502 Bad Gateway

```

```
HTTP/1.1 502 Bad Gateway
< Date: Tue, 26 Aug 2025 14:53:58 GMT
Date: Tue, 26 Aug 2025 14:53:58 GMT
< Content-Type: text/html
Content-Type: text/html
< Content-Length: 150
Content-Length: 150
< Connection: keep-alive
Connection: keep-alive
< Strict-Transport-Security: max-age=3600; includeSubDomains
Strict-Transport-Security: max-age=3600; includeSubDomains
< X-Frame-Options: SAMEORIGIN
X-Frame-Options: SAMEORIGIN
< X-Request-Start: t=1756220038.934
X-Request-Start: t=1756220038.934
< X-XSS-Protection: 1; mode=block
X-XSS-Protection: 1; mode=block
< allow-cross-namespace-resources: true
allow-cross-namespace-resources: true
< allow-snippet-annotations: true
allow-snippet-annotations: true
< annotations-risk-level: Critical
annotations-risk-level: Critical
<
* Connection #0 to host 198.51.100.33 left intact
```

Evaluation:

- Verify the SSL connection is established successfully.
- Verify certificate validation messages indicate success.
- Receiving an HTTP response confirms certificate-based authentication works.

Corrective Action: If certificate validation fails:

- Verify the device certificate is valid and not expired.
- For COP deployments, verify the correct `--use-certmgr` or `--use-certmgr-hpe` flag is used.
- Check that the server certificate is valid and not expired.

HPE Aruba Networking Central Activate Connectivity

The CX switch cannot establish connectivity with the Activate server. The Activate server is used to provision devices and provide the Central location when other sources are not configured.

Troubleshooting Workflow Diagram



Troubleshooting Workflow Steps

Follow the same troubleshooting steps as described in the Central connectivity section, with the following differences:

Server URL: Use <https://devices-v2.arubanetworks.com> when making HTTPS HEAD requests to test connectivity with the Activate server.

VRF Selection: Try each available VRF until successful DNS resolution and connectivity is established. To list available VRFs, use:

```
switch# start-shell
ovsdb-client dump VRF name

VRF table
name
-----
default
mgmt
```

Test connectivity using each VRF until a successful connection is made.

HPE Aruba Networking Central RCS

Users with admin roles cannot establish a remote console shell (RCS) session from Central to the switch. The RCS feature allows administrators to access the switch CLI directly through SSH from Central for troubleshooting purposes. Connection failures may occur due to network issues, SSH configuration problems, or incompatible security settings.

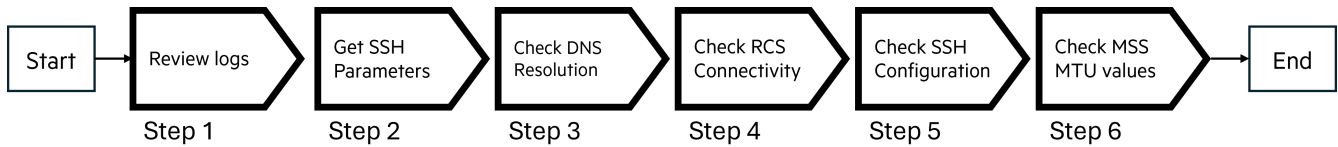
Prerequisites

Enable debug logs

Enable debug logging to capture detailed information about RCS connection attempts:

```
switch# debug sshutils all
switch# debug rest all
```

Troubleshooting Workflow Diagram



Troubleshooting Workflow Steps

Step 1: Review logs for SSH connectivity

Initiate a remote console shell session from Central and review the logs on the switch.

Diagnostic Command:

```
switch# start-shell
journalctl -af -u hpe-restd -u sshd* -u log-proxyd
```

Successful Connection - Expected Output:

```
debug|LOG_INFO|RSSH : id 1, user iap-test, port 443, ssh_port 33484,
IP rcs-test.central.arubanetworks.com, nsName swns, sourceIp 192.0.2.1.
debug|LOG_INFO|RSSH command: [/usr/bin/sshpass -d 3 ssh -tt -o
StrictHostKeyChecking=no -o UserKnownHostsFile=/dev/null -b 192.0.2.1 -R
33484:localhost:22
iap-test@rcs-test.central.arubanetworks.com -p 443 -vv]
Accepted password for admin from ::1 port 57504 ssh2
pam_unix(sshd:session): session opened for user admin by (uid=0)
from ::1
Event|5209|LOG_INFO|CDTR|1|User admin logged in from ::1 through SSH
session.
```

Failed Connection - Example Output:

```
debug|LOG_INFO|RSSH : id 1, user iap-test, port 443, ssh_port 33484, IP rcs-
test.central.arubanetworks.com, nsName swns, sourceIp 192.0.2.1.
debug|LOG_INFO|Using proxy ip 198.51.100.33, port 8080
debug|LOG_INFO|RSSH command: [/usr/bin/sshpass -d 3 ssh -tt -o
StrictHostKeyChecking=no -o UserKnownHostsFile=/dev/null -b 192.0.2.1 -R
33484:localhost:22
iap-test@rcs-test.central.arubanetworks.com -p 443 -vv]
Unable to negotiate with ::1 port 47750: no matching key exchange method
found. Their offer: curve25519-sha256,curve25519-sha256@libssh.org,ecdh-
sha2-nistp256,
ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-
sha256,diffie-hellman-group16-sha512,diffie-hellman-group18-sha512,diffie-
```

```
hellman-group14-sha256,  
ext-info-c,kex-strict-c-v00@openssh.com [preauth]  
debug|LOG_DEBUG|Wait returned for /usr/libexec/common/scripts/cgescape  
is exit status 255.  
debug|LOG_DEBUG|SSH connection logging | OpenSSH_9.6p1,  
OpenSSL 3.4.0 22 Oct 2024  
debug1: cannot process default engine config file: No such file or directory  
debug1: Reading configuration data /etc/ssh/ssh_config  
debug1: ssh_set_validator: ignore responder url  
debug1: Connecting to rcs-test.central.arubanetworks.com [203.0.113.12]  
port 443.  
debug1: ssh_create_socket: bound to 192.0.2.1  
debug1: Connection established.  
debug1: Local version string SSH-2.0-OpenSSH_9.6 PKIX[14.4.2]  
debug1: Remote protocol version 2.0, remote software version Go  
debug1: no match: Go  
debug1: x.509 compatibility rfc6187_missing_key_identifier=no:  
pattern '*' match 'Go'  
debug1: x.509 compatibility rfc6187_asn1_opaque_ecdsa_signature=no:  
pattern '*' match 'Go'  
debug1: x.509 compatibility broken list with accepted  
publickey algorithms=no: pattern '*' match 'Go'  
debug1: Authenticating to rcs-test.central.arubanetworks.com:443  
as 'iap-test'  
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts: No such  
file or directory  
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts2: No  
such file or directory  
debug1: SSH2_MSG_KEXINIT sent  
debug1: SSH2_MSG_KEXINIT received  
debug1: kex: algorithm: diffie-hellman-group14-sha1  
debug1: kex: host key algorithm: rsa-sha2-256  
debug1: kex: server->client cipher: aes128-ctr MAC:  
hmac-sha2-256-etm@openssh.com compression: none  
debug1: kex: client->server cipher: aes128-ctr MAC:  
hmac-sha2-256-etm@openssh.com compression: none  
debug1: expecting SSH2_MSG_KEX_ECDH_REPLY  
debug1: SSH2_MSG_KEX_ECDH_REPLY received  
debug1: Server host key: rsa-sha2-256  
SHA256:U2DN6AjUh24D475uVopix1ihDlnTX7/5zT9P764VhDE  
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts: No such  
file or directory  
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts2: No
```

```

such file or directory
debug1: checking without port identifier
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts: No such
file or directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts2: No
such file or directory
Warning: Permanently added '[rcs-test.central.arubanetworks.com]:443'
(RSA) to the list of known hosts.
debug1: ssh_packet_send2_wrapped: resetting send seqnr 3
debug1: rekey out after 67108864 blocks
debug1: SSH2_MSG_NEWKEYS sent
debug1: expecting SSH2_MSG_NEWKEYS
debug1: ssh_packet_read_poll2: resetting read seqnr 3
debug1: SSH2_MSG_NEWKEYS received
debug1: rekey in after 67108864 blocks
debug1: SSH2_MSG_EXT_INFO received
debug1: kex_input_ext_info: server-sig-algs=<ssh-
ed25519,sk-ssh-ed25519@openssh.com,sk-ecdsa-sha2-nistp256@openssh.com,ecdsa-
sha2-nistp256,ecdsa-sha2-nistp384,ecdsa-sha2-nistp521,rsa-sha2-256,rsa-
sha2-512,ssh-rsa,ssh-dss>
debug1: SSH2_MSG_SERVICE_ACCEPT received
debug1: Authentications that can continue: password,publickey
debug1: Next authentication method: publickey
debug1: Next authentication method: password
Authenticated to rcs-test.central.arubanetworks.com ([203.0.113.12]:443)
using "password".
debug1: Remote connections from LOCALHOST:33484 forwarded
to local address localhost:22
debug1: ssh_init_forwarding: expecting replies for 1 forwards
debug1: channel 0: new session [client-session] (inactive timeout: 0)
debug1: Entering interactive session.
debug1: pledge: network
debug1: remote forward success for: listen 33484, connect localhost:22
debug1: forwarding_success: all expected forwarding replies received
debug1: client_input_channel_open: ctype forwarded-tcpip rchan
0 win 2097152 max 32768
debug1: client_request_forwarded_tcpip: listen localhost port
33484, originator 127.0.0.1 port 35772
debug1: connect_next: start for host localhost (:::1):22)
debug1: connect_next: connect host localhost (:::1):22) in progress, fd=7
debug1: channel 1: new forwarded-tcpip [127.0.0.1] (inactive timeout: 0)
debug1: confirm forwarded-tcpip
debug1: channel 1: connected to localhost port 22

```



```

debug1: ssh_tty_make_modes: no fd or tio
debug1: pledge: network
debug1: fd 4 clearing O_NONBLOCK
You are connected to RCS Server
Type 'exit' or 'quit' to close the connection
# debug1: fd 5 clearing O_NONBLOCK
debug1: channel 0: free: client-session, nchannels 2
debug1: channel 1: free: 127.0.0.1, nchannels 1
Connection to rcs-test.central.arubanetworks.com closed.
Transferred: sent 2800, received 3976 bytes, in 0.1 seconds
Bytes per second: sent 21911.9, received 31114.9
debug1: Exit status -1

```

Evaluation: Review the logs for error messages indicating the cause of connection failure.

Step 2: Extract SSH parameters from logs

From the logs, identify the SSH connection parameters.

Parameters to Extract:

Parameter	Example Value	Notes
Server URL	rcs-test.central.arubanetworks.com	
Username	iap-test	
Port	443	
Source IP	192.0.2.1	May be empty if not configured
HTTP Proxy IP	198.51.100.33	May be empty if not configured
HTTP Proxy Port	8080	May be empty if not configured

Step 3: Verify DNS resolution

If the RCS server address is an FQDN, verify DNS resolution.

Diagnostic Command:

```

switch# start-shell
sudo start_vrf_shell <Central VRF>
nslookup <Server Address>

```

Example:

```

sudo start_vrf_shell default
nslookup rcs-test.central.arubanetworks.com

```

Evaluation: Verify the DNS query returns a valid IP address.

Corrective Action: If DNS resolution fails, verify DNS server configuration and connectivity.

Step 4: Test connectivity with RCS server

Test SSH connectivity to the RCS server.

Diagnostic Command (without HTTP Proxy):

```
switch# start-shell
sudo start_vrf_shell <Central VRF>
ssh -b <Source IP> <Username>@<Server Address> -p <Port> -vvv
```

Example:

```
switch# start-shell
sudo start_vrf_shell default
ssh -b 192.0.2.1 iap-test@rcs-test.central.arubanetworks.com -p 443
-vvv
```

Diagnostic Command (with HTTP Proxy):

```
switch# start-shell
sudo start_vrf_shell <Central VRF>
ssh -b <Source IP> -o ProxyCommand="/usr/bin/socat - PROXY:<HTTP
Proxy IP>:%h:%p,proxyport=<HTTP Proxy Port>" <Username>@<Server Address> -p
<Port> -vvv
```

Example:

```
sudo start_vrf_shell default
ssh -b 192.0.2.1 -o ProxyCommand="/usr/bin/
socat - PROXY:198.51.100.33:%h:%p,proxyport=8080" iap-test@rcs-
test.central.arubanetworks.com -p 443 -vvv
```

Evaluation: If the command reaches a password prompt, connectivity is successful.

Corrective Action: If connection fails, verify network connectivity, firewall rules and continue with the next steps.

Step 5: Verify SSH server configuration

Verify the SSH server configuration on the switch.

Diagnostic Command:

```
switch# show ssh server all-vrfs
```

Issue 5.1: SSH server not configured on Central VRF

Corrective Action:

```
switch# configure
switch(config)# ssh server vrf <VRF>
```

Issue 5.2: SSH server not listening on port 22

The RCS connection requires the SSH server to listen on TCP port 22.

Corrective Action:

```
switch# configure
switch(config)# ssh server port 22
```

Issue 5.3: SSH server allow list does not include loopback

If the SSH server is configured to allow certain IP addresses to connect, make sure to also allow the IPv6 loopback address (::1).

Corrective Action:

```
switch# configure
switch(config)# ssh server allow-list
switch(config-ssh-al)# ipv6 ::1
switch(config-ssh-al)# enable
```

Issue 5.4: Maximum SSH connections reached

The SSH server allows a maximum of 5 concurrent connections.

Diagnostic Command:

```
switch# show ssh server sessions vrf <VRF>
```

Corrective Action: Close unused SSH sessions to free up connection slots.

Issue 5.5: SSH ciphers not compatible with RCS server

Diagnostic Command:

```
switch# show running-config | include "ssh ciphers"
```

Evaluation: Verify configured ciphers are supported by the RCS server:

Cipher	RCS Support
aes128-ctr	Yes
aes256-ctr	Yes
aes128-gcm@openssh.com	Yes
aes256-gcm@openssh.com	Yes
chacha20-poly1305@openssh.com	Yes

Cipher	RCS Support
aes192-ctr	Yes
aes128-cbc	No
aes256-cbc	No
aes192-cbc	No

Corrective Action: Configure supported ciphers:

```
switch# configure
switch(config)# ssh ciphers <ciphers>
```

Issue 5.6: SSH host key algorithms not compatible with RCS server

Diagnostic Command:

```
switch# show running-config | include "ssh host-key-algorithms"
```

Evaluation: Verify configured algorithms are supported by the RCS server:

Algorithm	RCS Support
rsa-sha2-256	Yes
rsa-sha2-512	Yes
ssh-rsa	Yes
ecdsa-sha2-nistp256	No
ecdsa-sha2-nistp384	No
ecdsa-sha2-nistp521	No
ssh-ed25519	No

Corrective Action: Configure supported algorithms:

```
switch# configure
switch(config)# ssh host-key-algorithms <algorithms>
```

Issue 5.7: SSH key exchange algorithms not compatible with RCS server

Diagnostic Command:

```
switch# show running-config | include "ssh key-exchange-algorithms"
```

Evaluation: Verify configured algorithms are supported by the RCS server:

Algorithm	RCS Support
curve25519-sha256	Yes
curve25519-sha256@libssh.org	Yes
ecdh-sha2-nistp256	Yes
ecdh-sha2-nistp384	Yes
ecdh-sha2-nistp521	Yes
diffie-hellman-group14-sha256	Yes
diffie-hellman-group14-sha1	No
diffie-hellman-group-exchange-sha1	No
diffie-hellman-group-exchange-sha256	No
diffie-hellman-group16-sha512	No
diffie-hellman-group18-sha512	No

Corrective Action: Configure supported algorithms:

```
switch# configure
switch(config)# ssh key-exchange-algorithms <algorithms>
```

Step 6: Verify MSS and MTU values

If network issues are suspected, verify MSS and MTU values are consistent across the network path.

Diagnostic Command:

Review current MTU configuration and validate with network infrastructure to ensure all devices in the path support the configured values.

Corrective Action: Adjust MTU values if mismatches are identified in the network path.

HPE Aruba Networking Central Disconnection Timeout

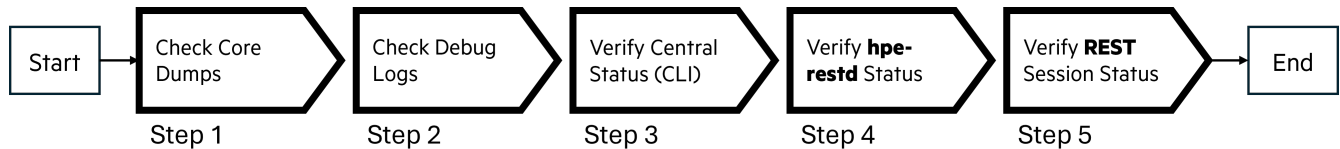
The switch experiences delayed disconnection from Central or the hpe-restd service becomes unresponsive during a disconnection event. This may occur during:

- Network interface flapping on the interface used for Central connectivity
- Central server upgrades or maintenance
- Manual disable/enable of the hpe-anw-central feature
- Disconnection from external clients (NetEdit or WebUI)

Symptoms include:

- hpe-restd does not respond to configuration changes
- Switch fails to reconnect to Central within the expected timeframe
- After 12 minutes, hpe-restd automatically restarts with a timeout message

Troubleshooting Workflow Diagram



Troubleshooting Workflow Steps

Step 1: Check for core dumps

Verify if the hpe-restd service has generated core dumps.

Diagnostic Command:

```
switch# show core-dump all
```

Evaluation: If core dumps are present, note the timestamp and proceed with the remaining steps.

Step 2: Review debug logs

Check if the disconnection timeout mechanism was triggered.

Diagnostic Command:

```
switch# show debug buffer module central | inc "hpe-restd did not  
close the connection to Central"  
after: 12 minutes, restarting service"
```

Evaluation: If this message is present, the timeout mechanism was triggered and the service should have automatically restarted to restore connectivity.

Step 3: Verify Central connection status via CLI

Check the current Central connection status.

Diagnostic Command:

```
switch# show hpe-anw-central
```

Evaluation: The Central connection status may show:

- connected: Connection appears established (verify with next steps).

- connection_failure: Connection has failed.
- N/A: Status is unavailable.

**NOTE**

The CLI status may not always reflect the actual connection state. Continue with the next steps to verify the internal status.

Step 4: Verify Central status in hpe-restd

Check the internal Central connection status in the hpe-restd service.

Diagnostic Command:

```
switch# start-shell
sudo ovs-appctl -t hpe-restd central/show --state
```

Evaluation: If the status shows connected when Central connectivity is not working, the service may be waiting for internal processes to complete the disconnection.

Step 5: Verify WebSocket connection status

Check if an active WebSocket connection exists.

Diagnostic Command:

```
switch# start-shell
sudo start_vrf_shell <Central VRF>
netstat -ntp
```

Evaluation: Look for active connections to the Central IP address. If the Central connection is present when it should be disconnected, the disconnection process is incomplete.

Step 6: Verify REST session status

Check if a persistent REST session exists for Central.

Diagnostic Command:

```
switch# start-shell
sudo start_vrf_shell <Central VRF>
```

Evaluation: Look for a row with the PERSISTENT column set to True. This indicates the Central session still exists and the disconnection is incomplete.

Step 7: Collect diagnostic information

If the issue persists, collect comprehensive diagnostic information.

Corrective Actions:

1. Save the output of all diagnostic commands from the previous steps.
2. Generate support files without rebooting the switch to preserve diagnostic state:

```
switch# copy support-files
```

3. Contact HPE support and provide:

- Collected command outputs.
- Support files.
- Description of when the issue occurred.
- Any recent configuration changes.

Workaround

If hpe-restd is unresponsive and has not automatically restarted, manually restart the service.

Corrective Action:

```
switch# https-server session close all
```

This command closes all HTTPS server sessions and triggers a restart of the **hpe-restd** service, which should restore normal operation.

HPE Aruba Networking Central commands

Select a command from the list in the left navigation menu.

Subtopics

[hpe-anw-central](#)

[hpe-anw-central support-mode](#)

[configuration-lockout central managed](#)

[diag-dump rest basic](#)

[disable](#)

[enable](#)

[location-override](#)

[location-override-alternative](#)

[show hpe-anw-central](#)

[show running-config current-context](#)

hpe-anw-central

Syntax

```
hpe-anw-central
no hpe-anw-central
```

Description

Creates or enters the HPE Aruba Networking Central configuration context (**config-hpe-anw-central**).

When a switch is able to connect to HPE Aruba Networking Central but is not registered in the HPE Aruba Networking Central inventory or does not have a proper license, the switch will get disconnected. If the HPE Aruba Networking Central feature is enabled using this command, the switch will then reconnect back to HPE Aruba Networking Central and will get disconnected again. This connect/disconnect process will continue until the switch is properly registered in HPE Aruba Networking Central. To avoid this unnecessary reconnection cycle, best practices is to disable HPE Aruba Networking Central until the switch is registered in HPE Aruba Networking Central, or a license is obtained for that device.

Example

Creating the HPE Aruba Networking Central configuration context:

```
switch(config) # hpe-anw-central
switch(config-hpe-anw-central) #
```

Command History

Release	Modification
10.15	Command syntax changed from aruba - central to hpe - anw - central .
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

hpe-anw-central support-mode

Syntax

```
hpe-anw-central support-mode
no hpe-anw-central support-mode
```

Description

Allows the device to be writable for all operations in HPE Aruba Networking Central lockout mode for troubleshooting. The no form of this command disables this activity.



NOTE

Support-mode is disabled by default when the switch is managed by HPE Aruba Networking Central. This command is only effective in the CLI session where it is executed.

Examples

Configuring the device to be writable for all operations in HPE Aruba Networking Central lockout mode:

```
switch# hpe-anw-central support-mode
```

Removing the configuration that allows the device to be writable for all operations in HPE Aruba Networking Central lockout mode:

```
switch# no hpe-anw-central support-mode
switch#
```

Command History

Release	Modification
10.15	Command syntax changed from aruba - central to hpe - anw - central .
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

configuration-lockout central managed

Syntax

```
configuration-lockout central managed
no configuration-lockout central managed
```

Description

Configures the device to only be writable from HPE Aruba Networking Central. HPE Aruba Networking Central will be the only agent that can add, modify, or delete configurations on the device. The no form of this command disables this feature.



NOTE

The no form of this command is only available when the device is disconnected from HPE Aruba Networking Central.

Usage

The AOS-CX switch connects to HPE Aruba Networking Central in either of two modes: monitor or managed. When the device is connected in monitor mode, HPE Aruba Networking Central monitors the configurations on the switch. When the device is connected in managed mode, the **configuration-lockout central managed** command does not allow configuration changes from other interfaces such as CLI or Web UI.

Examples

Configuring the device to only be writable from HPE Aruba Networking Central :

```
switch(config)# configuration-lockout central managed
switch# show configuration-lockout
configuration lockout
-----
central: managed
switch# sh hpe-anw-central
Central admin state           :enable
Central location              :20.0.0.2:8083
VRF for connection            :default
Central connection status     :connected
Central source                :cli
Central source connection status :connected
Central source last connected on :Tue Feb 9 17:53:13 UTC 2021
Activate Server URL           :devices-v2.arubanetworks.com
CLI location                   :20.0.2:8083
```

```
CLI VRF                                     :default
switch(config)# end
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

diag-dump rest basic

Syntax

```
diag-dump rest basic
```

Description

Display REST paths and diagnostics information for the REST feature. The output of this command includes diagnostics information for HPE Aruba Networking Central.

Example

The following example displays just the **Central Counters** portion of the output for this command.

```
switch# diag-dump rest basic
=====
[Start] Feature rest Time : Sun Sep 17 17:06:50 2023
=====
...
...
...
Central Counters:
-----
```

```

Logins: 1
Connections: 1
Disconnections: 0
-----
Pings: 50
Pongs: 50
-----
API Requests: 20
Response API Requests: 20
-----
v1 Notification Requests Received: 5
v1 Notifications Sent: 15
Initial Snapshots: 5
Notifications: 10
Errors: 0
-----
v2 Notification Requests Received: 10
v2 Notifications Sent: 24
Initial Snapshots: 10
Unsubscribes: 0
Notifications: 14
Errors: 0
-----
Views Notification Requests: 0
Views Notification Responses: 0
Views Unary Requests: 0
Views Unary Responses: 0
-----
Packet Capture Responses: 0
-----

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
all platforms	config	Administrators or local user group members with execution rights for this command.

disable

Syntax

```
disable
```

Description

Disables connection to HPE Aruba Networking Central server.

When the connection is disabled, the switch does not attempt to connect to the HPE Aruba Networking Central server or fetch central location from any of the three sources (CLI/HPE GreenLake/DHCP). It also disconnects any active connection to the HPE Aruba Networking Central server.

Example

```
switch(config-hpe-anw-central)# disable
switch(config-hpe-anw-central)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-hpe_anw- central	Administrators or local user group members with execution rights for this command.

enable

Syntax

```
enable
```

Description

Enables connection to HPE Aruba Networking Central server. When the connection is enabled, the switch attempts to download the location of the HPE Aruba Networking Central server in one of the following ways at startup and after the connection is lost:

- Using command-line interface (CLI).
- Connecting to the HPE GreenLake server.
- Using DHCP options provided during ZTP.

DHCP servers provide the options requested by the device to connect to Central, Central On-premise management, or the TFTP server.



NOTE

When a switch is able to connect to HPE Aruba Networking Central, but is not registered in the HPE Aruba Networking Central inventory or does not have a proper license, the switch will get disconnected. If the HPE Aruba Networking Central feature is enabled using this command, the switch will then reconnect back to HPE Aruba Networking Central and will get disconnected again. This connect/disconnect process will continue until the switch is properly registered in HPE Aruba Networking Central. To avoid this unnecessary reconnection cycle, best practices is to disable HPE Aruba Networking Central until the switch is registered in HPE Aruba Networking Central, or a license is obtained for that device.

Examples

```
switch(config-hpe-anw-central)# enable  
switch(config-hpe-anw-central)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-hpe_anw- central	Administrators or local user group members with execution rights for this command.

location-override

Syntax

```
location-override <location> [vrf <VRF-NAME>]
no location-override
```

Description

When **location** and **vrf** are configured, the switch overrides existing connections to HPE Aruba Networking Central. The switch attempts to establish connection to HPE Aruba Networking Central with the specified location and VRF with highest priority.

Location can take one of the following values:

- A fully qualified domain name (FQDN) along with an optional port number.
- An IPv4 address with an optional port number.
- An IPv6 address with an optional port number.

If the port number is not specified, then port 443 is used by default. If the command is executed without the VRF parameter, the switch uses

the 'default' VRF.

The **no** form of this command removes location override values from the HPE Aruba Networking Central configuration context.



NOTE

When you configure an IPv6 address with a port number, specify the address part inside square brackets, optionally followed by the port number, e.g. [2001:0db8:85a3:0000:0000:8a2e:0370:7334]:443.

Parameter	Description
<code><location></code>	Specifies one of these values: <code><FQDN></code> : a fully qualified domain name. <code><IPV4></code> : an IPv4 address. <code><IPV6></code> : an IPv6 address.
<code>vrf <VRF-NAME></code>	Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named default is used.

Examples

Configuring location override with location and VRF:

```
switch(config-hpe-anw-central)# location-override hpe-anw-central.com vrf
default
switch(config-hpe-anw-central)#
switch(config-hpe-anw-central)# location-override hpe-anw-central.com vrf
red
switch(config-hpe-anw-central)# location-override 10.0.0.1 vrf red
switch(config-hpe-anw-central)# location-override 10.0.0.1:443 vrf red
switch(config-hpe-anw-central)# location-override
2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf red
switch(config-hpe-anw-central)# location-override
[2001:0db8:85a3:0000:0000:8a2e:0370:7334]:443 vrf red
```

Configuring location override with location only:

```
switch(config-hpe-anw-central)# location-override hpe-anw-central.com
switch(config-hpe-anw-central)#
```

Removing location override values from the HPE Aruba Networking Central configuration context:

```
switch(config-hpe-anw-central)# no location-override
switch(config-hpe-anw-central)#
```

Command History

Release	Modification
10.13.1000	Command updated to reflect an OTP scenario.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-hpe_anw-central</code>	Administrators or local user group members with execution rights for this command.

location-override-alternative

Syntax

```
location-override-alternative <LOCATION> [vrf <VRF>]  
no location-override-alternative <LOCATION> [vrf <VRF>]
```

Description

Configures information about HPE Aruba Networking Central connection when the alternative location is used.

The **no** form of this command removes the **location-override-alternative** configuration.

Parameter	Description
<code><LOCATION></code>	Specifies the HPE Aruba Networking Central location.
<code>vrf <VRF></code>	Specifies the VRF used to connect to HPE Aruba Networking Central.

Usage

When the main and alternative HPE Aruba Networking Central server locations are specified, the switch attempts to connect to the main HPE Aruba Networking Central server. If there is connectivity failure with the main HPE Aruba Networking Central server location, it attempts to establish a connection with the alternative server location.

If the alternative location is configured without a main location, the user is prompted for confirmation. In this case, there is no redundancy and the switch attempts to connect to the alternative location.

Location can take one of the following values:

- A fully qualified domain name (FQDN) along with an optional port number.

- An IPv4 address with an optional port number.
- An IPv6 address with an optional port number.

If the port number is not specified, then port 443 is used by default. If the command is executed without the VRF parameter, the switch uses

the 'default' VRF.

An HPE Aruba Networking Central server location can only be a fully qualified domain name (FQDN) or a valid IP address. If the command is entered without the VRF parameter, the switch uses the default VRF.

Examples

Example of configuring with the hpe-anw-central.com location and VRF red:

```
switch(config-hpe-anw-central)# location-override-alternative hpe-anw-
central.com vrf red
switch(config-hpe-anw-central)#
```

Example of a configuration with location only:

```
switch(config-hpe-anw-central)# location-override-alternative hpe-anw-
central.com
switch(config-hpe-anw-central)#
```

Example of removing the override configuration:

```
switch(config-hpe-anw-central)# no location-override-alternative

switch(config-hpe-anw-central)# location-override-alternative 10.0.0.1 vrf
red
switch(config-hpe-anw-central)# location-override-alternative 10.0.0.1:443
vrf red
switch(config-hpe-anw-central)# location-override-alternative
2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf red
switch(config-hpe-anw-central)# location-override-alternative
[2001:0db8:85a3:0000:0000:8a2e:0370:7334]:443 vrf red
```

Command History

Release	Modification
10.13.1000	Command updated to reflect OTP scenario.
10.12.1000	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config-hpe_anw-central	Administrators or local user group members with execution rights for this command.

show hpe-anw-central

Syntax

```
show hpe-anw-central
```

Description

Shows information about HPE Aruba Networking Central connection and the status of the Activate server connection.

Examples

Example of a switch that has the HPE Aruba Networking Central connection:

```
switch# show hpe-anw-central
Central admin state           : enabled

Central location              : 10.0.0.1
VRF for connection            : mgmt

Shared token                  : N/A
Central connection status     : connected
Central connection reason     : N/A
Last Central disconnection reason : N/A

Central source                : activate
Central source connection status : connected
Central source last connected on : Wed Jun 28 23:07:25 UTC 2023

Main location                 : 10.0.0.1
Main VRF                      : mgmt

Alternative location           : N/A
Alternative VRF                : N/A
```

```
Activate Server URL           : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A

Source IP                     : N/A
Source IP Overridden          : False

Central support mode          : disabled
```

Example of a switch when the main CLI location is used:

```
switch# show hpe-anw-central
Central admin state           : enabled

Central location              : 10.0.0.1
VRF for connection            : mgmt

Shared secret                 : N/A
Central connection status     : connected
Central connection reason     : N/A
Last Central disconnection reason : N/A

Central source                : cli
Central source connection status : connected
Central source last connected on : Wed Jun 28 23:07:25 UTC 2023

Main location                 : 10.0.0.1
Main VRF                      : mgmt

Alternative location          : 20.0.0.1
Alternative VRF               : N/A

Activate server URL           : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A

Source IP                     : N/A
Source IP Overridden          : False

Central support mode          : disabled
```

Example of a switch when the alternative CLI location is used:

```
switch# show hpe-anw-central
Central admin state           : enabled

Central location              : 20.0.0.1
VRF for connection            : mgmt
```

```

Shared secret                : N/A
Central connection status    : connected
Central connection reason    : N/A
Last Central disconnection reason : N/A

Central source               : cli
Central source connection status : connected
Central source last connected on : Wed Jun 28 23:07:25 UTC 2023

Main location                : 10.0.0.1
Main VRF                     : mgmt

Alternative location         : 20.0.0.1
Alternative VRF              : default

Activate server URL          : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A

Source IP                    : N/A
Source IP Overridden         : False

Central support mode         : disabled

```

Example of a switch when the location is obtained from DHCP options:

```

switch# show hpe-anw-central
Central admin state          : enabled

Central location             : central-western-
us.arubanetworks.com
VRF for connection          : mgmt

Shared secret                : mysecret
Central connection status    : connected
Central connection reason    : N/A
Last Central disconnection reason : N/A

Central source               : DHCP
Central source connection status : connected
Central source last connected on : Fri Jun 30 20:22:33 UTC 2023

Main location                : central-western-
us.arubanetworks.com
Main VRF                     : mgmt

```

```

Alternative location          : N/A
Alternative VRF              : N/A

Activate server URL          : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A

Source IP                    : 100.0.0.1
Source IP Overridden         : False

Central support mode         : disabled

```

Example of a switch when HPE Aruba Networking Central is disabled:

```

switch# show hpe-anw-central
Central admin state          : disabled
Central location             : N/A
VRF for connection          : N/A
Shared secret                : N/A
Central connection status    : N/A
Central connection reason    : N/A
Last Central disconnection reason : N/A

Central source               : none
Central source connection status : N/A
Central source last connected on : N/A

Main location                : N/A
Main VRF                     : N/A
Alternative location          : N/A
Alternative VRF              : N/A

Activate server URL          : devices-v2.arubanetworks.com
System time synchronized from Activate : N/A

Source IP                    : N/A
Source IP Overridden         : False

Central support mode         : disabled

```

Example of a switch when the location is obtained from Activate:

```

switch# show hpe-anw-central
Central admin state          : enabled

Central location             : central-western-
us.arubanetworks.com
VRF for connection          : mgmt

```

```

Shared secret                : N/A
Central connection status    : connected
Central connection reason    : N/A
Last Central disconnection reason : N/A

Central source               : activate
Central source connection status : connected
Central source last connected on : Tue Nov 24 21:43:56 UTC 2020

Main location                : central-western-
us.arubanetworks.com

Main VRF                     : mgmt
Alternative location         : N/A
Alternative VRF              : N/A

Activate Server URL          : devices-v2.arubanetworks.com
System time synchronized from Activate : True

Source IP                   : 100.0.0.1
Source IP Overridden        : False

Central support mode        : disabled

```

Example of a switch when HPE Aruba Networking Central is not reachable:

```

switch# show hpe-anw-central
Central admin state          : enabled

Central location            : central-western-
us.arubanetworks.com
VRF for connection          : mgmt
Shared secret               : N/A
Central connection status    : no_route_to_server
Central connection reason    : Central server unreachable
Last Central disconnection reason : N/A

Central source              : cli
Central source connection status : connected
Central source last connected on : Tue Nov 24 21:43:56 UTC 2020

Main location               : central-western-
us.arubanetworks.com
Main VRF                    : mgmt
Alternative location         : N/A
Alternative VRF              : N/A

```



```

Activate Server URL           : devices-v2.arubanetworks.com
System time synchronized from Activate : False

Source IP                     : N/A
Source IP Overridden          : False

Central support mode          : disabled

```

Example of an unlicensed switch:

```

switch# show hpe-anw-central
Central admin state           : enabled

Central location              : central-western-
us.arubanetworks.com

VRF for connection            : mgmt
Shared secret                  : N/A
Central connection status     : connection_failure
Central connection reason     : N/A
Last Central disconnection reason : CX switch not licensed in Central

Central source                 : cli
Central source connection status : connected
Central source last connected on : Mon Sep 01 15:27:42 UTC 2025

Main location                  : central-western-
us.arubanetworks.com
Main VRF                       : mgmt
Alternative location           : N/A
Alternative VRF                 : N/A
Activate Server URL            : devices-v2.arubanetworks.com
System time synchronized from Activate : True

Source IP                     : 100.0.0.1
Source IP Overridden          : False

Central support mode          : disabled

```

Command History

Release	Modification
10.18	Command output updated.

Release	Modification
10.15	Command syntax changed from aruba - central to hpe - anw - central .
10.12.1000	Enhanced to support additional scenarios.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config current-context

Syntax

```
show running-config current-context
```

Description

Shows the running configuration for the current-context. If user is in the context of HPE Aruba Networking Central (**config-hpe-anw-central**), then the HPE Aruba Networking Central running configuration is displayed.

Examples

Shows the running configuration of HPE Aruba Networking Central:

```
switch(config-hpe-anw-central)# show running-config current-context
hpe-anw-central
  disable
```

Command History

Release	Modification
10.07 or earlier	- -

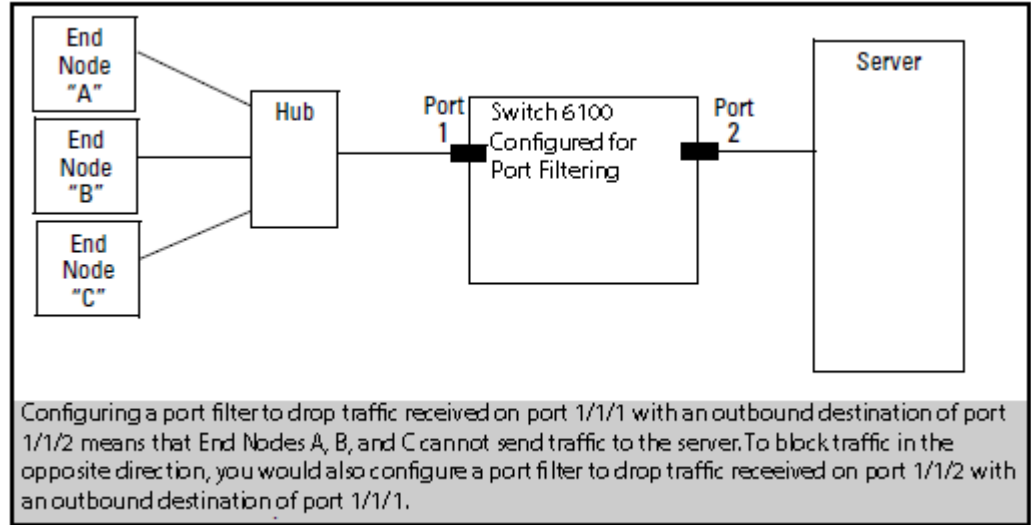
Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Port filtering

Port filtering is a feature in which packets that are ingressed through a source port can be blocked for egressing on a specific set of ports.

Figure 1. Port Filter Application



Subtopics

Port filtering commands

Port filtering commands

Select a command from the list in the left navigation menu.

Subtopics

- portfilter
- show portfilter

portfilter

Syntax

```
portfilter <INTERFACE-LIST>
no portfilter [<INTERFACE-LIST>]
```

Description

Configures the specified ports so they do not egress any packets that were received on the source port specified in interface context.

The **no** form of this command removes the port filter setting from one or more ingress ports/LAGs.



CAUTION

This configuration will disable flow tracking statistics collection.

Parameter

Description

<INTERFACE-LIST>

Specifies a list of ports/LAGs to be blocked for egressing. Specify a single interface or LAG, or a range as a comma - separated list, or both. For example: `1/1/1`, `1/1/3-1/1/6`, `lag2`, `lag1-lag4`. On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Usage

When a port filter configuration is applied on the same ingress physical port/LAG, the configuration is updated with the new sets of egress ports/LAGs that are to be blocked for egressing and that are not a part of its previous configuration. Duplicate updates on an existing port filter configuration are ignored.

When egress ports/LAGs are removed from the existing port filter configuration of an ingress port/LAG, egressing is allowed again on those egress ports/LAGs for all packets originating from the ingress port/LAG.

The **no portfilter [<IF-NAME-LIST>]** command removes port filter configurations from the egress ports/LAGs listed in the <IF-NAME-LIST> parameter only. All other egress ports/LAGs in the port filter configuration of the ingress port/LAG remain intact.

If no physical ports or LAGs are provided for the **no portfilter** command, the command removes the entire port filter configuration for the ingress port/LAG.

Examples

Creating a filter that prevents packets received on port **1/1/1** from forwarding to ports **1/1/3-1/1/6** and to LAGs **1** through **4**:

```
switch(config) # interface 1/1/1
switch(config-if) # portfilter 1/1/3-1/1/6,lag1-lag4
```

Creating a filter that prevents packets received on LAG **1** from forwarding to ports **1/1/6** and LAGs **2** and **4**:

```
switch(config) # interface lag 1
switch(config-lag-if) # portfilter 1/1/6,lag2,lag4
```

Removing filters from an existing configuration that allows back packets received on port **1/1/1** to forward to ports **1/1/6** and LAGs **3** and **4**:

```
switch(config) # interface 1/1/1
switch(config-if) # no portfilter 1/1/6,lag3,lag4
```

Removing all filters from an existing configuration that allows back packets received on LAG **1** to forward to all the ports and LAGs:

```
switch(config) # interface lag 1
switch(config-lag-if) # no portfilter
```

Command History

Release	Modification
10.14	Added information related to role based IPFIX.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6200	config-if config-lag-if	Administrators or local user group members with execution rights for this command.

show portfilter

Syntax

```
show portfilter [<IFNAME>]
```

Description

Displays filter settings for all interfaces or a specific interface.

Parameter	Description
<code><IFNAME></code>	Specifies the ingress interface name. Specifies one of these values: <code><FQDN></code> : a fully qualified domain name. <code><IPV4></code> : an IPv4 address. <code><IPV6></code> : an IPv6 address.

Examples

Displaying all port filter settings on the switch:

```
switch# show portfilter
Incoming    Blocked
Interface   Outgoing Interfaces
-----
---
1/1/1       1/1/3-1/1/6, lag1-lag2
1/1/3
1/1/1, 1/1/5, 1/1/7, 1/1/9, 1/1/11, 1/1/13, 1/1/15, 1/1/17, 1/1/19, 1/1/21,
          1/1/23, 1/1/25, 1/1/27, 1/1/29, 1/1/31, 1/1/33, 1/1/35
lag2        1/1/1, 1/1/3-1/1/6
```

Displaying the port filter settings for port **1/1/1**:

```
switch# show portfilter 1/1/1
Incoming    Blocked
Interface   Outgoing Interfaces
-----
---
1/1/1       1/1/3-1/1/6, lag1-lag2
```

Displaying the port filter settings for **LAG2**:

```
switch# show portfilter lag2
Incoming    Blocked
Interface   Outgoing Interfaces
-----
---
lag2        1/1/1, 1/1/3-1/1/6
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

DNS

The Domain Name System (DNS) is the Internet protocol for mapping domain names or hostnames into addresses. DNS allows users to enter more readily memorable and intuitive hostnames, rather than IP addresses, to identify devices connected to a network. It also allows a host to keep the same hostname even if it changes its IP address.

Hostname resolution can be either static or dynamic.

- In static resolution, a local table is defined on the switch that associates hostnames with their IP addresses. Static tables can be used to speed up the resolution of frequently queried hosts.
- Dynamic resolution requires that the switch query a DNS server located elsewhere on the network. Dynamic name resolution takes more time than static name resolution, but requires far less configuration and management.

Configuration

VxLAN IPv4 and IPv6 underlay support the DNS client. To ensure configuration, make sure the following are set up:

- VxLAN overlay
 - Must be configured to establish the VxLAN tunnel.
- VxLAN tunnel
 - Ensure the tunnel is established between the VTEPS via either static or EVPN using the following command: **show interface VxLAN VTEPS**.

-
-



NOTE

For more information, refer to the .

Subtopics

[DNS client](#)

[Configuring the DNS client](#)

[DNS client commands](#)

DNS client

The DNS client resolves hostnames to IP addresses for protocols that are running on the switch. When the DNS client receives a request to resolve a hostname, it can do so in one of two ways:

- Forward the request to a DNS name server for resolution.
- Reply to the request without using a DNS name server, by resolving the name using a statically defined table of hostnames and their associated IP addresses.

Configuring the DNS client

Procedure

1. Configure one or more DNS name servers with the command `ip dns server`.
2. To resolve DNS requests by appending a domain name to the requests, either configure a single domain name with the command `ip dns domain-name`, or configure a list of up to six domain names with the command `ip dns domain-list`.
3. To use static name resolution for certain hosts, associate an IP address to a host with the command `ip dns host`.
4. Review your DNS configuration settings with the command `show ip dns`.

Examples

This example creates the following configuration:

- Defines the domain **switch.com** to append to all requests.

- Defines a DNS server with IPv4 address of **1.1.1.1**.
- Defines a static DNS host named **myhost1** with an IPv4 address of **3.3.3.3**.
- DNS client traffic is sent on the default VRF (named **default**).

```
switch(config)# ip dns domain-name switch.com
switch(config)# ip dns server-address 1.1.1.1
switch(config)# ip dns host myhost1 3.3.3.3
switch(config)# exit
switch# show ip dns
VRF Name : vrf_mgmt
Host Name                                     Address
-----
----
VRF Name : vrf_default
Domain Name : switch.com
DNS Domain list :
Name Server(s) : 1.1.1.1
Host Name                                     Address
-----
----
myhost1
```

DNS client commands

Select a command from the list in the left navigation menu.

Subtopics

- [ip dns domain-list](#)
- [ip dns domain-name](#)
- [ip dns host](#)
- [ip dns server address](#)
- [ip dns fqdn-resolver refresh-interval](#)
- [ip dns fqdn-resolver force-refresh](#)
- [show ip dns](#)
- [Show FQDN Resolver](#)
- [Show IP DNS FQDN Resolver Detail](#)
- [Show IP DNS FQDN Resolver Refresh Interval](#)

ip dns domain-list

Syntax

```
ip dns domain-list <DOMAIN-NAME> [vrf
    <VRF-NAME>

]
no ip dns domain-list <DOMAIN-NAME> [vrf
    <VRF-NAME>

]
```

Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The **no** form of this command removes a domain from the list.

Parameter	Description
<code>list <DOMAIN-NAME></code>	Specifies a domain name. Up to six domains can be added to the list. Length: 1 to 256 characters.
<code>vrf <VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

This example defines a list with two entries: **domain1.com** and **domain2.com**.

```
switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com

switch(config)# ip dns domain-list domain2.com vrf mainvrf
switch(config)# ip dns domain-list domain5.com vrf mainvrf
```

This example removes the entry **domain1.com**.

```
switch(config)# no ip dns domain-list domain1.com
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns domain-name

Syntax

```
ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
no ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
```

Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command `ip dns domain-list`, the domain name defined with this command is ignored.

The **no** form of this command removes the domain name.

Parameter	Description
<code><DOMAIN-NAME></code>	Specifies the domain name to append to DNS requests. Length: 1 to 256 characters.
<code>vrf <VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

Setting the default domain name to `domain.com`:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name `domain.com`:

```
switch(config)# no ip dns domain-name domain.com
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

ip dns host

Syntax

```
ip dns host <HOST-NAME>
           <IP-ADDR> [ vrf <VRF-NAME> ]
no ip dns host <HOST-NAME>
           <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a static IP address associated with a hostname.

Parameter	Description
<code>host <HOST-NAME></code>	Specifies the name of a host. Length: 1 to 256 characters.

Parameter	Description
<code><IP-ADDR></code>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:x xxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.
<code>vrf <VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

This example defines an IPv4 address of **3.3.3.3** for **host1**.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of **b::5** for **host 1**.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for **host 1** with address **b::5**.

```
switch(config)# no ip dns host host1 b::5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns server address

Syntax

```
ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]  
no ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a name server from the list.

Parameter	Description
<code><IP-ADDR></code>	Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (<code>xxxx:xxxx:x xxx:xxxx:xxxx:xxxx:xxxx:xxxx</code>), where x is a hexadecimal number from 0 to F.
<code>vrf <VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

This example defines a name server at **1.1.1.1**.

```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at **a::1**.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at **a::1**.

```
switch(config)# no ip dns server-address a::1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns fqdn-resolver refresh-interval

Syntax

```
[no] ip dns fqdn-resolver refresh-interval <3600-86400>
```

Description

Sets the refresh interval for FQDN resolvers. The interval specifies the time in seconds after which the FQDN resolver entries are refreshed. The valid range for the interval is between 3600 seconds (1 hour) to 86400 seconds (1 day). For each FQDN entry refresh timer is applicable from the time entry is installed.

The **no** form of this command removes the refresh interval and sets it back to default value which is 1 day.

Parameter	Description
<3600-86400>	Refresh interval in seconds (default: 86400 seconds). Required

Examples

This example sets a custom refresh interval.

```
switch(config)# ip dns fqdn-resolver refresh-interval 6000  
switch(config)# ip dns fqdn-resolver refresh-interval 85400
```

Resetting the refresh interval to the default value.

```
switch(config)# no ip dns fqdn-resolver refresh-interval
```

Attempting to set an invalid interval:

```
switch(config)# ip dns fqdn-resolver refresh-interval 20
Invalid interval. Valid range: 3600-86400 seconds.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ip dns fqdn-resolver force-refresh

Syntax

```
ip dns fqdn-resolver force-refresh { fqdn WORD | source port-access | vrf
WORD | status (resolved | stale | unresolved) }
```

Description

This command initiates a force refresh for FQDN resolver entry/entries. It can refresh all FQDN entries, or entries filtered by FQDN, source, VRF or status. If no FQDN is specified, all entries are refreshed.

Parameter	Description
fqdn	Refresh entries associated with specific FQDN. Optional
source	Refresh entries based on the source of the resolution request. Optional
vrf	Refresh entries associated with a specific VRF. Optional

Parameter	Description
status	Refresh entries associated with a specific resolution status. Optional

Examples

Refreshing all FQDN entries:

```
switch(config) # ip dns fqdn-resolver force-refresh
```

Refreshing a specific FQDN:

```
switch(config) # ip dns fqdn-resolver force-refresh fqdn www.example.com
```

Refreshing all FQDNs from specific source:

```
switch(config) # ip dns fqdn-resolver force-refresh source port-access
```

Refreshing FQDN in a specific VRF:

```
switch(config) # ip dns fqdn-resolver force-refresh fqdn www.example.com vrf
red
```

Refreshing FQDN for a specific source and vrf:

```
switch(config) # ip dns fqdn-resolver force-refresh source classifier vrf red
```

When a FQDN entry is not found:

```
switch(config) # ip dns fqdn-resolver force-refresh fqdn www.nonexistent.com
No FQDN entries matching the given criteria.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show ip dns

Syntax

```
show ip dns [vrf <VRF-NAME>]
```

Description

Shows all DNS client configuration settings or the settings for a specific VRF.

Parameter	Description
vrf <VRF-NAME>	Specifies the VRF for which to show information. If no VRF is defined, the default VRF is used.

Usage

DNS client arbitration on the MGMT interface on a MGMT VRF can be updated via three different methods.

1. Using the **domain-name** <name> or **nameservers** <servers> commands in the command-line interface.
2. Using the **ip dns domain-name** <DOMAIN-NAME> **vrf** MGMT or **ip dns server-address** <SERVER> **vrf** MGMT commands in the command-line interface.
3. Using the **ip dhcp** command in the command-line interface (dynamic entries).

AOS-CX gives the following priority levels to the these three update methods.

- Priority 1 - standalone CLI configuration
- Priority 2 - static ip dns configuration
- Priority 3 - Dynamic config

Examples

These examples define DNS settings and then show how they are displayed with the **show ip dns** command.

```
switch(config)# ip dns domain-name domain.com
switch(config)# ip dns domain-list domain5.com
switch(config)# ip dns domain-list domain8.com
switch(config)# ip dns server-address 4.4.4.4
switch(config)# ip dns server-address 6.6.6.6
switch(config)# ip dns host host3 5.5.5.5
switch(config)# ip dns host host2 2.2.2.2
```

```

switch(config)# ip dns host host3 c::12
switch(config)# ip dns domain-name reddomain.com vrf
                    red
                    default
switch(config)# ip dns domain-list reddomain5.com vrf
                    red
                    default
switch(config)# ip dns domain-list reddomain8.com vrf
                    red
                    default
switch(config)# ip dns server-address 4.4.4.5 vrf
                    red
                    default
switch(config)# ip dns server-address 6.6.6.7 vrf
                    red
                    default
switch(config)# ip dns host host3 5.5.5.6 vrf
                    red
                    default
switch(config)# ip dns host host2 2.2.2.3 vrf
                    red
                    default
switch(config)# ip dns host host3 c::13 vrf

                    red

```

```

switch# show ip dns
VRF Name : default
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name      Address
-----
host2          2.2.2.2
host3          5.5.5.5
host3          c::12

VRF Name : red
Domain Name : reddomain.com
DNS Domain list : reddomain5.com, reddomain8.com
Name Server(s) : 4.4.4.5, 6.6.6.7
Host Name      Address

```

```

-----
host2          2.2.2.3
host3          5.5.5.6
host3          c::13

switch(config)# ip dns domain-name domain.com vrf red
switch(config)# ip dns domain-list domain5.com vrf red
switch(config)# ip dns domain-list domain8.com vrf red
switch(config)# ip dns server-address 4.4.4.4 vrf red
switch(config)# ip dns server-address 6.6.6.6 vrf red
switch(config)# ip dns host host3 5.5.5.5 vrf red
switch(config)# no ip dns host host2 2.2.2.2 vrf red
switch(config)# ip dns host host3 c::12 vrf red
switch# show ip dns vrf red
VRF Name : red
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name      Address
-----
host3          5.5.5.5
host3          c::12
switch(config)# ip dns domain-name domain.com
switch(config)# ip dns domain-list domain5.com
switch(config)# ip dns domain-list domain8.com
switch(config)# ip dns server-address 4.4.4.4
switch(config)# ip dns server-address 6.6.6.6
switch(config)# ip dns host host3 5.5.5.5
switch(config)# ip dns host host3 c::12
switch# show ip dns
VRF Name : default
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name      Address
-----
host3          5.5.5.5
host3          c::12

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Show FQDN Resolver

Syntax

```
show ip dns fqdn-resolver
```

Description

This command displays a summary table of all FQDN resolver entries, including the total count of FQDN entries, refresh time, source information, and a tabular display of all entries.

Examples

Show unresolved FQDN entries with ipv4 requested:

```
switch# show ip dns fqdn-resolver
Total FQDN: 1
Refresh Time: 86400 seconds
Source: port-access
VRF      FQDN      Status      Addr-Family  ip-
address
-----
default  host1.search.com  unresolved  ipv4         -
```

Show resolved FQDN entries:

```
switch# show ip dns fqdn-resolver
Total FQDN: 2
```

Refresh Time: 7200 seconds

Source: port-access

VRF	FQDN	Status	Addr-Family
ip-address			

default	www.example.com	resolved	ipv4
192.168.1.1, 192.168.1.2			
default	api.test.com	resolved	ipv4
10.0.0.1			
			ipv6
2001:db8::1, 2001:db8::2			
long_vrf_na	a_very_long_fqdn_name	resolved	ipv4
10.0.0.2			

When no FQDN entries exist:

```
switch(config)# show ip dns fqdn-resolver
No FQDN resolver entries found.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Show IP DNS FQDN Resolver Detail

Syntax

```
show ip dns fqdn-resolver detail {fqdn WORD | source (port-access) | vrf WORD | requested-address-family (ipv4|ipv6|both)}
```

Description

Displays detailed hierarchical information for FQDN resolver entries, including the FQDN, VRF, requested address family, resolved IPv4 and IPv6 addresses, resolution status, and remaining refresh timer. The command supports filtering by FQDN, source, VRF, and address family. If no filters are provided, all entries are displayed.

Examples

Show detailed view with unresolved entry:

```
switch# show ip dns fqdn-resolver detail
```

```
-----  
----  
Source: port-access  
-----  
----  
FQDN: host1.search.com  
VRF: default  
Requested Address Family: ipv4  
Status: unresolved  
-----  
----
```

Show detailed view with resolved entries:

```
switch# show ip dns fqdn-resolver detail
```

```
Source: port-access  
  FQDN: www.example.com  
  VRF: default  
  Status: resolved  
  Requested Address Family: both  
    IPv4 Addresses:  
      192.168.1.1  
      192.168.1.2  
    IPv6 Addresses:  
      2001:db8::1  
      2001:db8::2  
-----  
----  
  FQDN: www.hpe.com  
  VRF: default  
  Status: resolved  
  Requested Address Family: ipv4  
    IPv4 Addresses:  
      10.0.0.1
```


Show detailed view with filter:

```
switch# show ip dns fqdn-resolver detail fqdn www.example.com
Source: port-access
  FQDN: www.example.com
  VRF: default
  Status: resolved
  Requested Address Family: both
  IPv4 Addresses:
    192.168.1.1
    192.168.1.2
  IPv6 Addresses:
    2001:db8::1
    2001:db8::2
-----
-----
```

Show detailed view when no entries exist:

```
switch# show ip dns fqdn-resolver detail
No FQDN resolver entries found
```



NOTE
The **source** parameter is not supported in 8320, 8400 or 9300 switch series.



NOTE
For more information on features that use this command, refer to the Security Guide for your switch model.

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Show IP DNS FQDN Resolver Refresh Interval

Syntax

```
show ip dns fqdn-resolver refresh-interval
```

Description

Displays the current refresh interval for FQDN resolvers.

Examples

Show current refresh interval (default):

```
switch# show ip dns fqdn-resolver refresh-interval
Refresh Interval: 86400 seconds
```

Show refresh interval after configuration change:

```
switch(config)# ip dns fqdn-resolver refresh-interval 7200
switch(config)# exit
switch# show ip dns fqdn-resolver refresh-interval
Refresh Interval: 7200 seconds
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Device discovery and configuration

The switch supports automatic discovery and configuration of other devices on the network.

Subtopics

Device profiles

LLDP

Cisco Discovery Protocol (CDP)

Device profiles



NOTE

Device profiles rely on role configurations. For information on role configurations, see the Security Guide .

Device profiles are used to dynamically assign port attributes based on the type of devices connected, without having to create a RADIUS infrastructure. You can map device profiles to device groups. A device group contains various match criteria, which can be obtained from multiple sources, such as LLDP, CDP, and local MAC match. Device profiles contain port attributes to be assigned to the port when a connected device matches a device group.

Device profiles are supported on different scenarios. It can be applied on interfaces that are configured with security (802.1X or MAC authentication), or applied based on L2 port (LLDP, CDP), or applied on standalone ports with the block-until-profile-applied command enabled. All the methods are mutually exclusive of each other. The block-until-profile-applied mode must be configured only when there is a standalone port where no security has been configured and when you want the port to be offline until at least one client is onboarded based on the match and ignore criteria that you configure. Local MAC match is supported when you configure block-until-profile-applied command or device profile with security.



NOTE

Up to eight device profiles can be configured.

See the Security Guide for the following commands:

- The **port-access onboarding-method precedence** command—If you are configuring both security and device profile on the port, and you want to configure the order in which the methods will be executed.
- The **port-access fallback-role** command—If you want to configure a role that must be applied to devices when no other role exists or can be derived for that device.

If you configure a match criteria that matches across multiple device profiles, then the priority considered is LLDP, CDP, and then local MAC match. That is, LLDP precedes over CDP, which in turn precedes over local MAC match.

The following figure displays a simple configuration of device profile and AAA authentication with RADIUS server and HPE Aruba Networking ClearPass Policy Manager. Local MAC match feature is useful when you

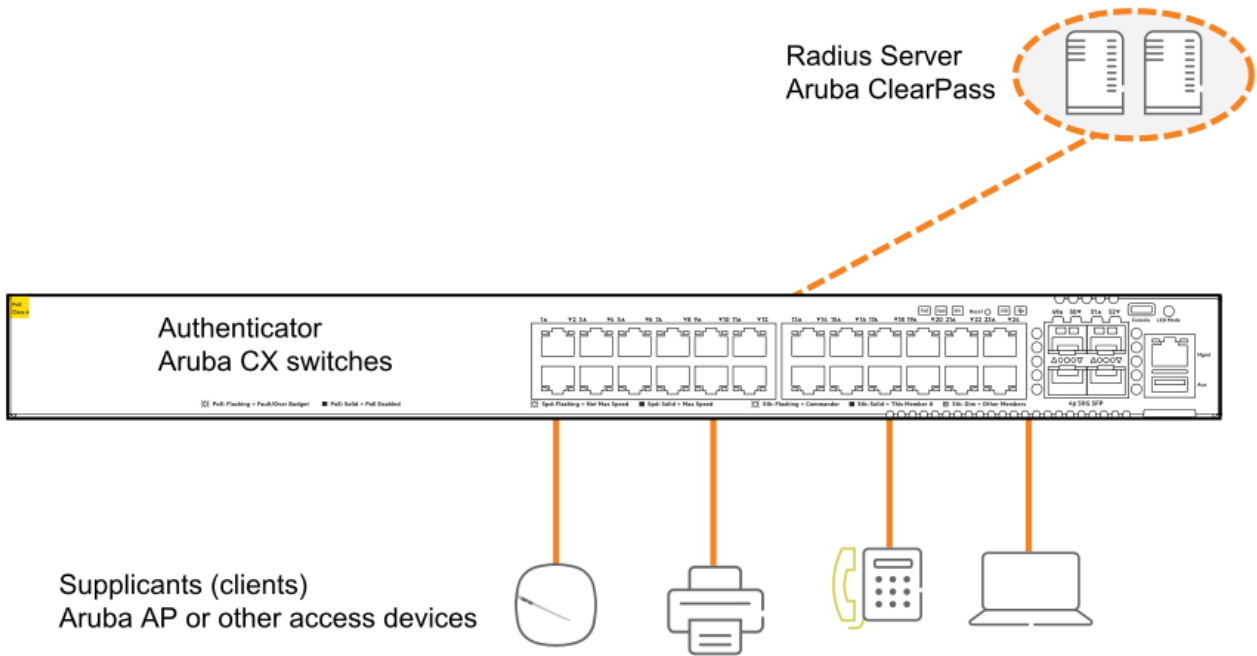
do not want to afford RADIUS infrastructure or when you want to use local authentication as a backup method in case the RADIUS server is unreachable.



NOTE

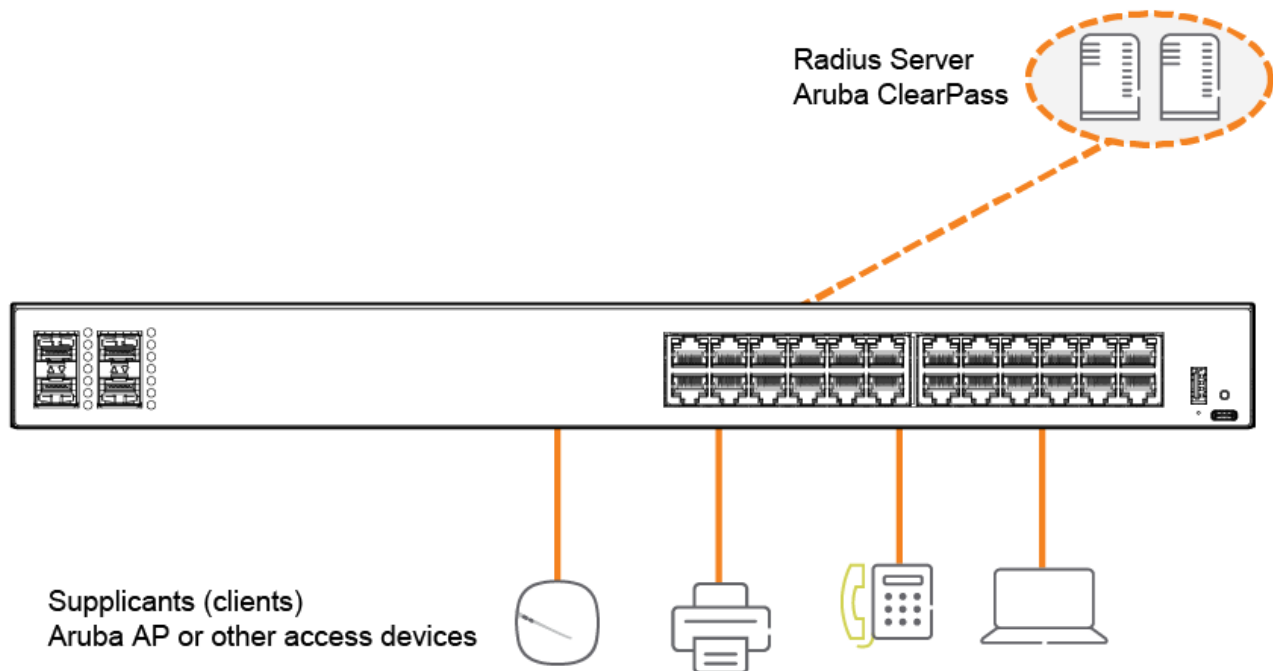
When using a device profile on a LAG/MCLAG port, best practices is to allow LLDP and CDP BPDUs using the **aaa authentication port-access allow-lldp-bpdu** and **aaa authentication port-access allow-cdp-bpdu** commands.

Figure 1. Example of device profile setup along with RADIUS infrastructure



;"/>

Figure 2. Example of device profile setup along with RADIUS infrastructure



;" />

Subtopics

[Configuring a device profile for LLDP](#)

[Configuring a device profile for CDP](#)

[Configuring a device profile for local MAC match](#)

[Device Profile Usage Considerations](#)

[Device profile commands](#)

Configuring a device profile for LLDP

Procedure

1. Create an LLDP group with the command `port-access lldp-group`.
2. Define rules for adding devices to an LLDP group with the command `match`.
3. Define rules for ignoring devices so that they are not added to an LLDP group with the command `ignore`.
4. Create a device profile with the command `port-access device-profile`.
5. Add the LLDP group with the command `associate lldp-group`.

6. Add a role to a device profile with the command `associate role`. Make sure that the role is already created. For information on how to create a role, see port access role information in the Security Guide.
7. Enable the device profile with the command `enable`.

Configuring a device profile for CDP

Procedure

1. Create a CDP group with the command `port-access cdp-group`.
2. Define rules for adding devices to a CDP group with the command `match`.
3. Define rules for ignoring devices so that they are not added to a CDP group with the command `ignore`.
4. Create a device profile with the command `port-access device-profile`.
5. Add a CDP group to a device profile with the command `associate cdp-group`.
6. Add a role to a device profile with the command `associate role`. Make sure that the role is already created. For information on how to create a role, see port access role information in the Security Guide.
7. Enable a device profile with the command `enable`.

Configuring a device profile for local MAC match

Procedure

1. Create a MAC group with the `mac-group` command.
2. Define rules for adding devices to a MAC group with the `match (for MAC groups)` command.
3. Define rules for ignoring devices so that they are not added to a MAC group with the `ignore (for MAC groups)` command.
4. Create a device profile with the `port-access device-profile` command.
5. Associate a MAC group with a device profile with the `associate mac-group` command.

6. Add a role to a device profile with the `associate role` command. Make sure that the role is already created. For information on how to create a role, see port access role information in the Security Guide .
7. Enable a device profile with the `enable` command.

Device Profile Usage Considerations

The following behaviors and limitations should be taken into consideration when configuring a device profile on Link Aggregation Group (LAG) or Multi-chassis Link Aggregation Group (MCLAG).

Device Profiles on LAG and MCLAG

- When using Device profile to onboard a device with multiple interfaces (such as a dual port AP) over a LAG, set the authentication mode in the role applied to the device to **device-mode**. The **multi-domain** authentication mode is not supported with a device profile on a LAG.
- When using Device profile to onboard a device on MCLAG port, set the authentication mode in the role applied to the device to **device-mode** or **proxy-mode**. The **multi-domain** and **client-mode** authentication mode is not supported with device profile on MCLAG.
- It is not recommended to connect different devices to the same LAG when using a device profile on the LAG.
- When updating a LAG membership, all the device profile clients on the LAG are logged off automatically from the switch and from both peers on MCLAG port.
- If you issue the command **aaa authentication port-access allow-cdp-proxy-logoff** on the LAG, a proxy-logoff TLV received on any member interface of the LAG will log off all clients onboarded on the LAG.
- Depending on resource availability or auth/role application status on each peer, the same MAC may be allowed in one peer and denied in another for a MAC learned on MCLAG port.

Security Considerations

- Each individual interface of a LAG must successfully match a Device profile for the corresponding interface to be unblocked on the data plane.
- An onboarded client on the LAG is authorized to send and receive packets on any of the member interface of the LAG that is unblocked on the data plane. This is true even for a different client connected to the same LAG.

- When using LLDP packets to authenticate a device using a device profile on a LAG, issue the command **aaa authentication port-access allow-lldp-auth mac chassis-mac** to set the authentication MAC address as the chassis MAC address
- LLDP BPDUs are blocked by default. Issue the command **aaa authentication port-access allow-lldp-bpdu** to allow the LLDP BPDUs.
- If a Device profile is used to onboard multiple devices connected to the same LAG, the client limit on the LAG must be configured to match the number of clients onboarding the LAG.
- When using LLDP packets to authenticate a Dual Port AP using a device profile on a LAG, issue the command **aaa authentication port-access allow-lldp-auth mac source-mac** to set the authentication MAC address as the port/interface MAC address.

Supported Role Attributes

The following port-access role attributes are supported for clients that onboard via device profile on a LAG.

- Authentication mode
- Client inactivity timeout
- Session timeout
- Access and Trunk VLANs
- Port-Access policy (not supported on MCLAG)
- MTU
- PoE Priority
- PoE Allocate-By
- Private-VLAN
- STP admin edge port
- Trust mode
- GBP role

Device profile commands

Select a command from the list in the left navigation menu.

Subtopics

[aaa authentication port-access allow-cdp-auth](#)
[aaa authentication port-access allow-cdp-bpdu](#)
[aaa authentication port-access allow-cdp-proxy-logoff](#)
[aaa authentication port-access allow-lldp-bpdu](#)
[associate cdp-group](#)
[associate lldp-group](#)
[associate mac-group](#)
[associate role](#)
[disable](#)
[enable](#)
[enable](#)
[ignore \(for CDP groups\)](#)
[ignore \(for LLDP groups\)](#)
[ignore \(for MAC groups\)](#)
[mac-group](#)
[match \(for CDP groups\)](#)
[match \(for LLDP groups\)](#)
[match \(for MAC groups\)](#)
[port-access cdp-group](#)
[port-access device-profile](#)
[port-access device-profile mode block-until-profile-applied](#)
[port-access lldp-group](#)
[show port-access device-profile](#)

aaa authentication port-access allow-cdp-auth

Syntax

```

aaa authentication port-access allow-cdp-auth
no aaa authentication port-access allow-cdp-auth

```

Description

Use this command to allow or block authentication on the CDP (Cisco Discovery Protocol) BPDU (Bridge Protocol Data Unit) . This is allowed by default. The **no** form of this command prevents authentication on CDP packets received on the port. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

Examples

Allowing authentication on a CDP CPDU on interface 1/1/1:

```

switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-cdp-auth

```

Allowing authentication on a CDP CPDU on a LAG port:


```
switch(config)# interface lag 1
switch(config-lag-if)# aaa authentication port-access allow-cdp-auth
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.
6200	config-lag-if	

aaa authentication port-access allow-cdp-bpdu

Syntax

```
aaa authentication port-access allow-cdp-bpdu
no aaa authentication port-access allow-cdp-bpdu
```

Description

Allows all packets related to the CDP (Cisco Discovery Protocol) BPDU (Bridge Protocol Data Unit) on a secure port or LAG. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts. The **no** form of this command blocks the CDP BPDU on a secure port. On a nonsecure port, the command has no effect.

Examples

Allowing a CDP BPDU on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-cdp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version AOS-CX 10.0X.0000
```

```

led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
aaa authentication port-access dot1x authenticator
    enable
interface 1/1/1
    no shutdown

    vlan access 1
    aaa authentication port-access allow-cdp-bpdu
    aaa authentication port-access mac-auth
        enable
    aaa authentication port-access dot1x authenticator
        enable
switch(config-if)# do show port-access device-profile interface all
Port 1/1/1, Neighbor-Mac 00:0c:29:9e:d1:20
    Profile Name      : access_switches
    LLDP Group        :
    CDP Group         : hpe-anw-ap_cdp
    Role              : test_ap_role
    Status            : In Progress
    Failure Reason    :

```

Blocking LLDP packet on secure port **1/1/1**:

```

switch(config)# interface 1/1/1
switch(config-if)# no aaa authentication port-access allow-cdp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version AOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
interface 1/1/1
    no shutdown
    vlan access 1
    aaa authentication port-access mac-auth
        enable

```

Allowing a CDP BPDU on LAG 1:

```
switch(config)# interface lag 1
switch(config-lag-if)# aaa authentication port-access allow-cdp-bpdu
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.
6200	config-lag-if	

aaa authentication port-access allow-cdp-proxy-logoff

Syntax

```
aaa authentication port-access allow-cdp-proxy-logoff
no aaa authentication port-access allow-cdp-proxy-logoff
```

Description

Allows a client to be logged off from the system via a special TLV in the CDP packet. By default, proxy logoff via CDP packet support is disabled. When **allow-cdp-proxy-logoff** is enabled, TLV received from CDP packets corresponding to logoff processing will be read and logoff is issued to the clients. This only works on client authentication enabled ports and **aaa authentication port-access allow-cdp-bpdu** must be enabled to process. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

Examples

Allowing a client to be logged off from the system via a special TLV in the CDP packet:

```

switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-cdp-proxy-logoff
switch(config-if)# show running-config interface 1/1/1
interface 1/1/1
    no shutdown

    vlan access 1
    aaa authentication port-access allow-cdp-bpdu
    aaa authentication port-access allow-cdp-proxy-logoff
    aaa authentication port-access allow client-limit 2
    aaa authentication port-access dot1x authenticator
        enable
    aaa authentication port-access mac-auth
        enable
    exit

```

The `aaa authentication port-access allow-cdp-proxy-logoff` command can also be issued from a LAG port context

```

switch(config)# interface lag 1
switch(config-lag-if)# aaa authentication port-access allow-cdp-proxy-logoff

```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.09.1000	Command introduced.

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.
6200	config-lag-if	

aaa authentication port-access allow-lldp-bpd

u

Syntax

```
aaa authentication port-access allow-lldp-bpdu
no aaa authentication port-access allow-lldp-bpdu
```

Description

Allows all packets related to the LLDP BPDU (Bridge Protocol Data Unit) on a secure port. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

The **no** form of this command blocks the LLDP BPDU on a secure port. On a nonsecure port, the command has no effect.

Examples

Allowing an LLDP BPDU on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-lldp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version AOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
interface 1/1/1
    no shutdown
    vlan access 1
aaa authentication port-access allow-lldp-bpdu
    aaa authentication port-access mac-auth
        enable
switch(config-if)# do show port-access device-profile interface all
Port 1/1/1, Neighbor-Mac 00:0c:29:9e:d1:20
    Profile Name      : access_switches
    LLDP Group        : 2920-grp
    CDP Group         :
    Role              : local_2920_role
    Status            : Profile Applied
    Failure Reason    :
```

Blocking LLDP BPDU on secure port **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no aaa authentication port-access allow-lldp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version AOS-CX 10.0X.0000led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
    enable
interface 1/1/1
    no shutdown
    vlan access 1
aaa authentication port-access mac-auth
    enable
```

Allowing an LLDP BPDU on a LAG port:

```
switch(config)# interface lag 1
switch(config-lag-if)#aaa authentication port-access allow-lldp-bpdu
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.
6200	config-lag-if	

associate cdp-group

Syntax

```
associate cdp-group <GROUP-NAME>
no associate cdp-group <GROUP-NAME>
```

Description

Associates a CDP (Cisco Discovery Protocol) group with a device profile. A maximum of two CDP groups can be associated with a device profile.

The **no** form of this command removes a CDP group from a device profile.

Parameter	Description
<code><GROUP-NAME></code>	Specifies the name of the CDP group to associate with this device profile. Range: 1 to 32 alphanumeric characters.

Examples

Associating the CDP group **my-cdp-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate cdp-group my-cdp-group
```

Removing the CDP group **my-cdp-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate cdp-group my-cdp-group
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-device-profile	Administrators or local user group members with execution rights for this command.

associate lldp-group

Syntax

```
associate lldp-group <GROUP-NAME>
no associate lldp-group <GROUP-NAME>
```

Description

Associates an LLDP group with a device profile. A maximum of two LLDP groups can be associated with a device profile.

The **no** form of this command removes an LLDP group from a device profile.

Parameter	Description
<code><GROUP-NAME></code>	Specifies the name of the LLDP group to associate with the device profile. Range: 1 to 32 alphanumeric characters.

Examples

Associating the LLDP group **my-lldp-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate lldp-group my-lldp-group
```

Removing the LLDP group **my-lldp-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate lldp-group my-lldp-group
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-device-profile	Administrators or local user group members with execution rights for this command.

associate mac-group

Syntax

```
associate mac-group <GROUP-NAME>
no associate mac-group <GROUP-NAME>
```

Description

Associates a MAC group with a device profile. A maximum of two MAC groups can be associated with a device profile.

The **no** form of this command removes a MAC group from a device profile.

Parameter	Description
<GROUP-NAME>	Specifies the name of the MAC group to associate with this device profile. Range: 1 to 32 alphanumeric characters.

Examples

Associating the MAC group **mac01-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate mac-group mac01-group
```

Removing the MAC group **mac01-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate mac-group mac01-group
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-device-profile	Administrators or local user group members with execution rights for this command.

associate role

Syntax

```
associate role <ROLE-NAME>  
no associate role <ROLE-NAME>
```

Description

Associates a role with a device profile. Only one role can be associated with a device profile. For information on how to configure a role, see the port access role information in the Security Guide .

The **no** form of this command removes a role from a device profile.

Parameter	Description
<ROLE-NAME>	Specifies the name of the role to associate with the device profile. Range: 1 to 64 alphanumeric characters.

Examples

Associating the role **my-role** with the device profile **profile01**:

```
switch(config) # port-access device-profile profile01
switch(config-device-profile) # associate role my-role
```

Removing the role **my-role** from the device profile **profile01**:

```
switch(config) # port-access device-profile profile01
switch(config-device-profile) # no associate role my-role
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-device-profile	Administrators or local user group members with execution rights for this command.

disable

Syntax

```
disable
no disable
```

Description

Disables a device profile.

The **no** form of this command enables a device profile.

Examples

Disabling a device profile:

```
switch(config) # port-access device-profile profile01
switch(config-device-profile) # disable
```

Enabling a device profile named profile01:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-device-profile	Administrators or local user group members with execution rights for this command.

enable

Syntax

```
enable
no enable
```

Description

Enables a device profile.

The **no** form of this command disables a device profile.

Examples

Enabling a device profile:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# enable
```

Disabling a device profile named profile01:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-device-profile	Administrators or local user group members with execution rights for this command.

enable

Syntax

```
enable
no enable
```

Description

Enables device profile on the port. By default, device profile is already enabled on the port.,
The **no** form of this command disables device profile on the port.

Examples

Enabling a device profile on port 1/1/1 and disabling on lag 1:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access device-profile
switch(config-if-deviceprofile)#
enable Enable device-profile on this port.
mode Configure device-profile mode for this port.
switch(config-if-deviceprofile)# enable
switch(config-if-deviceprofile)# end
switch(config)# interface lag 1
```

```

switch(config-lag-if)# port-access device-profile
switch(config-lag-if-deviceprofile)# no
enable    Enable device-profile on this port.
mode      Configure device-profile mode for this port.
switch(config-lag-if-deviceprofile)# no enable
switch(config-lag-if-deviceprofile)# end

```

Showing running configuration:

```

switch#
switch# show running-config
interface lag 1
    no shutdown
    no routing
    vlan access 1
    port-access device-profile
        no enable
interface 1/1/1
    no shutdown
    no routing
    vlan access 1
exit

```

Command History

Release	Modification
10.17	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config-device-profile	Administrators or local user group members with execution rights for this command.

ignore (for CDP groups)

Syntax

```
ignore [seq <SEQ-NUM>] {platform <PLATFORM> | sw-version <SWVERSION> |  
    voice-vlan-query <VLAN-ID>}  
no ignore [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |  
    voice-vlan-query <VLAN-ID>}
```

Description

Defines a rule to ignore devices for a CDP (Cisco Discovery Protocol) group. Up to 64 match/ignore rules can be defined for a group.

The **no** form of this command removes a rule for ignoring devices from a CDP group.

Parameter	Description
seq <SEQ-ID>	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.
platform <PLATFORM>	Specifies the hardware or model details of the neighbor. Range: 1 to 128 alphanumeric characters.
sw-version <SWVERSION>	Specifies the software version of the neighbor. Range: 1 to 128 alphanumeric characters.
voice-vlan-query <VLAN-ID>	Specifies the VLAN query value of the neighbor. Range: 1 to 65 535.

Examples

Adding a rule to the CDP group **grp01** that ignores a device that transmits **PLATFORM01** in the platform TLV:

```
switch(config)# port-access cdp-group grp01  
switch(config-cdp-group)# ignore platform PLATFORM01
```

Adding a rule to the CDP group **grp01** that ignores a device that transmits **SWVERSION** in software version TLV:

```
switch(config)# port-access cdp-group grp01  
switch(config-cdp-group)# ignore sw-version SWVERSION
```

Removing the rule that matches the sequence number **25** from the CDP group named **grp01**.

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# no ignore seq 25
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-cdp-group	Administrators or local user group members with execution rights for this command.

ignore (for LLDP groups)

Syntax

```
ignore [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
no ignore [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

Description

Defines a rule to ignore devices for an LLDP group. Up to 64 match/ignore rules can be defined for a group. The **no** form of this command removes a rule for ignoring devices from an LLDP group.

Parameter	Description
seq <SEQ-ID>	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.

Parameter	Description
sys-desc <SYS-DESC>	Specifies the LLDP system description type - length - value (TLV). Range: 1 to 256 alphanumeric characters.
sysname <SYS - NAME>	Specifies the LLDP system name TLV. Range: 1 to 64 alphanumeric characters.
vendor-oui <VENDOR-OUI>	Specifies the LLDP system vendor OUI TLV. Range: 1 to 6 alphanumeric characters.
type <KEY>	Specifies the vendor OUI subtype key. Optional.
value <VALUE>	Specifies the vendor OUI subtype value. Range: 1 to 256 alphanumeric characters.

Examples

Adding a rule to the LLDP group **grp01** that ignores a device that transmits **PLATFORM01** in the system description TLV:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# ignore sys-desc PLATFORM01
```

Removing the rule that matches the sequence number **25** from the LLDP group named **grp01**.

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# no match seq 25
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-lldp-group	Administrators or local user group members with execution rights for this command.

ignore (for MAC groups)

Syntax

```
[seq <SEQ-ID>] ignore {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}  
no [seq <SEQ-ID>] ignore {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

Description

Defines a rule to ignore devices for a MAC group based on the criteria of MAC address, MAC address mask, or MAC Organizational Unique Identifier (OUI). Up to 64 ignore rules can be defined for a group.

The **no** form of this command removes a rule for ignoring devices from a MAC group.

Parameter	Description
seq <SEQ-ID>	Specifies the entry sequence ID of the rule to create or modify a MAC group. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence. Range: 1 to 4294967295 .
mac <MAC-ADDR>	Specifies the MAC address of the device to ignore.
mac-mask <MAC-MASK>	Specifies the MAC address mask to ignore devices in that range. Supported MAC address masks: /32 and /40 .
mac-oui <MAC-OUI>	Specifies the MAC OUI to ignore devices in that range. Supports MAC OUI address of maximum length of 24 bits.

Usage

To achieve the required configuration of matches for devices, it is recommended to first ignore the devices that you do not want to add. Then match the criteria for the rest of the devices that you want to add to the MAC group.

For example, if you want to ignore a specific device but add all the other devices that belong to a MAC OUI, then you must first configure the ignore criteria with a lower sequence number. And then configure match criteria with a higher sequence number.

Examples

Adding a rule to the MAC group **grp01** to ignore a device based on MAC address, but match all other devices belonging to a MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac 1a:2b:3c:4d:5e:6f
switch(config-mac-group)# match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
!
!
ssh server vrf mgmt
    default
!
!
!
!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac 1a:2b:3c:4d:5e:6f
    seq 20 match mac-oui 1a:2b:3c
...

```

Adding a rule to the MAC group **grp01** to ignore devices based on MAC address mask, but match all other devices belonging to a MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac-mask 1a:2b:3c:4d/32
switch(config-mac-group)# match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!

```

```

!
!
!
ssh server vrf mgmt
    default
!
!
!
!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac-mask 1a:2b:3c:4d/32
    seq 20 match mac-oui 1a:2b:3c
...

```

Adding a rule to the MAC group **grp01** that ignores a device based on complete MAC address:

```

switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac 1a:2b:3c:4d:5e:6f

```

Adding a rule to the MAC group **grp02** that ignores devices based on MAC mask:

```

switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac-mask 1a:2b:3c:4d:5e/40
switch(config-mac-group)# ignore mac-mask 18:e3:ab:73/32

```

Adding a rule to the MAC group **grp03** that ignores devices based on MAC OUI:

```

switch(config)# mac-group grp03
switch(config-mac-group)# ignore mac-oui 81:cd:93

```

Adding a rule to the MAC group **grp01** that ignores devices with a sequence number and based on MAC address:

```

switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 ignore mac b2:c3:44:12:78:11
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default

```

```

led locator on
!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac b2:c3:44:12:78:11
...

```

Removing the rule from the MAC group **grp01** based on sequence number:

```

switch(config)# mac-group grp01
switch(config-mac-group)# no ignore seq 10
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp01
...

```

Adding a rule to the MAC group **grp01** that ignores devices with MAC entry sequence number and based on MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 ignore mac b2:c3:44:12:78:11
switch(config-mac-group)# seq 20 ignore mac-oui 1a:2b:3c
switch(config-mac-group)# seq 30 ignore mac-mask 71:14:89:f3/32
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on

```

```

!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac b2:c3:44:12:78:11
    seq 20 ignore mac-oui 1a:2b:3c
    seq 30 ignore mac-mask 71:14:89:f3/32
...

switch(config)# mac-group grp01
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 ignore mac b2:c3:44:12:78:11
switch(config-mac-group)# seq 20 ignore mac-oui 1a:2b:3c
switch(config-mac-group)# seq 30 ignore mac-mask 71:14:89:42/32
switch(config-mac-group)# exit
switch(config)#
switch(config)# ^Z
switch#
switch#
switch# show running-config
Current configuration:
!
!Version AOS-CX PL.10.06.0002
!export-password: default
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
mac-group grp01
    seq 10 ignore mac b2:c3:44:12:78:11
    seq 20 ignore mac-oui 1a:2b:3c
    seq 30 ignore mac-mask 71:14:89:42/32
interface 1/1/1
    no shutdown
    vlan access 1
interface 1/1/2
    no shutdown

```

```
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
```

```

interface vlan 1
    ip dhcp
!
!
!
!
!
https-server vrf default

```

Removing the rule from the MAC group **grp01** based on sequence number and MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# no seq 20 ignore mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp01
    seq 10 ignore mac b2:c3:44:12:78:11
    seq 30 ignore mac-mask 71:14:89:f3/32
...

```

Removing the rule that matches the sequence number **25** from the MAC group named **grp01**.

```

switch(config)# mac-group grp01
switch(config-mac-group)# no ignore seq 25

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-mac-group	Administrators or local user group members with execution rights for this command.

mac-group

Syntax

```
mac-group <MAC-GROUP-NAME>  
no mac-group <MAC-GROUP-NAME>
```

Description

Creates a MAC group or modifies an existing MAC group. A MAC group is used to classify connected devices based on the MAC address details, such as mask or OUI.

A maximum of 32 MAC groups can be configured on the switch. A maximum of 2 MAC groups can be associated with a device profile. Each group accepts 64 match or ignore commands.

The **no** form of this command removes a MAC group.

Parameter	Description
<MAC-GROUP-NAME>	Specifies the name of the MAC group to create or modify. The maximum number of characters supported is 32.

Examples

Creating a MAC group named **grp01**:

```
switch(config)# mac-group grp01  
switch(config-mac-group)# exit
```

Removing a MAC group named **grp01**:

```
switch(config)# no mac-group grp01
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

match (for CDP groups)

Syntax

```
match [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |  
    voice-vlan-query <VLAN-ID>}  
no match [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |  
    voice-vlan-query <VLAN-ID>}
```

Description

Defines a rule to match devices for a CDP group. A maximum of 32 CDP groups can be configured on the switch. Up to 64 match or ignore rules can be defined for each group.

The **no** form of this command removes a rule for adding devices to a CDP group.

Parameter	Description
seq <SEQ-ID>	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.
platform <PLATFORM>	Specifies the hardware or model details of the neighbor. Range: 1 to 128 alphanumeric characters.

Parameter	Description
<code>sw-version <SWVERSION></code>	Specifies the software version of the neighbor. Range: 1 to 128 alphanumeric characters.
<code>voice-vlan-query <VLAN-ID></code>	Specifies the VLAN query value of the neighbor. Range: 1 to 65 535.

Examples

Adding rules to match a Cisco device with a specific software version on VLAN **512** to the CDP group **grp01**:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match platform CISCO
switch(config-cdp-group)# match sw-version 11.2(12)P
switch(config-cdp-group)# match voice-vlan-query 512
switch(config-cdp-group)# match seq 50 platform cisco sw-version 11.2(12)P
voice-vlan-query 512
switch(config-cdp-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access cdp-group grp01
    seq 10 match platform CISCO
    seq 20 match sw-version 11.2(12)P
    seq 30 match voice-vlan-query 512
    seq 50 match platform cisco sw-version 11.2(12)P voice-vlan-query 512
```

Removing a rule that matches the sequence number **25** from the CDP group named **grp01**:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# no match seq 25
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-cdp-group	Administrators or local user group members with execution rights for this command.

match (for LLDP groups)

Syntax

```
match [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |  
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}  
no match [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |  
    vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

Description

Defines a rule to match devices for an LLDP group. Up to 64 match/ignore rules can be defined for a group.

The **no** form of this command removes a rule.

Parameter	Description
seq <SEQ-ID>	Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.
sys-desc <SYS-DESC>	Specifies the LLDP system description type - length - value (TLV). The sys - desc parameter be configured with a substring instead of a complete string to match. Range: 1 to 256 alphanumeric characters.
sysname <SYS-NAME>	Specifies the LLDP system name TLV. The sysname parameter be configured with a substring instead of a complete string to match. Range: 1 to 64 alphanumeric characters.
vendor-oui <VENDOR-OUI>	Specifies the LLDP system vendor OUI TLV. Range: 1 to 6 alphanumeric characters.

Parameter	Description
<code>type <KEY></code>	Specifies the vendor OUI subtype key.
<code>value <VALUE></code>	Specifies the vendor OUI subtype value. Range: 1 to 256 alphanumeric characters.

Examples

Adding rules that match the LLDP system description **HPE_ANW-Switch** and system name **HPE_ANW** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match sys-desc HPE_ANW-Switch
switch(config-lldp-group)# match sysname HPE_ANW
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access lldp-group grp01
    seq 10 match sys-desc HPE_ANW-Switch
    seq 20 match sysname HPE_ANW
```

Removing a rule that matches the sequence number **25** from an LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# no match seq 25
```

Adding a rule that matches the value of vendor-OUI **000b86** with type of **1** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match vendor-oui 000b86 type 1
```

Adding a rule that matches the value of vendor-OUI **000c34** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match vendor-oui 000c34
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-lddp-group	Administrators or local user group members with execution rights for this command.

match (for MAC groups)

Syntax

```
[seq <SEQ-ID>] match {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}  
no [seq <SEQ-ID>] match {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

Description

Defines a rule to match devices for a MAC group based on the criteria of MAC address, MAC address mask, or MAC Organizational Unique Identifier (OUI). Up to 64 match rules can be defined for a group.



NOTE

You must not configure the following special MAC addresses:

- Null MAC—For example, 00:00:00:00:00:00 or 00:00:00/32
- Multicast MAC
- Broadcast MAC—For example, ff:ff:ff:ff:ff:ff
- System MAC

Although the switch accepts these addresses, it will not process these addresses for the local MAC match feature.

The **no** form of this command removes a rule for adding devices to a MAC group.

The number of clients that can onboard based on the match criteria is configured in the **aaa authentication port-access client-limit** command. For information about this command, see the Security Guide for your switch.

Parameter	Description
<code>seq <SEQ-ID></code>	Specifies the entry sequence ID of the rule to create or modify a MAC group. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence. Range: 1 to 4294967295 .
<code>mac <MAC-ADDR></code>	Specifies the MAC address of the device.
<code>mac-mask <MAC-MASK></code>	Specifies the MAC address mask to add devices in that range. Supported MAC address masks: /32 and /40 .
<code>mac-oui <MAC-OUI></code>	Specifies the MAC OUI to add devices in that range. Supports MAC OUI address of maximum length of 24 bits.

Examples

Adding a device to the MAC group **grp01** based on complete MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# match mac 1a:2b:3c:4d:5e:6f
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp02** based on MAC mask:

```
switch(config)# mac-group grp01
switch(config-mac-group)# match mac-mask 1a:2b:3c:4d:5e/40
switch(config-mac-group)# match mac-mask 18:e3:ab:73/32
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp03** based on MAC OUI:

```
switch(config)# mac-group grp03
switch(config-mac-group)# match mac-oui 81:cd:93
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp01** with MAC entry sequence number and based on MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 match mac b2:c3:44:12:78:11
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
```

```

!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp01
    seq 10 match mac b2:c3:44:12:78:11
...

```

Removing devices from the MAC group **grp01** based on sequence number:

```

switch(config)# mac-group grp01
switch(config-mac-group)# no match seq 10
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp01
...

```

Adding devices to the MAC group **grp01** with MAC entry sequence number and based on MAC address, MAC address mask, and MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 match mac b2:c3:44:12:78:11
switch(config-mac-group)# seq 20 match mac-oui 1a:2b:3c
switch(config-mac-group)# seq 30 match mac-mask 71:14:89:f3/32
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001

```



```

!export-password: default
led locator on
!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp01
    seq 10 match mac b2:c3:44:12:78:11
    seq 20 match mac-oui 1a:2b:3c
    seq 30 match mac-mask 71:14:89:f3/32
...

```

Removing devices from the MAC group **grp01** based on MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# no seq 20 match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp01
    seq 10 match mac b2:c3:44:12:78:11
    seq 30 match mac-mask 71:14:89:f3/32
...

```

Adding devices to the MAC group **grp03** with MAC entry sequence number and based on MAC address mask:

```

switch(config)# mac-group grp03
switch(config-mac-group)# seq 10 match mac-mask 10:14:a3:b7:55/40
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!

```

```

!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp03
    seq 10 match mac-mask 10:14:a3:b7:55/40
...

```

Removing devices from the MAC group **grp03** based on MAC address mask:

```

switch(config)# mac-group grp03
switch(config-mac-group)# no seq 10 match mac-mask 10:14:a3:b7:55/40
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface mgmt

    no shutdown
    ip dhcp
mac-group grp03
...

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-mac-group	Administrators or local user group members with execution rights for this command.

port-access cdp-group

Syntax

```
port-access cdp-group <CDP-GROUP-NAME>
no port-access cdp-group <CDP-GROUP-NAME>
```

Description

Creates a CDP (Cisco Discovery Protocol) group or modifies an existing CDP group. A CDP Group is used to classify connected devices based on the CDP packet details advertised by the device. A maximum of 32 CDP groups can be configured on the switch. Each group accepts 64 match/ignore commands.

The **no** form of this command removes a CDP group.

Parameter	Description
<CDP-GROUP-NAME>	Specifies the name of the CDP group to create or modify. The maximum number of characters supported is 32. Required.

Examples

Creating a CDP group named grp01:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match platform CISCO
switch(config-cdp-group)# match sw-version 11.2(12)P
switch(config-cdp-group)# match voice-vlan-query 512
switch(config-cdp-group)# seq 50 match platform cisco sw-version 11.2(12)P
voice-vlan-query 512
switch(config-cdp-group)# exit
switch(config)# do show running-config
```

Current configuration:

```
!  
!Version AOS-CX Virtual.10.0X.000  
!export-password: default  
led locator on  
!  
!  
vlan 1  
port-access cdp-group grp01  
    seq 10 match platform CISCO  
    seq 20 match sw-version 11.2(12)P  
    seq 30 match voice-vlan-query 512  
    seq 50 match platform cisco sw-version 11.2(12)P voice-vlan-query 512
```

Removing a CDP group named grp01:

```
switch(config)# no port-access cdp-group grp01
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

port-access device-profile

Syntax

```
port-access device-profile <DEVICE-PROFILE-NAME>  
no port-access device-profile <DEVICE-PROFILE-NAME>
```

Description

Creates a new device profile and switches to the **config-device-profile** context. A maximum of 32 device profiles can be created. This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts.

The **no** form of this command removes a device profile.

Parameter	Description
<code><DEVICE-PROFILE-NAME></code>	Specifies the name of a device profile. Range: 1 to 32 alphanumeric characters.

Usage

When multiple device profiles are created on the switch using the **port-access device-profile <device-profile-name>** command, the switch goes through the list of device profiles names in alphabetical order to find the device profile to associate to a client. For example, if the switch had the following configuration:

```
port-access device-profile Switch
    associate lldp-group lldp1
port-access device-profile AP
    associate lldp-group lldp1
```

The order of the attempted match would be device profile **AP**, and then device profile **Switch**.

Examples

Creating a device profile named **profile01**:

```
switch(config) # port-access device-profile profile01
switch(config-device-profile) #
```

Removing a device profile named **profile01**:

```
switch(config) # no port-access device-profile profile01
```

Creating a device profile named **profile02** on a LAG port:

```
switch(config) # interface lag 1
switch(config-lag-if) # port-access device-profile profile01
switch(config-device-profile) #
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420	config	Administrators or local user group members with execution rights for this command.
6200	config-if	
	config-lag-if	

port-access device-profile mode block-until-profile-applied

Syntax



NOTE

You must configure this mode in device profile only on standalone ports where there is no security configured and when you not want the port to be offline until one client is onboarded.

```
port-access device-profile mode block-until-profile-applied
no port-access device-profile mode block-until-profile-applied
```

Description

Configures the switch to block the port until a profile match occurs for a device. This configuration is required when no security feature is enabled on the port.

You must enable this mode or security on the port for local MAC match feature to operate. You must not enable both features on the same port at the same time.



NOTE

You must not combine any other AAA configurations with the block-until-profile-applied mode.

This command can be issued from the interface (**config-if**) or Link Aggregation Group (**config-lag-if**) contexts. The **no** form of this command removes a rule for adding devices to a MAC group.

Example

Configuring block-until-profile applied mode on port 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access device-profile
switch(config-if-deviceprofile)# mode block-until-profile-applied
switch(config-if-deviceprofile)# end
```

Command History

Release	Modification
10.13	This command can be issued from a Link Aggregation Group (LAG) context.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-if config-if-deviceprofile config-lag-if	Administrators or local user group members with execution rights for this command.

port-access lldp-group

Syntax

```
port-access lldp-group <LLDP-GROUP-NAME>
no port-access lldp-group <LLDP-GROUP-NAME>
```

Description

Creates an LLDP group or modifies an existing LLDP group. An LLDP group is used to classify connected devices based on the LLDP type-length-values (TLVs) advertised by the device. A maximum of 32 LLDP groups can be configured on the switch. Each group accepts 64 match/ignore commands.

The **no** form of this command removes an LLDP group.

Parameter	Description
<code><LLDP-GROUP-NAME></code>	Specifies the name of the LLDP group to create or modify. The maximum number of characters supported is 32. Required.

Examples

Creating an LLDP group named grp01:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)#
```

Removing an LLDP group named grp01:

```
switch(config)# no port-access lldp-group grp01
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

show port-access device-profile

Syntax

```
show port-access device-profile [[interface {all | <INTERFACE-ID>}
[client-status <MAC-ADDR>]] | name <DEVICE-PROFILE-NAME>]
```

Description

Shows the client status for a specific MAC address or profile name.

Parameter	Description
interface {all <INTERFACE-ID>}	Select all for all interfaces or specify the name of an interface in the format: member/slot/port .
client-status <MAC-ADDR>	Specifies a MAC address (xx:xx:xx:xx:xx:xx), where x is a hexa decimal number from 0 to F.
name <DEVICE-PROFILE-NAME>	Specifies the name of the device profile.

Examples

Showing the applied state of the device profiles:

```
switch# show port-access device-profile
  Profile Name      : accesspoints
  LLDP Groups       : 2920-grp
  CDP Groups        :
  MAC Groups        : 2920-mac-grp1,2920-iot-grp2
  Role              : local_role_1
  State             : Enabled
  Profile Name      : access_switches
  LLDP Groups       : 2920-grp
  CDP Groups        :
  MAC Groups        :
  Role              : local_2920_role
  State             : Enabled
  Profile Name      : iot_devices
  LLDP Groups       :
  CDP Groups        :
  MAC Groups        : iot_camera-grp1,iot_sensors-grp1
  Role              : local_2920_role
  State             : Enabled
  Profile Name      : lobbyaps
  LLDP Groups       :
  CDP Groups        : lobby_ap_cdp_grp
  MAC Groups        :
  Role              : test_ap_role
  State             : Disabled
```

Showing the applied state of the device profile on interface 1/1/3:

```
switch# show port-access device-profile interface 1/1/3 client-status
00:0c:29:9e:d1:20
Port 1/1/3, Neighbor-Mac 00:0c:29:9e:d1:20
```

```

Profile Name      : lobbyaps
LLDP Group       :
CDP Group        : hpe-anw-ap_cdp
MAC Group        :
Role             : test_ap_role
Status           : Failed
Failure Reason    : Failed to apply MAC based VLAN

```

Showing the applied state of a specific device profile:

```

switch# show port-access device-profile name lldp-group
Profile Name      : lldp-group
LLDP Groups       :
CDP Groups        :
MAC Groups        : pc-behind-phone, lldp
Role              : auth_role
State             : Enabled

```

Showing when device-profile is configured but not enabled on the port:

```

Switch# show port-access device-profile interface all
No device-profile clients found.
Switch# show port-access device-profile interface 1/1/1
No device-profile clients found.
Switch# show port-access device-profile interface 1/1/1-1/1/5
No device-profile clients found.
Switch# show port-access device-profile interface 1/1/1 client-status
00:50:56:bd:50:43
No device-profile clients found.
Switch# show port-access device-profile interface 1/1/1-1/1/5 client-status
00:50:56:bd:50:43
No device-profile clients found.

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

LLDP

The IEEE 802.1AB Link Layer Discovery Protocol (LLDP) provides a standards-based method for network devices to discover each other and exchange information about their capabilities. An LLDP device advertises itself to adjacent (neighbor) devices by transmitting LLDP data packets on all interfaces on which outbound LLDP is enabled, and reading LLDP advertisements from neighbor devices on ports on which inbound LLDP is enabled. Inbound packets from neighbor devices are stored in a special LLDP MIB (management information base). This information can then be queried by other devices through SNMP.

LLDP information is used by network management tools to create accurate physical network topologies by determining which devices are neighbors and through which interfaces they connect. LLDP operates at layer 2 and requires an LLDP agent to be active on each interface that sends and receives LLDP advertisements. LLDP advertisements can contain a variable number of TLV (type, length, value) information elements. Each TLV describes a single attribute of a device such as: system capabilities, management IP address, device ID, port ID.

Packet boundaries

When multiple LLDP devices are directly connected, an outbound LLDP packet travels only to the next LLDP device. An LLDP-capable device does not forward LLDP packets to any other devices, regardless of whether they are LLDP-enabled.

An intervening hub or repeater forwards the LLDP packets it receives in the same manner as any other multicast packets it receives. Therefore, two LLDP switches joined by a hub or repeater handle LLDP traffic in the same way that they would if directly connected.

Any intervening 802.1D device or Layer-3 device that is either LLDP-unaware or has disabled LLDP operation drops the packet.

LLDP-MED

LLDP-MED (ANSI/TIA-1057/D6) extends the LLDP (IEEE 802.1AB) industry standard to support advanced features on the network edge for Voice Over IP (VoIP) endpoint devices with specialized capabilities and LLDP-MED standards-based functionality. LLDP-MED in the switches uses the standard LLDP commands described earlier in this section, with some extensions, and also introduces new commands unique to LLDP-MED operation. The show commands described elsewhere in this section are applicable to both LLDP and LLDP-MED operation. LLDP-MED enables:

- Configure Voice VLAN and advertise it to connected MED endpoint devices.
- Power over Ethernet (PoE) status and troubleshooting support via SNMP.

Subtopics

LLDP agent

LLDP MED support

LLDP EEE

Configuring the LLDP agent

LLDP commands

LLDP agent

When you enable LLDP on the switch, it is automatically enabled on all data plane interfaces. You can customize this behavior by manually enabling/disabling support on each interface.

Supported standards

The LLDP agent supports the following standards: IEEE 802.1AB-2005, Station, and Media Access Control Connectivity Discovery.

Supported interfaces

LLDP is supported on interfaces mapped to a physical port, and the Out-Of-Band Management (OOBM) port. It is not supported on logical interfaces, such as loopback, tunnels, and SVIs.

Operating modes

When LLDP is enabled, the switch periodically transmits an LLDP advertisement (packet) out each active port enabled for outbound LLDP transmissions and receives LLDP advertisements on each active port enabled to receive LLDP traffic.

The LLDP agent can operate in one of the following modes:

- **Transmit and receive (TxRx):** This is the default setting on all ports. It enables a given port to both transmit and receive LLDP packets and to store the data from received (inbound) LLDP packets in the switch's MIB.
- **Transmit only (Tx):** Enables a port to transmit LLDP packets that can be read by LLDP neighbors. However, the port drops inbound LLDP packets from LLDP neighbors without reading them. This prevents the switch from learning about LLDP neighbors on that port.
- **Receive only (Rx):** Enables a port to receive and read LLDP packets from LLDP neighbors and to store the packet data in the switch's MIB. However, the port does not transmit outbound LLDP packets. This prevents LLDP neighbors from learning about the switch through that port.
- **Disabled:** Disables LLDP packet transmissions and reception on a port. In this state, the switch does not use the port for either learning about LLDP neighbors or informing LLDP neighbors of its presence.

An LLDP agent operating in TxRx mode or Tx mode sends LLDP frames to its directly connected devices both periodically and when the local configuration changes.

Sending LLDP frames

Each time the LLDP operating mode of an LLDP agent changes, its LLDP protocol state machine reinitializes. A configurable reinitialization delay prevents frequent initializations caused by frequent changes to the operating mode. If you configure the reinitialization delay, an LLDP agent must wait the specified amount of time to initialize LLDP after the LLDP operating mode changes.

Receiving LLDP frames

An LLDP agent operating in TxRx mode or Rx mode confirms the validity of TLVs carried in every received LLDP frame. If the TLVs are valid, the LLDP agent saves the information and starts an aging timer. The initial value of the aging timer is equal to the TTL value in the Time To Live TLV carried in the LLDP frame. When the LLDP agent receives a new LLDP frame, the aging timer restarts. When the aging timer decreases to zero, all saved information ages out.

TLV support

By default, the agent sends and receives the following mandatory TLVs on each interface:

- Port ID
- Chassis ID
- TTL

By default, the following ANSI/TIA-1057 TLVs for LLDP Media Endpoint Discovery (MED) are enabled on an agent. Sending them depends on the configuration and reception of any MED TLVs:

- MAC/PHY status. Includes the bit rate and auto negotiation status of the link.
- Power Via MDI: Includes Power Over Ethernet related information for supported interfaces.
- Port description
- System name
- System description
- Management address
- System capabilities
- Port VLAN ID

By default, the agent sends and receives the following ANSI/TIA-1057 TLVs for LLDP Media Endpoint Discovery (MED):

- Capabilities: Indicates MED TLV capability.
- Power Via MDI: Includes Power Over Ethernet related information.
- Network Policy: Includes the VLAN configuration for voice application.
- Location: Location identification information.
- Extended Power Via MDI: Power Over Ethernet related information

TLV advertisements

The LLDP agent transmits the following:

- Chassis-ID: Base MAC address of the switch.
- Port-ID: Port number of the physical port.
- Time-to-Live (TTL): Length of time an LLDP neighbor retains advertised data before discarding it.
- System capabilities: Identifies the primary switch capabilities (bridge, router). Identifies the primary switch functions that are enabled, such as routing.
- System description: Includes switch model name and running software version, and ROM version.
- System name: Name assigned to the switch.
- Management address: Default address selection method unless an optional address is configured.
- Port description: Physical port identifier.
- Port VLAN ID: On an L2 port, contains access or native VLAN ID. On an L3 port, contains a value of 0. Trunk allowed VLANs information are not advertised as part of the Port VLAN ID TLV. (Not supported on the OOBM interface)
- Port VLAN Name: Access or native VLAN name. Enabled by default.
- Port MFS: Maximum Frame Size. Enabled by default.

LLDP MED support

LLDP-MED interoperates with directly connected IP telephony (endpoint) clients and provides the following features:

- Advertisement of the voice VLAN configured on the interface which is used by connected IP telephony (endpoint) clients.
- Advertisement of the configured location on the switch that can be used by the connected endpoint.
- Support for the fast-start capability



NOTE

LLDP-MED is intended for use with VoIP endpoints and is not designed to support links between network infrastructure devices (such as switch-to-switch or switch-to-router links).

LLDP EEE

Energy Efficient Ethernet (EEE) is a mechanism defined by IEEE 802.3az, which is an extension of 802.3 IEEE standards. EEE defines support for the device to operate in the Low Power Idle (LPI) mode during low link utilization. This allows both sides of a link to disable or turn off respective transmit/receive circuitry to save power. EEE uses LLDP for exchanging optimal set of parameters for both devices.

Guidelines

- An LLDP must not contain more than one EEE TLV.
- LLDP may or may not be enabled on the interfaces supporting EEE. If LLDP is not enabled, it might not be in the optimal operational mode.
- EEE TLV should not be exchanged until it negotiates from L1 and it detects peer EEE capabilities.
- EEE TLV is disabled by default and enabled only when EEE is active.
- EEE and EEE TLV exchange can be enabled or disabled at the interface level.

Configuring the LLDP agent

Procedure

1. By default, the LLDP agent is enabled on all active interfaces. If LLDP was disabled, enable it with the command `lldp`.
2. By default, the LLDP agent transmits and receive on all interfaces. To customize LLDP behavior on a specific interface, use the commands `lldp transmit` and `lldp receive`.
3. By default, the LLDP agent sets the management address in all TLVs in the following order:
 - a. LLDP management IP address.
 - b. Loopback interface IP.
 - c. ROP (L3 ports) or SVI (L2 ports).
 - d. OOBM (Management interface IP).
 - e. Base MAC.

On the OOBM port, the following order is used:

- a. LLDP management IP address,
- b. IP address of the management interface (OOBM port).

- c. IP address of the loopback interface.
- d. Base MAC address of the switch.

To specify a different address, use the commands `lldp management-ipv4-address` and `lldp management-ipv6-address`

- 4. By default, all supported TLVs are sent and received. To customize the list, use the command `lldp select-tlv`.
- 5. By default, support for the LLDP-MED TLV is enabled. To customize settings, use the commands `lldp med` and `lldp med-location`.
- 6. If required, adjust LLDP timer, holdtime, reinitialization delay, and transmit delay from their default values with the commands `lldp timer`, `lldp holdtime`, `lldp reinit`, and `lldp tx delay`.

Example

This example creates the following configuration:

- Enables LLDP support.
- Disables LLDP transmission on interface **1/1/1**.

```
switch(config) # lldp
switch(config) # interface 1/1/1
switch(config-copp) # no lldp transmit
```

LLDP commands

Select a command from the list in the left navigation menu.

Subtopics

[clear lldp neighbors](#)

[clear lldp statistics](#)

[lldp](#)

[lldp dot3](#)

[lldp dot3 eee](#)

[lldp dot3 mfs](#)

[lldp holdtime-multiplier](#)

[lldp management-address vlan](#)

[lldp management-ip-address](#)

[lldp management-ipv6-address](#)

- lldp med
- lldp med location
- lldp receive
- lldp reinit
- lldp select-tlv
- lldp timer
- lldp tlv-enable
- lldp transmit
- lldp txdelay
- lldp trap enable
- show lldp configuration
- show lldp configuration mgmt
- show lldp local-device
- show lldp neighbor-info
- show lldp neighbor-info detail
- show lldp neighbor-info mgmt
- show lldp neighbor-info json
- show lldp statistics
- show lldp statistics mgmt
- show lldp tlv

clear lldp neighbors

Syntax

```
clear lldp neighbors
```

Description

Clears all LLDP neighbor details.

Examples

Clearing all LLDP neighbor details:

```
switch# clear lldp neighbors
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

clear lldp statistics

Syntax

```
clear lldp statistics
```

Description

Clears all LLDP neighbor statistics.

Examples

Clearing all LLDP neighbor statistics:

```
switch# clear lldp statistics
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

lldp

Syntax

```
lldp
no lldp
```

Description

Enables LLDP support globally on all active interfaces. By default, LLDP is enabled.

The **no** form of this command disables LLDP support globally on all active interfaces. It does not remove any LLDP configuration settings.

Examples

Enabling LLDP:

```
switch(config) # lldp
```

Disabling LLDP:

```
switch(config) # no lldp
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp dot3

Syntax

```
lldp dot3 {poe | macphy}  
no lldp dot3 {poe | macphy}
```

Description

Sets the 802.3 TLVs to be advertised. By default, advertisement of both POE and MAC/PHY TLVs is enabled. Not supported on the OOBM interface.

The **no** form of this command disables advertisement of 802.3 TLVs.

Parameter	Description
poe	Specifies advertisement of power over Ethernet data link classification.
macphy	Specifies advertisement of media access control and physical layer information.

Examples

Enabling advertisement of the POE TLV:

```
switch(config-if) # lldp dot3 poe
```

Disabling advertisement of the POE TLV:

```
switch(config-if) # no lldp dot3 poe
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

lldp dot3 eee

Syntax

```
lldp dot3 eee
no lldp dot3 eee
```

Description

Sets the 802.3 TLVs for Energy-Efficient Ethernet (EEE) to be advertised. By default, advertisement of EEE TLVs is enabled. Not supported on the OOBM interface.

The **no** form of this command disables advertisement of 802.3 TLVs.

Parameter	Description
eee	Specifies advertisement of 802.3 TLVs for EEE.

Examples

Enabling advertisement of the EEE TLVs:

```
switch(config-if) # lldp dot3 eee
```

Disabling advertisement of the EEE TLVs:

```
switch(config-if) # no lldp dot3 eee
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-if	Administrators or local user group members with execution rights for this command.

lldp dot3 mfs

Syntax

```
lldp dot3 mfs
no lldp dot3 mfs
```

Description

Enables the 802.3 TLV list in LLDP to advertise for maximum frame size (MFS). Enabled by default.

The **no** form of this command disables the advertisement of maximum frame size TLVs.

Examples

Enabling advertisement of maximum frame size TLVs:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 mfs
```

Disabling advertisement of maximum frame size TLVs:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp dot3 mfs
```

Command History

Release	Modification
10.11	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp holdtime-multiplier

Syntax

```
lldp holdtime-multiplier <multiplier>
no lldp holdtime-multiplier
```

Description

Sets the holdtime TTL multiplier value that is used to calculate the LLDP Time-to-Live value. Time-to-Live defines the length of time that neighbors consider LLDP information sent by this agent as valid. When Time-to-Live expires, the information is deleted by the neighbor. Time-to-live is calculated by multiplying holdtime by the value of **lldp timer**.

The **no** form of this command sets the holdtime TTL multiplier to its default value of 4.

Parameter	Description
<pre><multiplier></pre>	Specifies the TTL multiplier in the range of 2 to 10. Default: 4.

Formula

TTL = Holdtime-multiplier x lldp timer

where:

TTL = Time-to-Live

Holdtime-multiplier = Multiplying holdtime value

lldp timer = Message transmission interval

Examples

Setting the holdtime to 8 times of the value of lldp timer:

```
switch(config)# lldp holdtime-multiplier 8
```

Setting the holdtime to the default value of 4 times of the value of lldp timer:

```
switch(config)# no lldp holdtime-multiplier
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

lldp management-address vlan

Syntax

```
lldp management-address vlan <VLAN-ID>
no lldp management-address vlan <VLAN-ID>
```

Description

Sets the VLAN whose IPv4 or IPv6 address is advertised as the LLDP management authority.

The **no** form of this command removes the VLAN whose IPv4 or IPv6 address is advertised as the LLDP management authority.

The following is the precedence for the management IP address TLV in the LLDP packet (in order):

- LLDP management-IP-address and management-ipv6-address, if configured.
- LLDP management VLAN's IPv4 and IPv6 address, if configured.
- Loopback IP address from the smallest configured loopback interface identifier.
- Route-only-port IP address (Layer-3 interface) or IP address of the SVI (Layer-2 interface).
- OOBM IP address.
- Base MAC address of the switch.

Parameter	Description
<code><VLAN-ID></code>	Specifies the VLAN ID.

Examples

Setting the management authority for VLAN 10:

```
switch(config)# lldp management-address vlan 10
```

Removing the management authority for VLAN 10:

```
switch(config)# no lldp management-address vlan 10
```

Command History

Release	Modification
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp management-ip-address

Syntax

```
lldp management-ip-address <IPV-ADDR>  
no lldp management-ip-address
```

Description

Defines the IP management address of the switch which is sent in the management address TLV. One IPv4 and one IPv6 management address can be configured.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of the port
- IP address of the management interface
- Base MAC address of the switch

The **no** form of this command removes the IPv4 management address of the switch.

Parameter	Description
<code><IPV4-ADDR></code>	Specifies the management address of the switch as an IPv4 for mat (x.x.x.x), where x is a decimal value from 0 to 255.

Examples

Setting the management address to **10.10.10.2**:

```
switch(config)# lldp management-ip-address 10.10.10.2
```

Removing the management address:

```
switch(config)# no lldp management-ip-address
```

Command History

Release	Modification
10.14	The management - ipv4 - address keyword is deprecated and replaced with management - ip - address .
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

lldp management-ipv6-address

Syntax

```
lldp management-ipv6-address <IPV6-ADDR>
no lldp management-ipv6-address
```

Description

Defines the IPv6 management address of the switch. The management address is encapsulated in the management address TLV.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of the port
- IP address of the management interface
- Base MAC address of the switch

The **no** form of this command removes the IPv6 management address of the switch.

Parameter	Description
<code><IPv6-ADDR></code>	Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:x xxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

Examples

Setting the management address to 2001:db8:85a3::8a2e:370:7334:

```
switch(config)# lldp management-ipv6-address 2001:0db8:85a3::8a2e:0370:7334
```

Removing the management address:

```
switch(config)# no lldp management-ipv6-address
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

lldp med

Syntax

```
lldp med [poe [priority-override] | capability | network-policy]
no med [poe [priority-override] | capability | network-policy]
```

Description

Configures support for the LLDP-MED TLV. LLDP-MED (media endpoint devices) is an extension to LLDP developed by TIA to support interoperability between VoIP endpoint devices and other networking end-devices. The switch only sends the LLDP MED TLV after receiving a MED TLV from and connected endpoint device.

Not supported on the OOBM interface.

The **no** form of this command disables support for the LLDP MED TLV.

Parameter	Description
<code>poe [priority-override]</code>	Specifies advertisement of power over Ethernet data link classification. The priority - override option overrides user - configured port priority for Power over Ethernet. When both lldp dot3 poe and lldp med poe are enabled, the lldp dot3 poe3 setting takes precedence. Default: enabled.
<code>capability</code>	Specifies advertisement of supported LLDP MED TLVs. The capability TLV is always sent with other MED TLVs, therefore it can not be disabled when other MED TLVs are enabled. Default: enabled.
<code>network-policy</code>	Network policy discovery lets endpoints and network devices advertise their VLAN IDs, and IEEE 802.1p (PCP and DSCP) values for voice applications. This TLV is only sent when a voice VLAN policy is present. Default: enabled.

Examples

Enabling advertisement of the network policy TLV:

```
switch(config-if) # lldp med network-policy
```

Disabling advertisement of the network policy TLV:

```
switch(config-if) # no lldp med network-policy
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

lldp med location

Syntax

```
lldp med location {civic-addr elin-addr }  
no med location {civic-addr elin-addr }
```

Description

Configures support for the LLDP-MED TLV. Supports only civic address and emergency location information number (ELIN). Coordinate-based location is not supported.

The **no** form of this command disables support for the LLDP MED TLV.

Parameter	Description
civic-addr	Configures the LLDP MED civic location TLV.
elin-addr	Configures support for the LLDP MED emergency location TLV. This feature is intended for use in ECS applications to support class 3 LLDP - MED VoIP telephones connected to a switch in an MLTS infrastructure. An ELIN is a valid NANP format telephone number assigned to MLTS operators in North America by the appropriate authority. The ELIN is used to route emergency (E911) calls to a PSAP. (Range: 1 - 15 numeric characters)

The **lldp med location civic-addr** command requires a minimum of one type/value pair, but typically includes multiple type/value pairs as needed to configure a complete set of data describing a given location.

- CA-TYPE: This is the first entry in a type/value pair and is a number defining the type of data contained in the second entry in the type/value pair (CA-VALUE.) Some examples of CA-TYPE specifiers include: 3=city 6=street (name) 25=building name (Range: 0 - 255)
- CA-VALUE: This is the second entry in a type/value pair and is an alphanumeric string containing the location information corresponding to the immediately preceding CA-TYPE entry. Strings are delimited by either blank spaces, single quotes (' ... '), or double quotes ("... "). Each string should represent a specific data type in a set of unique type/value pairs comprising the description of a location, and each string must be preceded by a CA-TYPE number identifying the type of data in the string.

The following LLDP-MED TLV values are supported. For details on these value types, refer to **RFC 4776**

- 1: national subdivisions (state, canton, region, province, prefecture)
- 2: county, parish, gun (JP), district (IN)
- 3: city, township, shi (JP)
- 4: city division, borough, city district, ward, chou (JP)
- 5: neighborhood, block
- 6: group of streets below the neighborhood level
- 16: leading street direction N
- 17: trailing street suffix SW
- 18: street suffix or type
- 19: house number
- 20: house number suffix
- 21: landmark or vanity
- 22: location
- 23: name
- 24: postal/zip code
- 25: building (structure)
- 26: unit (apartment, suite)
- 27: floor
- 28: room

- 29: type of place
- 30: postal community name
- 31: post office box
- 32: additional code 13203000003
- 33: seat (desk, cubicle workstation)
- 34: primary road name 35 road section
- 36: branch road name
- 37: sub-branch road name
- 38: street name pre-modifier
- 39: street name post-modifier

Examples

Enabling support for the LLDP MED emergency location TLV:

```
switch(config-if) # lldp med location elin-addr 408-555-1212
```

Disabling support for the LLDP MED emergency location TLV:

```
switch(config-if) # no lldp med location elin-addr 408-555-1212
```

Enabling support for the LLDP MED civic address TLV:

```
switch(config-if) # lldp med location civic-addr US 1 19 123 6 Fake 18 Street
```

Disabling support for the LLDP MED civic address TLV:

```
switch(config-if) # no lldp med location civic-addr US 1 19 123 6 Fake 18  
Street
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

lldp receive

Syntax

```
lldp receive
no lldp receive
```

Description

Enables reception of LLDP information on an interface. By default, LLDP reception is enabled on all active interfaces, including the OOBM interface.

The **no** form of this command disables reception of LLDP information on an interface.

Examples

Enabling LLDP reception on interface **1/1/1**:

```
switch(config) # interface 1/1/1
switch(config-if) # lldp receive
```

Disabling LLDP reception on interface **1/1/1**:

```
switch(config) # interface 1/1/1
switch(config-if) # no lldp receive
```

Enabling LLDP reception on the OOBM interface:

```
switch(config) # interface mgmt
switch(config-if) # lldp receive
```

Disabling LLDP reception on the OOBM interface:

```
switch(config) # interface mgmt
switch(config-if) # no lldp receive
```


Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

lldp reinit

Syntax

```
lldp reinit <TIME>
no lldp reinit
```

Description

Sets the amount of time (in seconds) to wait before performing LLDP initialization on an interface.

The **no** form of this command sets the reinitialization time to its default value of 2 seconds.

Parameter	Description
<TIME>	Specifies the reinitialization time in seconds. Range: 1 to 10. Default: 2 seconds.

Examples

Setting the reinitialization time to 5 seconds:

```
switch(config) # lldp reinit 5
```

Setting the reinitialization time to the default value of 2 seconds:

```
switch(config)# no lldp reinit
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp select-tlv

Syntax

```
lldp select-tlv <TLV-NAME>  
no lldp select-tlv <TLV-NAME>
```

Description

Selects a TLV that the LLDP agent will send and receive. By default, all supported TLVs are sent and received.

The **no** form of this command stops the LLDP agent from sending and receiving a specific TLV.



NOTE

LLDP supports Organization Unique Identifiers (OUIs) with the following Organization-specific TLVs:

- IEEE 802.1 (DOT1) (oui:0x00, 0x80, 0xc2)
- IEEE 802.3 (DOT3) (oui:0x00, 0x12, 0x0f)
- HPE Aruba Networking (oui:0x88, 0x3a, 0x30)

Parameter	Description
<code>select-tlv <TLV-NAME></code>	Specifies the TLV name to send. The following TLV names are supported: • management - address: Selection is based on priority in the following list (for example if first TLV name isn't selected, the next will be, progressing through this list until a selection is made): 1. IPv4 or IPV6 management address. 2. IP address of the lowest configured loopback interface. 3. If layer 3, then the route - only port IP address. If layer 2, the IP address of the SVI. 4. OOBM interface IP address. 5. Base MAC address of the switch. • port - description: Select port - description TLV. • port - vlan - id: Select port - vlan - id TLV. • port - vlan - name: Select port - vlan - name TLV. • system - capabilities: Select system - capabilities TLV. • system - description: Select system - description TLV. • system - name: Select system - name TLV.

Examples

Stopping the LLDP agent from sending the **port-description** TLV:

```
switch(config) # no lldp select-tlv port-description
```

Enabling the LLDP agent to send the **port-description** TLV:

```
switch(config) # lldp select-tlv oui
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp timer

Syntax

```
lldp timer <TIME>
no lldp timer
```

Description

Sets the interval (in seconds) at which local LLDP information is updated and TLVs are sent to neighboring network devices by the LLDP agent. The minimum setting for this timer must be four times the value of **lldp txdelay**.

For example, this is a valid configuration:

- **lldp timer** = 16
- **lldp txdelay** = 4

And, this is an invalid configuration:

- **lldp timer** = 5

- **lldp txdelay** = 2



NOTE

When copying a saved configuration to the running configuration, the value for **lldp timer** is applied before the value of **lldp txdelay**. This can result in a configuration error if the saved configuration has a value of **lldp timer** that is not four times the value of **lldp txdelay** in the running configuration.

For example, if the saved configuration has the settings:

- **lldp timer** = 16
- **lldp txdelay** = 4

And the running configuration has the settings:

- **lldp timer** = 30
- **lldp txdelay** = 7

Then you will see an error indicating that certain configuration settings could not be applied, and you will have to manually adjust the value of **lldp txdelay** in the running configuration.

The **no** form of this command sets the update interval to its default value of 30 seconds.

Parameter	Description
<i><TIME></i>	Specifies the update interval (in seconds). Range: 5 to 32768. Default: 30.

Examples

Setting the update interval to 7 seconds:

```
switch(config)# lldp timer 7
```

Setting the update interval to the default value of 30 seconds:

```
switch(config)# no lldp timer
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp tlv-enable

Syntax

```
lldp tlv-enable basic-tlv management-address-tlv {ip <X.X.X.X>}|{ipv6  
<X:X::X:X>}  
no lldp select-tlv <TLV-NAME>
```

Description

A **TLV** is an information element that contains the type, length, and value fields. This command configures the IPv4/IPv6 address to be advertised in the management address TLV. This configured value will take the highest precedence in the selection logic the switch uses to identify the IP address to be advertised in the management address TLV.

Parameter	Description
<code>ip <X.X.X.X></code>	Specify an IP address in IPv4 format (X.X.X.X), where X is a decimal number from 0 to 255.
<code>ipv6 <X:X::X:X></code>	Specify an IP address in IPv6 format (<code>X:X::X:X</code>), where X is a hexadecimal number from 0 to F .

Examples

Configuring an IP address in IPv4 format:

```
switch(config)# interface 1/3/1
switch(config-if)# lldp tlv-enable basic-tlv management-address-tlv ip
10.0.0.1
```

Configuring an IP address in IPv6 format:

```
switch(config)# interface 1/3/1
switch(config-if)# lldp tlv-enable basic-tlv management-address-tlv ipv6
2001:db8:85a3::8a2e:370:7334
```

Command History

Release	Modification
10.14.1000	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	config-if	Administrators or local user group members with execution rights for this command.

lldp transmit

Syntax

```
lldp transmit
no lldp transmit
```

Description

Enables transmission of LLDP information on specific interface. By default, LLDP transmission is enabled on all active interfaces, including the OOBM interface.

The **no** form of this command disables transmission of LLDP information on an interface.

Examples

Enabling LLDP transmission on interface **1/1/1**:

```
switch(config) # interface 1/1/1
switch(config-if) # lldp transmit
```

Disabling LLDP transmission on interface **1/1/1**:

```
switch(config) # interface 1/1/1
switch(config-if) # no lldp transmit
```

Enabling LLDP transmission on the OOBM interface:

```
switch(config) # interface mgmt
switch(config-if) # lldp transmit
```

Disabling LLDP transmission on the OOBM interface:

```
switch(config) # interface mgmt
switch(config-if) # no lldp transmit
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if	Administrators or local user group members with execution rights for this command.

lldp txdelay

Syntax

```
lldp txdelay <TIME>
no lldp txdelay
```


Description

Sets the amount of time (in seconds) to wait before sending LLDP information from any interface. The maximum value for **txdelay** is 25% of the value of **lldp tx timer**.

The **no** form of this command sets the delay time to its default value of 2 seconds.

Parameter	Description
<code><TIME></code>	Specifies the delay time in seconds. Range: 0 to 10. Default: 2.

Examples

Setting the delay time to 8 seconds:

```
switch(config)# lldp txdelay 8
```

Setting the delay time to the default value of 2 seconds:

```
switch(config)# no lldp txdelay
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

lldp trap enable

Syntax

```
lldp trap enable
no lldp trap enable
```

Description

Enables sending SNMP traps for LLDP related events from a particular interface. LLDP trap generation is enabled by default on all the interfaces and has to be disabled for interfaces on which traps are not required to be generated.

The **no** form of this command disables the LLDP trap generation.



NOTE

LLDP trap generation is disabled by default at the global level and must be enabled before any LLDP traps are sent.

Examples

Enabling LLDP trap generation on global level:

```
switch(config) # lldp trap enable
```

Enabling LLDP trap generation on interface level:

```
switch(config-if) # lldp trap enable
```

Disabling LLDP trap generation on global level:

```
switch(config) # no lldp trap enable
```

Disabling LLDP trap generation on interface level:

```
switch(config-if) # no lldp trap enable
```

Displaying LLDP global configuration:

```
switch# show lldp configuration
LLDP Global Configuration
=====
LLDP Enabled                : No
LLDP Transmit Interval      : 30
LLDP Hold Time Multiplier   : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval  : 2
LLDP Trap Enabled           : No
TLVs Advertised
=====
```

Management Address

Port Description

Port VLAN-ID

System Description

System Name

LLDP Port Configuration

=====

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED

1/1/1	Yes	Yes	Yes
1/1/2	Yes	Yes	Yes
1/1/3	Yes	Yes	Yes
1/1/4	Yes	Yes	Yes
1/1/5	Yes	Yes	Yes
1/1/6	Yes	Yes	Yes
.....			
.....			
mgmt	Yes	Yes	Yes

Displaying LLDP Configuration for the interface:

```
switch# show lldp configuration 1/1/1
```

LLDP Global Configuration

=====

LLDP Enabled : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled : No

LLDP Port Configuration

=====

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED

1/1/1	Yes	Yes	Yes

Displaying LLDP Configuration for the management interface:

```
switch# show lldp configuration mgmt
```

LLDP Global Configuration

=====

LLDP Enabled : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled : Yes

LLDP Port Configuration

=====

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED
------	------------	------------	-------------------

mgmt	Yes	Yes	Yes
------	-----	-----	-----

Command History

Release

Modification

10.07 or earlier

- -

Command Information

Platforms

Command context

Authority

All platforms	<code>config</code> and <code>config -if</code>	Administrators or local user group members with execution rights for this command.
---------------	---	--

show lldp configuration

Syntax

```
show lldp configuration [<INTERFACE-ID>]
```

Description

Shows LLDP configuration settings for all interfaces or a specific interface.

Parameter

Description

<INTERFACE-ID>

Specifies an interface. Format: **member/slot/port**.

Example

Showing configuration settings for all interfaces:

```
switch# show lldp configuration
```

```
LLDP Global Configuration
```

```
=====
```

```
LLDP Enabled           : No
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : No
TLVs Advertised
```

```
=====
```

```
Management Address
```

```
Port Description
```

```
Port VLAN-ID
```

```
System Description
```

```
System Name
```

```
LLDP Port Configuration
```

```
=====
```

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED
------	------------	------------	-------------------

1/1/1	Yes	Yes	Yes
1/1/2	Yes	Yes	Yes
1/1/3	Yes	Yes	Yes
1/1/4	Yes	Yes	Yes
1/1/5	Yes	Yes	Yes
1/1/6	Yes	Yes	Yes

```
.....
```

```
.....
```

mgmt	Yes	Yes	Yes
------	-----	-----	-----

This example shows configuration settings for interface **1/1/1**.

```
switch# show lldp configuration 1/1/1
```

```
LLDP Global Configuration
```

```
=====
```

```
LLDP Enabled           : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled      : No
```

```
LLDP Port Configuration
```

```
=====
```

```
Auto Flush On Link Down : Yes
```

Management IPv4 Address	:	10.1.1.1	
Management IPv6 Address	:	3001:1::1:1	
Med Priority Override	:	Yes	
Med Location Civic-addr	:	US 1 4 ret 6 tyu 7 tiyuo	
Med Location Elin-addr	:	gher	
PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED

1/1/1	Yes	Yes	Yes

Command History

Release	Modification
10.14.1000	When displaying the LLDP configuration for a specific interface, the output of this command displays any configured management IP addresses.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp configuration mgmt

Syntax

```
show lldp configuration mgmt
```

Description

Shows LLDP configuration settings for the OOBM interface.

Example

Showing configuration settings for all interfaces:

```

switch# show lldp configuration mgmt
LLDP Global Configuration
=====
LLDP Enabled                : Yes
LLDP Transmit Interval     : 30
LLDP Hold Time Multiplier  : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval : 2
LLDP Trap Enabled          : Yes
LLDP Port Configuration
=====
PORT          TX-ENABLED      RX-ENABLED      INTF-TRAP-ENABLED
-----
mgmt          Yes             Yes             Yes

```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
5420	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.
6200		

show lldp local-device

Syntax

```
show lldp local-device
```

Description

Shows global LLDP information advertised by the switch, as well as port-based data. If VLANs are configured on any active interfaces, the VLAN ID is only shown for trunk native or untagged VLAN IDs on access interfaces.

Example

Showing global LLDP information only (all ports including OOBM port are administratively down):

```
switch# show lldp local-device
Global Data
=====
Chassis-ID           : 1c:98:ec:e3:45:00
System Name          : switch
System Description    : HPE ANW JL375A 8400X XL.01.01.0001
Management Address    : 192.168.10.1
Capabilities Available : Bridge, Router
Capabilities Enabled   : Bridge, Router
TTL                   : 120
switch# show lldp local-device
Global Data
=====
Chassis-ID           : 88:3a:30:47:c1:c0
System Name          : 6100
System Description    : HPE ANW JL679A PL.10.06.0001-346-g56a12a8f4cf15
Management Address    : 88:3a:30:47:c1:c0
Capabilities Available : Bridge, Router
Capabilities Enabled   : Bridge, Router
TTL                   : 120
```

Showing all ports except **1/1/11** and OOBM as administratively down:

```
switch# show lldp local-device
Global Data
=====
Chassis-ID           : 1c:98:ec:e3:45:00
System Name          : switch
System Description    : HPE ANW
Management Address    : 192.168.10.1
Capabilities Available : Bridge, Router
Capabilities Enabled   : Bridge, Router
TTL                   : 120
Port Based Data
=====
Port-ID              : 1/1/11
Port-Desc            : "1/1/11"
Port Mgmt-Address    : 164.254.21.220
Port VLAN ID         : 1
Port-ID              : mgmt
Port-Desc            : "mgmt"
Port Mgmt-Address    : 164.254.21.220
```



```

switch# show lldp local-device
Global Data
=====
Chassis-ID          : 88:3a:30:47:c1:c0
System Name         : 6100
System Description   : HPE ANW JL679A  PL.10.06.0001-346-g56a12a8f4cf15
Management Address   : 11.11.11.11
Capabilities Available : Bridge, Router
Capabilities Enabled  : Bridge, Router
TTL                  : 120
Port Based Data
=====
Port-ID             : 1/1/11
Port-Desc           : "1/1/11"
Port Mgmt-Address   : 11.11.11.11
Port VLAN ID        : 1
Parent Interface    : interface 1/1/11
switch#
switch# show lldp local-device
Global Data
=====
Chassis-ID          : 88:3a:30:47:c1:c0
System Name         : 4100i
System Description   : HPE ANW JL817A  RL.10.08.0001
Management Address   : 11.11.11.11
Capabilities Available : Bridge, Router
Capabilities Enabled  : Bridge, Router
TTL                  : 120
Port Based Data
=====
Port-ID             : 1/1/11
Port-Desc           : "1/1/11"
Port Mgmt-Address   : 11.11.11.11
Port VLAN ID        : 1
Parent Interface    : interface 1/1/11
switch#

```

In this example, all the ports except **1/1/11** are administratively down, and VLAN ID 100 is configured on this access interface.

```

switch# show lldp local-device
Global Data
=====
Chassis-ID          : 1c:98:ec:e3:45:00
System Name         : switch

```

```
System Description      : HPE ANW
Management Address      : 192.168.10.1
Capabilities Available   : Bridge, Router
Capabilities Enabled     : Bridge, Router
TTL                     : 120
Port Based Data
=====
Port-ID                 : 1/1/11
Port-Desc                : "1/1/11"
Port VLAN ID            : 100
Parent Interface        : interface 1/1/11
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor-info

Syntax

```
show lldp neighbor-info [<INTERFACE-NAME>]
```

Description

Displays information about neighboring devices for all interfaces or for a specific interface. The information displayed varies depending on the type of neighbor connected and the type of TLVs sent by the neighbor.

Parameter	Description
<code><INTERFACE-NAME></code>	Specifies the interface for which to show information for neighboring devices. Use the format member/slot/port (for example, 1/3/1).

Examples

Showing LLDP information for all interfaces:

```
switch# show lldp neighbor-info
LLDP Neighbor Information
=====
Total Neighbor Entries           : 3
Total Neighbor Entries Deleted   : 0
Total Neighbor Entries Dropped   : 0
Total Neighbor Entries Aged-Out  : 0
LOCAL-PORT  CHASSIS-ID          PORT-ID      PORT-DESC     TTL      SYS-NAME
-----
----
1/1/1       70:72:cf:a4:7d:50    1/1/1        1/1/1         32       switch
1/1/2       48:0f:cf:af:73:80    1/1/2        1/1/2         120      switch
1/1/46      48:0f:cf:af:73:80    1/1/46       1/1/46        120      switch
mgmt        48:0f:cf:af:73:80    mgmt         mgmt          120      switch
switch# show lldp neighbor-info
LLDP Neighbor Information
=====
Total Neighbor Entries           : 1
Total Neighbor Entries Deleted   : 5
Total Neighbor Entries Dropped   : 0
Total Neighbor Entries Aged-Out  : 3
LOCAL-PORT  CHASSIS-ID          PORT-ID      PORT-DESC     TTL      SYS-NAME
-----
----
1/1/2       38:21:c7:5c:df:40    1/1/2        1/1/2         120      switch
1/1/3       f8:60:f0:c9:e0:a0    1/1/3        1/1/3         120      switch
```

Showing information for interface **1/3/1** when it has only one switch connected as a neighbor:

```
switch# show lldp neighbor-info 1/3/1
Port                : 1/1/1
Neighbor Entries     : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
```

```

Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name         : HP-3800-24G-PoEP-2XG
Neighbor Chassis-Description   : HP J9587A 3800-24G-PoE+-2XG Switch,
revision...
Neighbor Chassis-ID           : 10:60:4b:39:3e:80
Neighbor Management-Address    : 192.168.1.1
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge
Neighbor Port-ID              : 1/1/1
Neighbor Port-Desc            : 1/1/1
Neighbor Port VLAN ID         : 1
Neighbor Port VLAN Name       : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                           : 120
Switch2# show lldp neighbor-info 1/1/3
Port                          : 1/1/3
Neighbor Entries              : 1
Neighbor Entries Deleted      : 1
Neighbor Entries Dropped      : 0
Neighbor Entries Aged-Out     : 1
Neighbor Chassis-Name         : 6100
Neighbor Chassis-Description   : HPE ANW JL679A PL.10.15.0001
Neighbor Chassis-ID           : 88:3a:30:47:d1:c0
Neighbor Management-Address    : 88:3a:30:47:d1:c0
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID              : 1/1/3
Neighbor Port-Desc            : 1/1/3
Neighbor Port VLAN ID         : 1
Neighbor Port VLAN Name       : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                           : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : True
Neighbor Auto-neg Enabled      : True
Neighbor Auto-neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASE_TFD
Neighbor MAU Type              : 1000 BASE_TFD

```

Showing information for interface **1/3/10** when the neighbor sends a DOT3 power TLV:

```

switch# show lldp neighbor-info 1/3/10
Port                          : 1/3/10
Neighbor Entries              : 1
Neighbor Entries Deleted      : 0
Neighbor Entries Dropped      : 0

```

```

Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name         : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description  : ArubaOS (MODEL: 325), Version HPE_ANW IAP
Neighbor Chassis-ID           : 84:d4:7e:ce:5d:68
Neighbor Management-Address   : 169.254.41.250
Chassis Capabilities Available : Bridge, WLAN
Chassis Capabilities Enabled   : WLAN
Neighbor Port-ID              : 84:d4:7e:ce:5d:68
Neighbor Port-Desc            : eth0
TTL                           : 120
Neighbor Port VLAN ID         : 1
Neighbor Port VLAN Name       : DEFAULT_VLAN_1
Neighbor Port MFS             : 1500
Neighbor PoE information      : DOT3
Neighbor Power Type           : TYPE2 PD
Neighbor Power Priority        : Unkown
Neighbor Power Source         : Primary
PD Requested Power Value      : 25.0 W
PSE Allocated Power Value: 25.0 W
Neighbor Power Supported      : Yes
Neighbor Power Enabled        : Yes
Neighbor Power Class          : 5
Neighbor Power Paircontrol    : No
PSE Power Pairs               : Signal

```

Showing information for interface **1/1/1** when it has multiple neighbors (displays a maximum of four):

```

switch# show lldp neighbor-info 1/1/1
Port                               : 1/1/1
Neighbor Entries                   : 4
Neighbor Entries Deleted           : 0
Neighbor Entries Dropped           : 0
Neighbor Entries Aged-Out          : 0
Neighbor Chassis-Name              : switch
Neighbor Chassis-Description       : hpe_anw JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID                : 1c:98:ec:fe:25:00
Neighbor Management-Address        : 10.1.1.2
Chassis Capabilities Available     : Bridge, Router
Chassis Capabilities Enabled       : Bridge, Router
Neighbor Port-ID                   : 1/1/1
Neighbor Port-Desc                 : 1/1/1
Neighbor Port VLAN ID              : 1
Neighbor Port VLAN Name            : DEFAULT_VLAN_1
Neighbor Port MFS                   : 1500
TTL                                : 120

```

```

Neighbor Chassis-Name      : switch
Neighbor Chassis-Description : hpe_anw JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID        : 1c:98:ec:fe:25:01
Neighbor Management-Address : 10.1.1.3
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled  : Bridge, Router
Neighbor Port-ID           : 1/1/1
Neighbor Port-Desc         : 1/1/1
Neighbor Port VLAN ID      : 1
Neighbor Port VLAN Name    : DEFAULT_VLAN_1
Neighbor Port MFS          : 1500
TTL                        : 120
Neighbor Chassis-Name      : switch
Neighbor Chassis-Description : HPE ANW JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID        : 1c:98:ec:fe:25:02
Neighbor Management-Address : 10.1.1.4
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled  : Bridge, Router
Neighbor Port-ID           : 1/1/1
Neighbor Port-Desc         : 1/1/1
Neighbor Port VLAN ID      : 50
Neighbor Port VLAN Name    : VLAN_50
Neighbor Port MFS          : 1500
TTL                        : 120
Neighbor Chassis-Name      : switch
Neighbor Chassis-Description : hpe_anw JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID        : 1c:98:ec:fe:25:03
Neighbor Management-Address : 10.1.1.5
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled  : Bridge, Router
Neighbor Port-ID           : 1/1/1
Neighbor Port-Desc         : 1/1/1
Neighbor Port VLAN ID      : 100
Neighbor Port VLAN Name    : VLAN_100
Neighbor Port MFS          : 1500
TTL                        : 120

```

Showing neighbor information for interface 1/3/2 when it has EEE enabled and successfully auto-negotiated:

```

switch# show lldp neighbor-info 1/3/2
Port                : 1/3/2
Neighbor Entries     : 1
Neighbor Entries Deleted : 1
Neighbor Entries Dropped : 0

```

```

Neighbor Entries Aged-Out      : 1
Neighbor Chassis-Name         : BLDG01-F1-6300
Neighbor Chassis-Description   : hpe_anw JL668A  FL.10.15.0001
Neighbor Chassis-ID           : 88:3a:30:92:a5:c0
Neighbor Management-Address    : 10.6.9.15
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID              : 1/1/1
Neighbor Port-Desc            : 1/1/1
Neighbor Port VLAN ID         : 1
Neighbor Port VLAN Name       : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                           : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled     : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASE_TFD
Neighbor MAU type              : 1000 BASE_TFD
Neighbor EEE information       : DOT3
Neighbor TX Wake time          : 17 us
Neighbor RX Wake time          : 17 us
Neighbor Fallback time         : 17 us
Neighbor TX Echo time          : 17 us
Neighbor RX Echo time          : 17 us

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor-info detail

Syntax

```
show lldp neighbor-info detail
```

Description

Shows detailed LLDP neighbor information for all LLDP neighbor connected interfaces.

Examples

Showing detailed LLDP information for all interfaces:

```
switch# show lldp neighbor-info detail
LLDP Neighbor Information
=====
Total Neighbor Entries           : 6
Total Neighbor Entries Deleted   : 2
Total Neighbor Entries Dropped   : 0
Total Neighbor Entries Aged-Out  : 2
-----
----
Port                             : 1/1/1
Neighbor Entries                 : 1
Neighbor Entries Deleted         : 0
Neighbor Entries Dropped         : 0
Neighbor Entries Aged-Out        : 0
Neighbor Chassis-Name            : 6300
Neighbor Chassis-Description     : hpe_anw ...
Neighbor Chassis-ID              : 38:11:17:1a:d5:00
Neighbor Management-Address      : 38:11:17:1a:d5:00
Chassis Capabilities Available   : Bridge, Router
Chassis Capabilities Enabled     : Bridge, Router
Neighbor Port-ID                 : 1/1/4
Neighbor Port-Desc               : 1/1/4
Neighbor Port VLAN ID           : 1
Neighbor Port VLAN Name          : DEFAULT_VLAN_1
Neighbor Port MFS                 : 1500
TTL                              : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported      : true
Neighbor Auto-Neg Enabled       : true
Neighbor Auto-Neg Advertised    : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type                : 1000 BASETFD
```



```

-----
----
Port                               : 1/1/2
Neighbor Entries                   : 1
Neighbor Entries Deleted           : 0
Neighbor Entries Dropped           : 0
Neighbor Entries Aged-Out          : 0
Neighbor Chassis-Name              : 6300
Neighbor Chassis-Description       : hpe_anw...
Neighbor Chassis-ID                : 38:11:17:1a:d5:00
Neighbor Management-Address        : 38:11:17:1a:d5:00
Chassis Capabilities Available     : Bridge, Router
Chassis Capabilities Enabled       : Bridge, Router
Neighbor Port-ID                   : 1/1/5
Neighbor Port-Desc                 : 1/1/5
Neighbor Port VLAN ID              : 1
Neighbor Port VLAN Name            : DEFAULT_VLAN_1
Neighbor Port MFS                  : 1500
TTL                                : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported        : true
Neighbor Auto-Neg Enabled          : true
Neighbor Auto-Neg Advertised       : 1000 BASE_TFD, 100 BASE_T4, 10 BASE_TFD
Neighbor MAU type                  : 1000 BASE_TFD
-----
----
Port                               : 1/1/3
Neighbor Entries                   : 1
Neighbor Entries Deleted           : 0
Neighbor Entries Dropped           : 0
Neighbor Entries Aged-Out          : 0
Neighbor Chassis-Name              : 6300
Neighbor Chassis-Description       : hpe_anw ...
Neighbor Chassis-ID                : 38:11:17:1a:d5:00
Neighbor Management-Address        : 38:11:17:1a:d5:00
Chassis Capabilities Available     : Bridge, Router
Chassis Capabilities Enabled       : Bridge, Router
Neighbor Port-ID                   : 1/1/6
Neighbor Port-Desc                 : 1/1/6
Neighbor Port VLAN ID              : 1
Neighbor Port VLAN Name            : DEFAULT_VLAN_1
Neighbor Port MFS                  : 1500
TTL                                : 120

```

```

Neighbor Mac-Phy details
Neighbor Auto-neg Supported      : true
Neighbor Auto-Neg Enabled       : true
Neighbor Auto-Neg Advertised    : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type                : 1000 BASETFD

```

```

-----
----

```

```

Port                          : 1/1/46
Neighbor Entries              : 1
Neighbor Entries Deleted      : 0
Neighbor Entries Dropped      : 0
Neighbor Entries Aged-Out     : 0
Neighbor Chassis-Name         : 6300
Neighbor Chassis-Description  : hpe_anw ...
Neighbor Chassis-ID           : 38:11:17:1a:d5:00
Neighbor Management-Address   : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled  : Bridge, Router
Neighbor Port-ID              : 1/1/19
Neighbor Port-Desc            : 1/1/19
Neighbor Port VLAN ID         : 1
Neighbor Port VLAN Name       : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                           : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled     : true
Neighbor Auto-Neg Advertised  : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type              : 1000 BASETFD

```

```

-----
----

```

```

Port                          : 1/1/47
Neighbor Entries              : 1
Neighbor Entries Deleted      : 0
Neighbor Entries Dropped      : 0
Neighbor Entries Aged-Out     : 0
Neighbor Chassis-Name         : 6300
Neighbor Chassis-Description  : hpe_anw ...
Neighbor Chassis-ID           : 38:11:17:1a:d5:00
Neighbor Management-Address   : 38:11:17:1a:d5:00
Chassis Cap

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor-info mgmt

Syntax

```
show lldp neighbor-info mgmt
```

Description

Displays information about neighboring devices connected to the OOBM interface.

Examples

Showing LLDP information for the OOBM interface:

```
switch# show lldp neighbor-info mgmt
Port                               : mgmt
Neighbor Entries                   : 1
Neighbor Entries Deleted           : 0
Neighbor Entries Dropped           : 0
Neighbor Entries Aged-Out          : 0
Neighbor Chassis-Name              : HP-3800-24G-PoEP-2XG
Neighbor Chassis-Description       : HP J9587A 3800-24G-PoE+-2XG Switch,
revision...
Neighbor Chassis-ID                : 10:60:4b:39:3e:80
Neighbor Management-Address        : 192.168.1.1
Chassis Capabilities Available     : Bridge, Router
Chassis Capabilities Enabled       : Bridge
Neighbor Port-ID                   : mgmt
```

```
Neighbor Port-Desc      : mgmt
TTL                     : 120
```

Showing LLDP information for the OOBM interface when there are four neighbors:

```
switch# show lldp neighbor-info mgmt
Port                     : mgmt
Neighbor Entries        : 4
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name    : switch
Neighbor Chassis-Description : HPE ANW JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID      : 1c:98:ec:fe:25:00
Neighbor Management-Address : 10.1.1.2
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled  : Bridge, Router
Neighbor Port-ID         : 1/1/1
Neighbor Port-Desc       : 1/1/1
TTL                      : 120
Neighbor Chassis-Name    : switch
Neighbor Chassis-Description : HPE ANW JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID      : 1c:98:ec:fe:25:01
Neighbor Management-Address : 10.1.1.3
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled  : Bridge, Router
Neighbor Port-ID         : 1/1/1
Neighbor Port-Desc       : 1/1/1
TTL                      : 120
Neighbor Chassis-Name    : switch
Neighbor Chassis-Description : HPE ANW JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID      : 1c:98:ec:fe:25:02
Neighbor Management-Address : 10.1.1.4
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled  : Bridge, Router
Neighbor Port-ID         : 1/1/1
Neighbor Port-Desc       : 1/1/1
TTL                      : 120
Neighbor Chassis-Name    : switch
Neighbor Chassis-Description : HPE ANW JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID      : 1c:98:ec:fe:25:03
Neighbor Management-Address : 10.1.1.5
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled  : Bridge, Router
Neighbor Port-ID         : 1/1/1
```

Neighbor Port-Desc	: 1/1/1
TTL	: 120

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor-info json

Syntax

```
show lldp neighbor-info json
```

Description

Displays LLDP neighbor information in structured JSON format. This command provides the same information as the standard `show lldp neighbor-info` command, but it prints the output in JSON format.

Example

```
switch# show lldp neighbor-info json
{
  "total neighbor entries": 3,
  "total neighbor entries deleted": 0,
  "total neighbor entries dropped": 0,
  "total neighbor entries aged out": 0,
  "neighbors": [
    {
      "local port": "1/1/25",
      "chassis id": "90:20:c2:1c:b5:40",
```

```

        "port id": "1/1/26",
        "port description": "1/1/26",
        "ttl": "120",
        "sys name": "switch"
    },
    {
        "local port": "1/1/26",
        "chassis id": "90:20:c2:1c:b5:40",
        "port id": "1/1/25",
        "port description": "1/1/25",
        "ttl": "120",
        "sys name": "switch"
    },
    {
        "local port": "mgmt",
        "chassis id": "d4:e0:53:99:78:80",
        "port id": "25",
        "port description": "25",
        "ttl": "120",
        "sys name": "Uplinks-213"
    }
]
}

```

Command History

Release	Modification
10.18 or later	- -

show lldp statistics

Syntax

```
show lldp statistics [<INTERFACE-ID>]
```

Description

Shows global LLDP statistics or statistics for a specific interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port .

Example

Showing global statistics for all interfaces:

```
switch# show lldp statistics
LLDP Global Statistics
=====
Total Packets Transmitted      : 19
Total Packets Received        : 19
Total Packets Received And Discarded : 0
Total TLVs Unrecognized       : 0
LLDP Port Statistics
=====
```

PORT-ID	TX-PACKETS	RX-PACKETS	RX-DISCARDED	TLVS-UNKNOWN
1/1/1	7	7	0	0
1/1/2	7	7	0	0
1/1/3	0	0	0	0
1/1/4	0	0	0	0
1/1/5	0	0	0	0
...				
mgmt	5	5	0	0
...				

```
switch# show copp-policy statistics
switch# show lldp statistics
LLDP Global Statistics
=====
Total Packets Transmitted      : 25
Total Packets Received        : 20
Total Packets Received And Discarded : 0
Total TLVs Unrecognized       : 0
LLDP Port Statistics
=====
```

PORT-ID	TX-PACKETS	RX-PACKETS	RX-DISCARDED	TLVS-UNKNOWN
1/1/1	25	20	0	0
1/1/2	0	0	0	0

1/1/3	0	0	0	0
1/1/4	0	0	0	0
1/1/5	0	0	0	0
1/1/6	0	0	0	0
1/1/7	0	0	0	0
1/1/8	0	0	0	0
1/1/9	0	0	0	0
1/1/10	0	0	0	0
1/1/11	0	0	0	0
1/1/12	0	0	0	0
1/1/13	0	0	0	0
1/1/14	0	0	0	0
1/1/15	0	0	0	0
1/1/16	0	0	0	0
1/1/17	0	0	0	0
1/1/18	0	0	0	0
1/1/19	0	0	0	0
1/1/20	0	0	0	0
1/1/21	0	0	0	0
1/1/22	0	0	0	0
1/1/23	0	0	0	0
1/1/24	0	0	0	0
1/1/25	0	0	0	0
1/1/26	0	0	0	0
1/1/27	0	0	0	0
1/1/28	0	0	0	0

Showing statistics for interface **1/1/1**:

```
switch# show lldp statistics 1/1/1
LLDP Statistics
=====
Port Name                : 1/1/1
Packets Transmitted      : 159
Packets Received         : 163
Packets Received And Discarded : 0
Packets Received And Unrecognized : 0
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp statistics mgmt

Syntax

```
show lldp statistics mgmt
```

Description

Shows LLDP statistics for the OOBM interface.

Example

Showing LLDP statistics for the OOBM interface:

```
switch# show lldp statistics mgmt
LLDP Statistics
=====
Port Name                : mgmt
Packets Transmitted      : 20
Packets Received         : 23
Packets Received And Discarded : 0
Packets Received And Unrecognized : 0
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp tlv

Syntax

```
show lldp tlv
```

Description

Shows the LLDP TLVs that are configured for send and receive.

Example

```
switch# show lldp tlv
TLVs Advertised
=====
Management Address
Port Description
Port VLAN-ID
System Capabilities
System Description
System Name
VLAN Name
MFS
OUI
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Cisco Discovery Protocol (CDP)

Cisco Discovery Protocol (CDP) is a proprietary layer 2 protocol supported by most Cisco devices. It is used to exchange information, such as software version, device capabilities, and voice VLAN information, between directly connected devices, such as a VoIP phone and a switch.

Subtopics

[CDP support](#)

[CDP commands](#)

CDP support

By default, CDP is enabled on each active switch port. This is a read-only capability, which means the switch can receive and store information about adjacent CDP devices, but does not generate CDP packets (except when communicating with Cisco IP phones.)

The switch supports CDPv2 only and does not support SNMP MIB traps.

When a CDP-enabled port receives a CDP packet from another CDP device, it enters data for that device into the CDP Neighbors table, along with the port number on which the data was received. It does not forward the packet. The switch also periodically purges the table of any entries that have expired. (The holdtime for any data entry in the switch CDP Neighbors table is configured in the device transmitting the CDP packet and cannot be controlled in the switch receiving the packet.) A switch reviews the list of CDP neighbor entries every three seconds and purges any expired entries.

Support for legacy Cisco IP phones

Autoconfiguration of legacy Cisco IP phones for tagged voice VLAN support requires CDPv2.

On initial boot-up, and sometimes periodically, a Cisco phone queries the switch and advertises information about itself using CDPv2. When the switch receives the VoIP VLAN Query TLV (type 0x0f) from the phone, the switch immediately responds with the voice VLAN ID in a reply packet using the VoIP VLAN Reply TLV (type 0x0e). This enables the Cisco phone to boot properly and send traffic on the advertised voice VLAN ID.

The switch CDP packet includes these TLVs:

- CDP Version: 2
- CDP TTL: 180 seconds
- Checksum
- Capabilities (type 0x04): 0x0008 (is a switch)
- Native VLAN: The PVID of the port
- VoIP VLAN Reply (type 0xe): voice VLAN ID (same as advertised by LLDP-MED)
- Trust Bitmap (type 0x12): 0x00
- Untrusted port CoS (type 0x13): 0x00

CDP commands

Select a command from the list in the left navigation menu.

Subtopics

[cdp](#)

[clear cdp counters](#)

[clear cdp neighbor-info](#)

[show cdp](#)

[show cdp neighbor-info](#)

[show cdp traffic](#)

[show cdp voice-vlan mode](#)

cdp

Syntax

```
cdp
```

Description

Configures CDP support globally on all active interfaces or on a specific interface. By default, CDP is enabled on all active interfaces.

When CDP is enabled, the switch adds entries to its CDP Neighbors table for any CDP packets it receives from neighboring CDP devices.

When CDP is disabled, the CDP Neighbors table is cleared and the switch drops all inbound CDP packets without entering the data in the CDP Neighbors table.

The **no** form of this command disables CDP support globally on all active interfaces or on a specific interface.

Examples

Enabling CDP globally:

```
switch(config) # cdp
```

Disabling CDP globally:

```
switch(config) # no cdp
```

Enabling CDP on interface **1/1/1**:

```
switch(config) # interface 1/1/1s
switch(config-if) # cdp
```

Disabling CDP on interface **1/1/1**:

```
switch(config) # interface 1/1/1
switch(config-if) # no cdp
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config config-if	Administrators or local user group members with execution rights for this command.

clear cdp counters

Syntax

```
clear cdp counters
```

Description

Clears CDP counters.

Examples

Clearing CDP counters:

```
switch(config) clear cdp counters
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

clear cdp neighbor-info

Syntax

```
clear cdp neighbor-info
```

Description

Clears CDP neighbor information.

Examples

Clearing CDP neighbor information:

```
switch(config) clear neighbor-info
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show cdp

Syntax

```
show cdp
```

Description

Shows CDP information for all interfaces.

Examples

Showing CDP information:

```
switch(config)# show cdp
CDP Global Information
=====
CDP           : Enabled
CDP Mode      : Rx only
CDP Hold Time : 180 seconds
Port          CDP
-----
1/1/1         Enabled
1/1/2         Enabled
1/1/3         Enabled
1/1/4         Enabled
1/1/5         Enabled
1/1/6         Enabled
1/1/7         Enabled
```

1/1/8	Enabled
1/1/9	Enabled
1/1/10	Enabled

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show cdp neighbor-info

Syntax

```
show cdp neighbor-info <INTERFACE-ID>
```

Description

Shows CDP information for all neighbors or for CDP information on a specific interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port .

Examples

Showing all CDP neighbor information:


```
switch(config)# show cdp neighbor-info
Total Neighbor Entries   : 1
Port                     Device ID                     Platform                     Capability
-----
1/1/1                   HPE_ANW-3810M-24G-1-slot... HPE ANW Sw                     S
```

Showing CDP information for interface **1/1/1**:

```
switch(config)# show cdp neighbor-info 1/1/1
Local Port               : 1/1/1
MAC                      : 70:10:6f:86:78:7f
Device ID                : HPE_ANW-3810M-24G-1-slot(70106f-867800)
Address                  : 127.0.0.1
Platform                 : HPE ANW Sw
Duplex                   : half
Version                  : Revision KB.16.07.0002, ROM KB.16.01....
Capability                : switch
Native VLAN              : 1
Voice VLAN Support       : No
Neighbor Port-ID         : 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show cdp traffic

Syntax

```
show cdp traffic
```

Description

Shows CDP statistics for each interface.

Examples

Showing CDP traffic statistics:

```
switch(config)# show cdp traffic
CDP Statistics
=====
Port          Transmitted Frames    Received Frames        Discarded
Frames
-----
1/1/1          0                      4                      0
1/1/2          0                      0                      0
1/1/3          0                      2                      0
1/1/4          0                      0                      0
1/1/5          0                      0                      0
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show cdp voice-vlan mode

Syntax

```
show cdp voice-vlan mode
```

Description

Shows CDP voice-vlan and mode.

Examples

Showing CDP voice-vlan and mode:

```
switch(config)# show cdp voice-vlan mode
CDP voice VLAN mode
=====
Port          Voice VLAN      Mode
-----
1/1/1         N/A            Rx only
1/1/2         N/A            Rx only
1/1/3         N/A            Rx only
1/1/4         N/A            Rx only
1/1/5         N/A            Rx only
1/1/6         N/A            Rx only
1/1/7         N/A            Rx only
1/1/8         N/A            Rx only
1/1/9         N/A            Rx only
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Zero Touch Provisioning

Zero Touch Provisioning (ZTP) enables the auto-configuration of factory default switches without a network administrator onsite.

When a switch is booted from its factory default configuration, ZTP autoprovisions the switch by automatically downloading and installing a firmware file, a configuration file, or both. With ZTP, even a nontechnical user (for example: a store manager in a retail chain or a teacher in a school) can deploy devices at a site.

ZTP connection priorities

An HPE Aruba Networking CX switch supports ZTP provisioning on an Out-of-Band Management (OOBM) port, a data port (vlan 1), or an image and config file provided via USB.

Priority is given to ZTP options received on the OOBM port. If the switch receives ZTP configuration options on a data port first or from a DHCPv6 server connected to an OOBM port, the switch waits for an additional 30 seconds to see if higher-priority ZTP options will be received on an OOBM port also.

Priority is given to ZTPv4 options over ZTPv6 options, so connection priorities are as follows, from highest to lowest priority:

1. ZTP options received via USB.
2. ZTP options received from a DHCPv4 server on an OOBM port.
3. ZTP options received from a DHCPv4 server on a data port.
4. ZTP options received from a DHCPv6 server on an OOBM port.

Subtopics

ZTP support

Setting up ZTP on a trusted network

ZTP using USB

ZTP process during switch boot

ZTP VSF switchover support

ZTP commands

ZTP support

The switch supports standards-based Zero Touch Provisioning (ZTP) operations as follows:

- ZTP operations are supported over IPv4 and IPv6 connections.
- The switch must be running the factory default configuration.
- The configuration file is a text file or JSON file that becomes the startup and running configuration on the switch after the ZTP operation is complete. The configuration can be in CLI or in JSON format.
- When the switch is started using the factory default configuration, the ZTP operation is started automatically and is active until any running configuration of the switch is modified. There is no CLI command required to start the operation.

- You must configure the DHCP server to provide a standards-based ZTP server solution. Options and features that are specific to Network Management Solution (NMS) tools, such as AirWave, are not supported.
 - Aruba Central on-premise can manage AOS-CX switches on supported models through DHCP ZTP using two approaches:
 - On the DHCP server, configure DHCP option-60 as "ArubaInstantAP" 90 and provide the value in option-43 in the format <group-details>, <aruba-central-on-prem-ip-or-fqdn>, <shared-token>.
 - On the DHCP server, configure DHCP option-60 as HPE vendor VCI and provide the value in option-43 in the tag-length-value (TLV) format. Next, enter the Aruba Central on-premise fully qualified domain name (FQDN) or IPv4 address as the value for sub-option code 146 using the format: <group-details>, <aruba-central-on-prem-ip-or-fqdn>, <shared-token>.

Supported ZTP options from a DHCPv4 server are:

DHCP option	Description
43	Vendor Specific Information
43 suboption 144	Name of the configuration file
43 suboption 145	Name of the firmware image file
43 suboption 146	FQDN or IPv4 address of the Aruba Central on - p remise server
43 suboption 148	FQDN or IPv4 or IPv6 address of the HTTP Proxy
60	Vendor Class Identifier (VCI)
66	IPv4 or IPv6 address of the TFTP or HTTP server (Specifying a host name instead of an IP address is not supported.)
67	Name of the configuration file (Option 43 suboptio n 144 takes precedence over this option.)

Unique ZTP IPv6 options can be sent from the DHCPv6 Server based on MAC address of the switch or based on the DHCP Vendor class Identifier of the switch. Supported ZTP options from a DHCPv6 server are:

DHCP option	Description
59	IPv6 address of the TFTP/HTTP server. The image file name also can be given in this URL or in the VCA option - 17.
 NOTE the boot file URL will support TFTPv6 Server or HTTP Server option with IPv6 address or FQDN name.	
17 suboption 145	Name of the image file.
17 suboption 144	Name of the configuration file.

The switch supports the following standards:

- **RFC 2131**, Dynamic Host Configuration Protocol.
- **RFC 2132**, DHCP Options and BOOTP Vendor Extensions. Support is limited to the options listed in the table "Supported DHCP options for ZTP on AOS-CX."
- **RFC 5970**, DHCPv6 Options for Network Boot. Support is limited to the options listed in the table "Supported DHCP options for ZTP on AOS-CX."

Hewlett Packard Enterprise recommends that you implement ZTP in a secure and private environment. Any public access can compromise the security of the switch, as follows:

- ZTP is enabled only in the factory default configuration of the switch, DHCP snooping is not enabled. The Rogue DHCP server must be manually managed.
- The DHCP offer is in plain data without encryption.

Setting up ZTP on a trusted network

The following procedure is an overview of setting up a Zero Touch Provisioning (ZTP) environment to provision newly installed switches automatically. The procedure is intended for network administrators who are familiar with automatically provisioning switches in a network, and does not provide detailed information about configuring or managing switches.

Procedure

1. For each switch model to be provisioned using ZTP, do the following:

- a. Obtain the switch firmware image file.
 - b. Prepare the switch configuration file. The configuration file becomes the running configuration and the startup configuration on the switch.
2. Set up a TFTP or HTTP server and record its IPv4 or IPv6 address. The address is required when you set up the DHCP server. The switch must be able to reach the TFTP or HTTP server and DHCP server, either on the same subnet, or on a remote subnet via DHCP relay.
Switches support provisioning through a network connected to a data port or through a network connected to the management port.
 3. Publish the configuration files and image files to the TFTP or HTTP server. You need to know the locations of the files and the IPv4/IPv6 address of the TFTP or HTTP server when you set up the vendor class options on the DHCP server.
 4. On the DHCP server, set up vendor classes for each switch model you plan to provision. To do this you need the following information:

- The IPv4/IPv6 address of the TFTP or HTTP server. Using a host name is not supported.
- The path to the switch configuration and firmware image files on the TFTP or HTTP server.
- The vendor class identifier (VCI) for each switch model.

You can obtain the VCI by entering the **show dhcp client vendor-class-identifier** command from a switch CLI command prompt in the manager context. The VCI is the text string in the response that starts with **Aruba**.

For example:

```
switch# show dhcp client vendor-class-identifier
Vendor Class Identifier: Aruba xxxxx xxxx
```

Where x indicates the switch model number.

5. At the installation site, provide the switch installer with a Cat6 network cable connected to the network that includes the DHCP and TFTP or HTTP servers, and information about the switch port to use. The switch installer plugs the cable into the data port you specify.
The ZTP operation begins when power is applied to the switch after the network cable is installed.
6. Assuming the downloaded configuration includes a way to access the CLI of the switch, you can enter the following command to show the options offered by the DHCP server and the status of the ZTP operation:
show ztp information

ZTP using USB

About this task

AOS-CX supports EXT4, FAT-16, and FAT-32 USB file systems for ZTP. To perform zero-touch provisioning via USB, the image and config files should be in one of the following folders on the USB device.



NOTE

The file with extension **.swi** is recognized as an image file and the file with extension **.cfg** is recognized as a configuration file. These file extensions are case sensitive.

A serial-number specific folder:

- AOSCX/CNXXXXXXXXX/ZTP/<image.swi>
- AOSCX/CNXXXXXXXXX/ZTP/<config.cfg>

A vendor-class specific folder:

- AOSCX/PRODUCTNAME_PARTNUM/ZTP/<image.swi>
- AOSCX/PRODUCTNAME_PARTNUM/ZTP/<config.cfg> (for example: AOSCX/8320_JL479A/ZTP)

A generic configuration folder:

- AOSCX/ZTP/<image.swi>
- AOSCX/ZTP/<config.cfg>

If multiple paths are present, then the priority is given in the following order:

Procedure

1. The serial number specific folder with an image file or configuration file.
2. The vendor class specific folder with an image file or configuration file.
3. The generic configuration folder with an image file or configuration file.

Results



NOTE

You can identify the switch product name and part number you should use to create a ZTP folder by issuing the command **show dhcp client vendor-class-identifier**. For example:

```
switch# show dhcp client vendor-class-identifier
Vendor Class Identifier:  Aruba <xxxx> <yyyy>
```

Where **<xxxx>** is the part number and **<yyyy>** is the product name in the output of this command.

If multiple image or configuration files are present in these ZTP folders, the first file that shows up in the output of the **ls -1 <usb_ztp_path>** command will be used for ZTP provisioning

If the switch boots up with a factory default configuration and a USB device is connected to the switch, the switch looks for the presence of ZTP-related folders. The switch then will be provisioned with the image and configuration files provided via USB. If either an image or a configuration file is present in the ZTP related folders on USB, then the switch will be provisioned with the respective file provided from the USB device.

ZTP using USB will not be triggered and the switch will continue with ZTP using DHCP in the following scenarios:

- There is no USB device connected.
- The USB device does not have the ZTP-specific folder.
- The ZTP folder has no image or configuration files in the first 127 files listed in the folder.
- The USB drive containing ZTP specific files is inserted into the device after it has booted up. (A device reboot is required to initiate the ZTP process using the USB.)

To confirm that the ZTP process has been triggered, issue the command **show ztp information**.

Performing an upgrade from a USB device without ZTP

To perform a manual upgrade using an image and configuration file from a USB device, issue the following copy commands to download the image and config file from the USB device directly on to the switch.

```
copy usb:/file <running-config|startup-config|primary|secondary>
```

ZTP process during switch boot

Procedure

1. The switch boots up with the factory default configuration. If the ZTP operation detects that the switch configuration is different from the factory default configuration, the ZTP operation ends. The switch must be configured at the installation site.
2. The switch sends out a DHCP discovery from the management port .

On switches that support ZTP operations on data ports, the switch also sends out a DHCP discovery from all data ports in the default VLAN.

The switch waits to receive DHCP options indefinitely or until the running configuration is modified. If a DHCP IP address is received but no DHCP options are received, the switch waits an additional minute before ending the ZTP operation.

On switches that support ZTP operations on data ports, DHCP options received on the management port have priority over DHCP options received on data ports:

- If DHCP options are received on the management port before being received on a data port, the switch processes those options immediately.
- If DHCP options are received on a data port, the switch waits an additional 30 seconds for options to be received on the management port. If no DHCP options are received on the management port during those 30 seconds, the switch processes the DHCP options it received on the data port.

After the ZTP operation ends, there is no automatic retry. You can either attempt to boot the switch with the factory default configuration again, configure the switch at the installation site, or use the ZTP force-provision CLI to trigger the ZTP process, ignoring the present running configuration of the switch.

- Once force-provision is enabled, new DHCP requests are sent from the switch. Disabling force-provision does not stop the DHCP already in progress, but only changes the switch configuration status of force-provision.
- If ZTP fails while force-provision is enabled, there is no automatic retry. To retry, **ztp force-provision** should be disabled and re-enabled to clear the current ZTP state and send a new DHCP request. When **ztp force-provision** is already enabled on the switch, re-enabling it results in no operation.
- If the DHCP server is configured to provide both ZTP image and configuration options and there is a non-default startup configuration present on the switch, clearing the non-default startup configuration before triggering **ztp force-provision** is recommended. If an image is downloaded via ZTP, the switch reboots once the image download is complete and ZTP force-provision configuration is lost, causing ZTP to enter into a failed state. ZTP force-provision will need to be enabled again to continue the process.

3. The DHCP server responds with an offer containing the following:
 - The IPv4 or IPv6 address of the TFTP or HTTP server

- One or both of the following:
 - The name of the firmware image file
 - The name of the configuration file
 - Aruba Central Location (optional) including the shared token value for the on-premise server
 - HTTP Proxy Location (optional)
4. If a firmware image file is offered, the ZTP operation downloads the image file from the TFTP or HTTP server to the switch. If the current switch image and downloaded firmware image version do not match, then the switch boots with the downloaded image:
- If the image upgrade fails, the switch retains its original firmware image and the ZTP operation ends with a failed status.
 - If the image upgrade succeeds, the ZTP operation is started again after the switch reboots. Because the downloaded image file matches the image file installed on the switch, the ZTP operation continues, and checks if a configuration file is offered.
5. If a configuration file is offered, the ZTP operation downloads the configuration file copies the file to the running-config and then to the startup-config of the switch:
- If the startup configuration update fails, the switch retains its factory-default running configuration and the ZTP operation ends with a failed status.
 - If the copy operation fails, the ZTP operation ends with a failed status.
 - If the copy operation succeeds, the ZTP operation ends successfully.

ZTP VSF switchover support

ZTP status is not synced in the VSF stack. When the VSF stack is formed, configuration changes are applied on the conductor switch, which is then synced to standby switch. When the switchover is performed on the VSF stack, the standby becomes the new conductor switch.

As part of the switchover process, the ZTP daemon starts on the new conductor. The status of the ZTP is failed because there are configuration changes present.

ZTP commands

Select a command from the list in the left navigation menu.

Subtopics

[show ztp information](#)

[ztp force provision](#)

show ztp information

Syntax

```
show ztp information
```

Description

Shows information about Zero Touch Provisioning (ZTP) operations performed on the switch.

Usage

When a switch configured to use ZTP is booted from a factory default configuration, the switch contacts a DHCP server, which offers options for obtaining files used to provision the switch:

- The IPv4/IPv6 address of TFTP Server or the IPv6 address of HTTP server
- The name of the image file
- The name of the configuration file
- The HPE Aruba Networking Central FDQN or IPv4/IPv6 address
- The HTTP proxy FDQN or IPv4/IPv6 address

The **show ztp information** command shows the options offered by the DHCP server and the status of the ZTP operation.

The status of the ZTP operation is one of the following:

Success

The ZTP operation succeeded.

One of the following is true:

- Both the running configuration and the startup configuration were updated.
- The IP address of the TFTP/HTTP server was received, but the offer did not include a configuration file or a firmware image file.
- Any combination of vendor encapsulated DHCP options are received as configured, along with the firmware image and switch configuration file.

- Only vendor encapsulated DHCP options are configured and are received accordingly.

Failed - Custom startup configuration detected	The switch was booted from a configuration that is not the factory default configuration. For example, the administrator password has been set.
Failed - Timed out while waiting to receive ZTP options	Either the switch received the DHCP IPv4 or IPv6 address but no ZTP options were received within 1 minute or ZTP force - provision is triggered and no ZTP options are received within 3 minutes.
Failed - Detected change in running configuration	The running configuration was modified by a user while the ZTP operation was in progress.
Failed - TFTP server unreachable	The TFTP server is not reachable at the specified IP address.
Failed - TFTP server information unavailable	The image file name or config file name is provided without the TFTP server location to fetch the files from and ZTP enters failed state.
Failed - Invalid configuration file received	Either the file transfer of the configuration file failed, or the configuration file is invalid (an error occurred while attempting to apply the configuration).
Failed - Invalid image file received	Either the file transfer of the firmware image file failed, or the firmware image file is invalid (an error occurred while verifying the image).
Failed - HTTP server unreachable	The HTTP server is not reachable at the specified IP address.
Failed - Invalid configuration file from USB	The configuration file is invalid or it is not in the proper format.
Failed - Invalid image file from USB	The image file is invalid or it is not in the proper format.
Failed - Error while copying the configuration from USB	ZTP failed to copy the configuration file from the USB device due to a file permissions, file access or system error (for example, if the switch is out of memory), if the user interrupts the ZTP process, or if the file is deleted from the USB device. This message may also display in the event of an unknown error.
Failed - Error while copying the image from USB	ZTP failed to copy the image file from the USB device due to a file permissions, file access or system error (for example, if the switch is out of memory), if the user interrupts the ZTP process, or if the file is deleted from the USB device.

leted from the USB device. This message may also display in the event of an unknown error.



NOTE

In the case of reconnection, connect with a main or alternative location to the COP instance as a user. The current connection is shown in the **Central location** field.

Scenario 1: If the location the device is currently connected on is updated, the system reconnects in order to connect with the new location.

Scenario 2: If the location in which the device is not currently connected on is updated, the DUT does not go through the reconnection process.

Examples

Showing switch image and config download from an IPv4 TFTP server in progress after receiving ZTP options:

```
switch# show ztp information
TFTP Server                : 10.0.0.2
Image File                 : TL_10_02_0001.swi
Configuration File         : ztp.cfg
ZTP Status                 : In-progress - Image download and
verification
HPE ANW Central Location   : NA
Alternative HPE ANW Central Location : NA
HPE ANW Central Shared Token : NA
Force-Provision            : Enabled
HTTP Proxy Location        : NA
```

Showing switch image download failure from an IPv4 TFTP server because the server was unreachable

```
switch# show ztp information
TFTP Server                : 10.0.0.2
Image File                 : TL_10_02_0001.swi
Configuration File         : ztpv6.cfg
ZTP Status                 : Failed - TFTP server unreachable
HPE ANW Central Location   : NA
Alternative HPE ANW Central Location : NA
HPE ANW Central Shared Token : NA
Force-Provision            : Enabled
HTTP Proxy Location        : NA
```

Showing a successful switch configuration download from an IPv6 HTTP server

```
switch# show ztp information
HTTP Server                : http://[2001::50]
```

```

Image File                : TL_10_15_0001.swi
Configuration File        : config_file
ZTP Status                : Success
HPE ANW Central Location  : NA
Alternative HPE ANW Central Location : NA
HPE ANW Central Shared Token : NA
Force-Provision           : Enabled
HTTP Proxy Location       : NA

```

The switch received image and configuration files from a serial-number specific folder on a USB device, and ZTP was successful.

```

switch# show ztp information
TFTP Server                : NA
Image File                : usb:/AOSCX/TW88KHD005/
GL_10_16_00001.swi
Configuration File        : usb:/AOSCX/TW88KHD005/ztp.cfg
Status                    : Success
HPE ANW Central Location  : NA
Alternative HPE ANW Central Location : NA
HPE ANW Central Shared Token : NA
Force-Provision           : Disabled
HTTP Proxy Location       : NA

```

The switch received image and configuration files from a generic folder on a USB device, and ZTP was successful.

```

switch# show ztp information
TFTP Server                : NA
Image File                : usb:/AOSCX/ztp/ML_10_15_1000F.swi
Configuration File        : usb:/AOSCX/ztp/config.cfg
Status                    : Success
HPE ANW Central Location  : NA
Alternative HPE ANW Central Location : NA
HPE ANW Central Shared Token : NA
Force-Provision           : Disabled
HTTP Proxy Location       : NA

```

The switch received an image file from a vendor-class specific folder on a USB device, and ZTP was successful.

```

switch# show ztp information
TFTP Server                : NA
Image File                : usb:/AOSCX/8325_JL636A/
GL_10_16_1000A.swi
Configuration File        : NA
Status                    : Success
HPE ANW Central Location  : NA
Alternative HPE ANW Central Location : NA

```

HPE ANW Central Shared Token	: NA
Force-Provision	: Disabled
HTTP Proxy Location	: NA

Command History

Release	Modification
10.16	The output of this command can show the status of ZTP over USB.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

ztp force provision

Syntax

```
ztp force-provision
no ztp force-provision
```

Description

Starts on-demand ZTP via DHCP.

Usage

DHCP options received are processed independent of the current state of configuration on the switch. Previous ZTP options are cleared, such as the ZTP TFTP server, image file, configuration file, HPE Aruba Networking Central location, and HTTP proxy location, and the switch sends a DHCP request. ZTP using USB is not triggered by this command.

Examples

In the following example, force-provision is enabled.


```
switch# configure terminal
switch(config)# ztp force-provision
```

In the following example, force-provision status is checked while enabled.

```
switch# show ztp information
TFTP Server           : 10.0.0.2
Image File            : TL_10_02_0001.swi
Configuration File     : ztp.cfg
Status                : Success
HPE ANW Central Location      : NA
HPE ANW Central Shared Token  : NA
Force-Provision         : Enabled
HTTP Proxy Location      : NA
```

In the following example, force-provision is disabled.

```
switch# configure terminal
switch(config)# no ztp force-provision
```

In the following example, force-provision status is checked while disabled.

```
switch# show ztp information
TFTP Server           : 10.0.0.2
Image File            : TL_10_02_0001.swi
Configuration File     : ztp.cfg
Status                : Success
HPE ANW Central Location      : NA
HPE ANW Central Shared Token  : NA
Force-Provision         : Disabled
HTTP Proxy Location      : NA
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

Thermal Management

Thermal management consists of the **fand** and **tempd** daemons, that manage the fans and temperature sensors for the system respectively. They are responsible for monitoring temperatures, adjusting fan speeds to keep the system at nominal temperatures, initiating shutdown sequences for chassis, subsystems, or transceivers during over-temperature conditions, and reporting temperatures, sensor states, fan speeds, and fan states.

Subtopics

- [Thermal management support](#)
- [Transceiver Thermal Management](#)

Thermal management support

	Description	Platforms
Transceiver Thermal Management	When the transceiver thermal management capability is enabled, a temperature sensor will be available for each transceiver that supports thermal monitoring.	
Thermal Shutdown Behavior	Thermal shutdown is a critical safety mechanism designed to prevent hardware damage caused by overheating. When critical temperature thresholds are exceeded, thermal daemons log an event and initiate a shutdown sequence to power down the affected components gracefully.	All platforms
Chassis (Switch) Thermal Shutdown	If a temperature sensor exceeds its critical threshold and the associated subsystem does not support being powered off, the entire chassis will shut down for a five minute cool down period. If the critical temperature is detected again after reboot, the shutdown sequence will repeat.	All platforms

	Description	Platforms
Subsystem (Line Card, Fabric Card)) Thermal Shutdown	If a temperature sensor exceeds its critical threshold and the associated subsystem supports being powered off, that subsystem will shut down for a five minute cool down period. If the critical temperature is detected again after reboot, the shutdown sequence will repeat.	5420 6400 8400
Transceiver Thermal Shutdown	If a transceiver's temperature sensor exceeds its critical threshold and the transceiver is configured to shut down upon exceeding the critical threshold, it will power down for a five - minute cool down period. If the critical temperature is detected again after reboot, the shutdown sequence will repeat.	8325 8325H 8325P 9300 - 32C8D 9300 - 32D

Transceiver Thermal Management

Transceiver temperature sensors are visible on REST API calls, SNMP and CLI (show environment temperature). Events logs also are triggered upon thermal status changes, shutdown and recovery. If a transceiver exceeds the temperature threshold and it is hot swapped, the cool time timer is canceled and the transceiver is re-enabled. There is no limit to the number of times a transceiver is thermally disabled and re-enabled if it's continuously above the critical temperature threshold.

This is an example of the temperature information on the normal conditions.

```
switch# show environment temperature
Temperature information
-----
--
Mbr/Slot-Sensor           Module Type           Current
                        temperature      Status
-----
--
1/1-xcv-5                 line-card-module      66.45 C      normal
...
```

If the temperature raises to 70 degrees Celsius, the status changes to **overtemp**.

```
switch# show environment temperature
Temperature information
```

```

-----
--
Mbr/Slot-Sensor              Module Type      Current
                                temperature  Status
-----
--
1/1-xcv-5                    line-card-module  70.00 C    overtemp
...

```

The status change will generate this event log: **Temperature sensor 1/1-xcv-5 status is overtemp at 70.000000 degC**

If the temperature raises to 82 degrees Celsius, for example, the status changes to **emergency**. The transceiver is disabled and the cool down period begins. The status change and the disabling of the interface reflect in the event logs.

```

Temperature sensor 1/1-xcv-5 status is overtemp at 70.000000 degC
Temperature sensor 1/1-xcv-5 status is critical at 82.000000 degC
Disabling interface 1/1/5 transceiver for 5 minutes - temperature sensor
1/1-xcv-5 status is emergency at 82.000000 degC

```

This is an example of the **show interface** command when the interface is **disabled**:

```

switch# show int 1/1/5
Interface 1/1/5 is down
Admin state is down
State information: Transceiver disabled - Module disabled by system thermal
management
Link state: down for 1 hour (since Wed May 14:22:17:42 UTC 2025)
Link transitions: 0
Description:
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 1500
Type 40G-eSR4 / QSPF+ eSR4
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
L3 Counters: Rx Disabled, Tx Disabled
Rate collection interval: 300 seconds
Rates
-----
Mbits / sec          RX          TX          Total (RX+TX)
-----
KPkts / sec          0.00         0.00         0.00
  Unicast             0.00         0.00         0.00
  Multicast            0.00         0.00         0.00

```

Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00
Statistics	RX	TX	Total

Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
L3 Packets	0	0	0
L3 Bytes	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0
Other	0	0	0

Once the cool down period is over, the transceiver is enabled. Therefore, the status changes to normal and the corresponding event log is triggered:

```
Temperature sensor 1/1-xcv-5 status is overtemp at 70.000000 degC
Temperature sensor 1/1-xcv-5 status is critical at 82.000000 degC
Disabling interface 1/1/5 transceiver for 5 minutes - temperature sensor
1/1-xcv-5 status is emergency at 82.000000 degC
Enabling interface 1/1/5 transceiver - thermal disablement period has
expired
```

If the transceiver is hot swapped, the interface is re-enabled and the event logs are:

```
Temperature sensor 1/1-xcv-5 status is overtemp at 70.000000 degC
Temperature sensor 1/1-xcv-5 status is critical at 82.000000 degC
Disabling interface 1/1/5 transceiver for 5 minutes - temperature sensor
1/1-xcv-5 status is emergency at 82.000000 degC
Enabling interface 1/1/5 transceiver - thermal disablement cancelled due to
hot-swap of transceiver
```



NOTE

The examples above apply to the 9300-32D Switch Series. The output of these commands may vary in other platforms. The temperature thresholds shown here are for illustration purposes only.

Switch system and hardware commands

Subtopics

[Switch system and hardware commands](#)

Switch system and hardware commands

Select a command from the list in the left navigation menu..

Subtopics

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[show usb file-system](#)
[show version](#)
[system resource-utilization poll-interval](#)
[top cpu](#)
[top memory](#)
[usb](#)
[usb mount | unmount](#)

allow-non-failsafe-updates

Syntax

```
allow-non-failsafe-updates <num_minutes>
```

Description

Allow non-failsafe updates of programmable devices for the specified number of minutes, or **enter** 0 to disallow unsafe updates. The maximum permitted value is 120 minutes.

Example

The following example will enable non-failsafe updates of programmable devices for the next 30 minutes.

```
switch(config)# allow-non-failsafe-updates 30  
This command will enable non-failsafe updates of programmable devices for  
the next 30 minutes. You will first need to wait for all line and fabric  
modules to reach the ready state, and then reboot the switch to begin  
applying any needed  
updates. Ensure that the switch will not lose power, be rebooted again, or  
have any
```

```
modules removed until all updates have finished and all line and fabric
modules have
returned to the ready state
WARNING: Interrupting these updates may make the product unusable!
Continue [y/n] ? y
Unsafe updates      : allowed (less than 30 minute(s) remaining)
```

Command History

Release	Modification
10.15.1010	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

bluetooth disable

Syntax

```
bluetooth disable
no bluetooth disable
```

Description

Disables the Bluetooth feature on the switch. The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE). Bluetooth is enabled by default.

The **no** form of this command enables the Bluetooth feature on the switch.

Example

Disabling Bluetooth on the switch. <XXXX> is the switch platform and <NNNNNNNNNN> is the device identifier.

```
switch(config)# bluetooth disable
switch# show bluetooth
Enabled          : No
```



```
Device name      : <XXXX>-<NNNNNNNNNN>
switch(config)# show running-config
...
bluetooth disabled
...
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

bluetooth enable

Syntax

```
bluetooth enable
no bluetooth enable
```

Description

This command enables the Bluetooth feature on the switch. The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE).

Default: Bluetooth is enabled by default.

The **no** form of this command disables the Bluetooth feature on the switch.

Usage

The default configuration of the Bluetooth feature is `enabled`. The output of the `show running-config` command includes Bluetooth information only if the Bluetooth feature is disabled.

The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE).

The Bluetooth feature requires the USB feature to be enabled. If the USB feature has been disabled, you must enable the USB feature before you can enable the Bluetooth feature.

Examples

```
switch(config)# bluetooth enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	zconfig	Administrators or local user group members with execution rights for this command.

clear events

Syntax

```
clear events
```

Description

Clears up event logs. Using the `show events` command will only display the logs generated after the `clear events` command.

Examples

Clearing all generated event logs:

```
switch# show events
-----
show event logs
-----
2018-10-14:06:57:53.534384|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 27
```

```

2018-10-14:06:58:30.805504|lldpd|103|LOG_INFO|MSTR|1|Configured LLDP tx-
timer to 36
2018-10-14:07:01:01.577564|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 49
switch# clear events
switch# show events
-----
show event logs
-----
2018-10-14:07:03:05.637544|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 34

```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

clear ip errors

Syntax

```
clear ip errors
```

Description

Clears all IP error statistics.

Example

Clearing and showing ip errors:

```

switch# clear ip errors
switch# show ip errors
-----

```

Drop reason	Packets

Malformed packets	0
IP address errors	0
...	

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

console baud-rate

Syntax

```
console baud-rate <SPEED>
no console baud-rate <SPEED>
```

Description

Sets the console serial port speed.

The no form of this command resets the console port speed to its default of 115200 bps.

Parameter	Description
<SPEED>	Selects the console port speed in bps, either 9600 or 115200.

Usage

The speed change occurs immediately for the active console session. The console will be inaccessible until the client terminal settings are updated to match the console port speed that you set. After the command is executed you will be prompted to log in again.

Examples

Setting the console port speed to 9600 bps:

```
switch(config)# console baud-rate 9600  
This command will configure the baud rate immediately for the active serial  
console session. After the command is executed the user will be prompted to  
re-login. The serial console will be inaccessible until the terminal client  
settings are updated to match the baud rate of the switch.  
Continue (y/n)? y
```

Resetting the console port to its default speed 115200 bps:

```
switch(config)# no console baud-rate
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

copy support-files top-cpu-processes

Syntax

```
copy support-files top-cpu <REMOTE_URL> {vrf <VRF_NAME>} | <STORAGE_URL> |  
<local-file>
```

Description

Copies CPU high-utilization diagnostic bundles from /tmp/system_utilization_snapshots (files matching cpu_high_*.tgz). These archives are created automatically when the 1-minute average CPU exceeds the upper threshold and are maintained as a rolling set (maximum 5 files).

Table 1.

Parameter	Description
REMOTE_URL	URL to copy support-files. Use one of the following syntax: <pre>sftp://USER@{IP HOST}[:PORT]/ {DIRECTORY}/ tftp://{IP HOST}[:PORT] [;blocksize=VAL]/ scp://USER@{IP HOST}[:PORT]/ {DIRECTORY/}</pre> Required
STORAGE_URL	USB or local SFTP server storage to copy support-files top-cpu. Syntax: <pre>{usb local}:{[DIRECTORY/]}</pre> Optional
VRF_NAME	Specifies the VRF name. Optional Default: default VRF

Examples

Copying CPU high-utilization diagnostic bundles to a remote URL:

```
switch# copy support-files top-cpu sftp://user@10.0.0.12/cpu_diags/ vrf mgmt  
Attempting to copy CPU diag archives...  
Done
```

Copying CPU high-utilization diagnostic bundles to a storage URL:

```
switch# copy support-files top-cpu usb:/cpu_diags/  
Copying CPU diag archives to /mnt/usb/cpu_diags/  
Done
```

Copying CPU high-utilization diagnostic bundles to a local file:

```
switch# copy support-files top-cpu local-file
Preserving CPU diag archives into local-files...
support-files are preserved successfully in the local-files.
```



NOTE

- Archives copied are located under **/tmp/system_utilization_snapshots** and named **cpu_high_*.tgz**.
- Up to 5 archives are retained (oldest overwritten).
- Remote copies support SFTP, SCP, TFTP with optional VRF; storage copies support USB/local SFTP directories.

Command History

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
6200	Manager (#)	Administrators or local user group members with execution rights for this command.

domain-name

Syntax

```
domain-name <NAME>
no domain-name [<NAME>]
```

Description

Specifies the domain name of the switch.

The **no** form of this command sets the domain name to the default, which is no domain name.

Parameter	Description
<code><NAME></code>	Specifies the domain name to be assigned to the switch. The first character of the name must be a letter or a number. Length: 1 to 32 characters.

Examples

Setting and showing the domain name:

```
switch# show domain-name
switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Setting the domain name to the default value:

```
switch(config)# no domain-name
switch(config)# show domain-name
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

front-panel-security factory-reset

Syntax

```
front-panel-security factory-reset  
no front-panel-security factory-reset
```

Description

Enable the **front-panel factory-reset** feature to allow a factory reset via the reset button on the front panel of the switch, without console access. Once the feature is enabled, this setting will be saved across reboots, until a factory-reset is performed. The **no** form of this command disables the front-panel factory-reset feature after it has been enabled. This feature is disabled by default.

Usage

If you enable front-panel factory-reset feature:

- pressing the reset button and releasing it before 5 seconds will reboot the system.
- pressing the reset button and releasing it between 5 to 20 seconds will start a hard reset of the system
- pressing the reset button and releasing it between 20 to 25 seconds will start a hard reset of the system, and the system will undergo the factory reset process.
- pressing the reset button for more than 25 seconds will not trigger any action.

If the front-panel factory-reset feature is left at its default disabled setting in the running configuration :

- pressing the reset button and releasing it before 5 seconds will reboot the system.
- pressing the reset button and releasing it between 5 to 20 seconds mark will start a hard reset of the system.
- pressing the reset button for more than 20 seconds will not trigger any action.

For more information about the LED indicators on this switch, refer to the **Installation Guide** for your switch.

Example

```
switch(config)# front-panel-security factory-reset  
This command will enable front-panel factory reset capability, where user  
can  
trigger factory-reset via reset button. This feature will remain enabled  
until  
it is disabled, or a factory-reset is performed.  
Continue (y/n)?
```

Command History

Release	Modification
10.14.1000	Command supported on 4100 and 6200 Switch Series.
10.13.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

hostname

Syntax

```
hostname <HOSTNAME>  
no hostname [<HOSTNAME>]
```

Description

Sets the host name of the switch.

The **no** form of this command sets the host name to the default value, which is `switch`.

Parameter	Description
<HOSTNAME>	Specifies the host name. The first character of the host name must be a letter or a number. Length: 1 to 32 characters. Default: <code>switch</code>

Examples

Setting and showing the host name:

```
switch# show hostname  
switch
```

```
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
myswitch
```

Setting the host name to the default value:

```
myswitch(config)# no hostname
switch(config)#
```

Error messages:

```
switch(config)# hostname
ThisIsAVeryLongHostnameThatExceedsTheMaximumLengthLimitOf64Characters
The hostname cannot exceed 63 characters in length.
switch(config)#
switch(config)# hostname -abd-123-
The hostname must begin and end with a letter or number.
switch(config)#
switch(config)# hostname abc.123@12
To be compliant with RFC 1123, the hostname must contain only letters,
numbers and hyphens, and must not start or end with a hyphen.
switch(config)#
switch(config)# hostname 123_123
Hostname contains one or more non-RFC 1123 compliant characters. Some
features may not work as intended.
Do you want to continue (y/n)? y
123_123(config)#
switch(config)# hostname test.name
Hostname contains one or more non-RFC 1123 compliant characters. Some
features may not work as intended.
Do you want to continue (y/n)? y
test.name(config)#
```

Command History

Release	Modification
10.17	Enhanced hostname to support dot (.) and underscore (_) characters.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

mtrace

Syntax

```
mtrace <IPV4-SRC-ADDR>
      <IPV4-GROUP-ADDR> [lhr <IPV4-LHR-ADDR>] [ttl <HOPS>]
      [vrf <VRF-NAME>]
```

Description

Traces the specified IPv4 source and group addresses.

Parameter	Description
<i>IPV4-SRC-ADDR</i>	Specifies the source IPv4 address to trace.
<i>IPV4-GROUP-ADDR</i>	Specifies the group IPv4 address to trace.
lhr <IPV4-LHR-ADDR>	Specifies the last hop router address from which to start the trace.
ttl <HOPS>	Specifies the Time - To - Live duration in hops. Range: 1 to 255 hops. Default: 8 hops.
vrf <VRF-NAME>	Specifies the name of the VRF. If a name is not specified the default VRF will be used.

Examples

Tracing with source, group, and LHR addresses and TTL:

```
(switch)# mtrace 20.0.0.1 239.1.1.1 lhr 10.1.1.1 ttl 10
Type escape sequence to abort.
Mtrace from 10.0.0.1 for Source 20.0.0.1 via Group 239.1.1.1
From destination(?) to source (?)...
Querying ful reverse path...
0 10.0.0.1
-1 30.0.0.1 PIM 0 ms
-2 40.0.0.1 PIM 2 ms
-3 50.0.0.1 PIM 100 ms
-4 60.0.0.1 PIM 156 ms
-5 20.0.0.1 PIM 123 ms
```

Tracing with source and group addresses:

```
(switch)# mtrace 200.0.0.1 239.1.1.1
Type escape sequence to abort.
Mtrace from self for Source 200.0.0.1 via Group 239.1.1.1
From destination(?) to source (?)...
Querying ful reverse path...
0 10.0.0.1
-1 30.0.0.1 PIM 0 ms
-2 40.0.0.1 PIM 2 ms
-3 50.0.0.1 PIM 100 ms
-4 60.0.0.1 PIM 156 ms
-5 200.0.0.1 PIM 123 ms
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

power consumption-average-period

Syntax

```
power consumption-average-period <PERIOD-IN-SECONDS>
```

Description

Configures a time period for average power consumption in seconds.

Parameter	Description
<PERIOD-IN-SECONDS>	Specifies the period in seconds for average power consumed. Range: 60 - 3600. Default: 600

For more information on this feature, see the video [AOS-CX 10.14 Release Update: Sustainability Update](#).

Example

Configuring a time period of 60 seconds for average power consumption:

```
switch(config)# power consumption-average-period 60
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	config	Administrators or local user group members with execution rights for this command.

power input-average-period

Syntax

```
power input-average-period <period-in-minutes>
```

Description

Configures a period to monitor the average PSU input voltage.

Parameter	Description
<code><period-in-minutes></code>	The input voltage monitoring period, in minutes Range: 1 - 1440. Default: 15

Example

Monitoring the PSU average input voltage over a period of 30 minutes:

```
switch(config)# power input-average-period 30
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
5420	<code>config</code>	Administrators or local user group members with execution rights for this command.

Reboot reasons

The **show boot-history** command displays the following reboot reasons for the management module:

Reboot reasons for management module

Parameter	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short - press of the reset button.
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.
Console error	Console failed to start.
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over from an unresponsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in the allotted time.
Critical service fault (error)	A daemon critical to switch operation has stopped functioning. An extra error string may be present to describe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
VSF member renumbered	An user requested a renumber of a VSF member.
VSF switchover requested	An user requested a VSF switchover.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceeded.

Uncontrolled reboots

- ops-switchd crashed

- ovsdb-server crashed
- Reset
 - Software thermal reset
 - Power on reset
 - Watchdog reset
 - CPU request reset
 - cold reset
 - Long press reset
 - Jumper reset
- switchd_agent crashed
- Power down due to hibernation and wake-up due to RTC trigger
- Power down due to hibernation and wake-up due to Button Mode trigger
- Hardware thermal reset

show front-panel-security status

Syntax

```
show front-panel-security status
```

Description

Displays the configuration status of the front panel factory reset feature and the time stamp of the first occurrence of factory reset using the reset button.

Example

Displays the status of the front panel factory reset feature:

```
switch# show front-panel-security status
Front panel factory reset           : enabled
First occurrence of front-panel factory reset : Wed 2023-09-06 02:57:50 +08
```

Displays the status of the front panel factory reset feature:

```
switch# show front-panel-security status
Front panel factory reset           : enabled but unsupported
First occurrence of front-panel factory reset : NA
```

If the status of the feature is displayed as **enabled but unsupported**, then users should issue the **allow-unsafe-updates** command and reboot the switch.

```
switch# config
switch(config)# allow-unsafe-updates 30
This command will enable non-failsafe updates of programmable devices for
the next 30 minutes. You will first need to wait for all line and fabric
modules to reach the ready state, and then reboot the switch to begin
applying any needed updates. Ensure that the switch will not lose power,
be rebooted again, or have any modules removed until all updates have
finished and all line and fabric modules have returned to the ready state.
WARNING: Interrupting these updates may make the product unusable!
Continue (y/n)? y
    Unsafe updates      : allowed (less than 30 minute(s) remaining)
```

Command History

Release	Modification
10.14.1000	Command supported on 4100 and 6200 Switch Series.
10.13.1000	Command introduced

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

show bluetooth

Syntax

```
show bluetooth
```

Description

Shows general status information about the Bluetooth wireless management feature on the switch.

Usage

This command shows status information about the following:

- The USB Bluetooth adapter
- Clients connected using Bluetooth
- The switch Bluetooth feature.

The output of the **show running-config** command includes Bluetooth information only if the Bluetooth feature is disabled.

The device name given to the switch includes the switch serial number to uniquely identify the switch while pairing with a mobile device.

The management IP address is a private network address created for managing the switch through a Bluetooth connection.

Examples

Example output when Bluetooth is enabled but no Bluetooth adapter is connected. <XXXX> is the switch platform and <NNNNNNNNNN> is the device identifier.

```
switch# show bluetooth
Enabled           : Yes
Device name       : <XXXX>-<NNNNNNNNNN>
Adapter State     : Absent
```

Example output when Bluetooth is enabled and there is a Bluetooth adapter connected:

```
switch# show bluetooth
Enabled           : Yes
Device name       : <XXXX>-
Adapter State     : Ready
Adapter IP address : 192.168.99.1
Adapter MAC address : 480fcf-af153a
Connected Clients
-----
Name              MAC Address      IP Address      Connected Since
-----
Mark's iPhone    089734-b12000    192.168.99.10   2018-07-09 08:47:22 PDT
```

Example output when Bluetooth is disabled:

```
switch# show bluetooth
Enabled           : No
Device name       : <XXXX>-<NNNNNNNNNN>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show boot-history

Syntax

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

**NOTE**

To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
all	Optional. Shows boot information for the active management module.
vsf member <1-10>	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
Index	The position of the boot in the history file. Range: 0 to 3.
Boot ID	A unique ID for the boot . A system - generated 128 - bit string.
Current Boot, up for <time>	For the current boot, the show boot - history command shows the number of seconds the module has been running on the current software.
<Timestamp>: boot reason	For previous boot operations, the show boot - history command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: • <DAEMON - NAME> crash: The daemon identified by <DAEMON - NAME> caused the module to boot. • Kernel crash: The operating system software associated with the module caused the module to boot. • Uncontrolled reboot: The reason for the reboot is not known. • Reboot requested through database: The reboot occurred because of a request made through the CLI or other API.

Table 1. Description of reboots handled through the database

Boot History String	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short - press of the reset button
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.

Boot History String	Description
Configuration version migration	A configuration version migration occurred which required a reboot.
Console error	The console failed to start.
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over from an unresponsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in the allotted time.
Critical service fault (error)	A daemon critical to switch operation has stopped functioning. An extra error string may be present to describe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
VSF member renumbered	A user requested a renumber of a VSF member.
VSF switchover requested	A user requested a VSF switchover.
VSX software update	Reset triggered by a VSX software update.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis low critical temperature	Chassis temperature below the minimum operating threshold.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceeded.
ISSU SMM update	Standby management module reboot triggered by an In - Service Software Upgrade (ISSU).
ISSU switchover	Redundancy switchover triggered by an In - Service Software Upgrade.

Boot History String	Description
ISSU aborted	Standby management module reset triggered by failure during an In - Service Software Upgrade.
Rollback timer expired	Reset triggered by the ISSU rollback timer expiring.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====
Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs
Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version:
FL.10.14.0000-1619-ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2
Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1
Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 : Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 : Switch boot count is 3
25 Aug 23 10:29:02 : Primary overtemperature fault detected with 0x2 in
PSU 1/1
```

Showing the boot history of the active management module and all line modules:

```
switch#
Management module
=====
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
```

```

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
Management module
=====
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
switch# show boot-history
Management module
=====
Index : 2
Boot ID : a61ad00d10864c748bc7893a5d4af2e4
15 Dec 23 19:02:02 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:02:02 : Switch boot count is 0
15 Dec 23 19:02:17 : PSU 1/1: Fault detected
Index : 1
Boot ID : 30d831bbfdfa425baf50a629ee01b185
15 Dec 23 19:01:58 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:01:58 : Switch boot count is 0

```

The following example displays the boot history for the VSF member **2**.


```
switch# show boot-history vsf member 2
Member-2
=====
Index : 0
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs
Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show capacities

Syntax

```
show capacities <FEATURE>
```

Description

Shows system capacities and their values for all features or a specific feature.

Parameter	Description
<FEATURE>	Specifies a feature. For example, aaa .

Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

Examples

Showing capacities for L3 resources on a 6200 switch:

```
switch# show capacities l3-resources
System Capacities: Filter L3 Resources
Capacities Name
Value
-----
-----
Maximum number of DLB (dynamic load balancing) ECMP Groups
64
Maximum number of IP neighbors (IPv4+IPv6) supported in the system
120000
Maximum number of IP Directed Broadcast Neighbors supported in the system
256
Maximum number of IPv4 neighbors(# of ARP entries) supported in the system
120000
Maximum number of IPv6 neighbors(# of ND entries) supported in the system
52000
Maximum number of flows tracked in L3 Dynamic Flow Table
32768
Maximum number of L3 Groups for IP Tunnels and ECMP Groups
2000
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP
groups
2045
Maximum number of L3 Destinations for Nexthops in Tunnels and ECMP groups
32632
Maximum number of routes (IPv4+IPv6) on the system
29696
Maximum number of IPv4 routes on the system
28672
Maximum number of IPv6 routes with prefix 0-64
13312
Maximum number of IPv6 routes with prefix 65-127
510
switch# show capacities bgp
System Capacities: Filter BGP
Capacities Name
Value
```

```

-----
-----
Maximum number of AS numbers in as-path
attribute                                     32
...

```

Showing port-access related capacities on the 6200:

```

switch# show capacities port-access
System Capacities: Filter Port-Access
Capacities Name
Value
-----
-----
Maximum number of Clients that can be authenticated on a
port                                     256
Maximum number of Device Profiles allowed to be created on the
system                                  16
Maximum number of unique FQDN entries per port access
policy                                  128
Maximum number of MAC Address that can be authorized on a
port                                     32
Maximum number of Port Access Role VLAN IDs allowed to be created on the
system                                  256
Maximum number of Port Access Role VLAN names allowed to be created on the
system                                  50
Maximum number of Port Access Roles allowed to be created on the
system                                  512

```

```

switch# show capacities
System Capacities:
Capacities
Name                                     Value
-----
-----
Maximum number of Access Control Entries configurable in a
system                                  4096
Maximum number of Access Control Lists configurable in a
system                                  512
Maximum number of class entries configurable in a
system                                  4096
Maximum number of classes configurable in a
system                                  512
Maximum number of entries in an Access Control
List                                    1024
Maximum number of entries in a

```

class	1024
Maximum number of entries in a policy	64
Maximum number of classifier policies configurable in a system	512
Maximum number of policy entries configurable in a system	4096
Maximum number of clients supported for tracking the IP address in the system	128
Maximum number of dynamic VLANs that can be allowed using MVRP	256
Maximum number of nexthops per IP ECMP group	1
Maximum number of IP neighbors (IPv4+IPv6) supported in the system	1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the system	1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the system	512
Maximum number of L2 MAC addresses supported in the system	8192
Maximum number of L3 Groups for IP Tunnels and ECMP Groups	1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups	1024
Maximum number of configurable LAG ports	8
Maximum number of members supported by a LAG port	8
Maximum number of VLANs across ports allowed in loop-protect	3328
Maximum number of IGMP/MLD groups supported	512
Maximum number of IGMP/MLD snooping groups supported	512
Maximum number of Mirror Sessions configurable in a system	4
Maximum number of enabled Mirror Sessions in a system	4
Maximum number of mstp instances configurable in a system	16
Maximum number of Clients that can be authenticated on a port	32

Maximum number of Device Profiles allowed to be created on the system	8
Maximum number of Port Access Roles allowed to be created on the system	32
Maximum number of MAC Address that can be authorized on a port	32
Maximum number of Port Access Role VLAN IDs allowed to be created on the system	50
Maximum number of Port Access Role VLAN names allowed to be created on the system	50
Maximum number of RBAC rules per user group	1024
Maximum number of RPVST VLANs configurable on the system	16
Maximum number of RPVST VPORTs supported in a system	512
Maximum number of SVIs supported in the system	16
Maximum number of routes (IPv4+IPv6) on the system	512
Maximum number of IPv4 routes on the system	512
Maximum number of IPv6 routes on the system	512
Maximum number of VLANs supported in the system	512
System Capacities:	
Capacities	
Name	Value

Maximum number of Access Control Entries configurable in a system	4096
Maximum number of Access Control Lists configurable in a system	512
Maximum number of class entries configurable in a system	4096
Maximum number of classes configurable in a system	512
Maximum number of entries in an Access Control List	1024
Maximum number of entries in a class	1024

Maximum number of entries in a policy	64
Maximum number of classifier policies configurable in a system	512
Maximum number of policy entries configurable in a system	4096
Maximum number of clients supported for tracking the IP address in the system	128
Maximum number of dynamic VLANs that can be allowed using MVRP	256
Maximum number of nexthops per IP ECMP group	1
Maximum number of IP neighbors (IPv4+IPv6) supported in the system	1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the system	1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the system	512
Maximum number of L2 MAC addresses supported in the system	8192
Maximum number of L3 Groups for IP Tunnels and ECMP Groups	1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups	1024
Maximum number of configurable LAG ports	8
Maximum number of members supported by a LAG port	8
Maximum number of VLANs across ports allowed in loop-protect	3328
Maximum number of IGMP/MLD groups supported	512
Maximum number of IGMP/MLD snooping groups supported	512
Maximum number of Mirror Sessions configurable in a system	4
Maximum number of enabled Mirror Sessions in a system	4
Maximum number of mstp instances configurable in a system	16
Maximum number of Clients that can be authenticated on a port	32
Maximum number of Device Profiles allowed to be created on the	

```

system                        8
Maximum number of Port Access Roles allowed to be created on the
system                        32
Maximum number of MAC Address that can be authorized on a
port                          32
Maximum number of Port Access Role VLAN IDs allowed to be created on the
system                        50
Maximum number of Port Access Role VLAN names allowed to be created on the
system                        50
Maximum number of RBAC rules per user
group                          1024
Maximum number of RPVST VLANs configurable on the
system                        16
Maximum number of RPVST VPORTs supported in a
system                        512
Maximum number of SVIs supported in the
system                        16
Maximum number of routes (IPv4+IPv6) on the
system                        512
Maximum number of IPv4 routes on the
system                        512
Maximum number of IPv6 routes on the
system                        512
Maximum number of VLANs supported in the
system                        512
switch# show capacities
System Capacities:
Capacities
Name                                                                    Value
-----
Maximum number of Access Control Entries configurable in a
system                                                                    4096
Maximum number of Access Control Lists configurable in a
system                                                                    512
Maximum number of class entries configurable in a
system                                                                    4096
Maximum number of classes configurable in a
system                                                                    512
Maximum number of entries in an Access Control
List                                                                      1024
Maximum number of entries in a
class                                                                      1024

```

Maximum number of entries in a policy	64
Maximum number of classifier policies configurable in a system	512
Maximum number of policy entries configurable in a system	4096
Maximum number of clients supported for tracking the IP address in the system	128
Maximum number of dynamic VLANs that can be allowed using MVRP	256
Maximum number of nexthops per IP ECMP group	1
Maximum number of IP neighbors (IPv4+IPv6) supported in the system	1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the system	1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the system	512
Maximum number of L2 MAC addresses supported in the system	8192
Maximum number of L3 Groups for IP Tunnels and ECMP Groups	1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups	1024
Maximum number of configurable LAG ports	8
Maximum number of members supported by a LAG port	8
Maximum number of VLANs across ports allowed in loop-protect	3328
Maximum number of IGMP/MLD groups supported	512
Maximum number of IGMP/MLD snooping groups supported	512
Maximum number of Mirror Sessions configurable in a system	4
Maximum number of enabled Mirror Sessions in a system	4
Maximum number of mstp instances configurable in a system	16
Maximum number of Clients that can be authenticated on a port	32
Maximum number of Device Profiles allowed to be created on the	

system	8	
Maximum number of Port Access Roles allowed to be created on the system	32	
Maximum number of MAC Address that can be authorized on a port	32	
Maximum number of Port Access Role VLAN IDs allowed to be created on the system	50	
Maximum number of Port Access Role VLAN names allowed to be created on the system	50	
Maximum number of RBAC rules per user group		1024
Maximum number of RPVST VLANs configurable on the system	16	
Maximum number of RPVST VPORTs supported in a system	512	
Maximum number of SVIs supported in the system		16
Maximum number of routes (IPv4+IPv6) on the system	512	
Maximum number of IPv4 routes on the system		512
Maximum number of IPv6 routes on the system		512
Maximum number of VLANs supported in the system	512	
System Capacities:		
Capacities		
Name		Value

Maximum number of Access Control Entries configurable in a system	4096	
Maximum number of Access Control Lists configurable in a system	512	
Maximum number of class entries configurable in a system	4096	
Maximum number of classes configurable in a system	512	
Maximum number of entries in an Access Control List	1024	
Maximum number of entries in a class		1024
Maximum number of entries in a		

policy	64
Maximum number of classifier policies configurable in a system	512
Maximum number of policy entries configurable in a system	4096
Maximum number of clients supported for tracking the IP address in the system	128
Maximum number of dynamic VLANs that can be allowed using MVRP	256
Maximum number of nexthops per IP ECMP group	1
Maximum number of IP neighbors (IPv4+IPv6) supported in the system	1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the system	1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the system	512
Maximum number of L2 MAC addresses supported in the system	8192
Maximum number of L3 Groups for IP Tunnels and ECMP Groups	1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups	1024
Maximum number of configurable LAG ports	8
Maximum number of members supported by a LAG port	8
Maximum number of VLANs across ports allowed in loop-protect	3328
Maximum number of IGMP/MLD groups supported	512
Maximum number of IGMP/MLD snooping groups supported	512
Maximum number of Mirror Sessions configurable in a system	4
Maximum number of enabled Mirror Sessions in a system	4
Maximum number of mstp instances configurable in a system	16
Maximum number of Clients that can be authenticated on a port	32
Maximum number of Device Profiles allowed to be created on the system	8

Maximum number of Port Access Roles allowed to be created on the system	32
Maximum number of MAC Address that can be authorized on a port	32
Maximum number of Port Access Role VLAN IDs allowed to be created on the system	50
Maximum number of Port Access Role VLAN names allowed to be created on the system	50
Maximum number of RBAC rules per user group	1024
Maximum number of RPVST VLANs configurable on the system	16
Maximum number of RPVST VPORTs supported in a system	512
Maximum number of SVIs supported in the system	16
Maximum number of routes (IPv4+IPv6) on the system	512
Maximum number of IPv4 routes on the system	512
Maximum number of IPv6 routes on the system	512
Maximum number of VLANs supported in the system	512

Showing all available capacities for mirroring:

```
switch# show capacities mirroring
System Capacities: Filter Mirroring
Capacities Name
Value
-----
-----
Maximum number of Mirror Sessions configurable in a
system 4
Maximum number of enabled Mirror Sessions in a
system 4
```

Showing all available capacities for MSTP:

```
switch# show capacities mstp
System Capacities: Filter MSTP
Capacities Name
Value
-----
-----
```

```
Maximum number of mstp instances configurable in a
system                                     64
```

Showing all available capacities for VLAN count:

```
switch# show capacities vlan-count
System Capacities: Filter VLAN Count
Capacities Name
Value
-----
Maximum number of VLANs supported in the
system                                     4094
```

Command History

Release	Modification
10.18	Maximum number of Port Access Roles allowed for the 6200 Switch Series has been increased to 512.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show capacities-status

Syntax

```
show capacities-status <FEATURE>
```

Description

Shows system capacities status and their values for all features or a specific feature.

Parameter	Description
<code><FEATURE></code>	Specifies the feature, for example aaa or vrrp for which to display capacities, values, and status. Required.

Examples

Showing capacities for L3 resources on a 6200 switch:

```
switch# show capacities-status l3-resources
System Capacities Status: Filter L3 Resources
Capacities Status Name                                     Value
Maximum
-----
--
Number of IP neighbor (IPv4+IPv6) entries                  0
120000
Number of IP Directed Broadcast neighbor entries           0      256
Number of IPv4 neighbor(ARP) entries                       0
120000
Number of IPv6 neighbor(ND) entries                        0
52000
Number of L3 Groups for IP Tunnels and ECMP Groups currently
configured                                0      2000
Number of L3 Destinations for Routes, Nexthops in ECMP groups
and Tunnels currently configured                        0      2045
Number of L3 Destinations for Neighbors and Routes currently
configured                                             0      55036
```

Showing the system capacities status for all features:

```
switch# show capacities-status
System Capacities Status
Capacities Status Name                                     Value
Maximum
-----
--
Number of active gateway mac addresses in a system         0      16
Number of aspath-lists configured                          0      64
Number of community-lists configured                      0      64
...
switch# show capacities-status
System Capacities Status
```

```

Capacities Status Name
Value Maximum
-----
-----
Number of Access Control Entries currently configured
0      4096
Number of Access Control Lists currently configured
0      512
Number of class entries currently configured
0      4096
Number of classes currently configured
0      512
Number of policies currently configured
0      512
Number of policy entries currently configured
0      4096
Number of dynamic VLANs currently learnt using MVRP
0      256
Number of IP neighbor (IPv4+IPv6) entries
1      1024
Number of IPv4 neighbor(ARP) entries
1      1024
Number of IPv6 neighbor(ND) entries
0      512
Number of L3 Groups for IP Tunnels and ECMP Groups currently configured
0      1
Number of L3 Destinations for Routes, Nexthops in ECMP groups and
Tunnels currently configured
0      1024
Number of Mirror Sessions currently configured
0      4
Number of Mirror Sessions currently enabled
0      4
Number of mstp instances currently configured
0      16
Number of RPVST VLANs currently configured
0      16
Number of routes (IPv4+IPv6) currently configured
1      512
Number of IPv4 routes currently configured
1      512
Number of IPv6 routes currently configured
0      512

```

Number of VLANs currently configured

1 512

Showing the system capacities status for BGP:

```
switch# show capacities-status bgp
```

System Capacities Status: Filter BGP

Capacities Status Name	Value
------------------------	-------

Maximum

Number of aspath-lists configured	0 64
-----------------------------------	------

Number of community-lists configured	0 64
--------------------------------------	------

Number of neighbors configured across all VRFs	0 50
--	------

Number of peer groups configured across all VRFs	0 25
--	------

Number of prefix-lists configured	0 64
-----------------------------------	------

Number of route-maps configured	0 64
---------------------------------	------

Number of routes in BGP RIB	0 256000
-----------------------------	----------

Number of route reflector clients configured across all VRFs	0 16
--	------

Command History

Release

Modification

10.07 or earlier

- -

Command Information

Platforms

Command context

Authority

All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.
---------------	---------------	--

show console

Syntax

```
show console
```

Description

Shows the serial console port current speed.

Examples

Showing the console port current speed:

```
switch# show console
Baud Rate: 9600
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show core-dump

Syntax

```
show core-dump
    all
```

Description

Shows core dump information about the specified module. When no parameters are specified, shows only the core dumps generated in the current boot of the management module. When the **all** parameter is specified, shows all available core dumps.

Parameter	Description
all	Shows all available core dumps.

Usage

When no parameters are specified, the **show core-dump** command shows only the core dumps generated in the current boot of the management module. You can use this command to determine when any crashes are occurring in the current boot.

If no core dumps have occurred, the following message is displayed: **No core dumps are present**

To show core dump information for the standby management module, you must use the **standby** command to switch to the standby management module and then execute the **show core-dump** command.

In the output, the meaning of the information is the following:

Daemon Name

Identifies name of the daemon for which there is dump information.

Instance ID

Identifies the specific instance of the daemon shown in the **Daemon Name** column.

Present

Indicates the status of the core dump:

Yes

The core dump has completed and available for copying.

In Progress

Core dump generation is in progress. Do not attempt to copy this core dump.

Timestamp

Indicates the time the daemon crash occurred. The time is the local time using the time zone configured on the switch.

Build ID

Identifies additional information about the software image associated with the daemon.

Examples

Showing core dump information for the current boot of the active management module only:

```
switch# show core-dump
=====
=====
Daemon Name      | Instance ID | Present      | Timestamp                | Build
ID
=====
=====
hpe-fand         | 1399        | Yes          | 2017-08-04 19:05:34      | 1246d2a
hpe-sysmond      | 957         | Yes          | 2017-08-04 19:05:29      | 1246d2a
=====
=====
Total number of core dumps : 2
=====
```

```

=====
=====
=====
Daemon Name      | Instance ID | Present      | Timestamp              | Build
ID
=====
=====
hpe-fand          1399         Yes           2017-08-04 19:05:34     1246d2a
hpe-sysmond       957          Yes           2017-08-04 19:05:29     1246d2a
=====
=====
Total number of core dumps : 2
=====
=====

```

Showing all core dumps:

```

switch# show core-dump all
=====
=
Management Module core-dumps
=====
=
Daemon Name      | Instance ID | Present      | Timestamp              | Build ID
=====
=
hpe-sysmond       513          Yes           2017-07-31 13:58:05     e70f101
hpe-tempd         1048         Yes           2017-08-13 13:31:53     e70f101
hpe-tempd         1052         Yes           2017-08-13 13:41:44     e70f101
Line Module core-dumps
=====
=
Line Module : 1/1
=====
=
dune_agent_0      18958        Yes           2017-08-12 11:50:17     e70f101
dune_agent_0      18842        Yes           2017-08-12 11:50:09     e70f101
=====
=
Total number of core dumps : 5
=====
=

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show deprecated commands

Syntax

```
show deprecated-commands [<feature>]
```

Description

Shows the list of CLI commands that will be deprecated in a future release along with the new form of the same command which is recommended for use.

Both the command options will be supported until a certain release, after which only the newer replacement command will be supported.

Parameter	Description
feature	Optional. Specify feature name. The list of features for which you can view deprecated commands are:

Examples

Check the deprecated CLI commands for a specific feature:

```
switch# show deprecated-commands
```

The following commands with ipv4 keyword will be replaced with ip

Deprecated: vrrp <1-255> address-family (ipv4 | ipv6)

Replacement: vrrp <1-255> address-family (ip | ipv6)

Deprecated: show bgp ipv4 unicast

Replacement: show bgp ip unicast

...

switch# show deprecated-commands vrrp

The following commands with ipv4 keyword will be replaced with ip

Deprecated: vrrp <1-255> address-family (ipv4 | ipv6)

Replacement: vrrp <1-255> address-family (ip | ipv6)

Deprecated: show vrrp (ipv4 | ipv6 | brief | detail)(<1-255>)

Replacement: show vrrp (ip | ipv6 | brief | detail)(<1-255>)

...

switch# show deprecated-commands vsf

Feature vsf has no deprecated commands.

Command History

Release	Modification
10.14	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show domain-name

Syntax

```
show domain-name
```

Description

Shows the current domain name.

Usage

If there is no domain name configured, the CLI displays a blank line.

Example

Setting and showing the domain name:

```
switch# show domain-name
switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show environment fan

Syntax

```
show environment fan [vsf]
```

Description

Shows the status information for all fans and fan trays (if present) in the system.

Parameter	Description
<code>vsf</code>	Shows output from the VSF member - id on switches that support VSF.

Usage

For fan trays, **Status** is one of the following values:

- **ready**:The fan tray is operating normally.
- **fault**:The fan tray is in a fault event. The status of the fan tray does not indicate the status of fans.
- **empty**:The fan tray is not installed in the system.

For fans:

Speed :Indicates the relative speed of the fan based on the nominal speed range of the fan. Values are:

Slow:The fan is running at less than 25% of its maximum speed.

Normal:The fan is running at 25-49% of its maximum speed.

Medium:The fan is running at 50-74% of its maximum speed.

Fast:The fan is running at 75-99% of its maximum speed.

Max:The fan is running at 100% of its maximum speed.

N/A:The fan is not installed.

front-to-back:Air flows from the front of the system to the back of the system.

N/A:The fan is not installed.

Status: Fan status. Values are:

uninitialized:The fan has not completed initialization.

ok: The fan is operating normally.

fault: The fan is in a fault state.

empty: The fan is not installed.

Examples

```
switch# show environment fan
```

```
Fan tray informationFan tray information
```

```
-----
--
Name   Description                               Status      Serial Number  Fans
-----
--
```

```

1/1    JL669A  X751 FB Fan Tray      ready      CN04KN9913    2
1/2    JL669A  X751 FB Fan Tray      ready      CN04KN97XJ    2

```

Fan information

```

-----
--
Location-      Product  Serial Number  Speed  Direction      Status  RPM
Mbr/Slot/Fan   Name
-----
--
Tray-1/1/1     N/A      N/A            slow   front-to-back  ok      4024
Tray-1/1/2     N/A      N/A            slow   front-to-back  ok      4041
Tray-1/2/1     N/A      N/A            slow   front-to-back  ok      4030
Tray-1/2/2     N/A      N/A            slow   front-to-back  ok      4028

```

switch# **show environment fan**

Fan information

```

-----
Mbr/Fan      Product  Serial Number  Speed  Direction      Status  RPM
              Name
-----
1/1           N/A      N/A            N/A    left-to-right  ok      N/A

```

Showing output for a system without a fan tray:

switch# **show environment fan**

Fan information

```

-----
Fan   Serial Number  Speed  Direction      Status      RPM
-----
1     SGXXXXXXXXXX    slow   front-to-back  ok          6000
2     SGXXXXXXXXXX    normal front-to-back  ok          8000
3     SGXXXXXXXXXX    medium front-to-back  ok          11000
4     SGXXXXXXXXXX    fast   front-to-back  ok          14000
5     SGXXXXXXXXXX    max    front-to-back  fault       16500
6     N/A            N/A    N/A            empty
...

```

switch# **show environment fan**

switch# **show environment fan**

Fan tray information

```

-----
--
Name  Description      Status      Serial Number
Fans
-----
--
1/1   JL630A HPE ANW F2B Fan  ready      N/A          2

```

1/2	JL630A	HPE	ANW	F2B	Fan	ready	N/A	2
1/3	JL630A	HPE	ANW	F2B	Fan	ready	N/A	2
1/4	JL630A	HPE	ANW	F2B	Fan	ready	N/A	2
1/5	JL630A	HPE	ANW	F2B	Fan	ready	N/A	2
1/6	JL630A	HPE	ANW	F2B	Fan	ready	N/A	2
Fan information								

--								
Location-	Product	Serial	Number	Speed	Direction	Status	RPM	
Mbr/Slot/Fan	Name							

--								
Tray-1/1/1	JL630A	N/A		slow	front-to-back	ok	6400	
Tray-1/1/2	JL630A	N/A		slow	front-to-back	ok	5400	
Tray-1/2/1	JL630A	N/A		slow	front-to-back	ok	6300	
Tray-1/2/2	JL630A	N/A		slow	front-to-back	ok	5400	
Tray-1/3/1	JL630A	N/A		slow	front-to-back	ok	6500	
Tray-1/3/2	JL630A	N/A		slow	front-to-back	ok	5500	
Tray-1/4/1	JL630A	N/A		slow	front-to-back	ok	6300	
Tray-1/4/2	JL630A	N/A		slow	front-to-back	ok	5400	
Tray-1/5/1	JL630A	N/A		slow	front-to-back	ok	6000	
Tray-1/5/2	JL630A	N/A		slow	front-to-back	ok	5200	
Tray-1/6/1	JL630A	N/A		slow	front-to-back	ok	6200	
Tray-1/6/2	JL630A	N/A		slow	front-to-back	ok	5400	

Command History

Release	Modification
10.18	Introduced support for 8325 and 8325P Switch series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show environment led

Syntax

```
show environment led
      <MEMBER-ID>
```

Description

Shows state and status information for all the configurable LEDs in the system.

Parameter	Description
<i><MEMBER-ID></i>	Shows output from the specified VSF member ID on switches that support VSF.

Example

Showing state and status for LED:

```
switch# show environment led
Name           State      Status
-----
locator        flashing   ok
switch# show environment led
Mbr/Name       State      Status
-----
1/locator      off        ok
switch# show environment led
Mbr/Name       State      Status
-----
1/locator      off        ok
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show environment power-consumption power-over-ethernet

Syntax

```
show environment power consumption power-over-ethernet
show environment power-consumption power-over-ethernet detail <DETAIL>
show environment power-consumption power-over-ethernet member <MEMBER-ID>
show environment power-consumption power-over-ethernet interface <INTERFACE>
```

Description

Displays the PoE power consumption information.

Parameter	Description
<DETAIL>	Displays detailed PoE power consumption information.
<ARG-1>	For VSF supported platforms only. Displays the VSF member command.
<ARG-2>	For VSF supported platforms only. Displays the power consumption information for the specified VSF member. Range: 1 - 10.
INTERFACE	Displays the interface name with PoE capability.
<MEMBER>	For VSF supported platforms only. Displays the PoE power consumption for a stack member.

Usage

For each PoE capable module, the PoE power consumed is displayed. PoE power consumed values are collected once every sampling period and the average value is calculated over the average period. The total power consumed is the total power of the PoE devices in a chassis. The average PoE power consumed is

calculated from the total PoE power consumed as a running average over a period of time. The default average period is 10 minutes. The minimum average period can be set to 60 seconds and maximum to 3600 seconds. The average period can be configured using **power consumption-average-period**. For VSF platforms, this command displays the PoE power consumed for all members.

The following information is provided for summary of power consumption:

- line module - PoE power used by line module
- chassis module - PoE power used by chassis module
- PoE Power total - total PoE power consumption
- PoE Power average - average for total PoE power consumption that is calculated over a given period
- average period - time to calculate power average

Example

Showing the POE instantaneous power consumption for a switch, in brief:

```
switch# show environment power-consumption power-over-ethernet
Power Consumption Averaging Period : 600 seconds
                                PoE Average
Name Description                Usage (W)
-----
1      6300M 48SR10 CL8 PoE 4p100G Sw      401.41
```

Showing the POE power consumption for a VSF stack, in brief:

```
switch# show environment power-consumption power-over-ethernet
Power Consumption Averaging Period : 120 seconds
                                PoE Average
Name Description                Usage (W)
-----
1      6300M 24SR5 CL6 PoE 4SFP56 Swc      23.86
2      6300M 48G CL4 PoE 4SFP56 Swch       11.27
3      6300F 24G CL4 PoE 4SFP56 Sw       197.60
```

Showing the POE power consumption for a chassis, in brief:

```
switch# show environment power-consumption power-over-ethernet
Power Consumption Averaging Period : 600 seconds
                                PoE Average
Name Description                Usage (W)
-----
1      6405 Chassis              120.11
```

Showing the POE power consumption for a standalone switch, in detail:

```
switch# show environment power-consumption power-over-ethernet detail
Power Consumption Averaging Period : 600 seconds
```

						PoE Average
Name	Description					Usage (W)

1	6300M	48SR10	CL8	PoE	4p100G Sw	432.23
PoE	Instantaneous			Average		
Port	Power (W)		Power (W)			

1/1/1	0.00		0.00			
1/1/2	0.00		0.00			
...						
1/1/47	16.27		16.27			
1/1/48	16.76		16.76			

Showing the POE power consumption for a chassis, in detail:

```
switch# show environment power-consumption power-over-ethernet detail
Power Consumption Averaging Period : 600 seconds

Name Description                                PoE Average Usage (W)
-----
1      6405 Chassis                                542.49
PoE      Instantaneous      Average
Port      Power (W)      Power (W)
-----
1/3/1      0.00      0.00
1/3/2      15.56      15.56
...
1/7/23      1.59      1.59
1/7/24      0.00      0.00
```

Showing the POE power consumption for 3 member VSF stack, in detail:

```
switch# show environment power-consumption power-over-ethernet detail
Power Consumption Averaging Period : 120 seconds

Name Description                                PoE Average Usage (W)
-----
1      6300M 24SR5 CL6 PoE 4SFP56 Swc            24.09
2      6300M 48G CL4 PoE 4SFP56 Swch             11.26
3      6300F 24G CL4 PoE 4SFP56 Sw              197.60
PoE      Instantaneous      Average
Port      Power (W)      Power (W)
-----
1/1/1      1.54      1.54
1/1/2      0.00      0.00
...
```

2/1/1	3.21	3.21
2/1/2	0.00	0.00

Showing the POE power consumption for VSF member 3:

```
switch# show environment power-consumption power-over-ethernet member 3
Power Consumption Averaging Period : 120 seconds
PoE Average
Name Description Usage (W)
-----
3 6300F 24G CL4 PoE 4SFP56 Sw 197.60
PoE Instantaneous Average
Port Power (W) Power (W)
-----
3/1/1 14.07 14.07
3/1/2 14.44 14.44
...
3/1/23 0.00 0.00
3/1/24 0.00 0.00
```

Showing the POE power consumption for VSF invalid member:

```
switch# show environment power-consumption power-over-ethernet member 5
Member 5 is not present
```

Showing the interface specifics:

```
switch# show environment power-consumption power-over-ethernet interface
3/1/1
Power Consumption Averaging Period : 60 seconds
PoE Instantaneous Average
Port Power (W) Power (W)
-----
3/1/1 14.07 14.07
```

Showing the range of interfaces:

```
switch# show environment power-consumption power-over-ethernet interface
3/1/1-3/1/3,3/1/15
Power Consumption Averaging Period : 60 seconds
PoE Instantaneous Average
Port Power (W) Power (W)
-----
3/1/1 14.07 14.07
3/1/2 14.44 14.44
3/1/3 14.09 14.09
3/1/15 14.44 14.44
```

Command History

Release	Modification
10.14.1000	Command introduced.

Command Information

Platforms	Command context	Authority
6200	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show environment power-consumption

Syntax

```
show environment power-consumption [detail]
show environment power-consumption member <MEMBER-ID>
```

Description

Displays power consumption information.

Parameter	Description
[detail]	Displays detailed power consumption information.
<MEMBER-ID>	For VSF supported platforms only. Displays the power consumption information for the specified VSF member. Range: 1 - 10.

Usage

Power consumed values are updated every 5 seconds. The total power consumed is the total power used in a chassis. The power consumed average is calculated from the total power consumed as a running average over a period of time. The average period has a default of 10 minutes. The period can be configured using **power-consumption-average-period**.

On chassis where power input is not supported, power consumed is displayed as Power Usage and PSU Output Power is displayed as N/A.

The following information is provided for a summary of power consumption:

- line module: power used by line module
- management module: power used by management module
- fabric module: power used by fabric module
- chassis module: power used by chassis module
- fan module: power used by fan module
- power total: total power consumption
- average power: average for total power consumption that is calculated over a given period
- average period: time to calculate power average

Example

Showing the power consumption for a 1RU switch:

```
switch# show environment power-consumption
Power Consumption Averaging Period : 600 seconds
```

Name	Description	Power Usage (W)		PSU Output (W)	
		Instant	Average	Instant	Average
1	8360-32Y4C v2 Switch	120.00	111.16	111.00	102.62

Showing the power consumption for a VSF stack:

```
switch# show environment power-consumption
Power Consumption Averaging Period : 600 seconds
```

Name	Description	Power Usage (W)		PSU Output (W)	
		Instant	Average	Instant	Average
1	6200M 24G 4SFP+ Sw	350.00	349.62	300.00	311.50
2	6200M 24G 4SFP+ Sw	300.00	299.50	280.00	275.50

Showing the power consumption for a switch where input reading is not supported:

```
switch# show environment power-consumption
Power Consumption Averaging Period : 600 seconds
```

Name	Description	Power Usage (W)		PSU Output (W)	
		Instant	Average	Instant	Average
1	6200M 24G 4SFP+ Sw	321.00	330.30	N/A	N/A

Showing the power consumption for a switch with multiple line cards, in brief:

```
switch# show environment power-consumption
Power Consumption Averaging Period : 600 seconds
```

		Power Usage (W)		PSU Output (W)	
--	--	-----------------	--	----------------	--

Name	Description	Instant	Average	Instant	Average
1	6410 v2 Chassis	900.00	905.62	798.00	803.23

Showing the power consumption for a switch with multiple line cards, in detail:

```
switch# show environment power-consumption detail
```

Power Consumption Averaging Period : 600 seconds

Name	Module	Type	Power Usage (W)		PSU Output (W)	
			Instant	Average	Instant	Average

1	chassis	total	1495.00	1499.99	1307.00	1300.67
1/3	line		97.00			
1/4	line		95.00			
1/5	line		90.00			
1/6	line		58.00			
1/7	line		90.00			
1/8	line		93.00			
1/9	line		96.00			
1/10	line		92.00			
1/11	line		100.00			
1/12	line		99.00			
1/1	management		14.00			
1/2	management		13.00			
	other		370.00			

Showing the power consumption for 4 member stack, in detail:

```
switch# show environment power-consumption detail
```

Power Consumption Averaging Period : 600 seconds

Name	Module	Type	Power Usage (W)		PSU Output (W)	
			Instant	Average	Instant	Average

1	chassis	total	930.00	905.17	830.00	803.40
2	chassis	total	900.00	920.00	875.00	880.25
3	chassis	total	850.00	857.25	802.00	805.00
4	chassis	total	1200.00	1100.56	921.00	950.00
Total Power Consumption			3428.00			

Showing the power consumption for VSF member 2:

```
switch# show environment power-consumption member 2
```

Power Consumption Averaging Period : 600 seconds

Name	Description	Power Usage (W)		PSU Output (W)	
		Instant	Average	Instant	Average

2	6200M 24G 4SFP+ Sw	300.00	299.50	280.00	275.50
---	--------------------	--------	--------	--------	--------

Showing the power consumption for VSF invalid member:


```
switch# show environment power-consumption member 5
Member 5 is not present
```

Command History

Release	Modification
10.14.1000	Introduced enhancement to power consumption and power consumption average.
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show environment power-redundancy

Syntax

```
show environment power-consumption [vsx-peer]
```

Description

Displays power consumption information.

Parameters

Parameter	Description
vsx-peer	Optional. Display switch power redundancy information for a VSX peer..

Example

Showing the power redundancy is not configured

```
switch# # show env power-redundancy
Configured   Operational
Redundancy   Redundancy
-----
n+n          none
```

Release	Modification
10.16	Command Introduced

Command Information

Platforms	Command context	Authority
5420	Operator (>) or Manager (#	Administrators or local user group members with execution rights for this command.

show environment power-supply

Syntax

```
show environment power-supply
  input-voltage [events] [vsf member <MEMBER-ID>]
  vsf member <MEMBER-ID>
  vsx-peer
```

Description

Shows status information about all power supplies in the switch.

Parameter	Description
input-voltage	Display the statistic and status of the Power Supply Unit (PSU) input voltage. This parameter is supported on 5420, 6200M, 6300M, 6400, 8320, 8325/8325H/8325P, 8100, 8360, 8400, 9300/9300S and 10000 Switch series.
events	Include the optional events parameter to display events for the power supply's input voltage. the output of the command displ

Parameter	Description
	ays information for the power supplies of the entire chassis or system (including the PSUs within a VSF member.)
<code>vsf member <MEMBER-ID></code>	For switches that support VSF, include the optional vsf member <MEMBER - ID> parameters after the input - voltage parameter to display input voltage information for a specific VSF member.
<code>vsf member <MEMBER-ID></code>	For switches that support VSF, include the vsf member <MEMBER - ID> parameter after the power - supply parameter to display information for power supplies on a specific VSF member.

Usage

The following information is provided for each power supply:

Parameter	Description
<code>Mbr/PSU</code>	Shows the member and slot number of the power supply.
<code>Product Number</code>	Shows the product number of the power supply.
<code>Serial Number</code>	Shows the serial number of the power supply, which uniquely identifies the power supply.
<code>PSU Status</code>	The status of the power supply. Values are: Here is your list with the numbers removed: • OK: power supply is operating normally • OK*: power supply is operating normally, although there is insufficient power to fully power the chassis. • Empty: no power supply is installed • Absent: no power supply in installed when one is expected • Input Fault: there is a fault condition on the input of the power supply • Output Fault: there is a fault condition on the output of the power supply • PoE Fault: there is a fault condition on the PoE output of the power supply • Airflow Fault: airflow direction does not match the expected chassis airflow direction • Redundancy Fault: type of capacity does not match the expected redundancy type and capacity

Parameter	Description
	• Mixed Input Fault: the input type of the inserted power supplies does not match. AC and DC power supplies cannot be mixed in the same switch. • Warning: there is an internal fault condition but the power supply is still operational • Non - recoverable Fault: there is an internal fault condition that the power supply cannot recover from • Unsupported: power supply is not of a known supported type • Unknown: power supply has experienced an unknown fault condition • Alert: power supply has experienced an abnormal event but is still operational • Undervoltage: Power supply's input voltage dips below the minimum operating voltage
input type	Power supply's input type Identification. It can be AC or DC .
Voltage Range	Valid operating voltage range for the power supply to support the current wattage maximum.
Wattage Maximum	Maximum amount of wattage that the power supply can provide

If you include the optional **input-voltage** parameter, the output includes the following information:

Output Column	Description
Mbr/PSU	Shows the member and slot number of the power supply.
Product Number	Shows the product number of the power supply.
Input Type	Power supply's input type Identification. It can be AC or DC .
Normal Input (V)	Normal operating voltage range for the power supply to support current wattage maximum - according to regulatory requirements.
Present Input (V)	Instantaneous input voltage reading acquired from power supply
Average Input	The average input voltage over the last 15 minutes (or since power was applied if less than 15 minutes ago).

Output Column	Description
Input voltage Violations: Count	Number of occurrences for which PSU's input voltage crosses outside of the normal input.
Input voltage Violations: Most Recent Event	Timestamp of the most recent violation where the PSU input voltage crosses outside of the normal input.
Recovery Count	Number of occurrences when the PSU's input voltage recovers to within the normal input range.
Recovery Time	Timestamp of the most recent input voltage's recovery.

Examples

```
switch# show environment power-supply
```

```
-----
-----
      Product  Serial          PSU          Input Voltage  Wattage
Mbr/PSU  Number  Number          Status          Type    Range
Maximum
-----
-----
1/1      JL372A   M031R002JAFC      OK*              AC      200V-240V  2700
1/2      JL372A   M031R003JAFC      Output Fault     AC      200V-240V  2430
1/3      N/A        N/A                Absent            --      --          --
1/4      N/A        N/A                Absent            --      --          --
```

```
switch# show environment power-supply
```

```
-----
-----
      Product  Serial          PSU          Input Voltage
Wattage
Mbr/PSU  Number  Number          Status          Type    Range
Maximum
-----
-----
1/1      R0X36A   CN91KMM2H3        OK              AC      200V-240V
3000
1/2      N/A        N/A                Absent            --      --          0
1/3      N/A        N/A                Absent            --      --          0
1/4      N/A        N/A                Absent            --      --          0
```

Showing example output when only one power supply is installed in a switch that supports modular power supplies:

```
switch# show environment power-supply
```


	Product	Serial	PSU		Input	Voltage
Wattage						
Mbr/PSU	Number	Number	Status		Type	Range
Maximum						

1/1	JL670A	CN97KN803F	OK		AC	200V-240V
1600						
1/2	N/A	N/A	Absent		--	-- 0

Fixed power supply switches that do not support modular power supplies will not have power supply information displayed in the output of this command, as shown in the example below.

```
switch# show environment power-supply
```


	Product	Serial	PSU		Input	Voltage
Wattage						
Mbr/PSU	Number	Number	Status		Type	Range
Maximum						

1/1	N/A	N/A	OK		AC	100V-240V --

The output of the following command shows input voltage data for a power supply. In the following example, the output of the **show environment power-supply input-voltage** command has been broken into two sections to better fit in this document. In the product command-line interface, the output appears as a single, wide table.

```
switch# show environment power-supply input-voltage
```

```
Input Voltage Averaging Period : 15 minutes
```

--						
Mbr/	Product	Input	Normal	Present	Average	Input Voltage
Violations						
PSU	Number	Type	Input (V)	Input (V)	Input (V)	Count

1/1	JL670A	AC	180-264	231.7	231.2	1
1/2	JL670A	AC	180-264	--	--	0

			Recovery	Recovery		

Most Recent Event	Count	Time
Mon Jun 23 02:45:28 UTC 2025	1	Mon Jun 23 02:45:41 UTC 2025
N/A	0	N/A

For switches that support VSF, you can include the **vsf member <MEMBER-ID>** parameter to display input voltage information for a specific VSF member. In the following example, the output of the **show environment power-supply input-voltage vsf member 1** command has been broken into separate sections to better fit in this document. In the product command-line interface, the output appears as a single, wide table.

```
switch# show environment power-supply input-voltage vsf member 2
Input Voltage Averaging Period : 15 minutes
Mbr      Product Input      Normal
/PSU     Number  Type       Input (V)
-----
2/1 JL086A  AC        90-264
2/2 JL758A  DC        40-60
Present   AverageE
Input (V)   Input (V)
-----
123.5      122.4
42.5       48.1
Input Voltage Violations:      Recovery  Recovery
Count    Most Recent Event      Count    Time
-----
--
0      N/A                      0      N/A
1      Mon Nov 18 06:37:53 UTC 2024      1      Mon Nov 18 09:01:44 UTC
2024
```

The output of the following command shows input voltage events data for a power supply in a VSF member. In the following example, the output of the **show environment power-supply input-voltage events vsf member 2** command has been broken into two sections to better fit in this document. In the product command-line interface, the output appears as a single, wide table.

```
switch# show env power-supply input-voltage events vsf member 2
Mbr/PSU      : 2/1
Product Number : JL670A
Serial Number  : CN1AL6H326
Status        : OK
Input Type     : AC
Normal Input (V) : 180-264
Present Input   : 201.8 V
Average Input   : 200.2 V
Averaging Period : 15 minutes
Max Input      : 236.0 V
Min Input      : 176.1 V
```

ID	Type	Voltage	Timestamp
0	violation	177.8	Tue Jun 11 13:20:23 UTC 2025
1	recovered	180.1	Tue Jun 17 06:51:45 UTC 2025

Mbr/PSU : 2/2
 951 Product Number : N/A
 Serial Number : N/A
 Status : Absent
 Input Type : --
 Normal Input (V) : --
 Present Input : --
 Average Input : --
 Averaging Period : 15 minutes
 Max Input : --
 Min Input : --

ID	Type	Voltage	Timestamp
No Input Voltage Violation/Recovery events			
...			



NOTE

Showing individual input voltage events for a switch power supply:

```

switch# show environment power-supply input-voltage events
Mbr/PSU : 1/2
Product Number : JL758A
Serial Number : CN1AL6H326
Status : OK
Input Type : DC
Normal Input (V) : 40-60
Present Input : 44.8 V
Average Input : 43.1 V
Averaging Period : 15 minutes
Max Input : 48.9 V
Min Input : 36.9 V

```

ID	Type	Voltage	Timestamp
0	violation	37.8	Mon Jul 21 06:37:53 UTC 2025
1	recovered	40.1	Mon Jul 21 09:01:44 UTC 2025

```

Mbr/PSU : 1/2
Product Number : JL670A
Serial Number : CN1AL6H326
Status : OK

```



```

Input Type      : AC
Normal Input (V) : 99-140
Present Input   : 100.8 V
Average Input    : 104.2 V
Averaging Period : 15 minutes
Max Input       : 119.8 V
Min Input       : 95.5 V
ID      Type      Voltage      Timestamp
-----
0      violation    96.5      Wed Jul 02 08:51:07 UTC 2025
1      recovered    99.1      Wed Jul 02 08:54:55 UTC 2025
2      violation    96.6      Wed Jul 02 09:11:14 UTC 2025
3      recovered    99.3      Wed Jul 02 10:24:19 UTC 2025
...

```

Command History

Release	Modification
10.16.1000	The input - voltage parameter is introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show environment temperature

Syntax

```

show environment temperature [detail]
                             [vsf]

```

Description

Shows the temperature information from sensors in the switch that affect fan control.

Parameter	Description
detail	Shows detailed information from each temperature sensor.
vsf	Shows output from the VSF member - id on switches that support VSF

Usage

Temperatures are shown in Celsius.

Valid values for status are the following: \

Parameter	Description
normal	Sensor is within nominal temperature range.
max	Highest temperature from this sensor.
low_critical	Lowest threshold temperature for this sensor.
critical	Highest threshold temperature for this sensor.
fault	Fault event for this sensor.
emergency	Over temperature event for Over temperature event for this sensor.

Examples

```
switch# show environment temperature
```

```
Temperature information
```

```
-----
--
Mbr/Slot-Sensor           Module Type           Current
                           temperature   Status
-----
--
1/1-PCIE-Switch           line-card-module      95.82 C    normal
1/1-Processor             line-card-module      0.00 C    fault
1/1-Switch-ASIC           line-card-module      116.36 C   emergency
1/1-Switch-ASIC-Internal  line-card-module      108.25 C   critical
```

```
1/2-PCIE-Switch          line-card-module    95.82 C    normal
```

```
...
```

```
switch# show environment temperature
```

```
Temperature information
```

```
-----  
--
```

Mbr/Slot-Sensor	Module Type	Current temperature	Status
-----------------	-------------	------------------------	--------

```
-----  
--
```

1/1-PHY-01-04	line-card-module	45.00 C	normal
1/1-PHY-05-08	line-card-module	45.00 C	normal
1/1-PHY-09-12	line-card-module	46.00 C	normal
1/1-PHY-13-16	line-card-module	47.00 C	normal
1/1-PHY-17-20	line-card-module	47.00 C	normal
1/1-PHY-21-24	line-card-module	50.00 C	normal
1/1-PHY-25-28	line-card-module	45.00 C	normal
1/1-PHY-29-32	line-card-module	47.00 C	normal
1/1-PHY-33-36	line-card-module	48.00 C	normal
1/1-PHY-37-40	line-card-module	47.00 C	normal
1/1-PHY-41-44	line-card-module	48.00 C	normal
1/1-PHY-45-48	line-card-module	49.00 C	normal
1/1-Switch-ASIC-Internal	line-card-module	56.25 C	normal
1/1-CPU-Zone-0	management-module	50.00 C	normal
1/1-CPU-Zone-1	management-module	50.00 C	normal
1/1-CPU-Zone-2	management-module	50.00 C	normal
1/1-CPU-Zone-3	management-module	51.00 C	normal
1/1-CPU-Zone-4	management-module	51.00 C	normal
1/1-CPU-diode	management-module	53.12 C	normal
1/1-DDR	management-module	45.25 C	normal
1/1-Inlet-Air	management-module	24.88 C	normal
1/1-MB-IBC	management-module	45.62 C	normal
1/1-Switch-ASIC-diode	management-module	58.06 C	normal

```
switch# show environment temperature
```

```
Temperature information
```

```
-----  
--
```

Mbr/Slot-Sensor	Module Type	Current temperature	Status
-----------------	-------------	------------------------	--------

```
-----  
--
```

1/1-COME-Daughter-Boar	line-card-module	66.45 C	normal
1/1-PCIE-Switch	line-card-module	95.82 C	normal

```

1/1-Processor          line-card-module    00.00 C    fault
1/1-Switch-ASIC        line-card-module    116.36 C   emergency
1/1-Switch-ASIC-Internal line-card-module    108.25 C   critical
1/2-COMe-Daughter-Boar line-card-module    67.29 C    normal
1/2-PCIE-Switch        line-card-module    95.82 C    normal
1/2-Processor-1        line-card-module    72.92 C    normal
1/2-Processor-2        line-card-module    73.05 C    normal
1/2-Switch-ASIC        line-card-module    97.41 C    normal
1/2-Switch-ASIC-Internal line-card-module    97.62 C    normal
...

```

```
switch# show environment temperature detail
```

```
Detailed temperature information
```

```

-----
Mbr/Slot-Sensor      : 1/1-PCIE-Switch
Module Type          : line-card-module
Module Description    : JL363A 8400X 32P 10G SFP/SFP+ Msec Mod
Status                : normal
Fan-state             : normal
Current temperature   : 95.82 C
Minimum temperature   : 93.52 C
Maximum temperature   : 96.15 C
...

```

```
Detailed temperature information
```

```

-----
Mbr/Slot-Sensor      : 1/1-PCIE-Switch
Module Type          : line-card-module
Module Description    : JL363A 8400X 32P 10G SFP/SFP+ Msec Mod
Status                : normal
Fan-state             : normal
Current temperature   : 95.82 C
Minimum temperature   : 93.52 C
Maximum temperature   : 96.15 C
...

```

Showing detailed temperature information for a 5420 or 6200 switch:

```
switch# show environment temperature
```

```
Temperature information
```

```

-----
--
Mbr/Slot-Sensor      Module Type          Current
                        temperature      Status
-----
--
1/1-PHY-01-08        line-card-module    51.00 C    normal

```

1/1-PHY-09-16	line-card-module	48.00 C	normal
1/1-PHY-17-24	line-card-module	50.00 C	normal
1/1-PHY-25-32	line-card-module	52.00 C	normal
1/1-PHY-33-40	line-card-module	53.00 C	normal
1/1-PHY-41-48	line-card-module	54.00 C	normal
1/1-Switch-ASIC-Internal	line-card-module	63.25 C	normal
1/1-CPU	management-module	47.56 C	normal
1/1-CPU-Zone-0	management-module	46.00 C	normal
1/1-CPU-Zone-1	management-module	46.00 C	normal
1/1-CPU-Zone-2	management-module	46.00 C	normal
1/1-CPU-Zone-3	management-module	47.00 C	normal
1/1-CPU-Zone-4	management-module	47.00 C	normal
1/1-DDR	management-module	37.75 C	normal
1/1-DDR-Inlet	management-module	30.69 C	normal
1/1-Inlet-Air	management-module	22.00 C	normal
1/1-Switch-ASIC-Remote	management-module	64.06 C	normal

switch# **show environment temperature detail**

Detailed temperature information

```

Mbr/Slot-Sensor      : 1/1-PHY-01-04
Module Type          : line-card-module
Module Description    : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status                : normal
Fan-state             : normal
Current temperature   : 45.00 C
Minimum temperature   : 41.00 C
Maximum temperature   : 50.00 C
Mbr/Slot-Sensor      : 1/1-PHY-05-08
Module Type          : line-card-module
Module Description    : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status                : normal
Fan-state             : normal
Current temperature   : 45.00 C
Minimum temperature   : 41.00 C
Maximum temperature   : 50.00 C

```

...

Detailed temperature information

```

Mbr/Slot-Sensor      : 1/1-PHY-01-04
Module Type          : line-card-module
Module Description    : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status                : normal
Fan-state             : normal

```

```

Current temperature : 45.00 C
Minimum temperature : 41.00 C
Maximum temperature : 50.00 C
Mbr/Slot-Sensor    : 1/1-PHY-05-08
Module Type        : line-card-module
Module Description  : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status             : normal
Fan-state          : normal
Current temperature : 45.00 C
Minimum temperature : 41.00 C
Maximum temperature : 50.00 C
...

```

Command History

Release	Modification
10.18	Adds support for 8325 Switch series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show events

Syntax

```

show events [ -e <EVENT-ID> |
    -s {emergency | alert | critical | error | warning | notice | info |
debug} |
    -r |
    -a |
    -n <COUNT> |
    -i <MEMBER-SLOT> |

```

```
-m {active | standby} |
-c {lldp | ospf | ...} |
-d {lldpd | bgpd | fand | ...}]
```

Description

Shows event logs generated by the switch modules since the last reboot.

Parameter	Description
<code>-e <EVENT-ID></code>	Shows the event logs for the specified event ID. Event ID range: 101 through 99999.
<code>-s {emergency alert critical error warning notice info debug}</code>	Shows the event logs for the specified severity. Select the severity from the following list: • emergency: Displays event logs with severity emergency only. • alert: Displays event logs with severity alert and above. • critical: Displays event logs with severity critical and above. • error: Displays event logs with severity error and above. • warning: Displays event logs with severity warning and above. • notice: Displays event logs with severity notice and above. • info: Displays event logs with severity info and above. • debug: Displays event logs with all severities.
<code>-r</code>	Shows the most recent event logs first.
<code>-a</code>	Shows all event logs, including those events from previous boots.
<code>-n <COUNT></code>	Displays the specified number of event logs.
<code>-i <MEMBER-SLOT></code>	On a 5420 or 6200: Shows the event logs for the specified VSF member ID.
<code>-m {active standby}</code>	On a 5420 or 6200: Shows the event logs for the specified role. Selecting <code>active</code> displays the event log for the VSF conductor role and <code>standby</code> displays event logs for the VSF standby role.
<code>-c {lldp ospf ...}</code>	Shows the event logs for the specified event category. Enter show event - c for a full listing of supported categories with descriptions.

Parameter	Description
<code>-d {lldpd bgpd fand ...}</code>	Shows the event logs for the specified process. Enter <code>show event -d</code> for a full listing of supported daemons with descriptions.

Examples

Showing event logs:

```
switch# show events
-----
show event logs
-----
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
in Hardware
```

Showing the most recent event logs first:

```
switch# show events -r
-----
show event logs
-----
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
in Hardware
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
up for bridge_normal interface
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
```

Showing all event logs:

```
switch# show events -a
-----
show event logs
-----
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
in Hardware
```

Showing event logs related to LACP:


```
switch# show events -c lacp
```

```
-----  
show event logs  
-----
```

```
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to  
70:72:cf:51:50:7c
```

```
switch# show events -m active
```

```
-----  
show event logs  
-----
```

```
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to  
70:72:cf:51:50:7c
```

```
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to  
up for bridge_normal interface
```

```
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1  
in Hardware
```

Showing event logs as per the specified member/slot ID:

```
switch# show events -i 1/1
```

```
-----  
show event logs  
-----
```

```
2017-08-17:22:32:25.743991|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource  
utilization poll interval is changed to 313
```

```
2017-08-17:22:33:01.692860|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource  
utilization poll interval is changed to 23
```

```
2017-08-17:22:33:06.181436|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource  
utilization poll interval is changed to 512
```

```
2017-08-17:22:33:06.181436|systemd-coredump|1201|LOG_CRIT|LC|1/1|hpe-sysmond  
crashed due to signal:11
```

Showing event logs as per the specified process:

```
switch# show events -d lacpd
```

```
-----  
show event logs  
-----
```

```
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to  
70:72:cf:51:50:7c
```

Displaying the specified number of event logs:

```
switch# show events -n 5
```

```
show event logs
```

```
-----  
2018-03-21:06:12:15.500603|arpmgrd|6101|LOG_INFO|AMM|-|ARPMGRD daemon has  
started  
2018-03-21:06:12:17.734405|lldpd|109|LOG_INFO|AMM|-|Configured LLDP tx-  
delay to 2  
2018-03-21:06:12:17.740517|lacpd|1307|LOG_INFO|AMM|-|LACP system ID set to  
70:72:cf:d4:34:42  
2018-03-21:06:12:17.743491|vrfgmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity  
42cc3df7-1113-412f-b5cb-e8227b8c22f2  
2018-03-21:06:12:17.904008|vrfgmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity  
4409133e-2071-4ab8-adfe-f9662c06b889
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

show hostname

Syntax

```
show hostname
```

Description

Shows the current host name.

Example

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show images

Syntax

```
show images
```

Description

Shows information about the software in the primary and secondary images.

Example

```
switch# show images
-----
AOS-CX Primary Image
-----

Version : XL.xx.xx.xxxx
Size    : 141 MB
Date    : 2017-06-30 14:02:34 PDT
```

SHA-256 : 2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824

AOS-CX Secondary Image

Version : XL.xx.xx.xxxx

Size : 143 MB

Date : 2017-06-30 14:02:34 PDT

SHA-256 : 2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824

Default Image : secondary

Management Module 1/5 (Active)

Active Image : primary

Service OS Version : GT.01.01.0001

BIOS Version : GT-01-0013

Management Module 1/6 (Standby)

Active Image : secondary

Service OS Version : GT.01.01.0001

BIOS Version : GT-01-0013

switch(config)# **show images**

AOS-CX Primary Image

Version : FL.xx.xx.xxxx

Size : 722 MB

Date : 2019-10-22 17:00:46 PDT

SHA-256 : 4c84e49c0961fc56b5c7eab064750a333f1050212b7ce2fab587d13469d24cfa

Primary Image

Version : FL.xx.xx.xxxx

Size : 722 MB

Date : 2019-10-22 17:00:46 PDT

SHA-256 : 4c84e49c0961fc56b5c7eab064750a333f1050212b7ce2fab587d13469d24cfa

AOS-CX Secondary Image

Version : FL.xx.xx.xxxx

Size : 722 MB

```
Date      : 2019-10-22 17:00:46 PDT
SHA-256   : 4c84e49c0961fc56b5c7eab064750a333f1050212b7ce2fab587d13469d24cfa
Default Image : secondary
```

```
-----
Management Module 1/1 (Active)
-----
```

```
Active Image      : secondary
Service OS Version : FL.01.05.0001-internal
BIOS Version      : FL.01.0001
```

```
switch(config)# show images
```

```
-----
AOS-CX Primary Image
-----
```

```
Version : FL.xx.xx.xxxxQ-2710-gd4ac39f30c9
Size     : 766 MB
Date     : 2019-10-30 17:22:01 PDT
SHA-256  : e560ca9141f425d19024d122573c5ff730df2a9a726488212263b45ea00382cf
```

```
-----
AOS-CX Secondary Image
-----
```

```
Version : FL.xx.xx.xxxx
Size     : 722 MB
Date     : 2019-10-21 19:36:26 PDT
SHA-256  : 657e28adc1b512217ce780e3523c37c94db3d3420231deac1ab9aaa8324dc6b9
Default Image : secondary
```

```
-----
Management Module 1/1 (Active)
-----
```

```
Active Image      : secondary
Service OS Version : FL.01.05.0001-internal
BIOS Version      : FL.01.0001
```

Showing the primary and secondary images on a 6200 switch:

```
switch(config)# show images
```

```
-----
AOS-CX Primary Image
-----
```

```
Version : ML.10.06.0001AAE-54-g89a7618
Size     : 446 MB
Date     : 2020-04-27 16:50:39 PDT
SHA-256  :
2e6261e26591f9a6b94be28b0890de521f0869a9dbc378659fc33d7c40fd12fc
```

```
-----
AOS-CX Secondary Image
```

```

-----
Version : ML.10.06.0001AAE-54-g89a7618
Size    : 446 MB
Date    : 2020-04-27 16:50:39 PDT
SHA-256 :
2e6261e26591f9a6b94be28b0890de521f0869a9dbc378659fc33d7c40fd12fc
Default Image : primary
-----
Management Module 1/1 (Active)
-----
Active Image      : primary
Service OS Version : ML.01.05.0001-internal
BIOS Version      : FL.01.0003

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ip errors

Syntax

```
show ip errors [{vrf <vrf-name>}|all-vrfs]
```

Description

Shows hardware and software IP error statistics for packets received by the switch since the switch was last booted. If none of the optional VRF parameters are included, the **show ip errors** command displays the current system-wide hardware IP error counters and the default VRF's software IP error counters on the switch

Parameter	Description
<code>vrf <vrf-name></code>	Optional. Display hardware and software IP error counters for a specific VRF.
<code>all-vrfs</code>	Optional. Display hardware and software IP error counters for all VRFs

Usage

IP error info about received packets is collected from each active line card on the switch and is preserved during failover events. Error counts are cleared when the switch is rebooted.

Drop reasons may include the following:

- **Malformed packet**

The packet does not conform to TCP/IP protocol standards such as packet length or internet header length.

A large number of malformed packets can indicate that there are hardware malfunctions such as loose cables, network card malfunctions, or that a DOS (denial of service) attack is occurring.

- **IP address error**

The packet has an error in the destination or source IP address. Examples of IP address errors include the following:

- The source IP address and destination IP address are the same.
- There is no destination IP address.
- The source IP address is a multicast IP address.
- The forwarding header of an IPv6 address is empty.
- There is no source IP address for an IPv6 packet.

- **Invalid TTLs**

The TTL (time to live) value of the packet reached zero. The packet was discarded because it traversed the maximum number of hops permitted by the TTL value.

TTLs are used to prevent packets from being circulated on the network endlessly.

Example

Showing the current system-wide hardware IP error counters and the default VRF's software IP error counters on the switch:

```
switch# show ip errors
System-wide (hardware) IP errors
-----
```

```

Drop reason                                Packets
-----
Malformed packets                          1
IP address errors                          10
Per-VRF (software) IP errors
-----
VRF
Drop reason                                Packets
-----
default
IPv4 maximum transmission unit              3
IPv4 time to live                          0
IPv6 maximum transmission unit              0
IPv6 hop limit                             6
-----
Malformed packets                          1
...

```

Showing the current system-wide hardware IP error counter and the software IP error counter for all VRFs on the switch.

```

switch# show ip errors all-vrfs
System-wide (hardware) IP errors
-----
Drop reason                                Packets
-----
Malformed packets                          1
IP address errors                          10
Per-VRF (software) IP errors
-----
VRF
Drop reason                                Packets
-----
default
IPv4 maximum transmission unit              3
IPv4 time to live                          0
IPv6 maximum transmission unit              0
IPv6 hop limit                             6
red
IPv4 maximum transmission unit              3
IPv4 time to live                          7
IPv6 maximum transmission unit              0
IPv6 hop limit                             6
blue
IPv4 maximum transmission unit              0

```


IPv4 time to live	1
IPv6 maximum transmission unit	0
IPv6 hop limit	6
...	

Command History

Release	Modification
10.14.1000	Command updated to display both hardware and software IP errors in the comm and output.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show module

Syntax

```
show module
```

Description

Shows information about installed line modules and management modules.



NOTE

Although this switch does not have removable modules, this command will still return information about the switch, referring to management modules and line modules.

Usage

Identifies and shows status information about the line modules and management modules that are installed in the switch.

To show the configuration information—if any—associated with that line module slot, use the **show running-configuration** command.

Status is one of the following values:

Active

This switch is the active management module.

Standby

This switch is the standby management module.

Deinitializing

The switch is being deinitialized.

Diagnostic

The switch is in a state used for troubleshooting.

Down

The switch is physically present but is powered down.

Empty

The switch hardware is not installed in the chassis.

Failed

The switch has experienced an error and failed.

Failover

This switch is a fabric module or a line module, and it is in the process of connecting to the new active management module during a management module failover event.

Initializing

The switch is being initialized.

Present

The switch hardware is installed in the chassis.

Ready

The switch is available for use.

Updating

A firmware update is being applied to the switch.

Examples

```
switch# show module
Management Modules
=====
Product                               Serial
```

```

Slot Number  Description                               Number      Status
-----
1/5  JL368A  8400 Mgmt Mod                               SG00000000  Active
1/6  JL368A  8400 Mgmt Mod                               SG00000000  Standby
Line Modules
=====
      Product                               Serial
Slot Number  Description                               Number      Status
-----
1/1  JL363A  8400X 32P 10G SFP/SFP+ Msec Mod          SG00000000  Down
1/3  JL365A  8400X 8P 40G QSFP+ Adv Mod              SG00000000  Ready
1/4  JL366A  8400X 6P 40G/100G QSFP28 Adv Mod         SG00000000  Initializing
1/8  JL365A  8400X 8P 40G QSFP+ Adv Mod              SG00000000  Ready
1/10 JL365A  8400X 8P 40G QSFP+ Adv Mod              SG00000000  Ready
...

```

```
switch(config)# show module
```

```
Management Modules
```

```

=====
      Product                               Serial
Name Number  Description                               Number      Status
-----
-----
1/1  JL581A  8320 Mgmt Mod                               TW87KCW00X  Ready
Line Modules
=====

```

```

      Product                               Serial
Name Number  Description                               Number      Status
-----
-----
1/1  JL581A  8320                               TW87KCW00X  Ready

```

```
switch(config)# show module
```

```
Management Modules
```

```

=====
      Product                               Serial
Name Number  Description                               Number      Status
-----
-----
1/1  JL659A  6300M 48SR5 CL6 PoE 4SFP56 Swch         ID9ZKHN090  Active
(local)
Line Modules
=====

```

```

      Product                               Serial
Name Number  Description                               Number      Status

```

```

-----
-----
1/1  JL659A  6300M 48SR5 CL6 PoE 4SFP56 Swch      ID9ZKHN090 Ready
Management Modules
=====
      Product                               Serial
Name Number  Description                    Number      Status
-----
-----
1/1  JL659A  6300M 48SR5 CL6 PoE 4SFP56 Swch      ID9ZKHN090 Active
(local)
Line Modules
=====
      Product                               Serial
Name Number  Description                    Number      Status
-----
-----
1/1  JL659A  6300M 48SR5 CL6 PoE 4SFP56 Swch      ID9ZKHN090 Ready

```

Showing all installed modules on a 6200 switch:

```

switch# show module
Management Modules
=====
      Product                               Serial
Name Number  Description                    Number      Status
-----
-----
1/1  JL727A  6200F 48G CL4 4SFP+370W Swch      SG0ZKW600P Active
(local)
Line Modules
=====
      Product                               Serial
Name Number  Description                    Number      Status
-----
-----
1/1  JL727A  6200F 48G CL4 4SFP+370W Swch      SG0ZKW600P Ready
switch# show module
Management Modules
=====
      Product                               Serial
Name Number  Description                    Number      Status
-----
-----
1/1  JL678A  6100F 24G 4SFP+ Swch                SG0ZKW600P Active

```

```
(local)
Line Modules
=====
      Product                               Serial
Name Number  Description                    Number      Status
-----
-----
1/1  JL678A  6100F 24G 4SFP+ Swch          SG0ZKW600P Ready
Management Modules
```

```
=====
      Product                               Serial
Name Number  Description                    Number      Status
-----
-----
1/1  JL678A  6100F 24G 4SFP+ Swch          SG0ZKW600P Active
```

```
(local)
Line Modules
=====
      Product                               Serial
Name Number  Description                    Number      Status
-----
-----
1/1  JL678A  6100F 24G 4SFP+ Swch          SG0ZKW600P Ready
```

```
switch# show module 1/5
Management module 1/5 is active
Admin state: Up
Description: 8400 Mgmt Mod
Full Description: 8400 Management Module
Serial number: SG00000000
Product number: JL368A
```

```
switch# show module 1/3
Line module 1/3 is ready
Admin state: Up
Description: 6400 24p 10GT 4SFP56 Mod
Full Description: 6400 24-port 10GBASE-T and 4-port SFP56 Module
Serial number: SG9ZKMS045
Product number: R0X42A
Power priority: 128
```

```
switch(config)# show module 1/3
Module 1/3 is not physically present
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show process monitor statistics

Syntax

```
show process monitor statistics [standby | member <SLOT-ID>]
```

Description

Shows the current status of all processes running on the switch. The status of the process information is available in the daemon table, the CLI reads each process status from the daemon table.

Parameter	Description
standby	Shows only the resource utilization data for the standby management module.
member <MEMBER-NUM>	Shows only the resource utilization data for the VSF member id identified by <MEMBER-NUM> .

Usage

For more details on the type of information displayed in the output of this command, refer to the video on the [HPE Aruba Networking Airheads Broadcasting Channel](#).

Examples

Showing process monitor statistics:

```
switch# show process monitor statistics
Process Monitor Statistics
-----
Name                               State                               Restarts Allowed
Restarts  Timeout Value           Time Started
-----
-----
irq/102-uio_vcb                    PROCESS_RUNNING                    0
0      --                        Tue May 14 12:51:46 2024
tempd                              PROCESS_RUNNING                    0
0      --                        Tue May 14 12:51:48 2024
irq/21-mmc0                        PROCESS_RUNNING                    0
0      30                       Tue May 14 12:51:51 2024
acctd                              PROCESS_RUNNING                    0
0      60                       Tue May 14 12:52:01 2024
khungtaskd                        PROCESS_RUNNING                    0
0      --                        Tue May 14 12:52:02 2024
dbus-daemon                       PROCESS_RUNNING                    3
3      --                        Tue May 14 12:52:02 2024
irq/105-uio_vcb                   PROCESS_RUNNING                    3
2      --                        Tue May 14 12:52:02 2024
kworker/0:1-events                PROCESS_RUNNING                    3
0      240                      Tue May 14 12:52:02 2024
dnsmasq                           PROCESS_RUNNING                    3
0      240                      Tue May 14 12:52:02 2024
```

Showing process monitor statistics with the standby parameter:

```
switch# show process monitor statistics standby
Process Monitor Statistics
-----
Name                               State                               Restarts Allowed
Restarts  Timeout Value           Time Started
-----
-----
irq/102-uio_vcb                    PROCESS_RUNNING                    0
0      --                        Tue May 14 12:51:46 2024
tempd                              PROCESS_RUNNING                    0
0      --                        Tue May 14 12:51:48 2024
irq/21-mmc0                        PROCESS_RUNNING                    0
0      30                       Tue May 14 12:51:51 2024
acctd                              PROCESS_RUNNING                    0
0      60                       Tue May 14 12:52:01 2024
khungtaskd                        PROCESS_RUNNING                    0
0      --                        Tue May 14 12:52:02 2024
```

```

dbus-daemon          PROCESS_RUNNING    3
3                    --          Tue May 14 12:52:02 2024
irq/105-uio_vcb      PROCESS_RUNNING    3
2                    --          Tue May 14 12:52:02 2024
kworker/0:1-events   PROCESS_RUNNING    3
0                    240         Tue May 14 12:52:02 2024
dnsmasq              PROCESS_RUNNING    3
0                    240         Tue May 14 12:52:02 2024

```

Showing process monitor statistics with the member parameter:

```

switch# show process monitor statistics member 3
Process Monitor Statistics
-----
Name                               State           Restarts Allowed
Restarts  Timeout Value  Time Started
----
-----
irq/102-uio_vcb                    PROCESS_RUNNING    0
0                    --          Tue May 14 12:51:46 2024
tempd                             PROCESS_RUNNING    0
0                    --          Tue May 14 12:51:48 2024
irq/21-mmc0                        PROCESS_RUNNING    0
0                    30         Tue May 14 12:51:51 2024
acctd                             PROCESS_RUNNING    0
0                    60         Tue May 14 12:52:01 2024
khungtaskd                        PROCESS_RUNNING    0
0                    --          Tue May 14 12:52:02 2024
dbus-daemon                       PROCESS_RUNNING    3
3                    --          Tue May 14 12:52:02 2024
irq/105-uio_vcb                    PROCESS_RUNNING    3
2                    --          Tue May 14 12:52:02 2024
kworker/0:1-events                 PROCESS_RUNNING    3
0                    240         Tue May 14 12:52:02 2024
dnsmasq                           PROCESS_RUNNING    3
0                    240         Tue May 14 12:52:02 2024

```

```

switch# show process monitor statistics module 1/4
Process Monitor Statistics
-----
Name                               State           Restarts Allowed
Restarts  Timeout Value  Time Started
----
-----
irq/102-uio_vcb                    PROCESS_RUNNING    0
0                    --          Tue May 14 12:51:46 2024

```



```

tempd                PROCESS_RUNNING    0
0                    --                Tue May 14 12:51:48 2024
irq/21-mmc0          PROCESS_RUNNING    0
0                    30                Tue May 14 12:51:51 2024
acctd                PROCESS_RUNNING    0
0                    60                Tue May 14 12:52:01 2024
khungtaskd           PROCESS_RUNNING    0
0                    --                Tue May 14 12:52:02 2024
dbus-daemon          PROCESS_RUNNING    3
3                    --                Tue May 14 12:52:02 2024
irq/105-uio_vcb      PROCESS_RUNNING    3
2                    --                Tue May 14 12:52:02 2024
kworker/0:1-events   PROCESS_RUNNING    3
0                    240                Tue May 14 12:52:02 2024
dnsmasq              PROCESS_RUNNING    3
0                    240                Tue May 14 12:52:02 2024

```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show process top cpu

Syntax

```

show process top-cpu {standby | member <MEMBER-NUMBER>}
show process top-cpu {standby | module <SLOT-NUMBER>}

```

Description

Displays the current top CPU consuming processes for the selected subsystem (management module, standby, VSF member, or line card module).



NOTE

Only the current top 10 CPU consuming processes are captured in the output. CPU process data is only refreshed when the respective utilization thresholds are crossed or updated by the daemon. If the daemon is down, the lists may be empty.

Parameters

Parameter	Description
<MEMBER-NUMBER>	Specifies the VSF member.
<SLOT-NUMBER>	Specifies the slot where the module is present.

Examples

Displaying the current top CPU consuming processes (6300 Switch series):

```
switch# show process top-cpu
Top CPU Processes
-----
Last Updated: Wed Dec 03 11:30:41 2025

PID      Name                                CPU (%)
-----
2791     dhcp-server-ada                     115.00
3296     hpe-config                          100.00
1125     ovssdb-server                       80.00
3602     hpe-policyd                         30.00
3383     hpe-tsdbd                           20.00
2682     arcd                                20.00
2862     hpe-mgmdd                           5.00
2866     hpe-sysmond                         5.00
1356     ops-switchd                         5.00
3319     containerd                          5.00
```

Displaying the current top CPU consuming processes for the standby management module:

```
switch# show process top-cpu standby
Top CPU Processes
-----
```

Last Updated: Wed Dec 03 11:30:41 2025

PID	Name	CPU (%)
2791	dhcp-server-ada	115.00
3296	hpe-config	100.00
1125	ovsdb-server	80.00

Displaying the current top CPU consuming processes for a specific member:

```
switch# show process top-cpu member 3
```

Top CPU Processes

Last Updated: Wed Dec 03 11:30:41 2025

PID	Name	CPU (%)
40825	stress-ng-cpu	70.00
40814	stress-ng-vm	65.00

Examples

Displaying the current top CPU consuming processes for a specific module:

```
switch# show process top-cpu module 1/1
```

Top CPU Processes

Last Updated: Wed Dec 03 11:30:41 2025

PID	Name	CPU (%)
40825	stress-ng-cpu	60.00

Command History

Table 1.

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Administrators or local user group members with execution rights for this command.

show process top-memory

Syntax

```
show process top-memory {standby | member <MEMBER-NUMBER>}  
show process top-memory {standby | module <SLOT-NUMBER>}
```

Description

Displays the current top memory consuming processes. Columns include percentage within captured set, user memory (KB), and kernel memory (KB) if reported.



NOTE

Memory process data is only refreshed when the respective utilization thresholds are crossed or updated by the daemon. If the daemon is down, the lists may be empty.

Parameters

Table 1.

Parameter	Description
<MEMBER-NUMBER>	Specifies the VSF member.
<SLOT-NUMBER>	Specifies the slot where the module is present.

Examples

Displaying the current top memory consuming processes:

```
switch# show process top-memory  
Top Memory Processes  
-----  
Last Updated: Wed Dec 03 11:30:41 2025
```

PID	Name	Mem (%)	UserMem (KB)	KernelMem (KB)
2791	dhcp-server-ada	72.50	120000	0
3296	hpe-config	60.00	95000	0

Displaying the current top memory consuming processes for a specific module:

```
switch# show process top-memory module 1/1
```

```
Top Memory Processes
```

```
-----
```

```
Last Updated: Wed Dec 03 11:30:41 2025
```

```

PID  Name                Mem (%)  UserMem (KB)  KernelMem (KB)
----  -
3576 prometheus          65.19    238988        0
...

```

show running-config

Syntax

```
show running-config [<FEATURE>] [all]
```

Description

Shows the current nondefault configuration running on the switch. No user information is displayed.

Parameter	Description
<code><FEATURE></code>	Specifies the name of a feature. For a list of feature names, enter the <code>show running-config</code> command, followed by a space, followed by a question mark (?).
<code>all</code>	Shows all default values for the current running configuration.

Examples

Showing the current running configuration:

```
switch> show running-config
Current configuration:
!
!Version AOS-CX 10.0X.XXXX
!
lldp enable
linecard-module LC1 part-number JL363A

                vrf default
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf green
    area 0.0.0.0
router pim vrf green
    enable
    rp-address 30.0.0.4
vlan 1
    no shutdown
vlan 20
    no shutdown
vlan 30
    no shutdown
interface 1/1/1
    no shutdown
    no routing
    vlan access 30
interface 1/1/32
    no shutdown
    no routing
    vlan access 20
interface bridge_normal-1
```

```

    no shutdown
interface bridge_normal-2
    no shutdown
interface vlan20
    no shutdown
    vrf attach green
    ip address 20.0.0.44/24
    ip ospf 1 area 0.0.0.0
    ip pim-sparse enable
interface vlan30
    no shutdown
    vrf attach green
    ip address 30.0.0.44/24
    ip ospf 1 area 0.0.0.0
    ip pim-sparse enable
    ip pim-sparse hello-interval 100

```

Showing the current running configuration in json format:

```

switch> show running-config json
Running-configuration in JSON:
{
  "Monitoring_Policy_Script": {
    "system_resource_monitor_mm1.1.0": {
      "Monitoring_Policy_Instance": {
        "system_resource_monitor_mm1.1.0/
system_resource_monitor_mm1.1.0.default": {
          "name": "system_resource_monitor_mm1.1.0.default",
          "origin": "system",
          "parameters_values": {
            "long_term_high_threshold": "70",
            "long_term_normal_threshold": "60",
            "long_term_time_period": "480",
            "medium_term_high_threshold": "80",
            "medium_term_normal_threshold": "60",
            "medium_term_time_period": "120",
            "short_term_high_threshold": "90",
            "short_term_normal_threshold": "80",
            "short_term_time_period": "5"
          }
        }
      }
    },
    ...
    ...

```

```
...  
...
```

Show the current running configuration without default values:

```
switch(config)# show running-config  
Current configuration:  
!  
!Version AOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty  
led locator on  
!  
!  
!  
!  
!  
!  
!  
!  
!  
vlan 1  
switch(config)# show running-config all  
Current configuration:  
!  
!Version AOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty  
led locator on  
!  
!  
!  
!  
!  
!  
!  
!  
!  
vlan 1  
switch(config)#  
switch(config)# show running-config  
Current configuration:  
!  
!Version AOS-CX PL.10.06.0001-346-g56a12a8f4cf15  
!export-password: default  
!  
!  
!  
!
```



```
ssh server vrf default
vlan 1
spanning-tree
interface 1/1/1
    no shutdown
    vlan access 1
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
```

```

    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
!
!
!
!
!
https-server vrf default
switch#
switch#
g56a12a8f4cf15
!export-password: default
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface 1/1/1
    no shutdown
    vlan access 1
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6

```

PL.10.06.0001-346-

```

        no shutdown
        vlan access 1
interface 1/1/7
        no shutdown
        vlan access 1
interface 1/1/8
        no shutdown
        vlan access 1
interface 1/1/9
        no shutdown
        vlan access 1
interface 1/1/10
        no shutdown
        vlan access 1
interface 1/1/11
        no shutdown
        vlan access 1
interface 1/1/12
        no shutdown
        vlan access 1
interface 1/1/13
        no shutdown
        vlan access 1
interface 1/1/14
        no shutdown
        vlan access 1
interface 1/1/15
        no shutdown
        vlan access 1
interface 1/1/16
        no shutdown
        vlan access 1
interface vlan 1
        ip dhcp
!
!
!
!
!
https-server vrf default
switch#
switch#

```

Show the current running configuration with default values:

```

switch(config)# snmp-server vrf mgmt
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
snmp-server vrf mgmt
!
!
!
!
!
vlan 1
switch(config)#
switch(config)#
switch(config)# show running-config all
Current configuration:
!
!Version AOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
snmp-server vrf mgmt
snmp-server agent-port 161
snmp-server community public
!
!
!
!
!
vlan 1
switch(config)#

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config current-context

Syntax

```
show running-config current-context
```

Description

Shows the current non-default configuration running on the switch in the current command context.

Usage

You can enter this command from the following configuration contexts:

- Any child of the global configuration (**config**) context. If the child context has instances—such as interfaces—you can enter the command in the context of a specific instance. Support for this command is provided for one level below the **config** context. For example, entering this command for a child of a child of the `config` context not supported. If you enter the command on a child of the **config** context, the current configuration of that context and the children of that context are displayed.
- The global configuration (**config**) context. If you enter this command in the global configuration (**config**) context, it shows the running configuration of the entire switch. Use the **show running-configuration** command instead.

Examples

Showing the running configuration for the current interface:

```
switch(config-if) # show running-config current-context
interface 1/1/1
```

```
no shutdown
description Example interface
vlan access 1
exit
```

Showing the current running configuration for the management interface:

```
switch(config-if-mgmt)# show running-config current-context
interface mgmt
    no shutdown
    ip static 10.0.0.1/24
    default-gateway 10.0.0.8
    nameserver 10.0.0.1
switch(config)# interface vlan 1
switch(config-if-vlan)#description IN-BAND Management Interface
switch(config-if-vlan)#ip dhcp
switch(config-if-vlan)#no shutdown
switch(config-if-vlan)#end
switch#
switch(config)# interface vlan 1
switch(config-if-vlan)#description IN-BAND Management Interface
switch(config-if-vlan)#no ip dhcp
switch(config-if-vlan)#ip address 192.168.10.1/24
switch(config-if-vlan)#no shutdown
switch(config-if-vlan)#end
switch#
```

Showing the running configuration for the external storage share named **nasfiles**:

```
switch(config-external-storage-nasfiles)# show running-config current-context
external-storage nasfiles
    address 192.168.0.1
    vrf default
    username nasuser
    password ciphertext
    AQBapalKj+XMsZumHEwIc9OR6YcOw5Z6Bh9rV+9ZtKDKzvbaBAAAAB1CTrM=
    type scp
    directory /home/nas
    enable
switch(config-external-storage-nasfiles)#
switch(config-vsx)# show run current-context
vsx
    inter-switch-link 1/1/1
    role secondary
    vsx-sync sflow time
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code> or a child of <code>config</code> . See Usage.	Administrators or local user group members with execution rights for this command.

show startup-config

Syntax

```
show startup-config [json]
```

Description

Shows the contents of the startup configuration.



NOTE

Switches in the `factory-default` configuration do not have a startup configuration to display.

Parameter	Description
<code>json</code>	Display output in JSON format.

Examples

```
switch# show startup-config
Startup configuration:
!
```

```

!Version AOS-CX XL.xx.xx.xxxx
!export-password: default
hostname BLDG03-F1
user admin group administrators password ciphertext AQBapfYzQNiCPJ/
JNSu7YNfaCzlWEnWFlwLfkARd8OG6yPpcYgAAABXRe9joTRtF1S1b4b09teMfMN3POKdQk+
br6SjSXGG40BB3ilZ8ym9qqSRkr84FEdq6w5uR2IGciVC5tBnZMrWCim0KR20XcQ5rFj/
TMBvTkHq8bJpSYG8nMh
module 1/3 product-number j1365a
module 1/1 product-number j1363a
module 1/2 product-number j1363a
cli-session
    timeout 0
!
!
!
ssh server vrf mgmt
ssh certified-algorithms-only
!
!
!
!
!
vlan 1,195,197-200,4000
interface mgmt
    no shutdown
    ip static 10.6.9.24/24
    default-gateway 10.6.9.1
interface 1/1/1
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed 1,197-200
interface vlan200
    ip address 10.3.200.3/24
ip route 0.0.0.0/0 10.3.200.1
https-server rest access-mode read-write
https-server vrf mgmt
Leaf2(config)# show startup-config
Startup configuration:
!
!Version AOS-CX TL.xx.xx.xxxx
hostname Leaf2
user admin group administrators password ciphertext

```



```

AQBapaGi+KZp4g8gw63UqK+zCtvO5zigFLv2DFBEH+lztqjdYgAAABwrJ+5GayUWArgv9tVfo9Az
MY6gmI7x/
KBekGBJDXjpFson2qM83CXBUI673qWHDQ0pEIZXeug0XogCVuId4oZiQVZlOe2MfxnqZL+E9hXa
MNVowBwbD0
cli-session
    timeout 0
!
!
!
ssh server vrf mgmt
switch(config)# show startup-config
Startup configuration:
!
!Version AOS-CX FL.xx.xx.xxxx
!export-password: default
hostname BLDG01-F1
user admin group administrators password ciphertext
AQBapWl8I2ZunZ43NE/8KlbQ7zYC4gTT6uSFYi6n6wyY9PdBYgAAACONCR/
3+AcNvzRBch0DoG7W9z84LpJA+6C9SKfNwCqi5/
nUPk/ZOvN91/
EQXvPNkHtBtQWYyZqfkebbEH78VVRHfWZjApv4II9qmQfxpA79wEvzshdzZmuAKrm
user ateam group administrators password ciphertext
AQBapcPqMXoF+H10NKrqAedXLvlSRwf4wUEL22hXGD6ZBhicygAAAGsbh70DKglu+ZelwxgmDXjk
GO3bseYiR3LKQg66vrfrqR/
M3oLlllPdZDnq9XMMvCL+7jBbYhYes8+uDxuSTh8kdkd/
qj3lo5FUuC5fENgCjU0YI117qtU+YEnsJ
!
!
!
!
radius-server host 10.10.10.15
!
radius dyn-authorization enable
ssh server vrf default
ssh server vrf mgmt
!
!
!
!
!
!
router ospf 1
    router-id 1.63.63.1
    area 0.0.0.0

```

```

vlan 1
vlan 66
    name vlan66
vlan 67
    name vlan67
vlan 999
    name vlan999
vlan 4000
spanning-tree
interface mgmt
    no shutdown
    ip static 10.6.9.15/24
    default-gateway 10.6.9.1

```

Showing the startup-configuration in non-JSON format for a 6200 switch:

```

switch# show startup-config
Startup configuration:
!
!Version AOS-CX ML.10.05.0001L
!export-password: default
no lldp
user admin group administrators password ciphertext
AQBapeGVVYk45m4sYxZDE6ufzB2CWmH2wDncy5Can9iEFZjmYgAAAMl3lOWIxEExNwi3xahHktOL6
81amYg/yg2ezRrlbMUtlU7fVlASiVbmIq8gVUj0lQ4STp9/
su3pnopopuWPxwk765zqofKyhL0E0Gj9yhoxCeZfyNeNpNbm6upKjC5LrfZt9
cli-session
    timeout 0
!
!

```

Showing the startup-configuration in non-JSON format for a 5420 switch:

```

switch# show startup-config
Startup configuration:
!
!Version AOS-CX BL.10.15.0001L
!export-password: default
no lldp
user admin group administrators password ciphertext
Pxwk76ofKyhm4sYxZDE6ufzBCWmH2wDncyCan9iEFZjmYgAAAMl3lOWIxEExNwi3xahHktOL681am
Yg/yg2ezRrlbMUjmYgAlASiVbmIq8gVUj0lQ4STp9/
su3pnopopuWPxwk765zqLE0Gj9yhoxCeZfyNeNpNbm6upKjC5LrfZt9
cli-session
    timeout 0
!
!

```

Showing the startup-configuration in JSON format:

```
switch# show startup-config json
Startup configuration:
{
  "AAA_Server_Group": {
    "local": {
      "group_name": "local"
    },
    "none": {
      "group_name": "none"
    }
  },
  ...
}
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show system

Syntax

```
show system [serviceos password-prompt]
```

Description

Shows general status information about the system.

Parameter	Description
serviceos password-prompt	Shows the Service OS password prompt status.

Usage

CPU utilization represents the average utilization across all the CPU cores.

System Description, System Contact, and System Location can be set with the `snmp-server` command.

Examples

Showing system information:

```
switch# show system
Hostname           : switch
System Description  : switch description
System Contact     : contact
System Location    : location
Vendor            : Aruba
Product Name       : XXXXXX ...
Chassis Serial Nbr : XXXXXXXXXXXX
Base MAC Address   : xxxxxxx-xxxxxx
AOS-CX Version     : XX.99.99.9999
Time Zone          : UTC
Up Time            : 1 week, 5 hours, 28 minutes
CPU Util (%)       : 5
CPU Util (% avg 1 min) : 11
CPU Util (% avg 5 min) : 10
Memory Usage (%)   : 35
```

Showing the Service OS password prompt status:

```
switch# show system serviceos password-prompt
password-prompt: disabled
switch# show system
Hostname           : vsx-primary
System Description  : switch description
System Contact     : contact
System Location    : location
Vendor            : Aruba
Product Name       : XXXXXX ...
Chassis Serial Nbr : XXXXXXXXXXXX
Base MAC Address   : xxxxxxx-xxxxxx
Hostname           : vsx-primary
System Description  : switch description
```

```

System Contact      : contact
System Location     : location
Vendor              : Aruba
Product Name        : XXXXXXXX ...
Chassis Serial Nbr  : XXXXXXXXXXXX
Base MAC Address    : xxxxxxxx-xxxxxxx
AOS-CX Version      : XX.99.99.9999
Time Zone           : UTC
Up Time             : 1 week, 2 hours, 15 minutes
CPU Util (%)        : 15
CPU Util (% avg 1 min) : 12
CPU Util (% avg 5 min) : 8
Memory Usage (%)    : 37
switch# show system vsx-peer
Hostname            : vsx-secondary
System Description   : switch description
System Contact       : contact
System Location      : location
Vendor              : Aruba
Product Name        : XXXXXXXX ...
Chassis Serial Nbr  : XXXXXXXXXXXX
Base MAC Address    : xxxxxxxx-xxxxxxx
AOS-CX Version      : XX.99.99.9999
Time Zone           : UTC
CPU Util (%)        : 7
CPU Util (% avg 1 min) : 13
CPU Util (% avg 5 min) : 9
Memory Usage (%)    : 32

```

Command History

Release	Modification
10.12	Added CPU Util (% avg 1 min) and CPU Util (% avg 5 min).
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show system resource-utilization

Syntax

```
show system resource-utilization [all | daemon <DAEMON-NAME>] |  
standby | member <MEMBER-NUM>]
```

Description

Shows the system resource utilization data.

Parameter	Description
all	Shows the resource utilization data for the entire switch.
daemon <DAEMON-NAME>	Shows only the resource utilization data for the process identified by <DAEMON-NAME> .
standby	Shows only the resource utilization data for the standby management module.
member <MEMBER-NUM>	Shows only the resource utilization data for the VSF member identified by <MEMBER-NUM> .

Usage

For a list of daemons that log events, enter `show events -d ?` from a switch prompt in the manager (#) context.

Examples

Showing system resource utilization data:

```

switch# show system resource-utilization
System Resources:
Processes                          : 144
CPU usage(%)                       : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%)                    : 22
Open FD's                          : 1358
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal
Data written to various partitions since boot
Nos      : 5 MB
Log       : 1 MB
Coredump : 23 MB
Security  : 2 MB
Selftest  : 405 KB
Swap      : 14 MB
Storage partition usage(%)
Nos      : 5
Log       : 60
Coredump : 23
Security  : 2
Selftest  : 1
Swap      : 0
Process          CPU Usage(%)   Memory Usage(%)   Open FD's
-----
hpe-sysmond      1             2             11
hpe-mgmdd        0             1             5
...

```

```

switch# show system resource-utilization
System Resources:
Processes                          : 144
CPU usage(%)                       : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%)                    : 22
Open FD's                          : 1358
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal
Data written to various partitions since boot
Nos      : 5 MB
Log       : 1 MB
Coredump : 23 MB

```

```

Security :    2 MB
Selftest : 405 KB
Swap      :   14 MB
Storage partition usage(%)
Nos       :    5
Log       :   60
Coredump  :   23
Security  :    2
Selftest  :    1
Swap      :    0

```

Attempting to show resource utilization data when system resource utilization polling is disabled:

```

switch# show system resource-utilization
System resource utilization data poll is currently disabled

```

Showing the resource utilization data for a particular process:

```

switch# show system resource-utilization daemon hpe-sysmond
Process                CPU Usage(%)    Memory Usage(%)    Open FD's
-----
hpe-sysmond             1                2                11
switch# show system resource-utilization all
-----
Resource utilization data for Management Module
-----

System Resources:
Processes                :   244
CPU usage(%)             :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)          :   11
Open FD's                : 1854
Storage 1: Endurance utilization = 0-10% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal
Data written to various partitions since boot
Nos       :   15 MB
Log       :    1 MB
Coredump  :   23 MB
Security  :    2 MB
Selftest  : 405 KB
Swap      :    0 KB
Storage partition usage(%)
Nos       :    5
Log       :   60
Coredump  :   23
Security  :    2

```



```

Selftest : 1
Swap : 0
Process CPU Usage(%) Memory Usage(%) Open FD's
-----
(sd-pam) 0 0 7
aautilspamcfg 0 1 10
...
switch# show system resource-utilization all
-----
Resource utilization data for Management Module
-----
System Resources:
Processes : 244
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 1854
Storage 1: Endurance utilization = 0-10% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal
Data written to various partitions since boot
Nos : 15 MB
Log : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 405 KB
Swap : 0 KB
Storage partition usage(%)
Nos : 5
Log : 60
Coredump : 23
Security : 2
Selftest : 1
Swap : 0

```

Showing resource utilization data for all VSF members:

```
aaa
```

Showing resource utilization data for a particular VSF member:

```

switch# show system resource-utilization member 2
-----
Resource utilization data for vsf member 2
-----
System Resources:
Processes : 244

```

```

CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 1854
Storage 1: Endurance utilization = 0-10% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal
Data written to various partitions since boot
Nos : 15 MB
Log : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 0 KB
Swap : 0 MB
Storage partition usage(%)
Nos : 5
Log : 60
Coredump : 23
Security : 2
Selftest : 1
Swap : 0
Process CPU Usage(%) Memory Usage(%) Open FD's
-----
(sd-pam) 0 0 7
agetty 0 0 4
ata_sff 0 0 0
...
switch# show system resource-utilization all
-----
Resource utilization data for vsf member 1
-----
System Resources:
Processes : 244
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 1854
Storage 1: Endurance utilization = 0-10% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal
Data written to various partitions since boot
Nos : 15 MB
Log : 1 MB

```

```

Coredump : 23 MB
Security : 2 MB
Selftest : 405 KB
Swap : 14 MB
Storage partition usage(%)
Nos : 5
Log : 60
Coredump : 23
Security : 2
Selftest : 1
Swap : 0
Process CPU Usage(%) Memory Usage(%) Open FD's
-----
(sd-pam) 0 0 7
aaa
utilspamcfg 0 1 10
-----
Resource utilization data for vsf member 2
-----
System Resources:
Processes : 244
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 1854
Storage 1: Endurance utilization = 0-10% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal
Data written to various partitions since boot
Nos : 15 MB
Log : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 0 KB
Swap : 0 MB
Storage partition usage(%)
Nos : 5
Log : 60
Coredump : 23
Security : 2
Selftest : 1
Swap : 0
Process CPU Usage(%) Memory Usage(%) Open FD's

```

```

-----
(sd-pam)                0                0                7
agetty                  0                0                4
ata_sff                 0                0                0
switch# show system resource-utilization standby
System Resources:
Processes                : 244
CPU usage(%)             : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%)          : 11
Open FD's                : 1854
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b)
Health = normal
Data written to various partitions since boot
Nos      : 15 MB
Log      : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 405 KB
Swap     : 14 MB
Storage partition usage(%)
Nos      : 5
Log      : 60
Coredump : 23
Security : 2
Selftest : 1
Swap     : 0
Process          CPU Usage(%)   Memory Usage(%)   Open FD's
-----
hpe-sysmond      1             2             11
hpe-mgmdd        0             1             5
...
switch# show system resource-utilization module 1/5
-----
System Resource utilization for line card module: 1/5
-----
CPU usage(%)                : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%)              : 11
Open FD's                   : 754

```

```

switch# show system resource-utilization all
-----
Resource utilization data for Management Module
-----

System Resources:
Processes                : 244
CPU usage(%)             : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%)          : 11
Open FD's                : 1854
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),
Health = normal
Data written to various partitions since boot
Nos      : 15 MB
Log      : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 405 KB
Swap     : 14 MB
Storage partition usage(%)
Nos      : 5
Log      : 60
Coredump : 23
Security : 2
Selftest : 1
Swap     : 0
Process          CPU Usage(%)   Memory Usage(%)   Open FD's
-----
(sd-pam)         0             0                 7
aaa
utilspamcfg      0             1                 10
-----

Resource utilization data for Standby Management Module
-----

System Resources:
Processes                : 244
CPU usage(%)             : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%)          : 11
Open FD's                : 1854
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),

```

```

    Health = normal
Data written to various partitions since boot
Nos      : 15 MB
Log      : 1 MB
Coredump : 23 MB
Security : 2 MB
Selftest : 405 KB
Swap     : 0 KB
Storage partition usage(%)
Nos      : 5
Log      : 60
Coredump : 23
Security : 2
Selftest : 1
Swap     : 0
-----
System Resource utilization for line card module: 1/7
-----
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 854
-----
System Resource utilization for line card module: 1/8
-----
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 980
switch# show system resource-utilization standby
System Resources:
Processes : 244
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 1854
Storage 1: Endurance utilization = 18% (ssd), Health = normal
Data written to various partitions since boot
Nos      : 15 MB
Log      : 1 MB

```

```

Coredump : 23 MB
Security : 2 MB
Selftest : 405 KB
Swap : 14 MB
Storage partition usage(%)
Nos : 5
Log : 60
Coredump : 23
Security : 2
Selftest : 1
Swap : 0
Process CPU Usage(%) Memory Usage(%) Open FD's
-----
hpe-sysmond 1 2 11
hpe-mgmdd 0 1 5
...

```

```
switch# show system resource-utilization module 1/1
```

```
-----
System Resource utilization for line card module: 1/1
-----
```

```

CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 754
Data written to various partitions since boot
Coredump : 23 MB
Storage partition usage(%)
Coredump : 45

```

```
switch# show system resource-utilization all
```

```
-----
Resource utilization data for Management Module
-----
```

```

System Resources:
Processes : 244
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 1854
Storage 1: Endurance utilization = 17% (ssd), Health = normal
Data written to various partitions since boot
Nos : 15 MB

```

```

Log      :    1 MB
Coredump :   23 MB
Security :    2 MB
Selftest :  405 KB
Swap     :    0 KB
Storage partition usage(%)
Nos      :    5
Log      :   60
Coredump :   23
Security :    2
Selftest :    1
Swap     :    0
Process          CPU Usage(%)   Memory Usage(%)   Open FD's
-----
(sd-pam)          0             0             7
aaa
utilspamcfg       0             1             10
-----
Resource utilization data for Standby Management Module
-----
System Resources:
Processes          :   244
CPU usage(%)       :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)    :   11
Open FD's          :  1854
Storage 1: Endurance utilization = 17% (ssd), Health = normal
Data written to various partitions since boot
Nos      :   15 MB
Log      :    1 MB
Coredump :   23 MB
Security :    2 MB
Selftest :  405 KB
Swap     :    0 KB
Storage partition usage(%)
Nos      :    5
Log      :   60
Coredump :   23
Security :    2
Selftest :    1
Swap     :    0
-----

```


System Resource utilization for line card module: 1/7

```
-----
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 480
Data written to various partitions since boot
Coredump : 23 MB
Storage partition usage(%)
Coredump : 23
-----
```

System Resource utilization for line card module: 1/8

```
-----
CPU usage(%) : 10
CPU usage(% average over 1 minute) : 11
CPU usage(% average over 5 minute) : 15
Memory usage(%) : 11
Open FD's : 485
Data written to various partitions since boot
Coredump : 23 MB
Storage partition usage(%)
Coredump : 47
-----
```

Showing memory usage data for all switches:

Memory usage in kilobytes:

```
Physical Memory : 7274948 KB
Reserved Memory : 1580224 KB
Used Memory : 1614416 KB
Free Memory : 5660532 KB
```

Command History

Release	Modification
10.17.1000	Added the following new fields: physical memory, reserved memory, used memory, and free memory.
10.12	The output of this command includes CPU usage(% average over 1 minute) and CPU usage(% average over 5 minute) .
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show tech

Syntax

```
show tech [basic | <FEATURE>] [local-file]
```

Description

Shows detailed information about switch features by automatically running the `show` commands associated with the feature. If no parameters are specified, the `show tech` command shows information about all switch features. Technical support personnel use the output from this command for troubleshooting.

Parameter	Description
<code>basic</code>	Specifies showing a basic set of information.
<code><FEATURE></code>	Specifies the name of a feature. For a list of feature names, enter the <code>show tech</code> command, followed by a space, followed by a question mark (?).
<code>local-file</code>	Shows the output of the <code>show tech</code> command to a local text file.

Usage

To terminate the output of the `show tech` command, enter **Ctrl+C**.

If the command was not terminated with **Ctrl+C**, at the end of the output, the `show tech` command shows the following:

- The time consumed to execute the command.
- The list of failed `show` commands, if any.

To get a copy of the local text file content created with the show tech command that is used with the local-file parameter, use the `copy show-tech local-file` command.

Example

Showing the basic set of system information:

```
switch# show tech basic
=====
Show Tech executed on Wed Sep  6 16:50:37 2017
=====
[Begin] Feature basic
=====
*****
Command : show core-dump all
*****
no core dumps are present
...
[End] Feature basic
=====
1 show tech command failed
=====
Failed command:
  1. show boot-history
=====
Show tech took 3.000000 seconds for execution
```

Directing the output of the **show tech basic** command to the local text file:

```
switch# show tech basic local-file
Show Tech output stored in local-file. Please use 'copy show-tech local-
file'
to copy-out this file.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show usb

Syntax

```
show usb
```

Description

Shows the USB port configuration and mount settings.

Examples

If USB has not been enabled:

```
switch> show usb
Enabled: No
Mounted: No
```

If USB has been enabled, but no device has been mounted:

```
switch> show usb
Enabled: Yes
Mounted: No
```

If USB has been enabled and a device mounted:

```
switch> show usb
Enabled: Yes
Mounted: Yes
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show usb file-system

Syntax

```
show usb file-system [<PATH>]
```

Description

Shows directory listings for a mounted USB device. When entered without the <PATH> parameter the top level directory tree is shown.

Parameter	Description
<PATH>	Specifies the file path to show. A leading "/" in the path is optional.

Usage

Adding a leading "/" as the first character of the <PATH> parameter is optional.

Attempting to enter '.' as any part of the <PATH> will generate an invalid path argument error. Only fully-qualified path names are supported.

Examples

Showing the top level directory tree:

```
switch# show usb file-system
/mnt/usb:
'System Volume Information'  dir1
/mnt/usb/System Volume Information':
IndexerVolumeGuid  WPSettings.dat
/mnt/usb/dir1:
dir2  test1
```

```
/mnt/usb/dir1/dir2:
test2
```

Showing available path options from the top level:

```
switch# show usb file-system /
total 64
drwxrwxrwx 2 32768 Jan 22 16:27 'System Volume Information'
drwxrwxrwx 3 32768 Mar  5 15:26 dir1
```

Showing the contents of a specific folder:

```
switch# show usb file-system /dir1
total 32
drwxrwxrwx 2 32768 Mar  5 15:26 dir2
-rwxrwxrwx 1      0 Feb  5 18:08 test1
switch# show usb file-system dir1/dir2
total 0
-rwxrwxrwx 1 0 Feb  6 05:35 test2
```

Attempting to enter an invalid character in the path:

```
switch# show usb file-system dir1/../../../../
Invalid path argument
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show version

Syntax

```
show version
```

Description

Shows version information about the network operating system software, service operating system software, and BIOS.

Example

```
DocTeam-4100i# show version
-----
-
ArubaOS-CX
(c) Copyright 2017-2023 Hewlett Packard Enterprise Development LP
-----
-
Version      : RL.10.13.1000S
Build Date   : 2023-11-15 15:29:10 UTC
Build ID     : ArubaOS-CX:RL.10.13.1000S:ac4c3941a38c:202311151508
Build SHA    : ac4c3941a38c5d66f8f0212706af8f35fa01adfd
Hot Patches  :
Active Image : secondary
Service OS Version : RL.01.09.0003
BIOS Version   : RL.01.0003
```

Showing version information for a 6200 switch:

```
switch(config)# show version
-----
-
AOS-CX
(c) Copyright 2017-2022 Hewlett Packard Enterprise Development LP
-----
-
Version      : ML.xx.xx.xxxx
Build Date   : 2022-05-27 17:00:46 PDT
Build ID     : AOS-CX:ML.xx.xx.xxxx:85c3c2f3d59e:201910222335
Build SHA    : 85c3c2f3d59ec8318ba97178fad387aecb671b33
Active Image : secondary
Service OS Version : ML.01.05.0001
BIOS Version   : ML.01.0001
```

Showing version information for a 5420 switch:

```
switch(config)# show version
-----
```

```
-
AOS-CX
(c) Copyright 2024 Hewlett Packard Enterprise Development LP
-----
-
Version      : BL.xx.xx.xxxx
Build Date   : 2024-11-11 17:00:46 PDT
Build ID     : AOS-CX:BL.xx.xx.xxxx:85c3c2f3d59e:201910222335
Build SHA    : 97178fd59ec8318ba97178fad387aecb671b33
Active Image : secondary
Service OS Version : BL.0X.XX.XXXX
BIOS Version      : BL.XX.XXXXXX
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

system resource-utilization poll-interval

Syntax

```
system resource-utilization poll-interval <SECONDS>
```

Description

Configures the polling interval for system resource information collection and recording such as CPU and memory usage.

Parameter	Description
<SECONDS>	Specifies the poll interval in seconds. Range: 10 - 3600. Default : 10.

Example

Configuring the system resource utilization poll interval:

```
switch(config)# system resource-utilization poll-interval 20
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

top cpu

Syntax

```
top cpu
```

Description

Shows CPU utilization information.

Example

Showing top CPU information:

```
switch# top cpu
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks: 76 total, 2 running, 74 sleeping, 0 stopped, 0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total, 2487508 free, 897040 used, 661948 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 2859196 avail Mem
  PID USER      PRI  NI   VIRT   RES   SHR S  %CPU  %MEM     TIME+ COMMAND
...
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

top memory

Syntax

```
top memory
```

Description

Shows memory utilization information.

Example

Showing top memory:

```
switch> top memory
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks: 76 total, 2 running, 74 sleeping, 0 stopped, 0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total, 2487508 free, 897040 used, 661948 buff/cache
```

```
KiB Swap:      0 total,      0 free,      0 used, 2859196 avail Mem
  PID USER      PRI  NI   VIRT   RES   SHR  S   %CPU  %MEM     TIME+  COMMAND
...
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

usb

Syntax

```
usb
no usb
```

Description

Enables the USB ports on the switch. This setting is persistent across switch reboots and management module failovers. Both active and standby management modules are affected by this setting.

The **no** form of this command disables the USB ports.

Example

Enabling USB ports:

```
switch(config) # usb
```

Disabling USB ports when a USB drive is mounted:

```
switch(config) # no usb
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

usb mount | unmount

Syntax

```
usb {mount | unmount}
```

Description

Enables or disables the inserted USB drive.

Parameter	Description
mount	Enables the inserted USB drive.
unmount	Disables the inserted USB drive in preparation for removal.

Usage

If USB has been enabled in the configuration, the USB port on the active management module is available for mounting a USB drive. The USB port on the standby management module is not available.

An inserted USB drive must be mounted each time the switch boots or fails over to a different management module.

A USB drive must be unmounted before removal.

The supported USB file systems are FAT16 and FAT32.

Examples

Mounting a USB drive in the USB port:

```
switch# usb mount
```

Unmounting a USB drive:

```
switch# usb unmount
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Support and Other Resources

Access HPE Aruba Networking support and updates, and view warranty and regulatory information

Subtopics

- [Accessing HPE Aruba Networking Support](#)
- [Accessing Updates](#)
- [Warranty Information](#)
- [Regulatory Information](#)
- [Documentation Feedback](#)

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.hpe.com/us/en/networking/hpe - aruba - networking - support - services.html
AOS - CX Switch Software Documentation Portal	https://arubanetworking.hpe.com/techdocs/ArubaDocPortal/content/new-portal/aoscx.html
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	https://www.hpe.com/psnow/doc/a50011948enw

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/

AOS - CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release : https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End - of - Life information	https://networkingsupport.hpe.com/end - of - life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at [**https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**](https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts)

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see [**https://www.arubanetworks.com/company/about-us/environmental-citizenship/**](https://www.arubanetworks.com/company/about-us/environmental-citizenship/).

Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback ([**docsfeedback-switching@hpe.com**](mailto:docsfeedback-switching@hpe.com)). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.